GREEN
CLIMATE
FUND

Meeting of the Board
10 – 13 July 2023
Songdo, Incheon, Republic of Korea
Provisional agenda item 12

GCF/B.36/02/Add.01

19 June 2023

Consideration of funding proposals – Addendum I

Funding proposal package for FP206

Summary

This addendum contains the following seven parts:

a) A funding proposal titled "Resilient Homestead and Livelihood support to the vulnerable coastal people of Bangladesh (RHL)";
b) No-objection letter issued by the national designated authority(ies) or focal point(s);
c) Environmental and social report(s) disclosure;
d) Secretariat's assessment;
e) Independent Technical Advisory Panel's assessment;
f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
g) Gender documentation.

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Funding Proposal

Project/Programme title:	Resilient Homestead and Livelihood support to the vulnerable coastal people of Bangladesh (RHL).
Country(ies):	Bangladesh
Accredited Entity:	Palli Karma-Sahayak Foundation (PKSF).
Date of first submission:	2018/03/18
Date of current submission	[2023/06/06]
Version number	V.7

GREEN CLIMATE FUND

Contents

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the GCF Information Disclosure Policy, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-RHL-PKSF-Bangladesh-2023/06/06”

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Project	**A.2. Public or private sector**	Public	
A.3. Request for Proposals (RFP)	Not applicable			
A.4. Result area(s)			**GCF contribution**	**Co-financers' contribution[1]**
	Mitigation total		Enter number%	Enter number%
	☐Energy generation and access		Enter number%	Enter number%
	☐Low-emission transport		Enter number%	Enter number%
	☐Buildings, cities, industries and appliances		Enter number%	Enter number%
	☐Forestry and land use		Enter number%	Enter number%
	Adaptation total		Enter number per cent	Enter number %
	☒ Most vulnerable people and communities		18.99 %	0 %
	☒ Health and well-being, and food and water security		36.64 %	100 %
	☒ Infrastructure and built environment		44.37 %	Enter number %
	☐ Ecosystems and ecosystem services		Enter number %	Enter number %
A.5. Expected mitigation outcome *(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)*		**A.6. Expected adaptation outcome** *(Core indicator 2: direct and indirect beneficiaries reached)*	*Indicate total number of direct and indirect beneficiaries*	
			About 362,475 direct beneficiaries. Male: 181,238 Female: 181,237	*770,050 indirect beneficiaries. Male: 385025 Female: 385,025*
			0.002% of total population in Bangladesh	*0.005%t of total population in Bangladesh*
A.7. Total financing (GCF + co-finance[2])	49,989,200USD	**A.9. Project size**	Small (Upto USD 50 million)	
A.8. Total GCF funding requested	42,201,200USD			

[1]Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e., GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.
[2] Refer to the Policy of Co-financing of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	☒ Grant 42,201,200 USD ☐ Loan Enter number ☐ Guarantee Enter number	☐ Equity Enter number ☐ Results-based payment Enter number	
A.11. Implementation period	*5 years*	**A.12. Total lifespan**	*20 years*
A.13. Expected date of AE internal approval	*5/15/2023*	**A.14. ESS category**	B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	**A.16. Has Readiness or PPF support been used to prepare this FP?**	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	**A.18. Is this FP included in the country programme?**	Yes ☒ No ☐
A.19. Complementarity and coherence	Yes ☒ No ☐		
A.20. Executing Entity information	*PKSF is the Executing Entity of the Project.*		

A.21. Executive summary (max. 750 words, approximately 1.5 pages)

Climate Context: The geographical location and low elevation of the coastal zone of Bangladesh make it susceptible to disasters, whereas climate change asserts a new depressing effect on the lives and livelihoods in the region[3]. The most recent IPCC report (AR6) argues that global sea level will elevate 0.44-0.76 m by 2100 under the intermediate GHG emissions scenario (SSP2-4.5). The NASA sea level tool (see Section B.1) confirms these projections for the entire coastline, but with an even higher maximum value of 0.91 m to the west of the country (by 2100 under SSP2-4.5.Even in the low GHG emission scenario (SSP1-2.6) sea level will rise 0.32-0.62 m by the end of the 21st century.[4] It is predicted for Bangladesh that a 45 cm rise in sea level may inundate 10-15per cent of the land by the year 2050, resulting in over 35 million climate migrants from the coastal districts [5]. The proposed project areas, which include Khulna, Bagerhat, Satkhira, Barguna, Patuakhali, Bhola and Cox's Bazar districts, are particularly vulnerable to sea level rise, salinity intrusion, coastal flooding, cyclones and storm surges because of their low elevation, ranging from 1-2 meters above average sea level.

Hence, the vulnerability of coastal people is characterized in three ways: i.e., 1) poor human settlement in low-lying areas 2) climate-sensitive livelihood, and 3) scarcity of safe drinking water. The majority of the coastal population is poor, small, and marginal farm families and shrimp workers. The coastal poor community builds their houses in low-lying areas that are subject to coastal flooding. Most of the houses are built with mud and *goal pata* (leaves of an indigenous coastal plant) which are severely affected by cyclones, storm surges, and high tides. These people have to spend a significant amount of their earnings repairing

[3] CCC (2016), "Assessment of Sea Level Rise on Bangladesh Coast through Trend Analysis", Climate Change Cell (CCC), Department of Environment, Ministry of Environment and Forests, Bangladesh.
[4] IPCC, 2021: Summary for Policymakers. In: Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Masson-Delmotte, V., P. Zhai, A. Pirani, S. L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M. I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J.B.R. Matthews, T. K. Maycock, T. Waterfield, O. Yelekçi, R. Yu and B. Zhou (eds.)]. Cambridge University Press. In Press.
[5] National Adaptation Programme of Action (NAPA), 2009. The Ministry of Environment and Forests, Government of Bangladesh, Dhaka

houses each year. Besides, the coastal communities primarily depend on seasonal subsistence agriculture and agricultural wage labour, which are highly climate-sensitive. Safe drinking water is highly vulnerable to sea level rise and salinity in the coastal zone of the country. The communities and local-level organisations have a lack of understanding of the present and future climate change impacts, which is one of the key barriers to promoting climate-resilient development in the coastal zone.

To respond to these challenges, the **primary goal** of the project is to develop climate-adaptive coastal communities in Bangladesh through the adoption of climate-resilient housing and livelihood technologies. The project will also enhance the capacity of communities and organisations to address climate change impacts in their localities.

The specific objectives are:

1. To develop climate-resilient homesteads for vulnerable communities in the southwest coastal zone of Bangladesh;
2. To develop climate-adaptive livelihoods for vulnerable coastal communities; and
3. To enhance the capacity of vulnerable communities and institutions so that they are able to plan and implement adaptation interventions.

The project is expected to achieve the following impacts: (i) reduced vulnerability of the coastal community to the adverse impacts of climate change, and (ii) enhanced capacity of the coastal vulnerable community to address climate change issues. The expected outcomes/results are: (i) decreased risk of loss of assets and lives from extreme weather events; (ii) livelihood resilience to SLR / storm surge and salinity; and (iii) improved climate planning and implementation by communities and local-level institutions.

Aligning with the stated specific objectives, the project will construct climate-resilient houses for the most vulnerable households, climate-adaptive farming technologies, (e.g., crab farming along with mangrove tree plantation, sheep or goat rearing, household-based agriculture, etc.), with technical, financial, and capacity building, supply chain development and market linkage support. The concept of a climate-resilient homesteads in the coastal areas of the country includes a raised homestead area, a cyclonic storm-resistant house structure, homestead-based vegetable cultivation, a sanitary latrine, a rainwater harvesting system and saline-tolerant fruit trees and mangrove species plantations in and around the raised homestead area. This integrated climate-resilient homestead development will protect them from coastal inundation and cyclones as well as help increase households' income. These homesteads will also provide some safe space for livestock farming for beneficiaries.

The project follows the GCF agriculture and food security sector guidelines while designing climate-resilient livelihoods. The proposed livelihood interventions build on the first paradigm shift pathway, i.e., promoting resilient agro-ecology. The project will promote ecologically suited, technologically feasible, and socio-culturally accepted high-value crops and farm animals while providing the required skills and technology upscaling. The project will promote an integrated fruit-fish-fiber model in the southwest coastal areas of Bangladesh.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context(max. 1000 words, approximately2 pages)

1. Bangladesh's coastal zone consists of 19 districts along a coastline of 710 km. The coastal zone extends over 47,150 sq. km of area and has a population of 38.52 million (BBS, 2011). The coastal zone is quite distinct from the rest of the country. The coastal zone has been delineated based on three characteristics, namely the level of tidal fluctuations; salinity conditions (both soil, surface and ground water); and risks of cyclones and storm surges[6]. The zone is characterized by a vast network of rivers and channels, enormous discharge of water with a huge amount of sediment, many islands, the Swatch of No Ground (underwater canyon located 45 km south of the Sundarbans in Bangladesh), the shallow northern Bay of Bengal, strong tidal influence and wind actions, tropical cyclones and storm surges. The average elevation in the southwest coastal zone ranges from 1-2 m and in the southeast coastal zone from 4-5 m (Figure 1). The flat topography, active delta, and dynamic morphology, along with changes in precipitation and temperature, play a significant role in its vulnerability to sea level change, salinity intrusion and cyclone and storm surge.

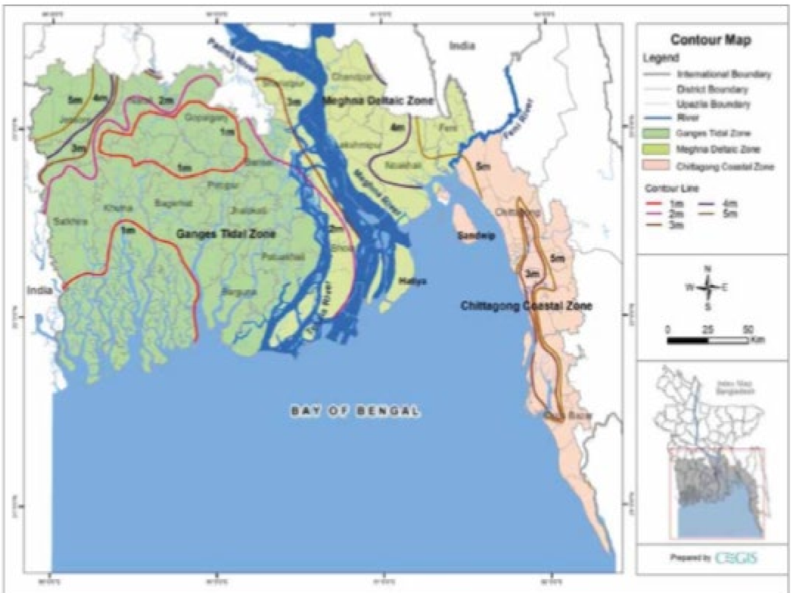

Figure 1: Topography of Bangladesh Coast

The proposed project area

2. The proposed project has selected seven exposed districts of Bangladesh's coastal zone. These are Satkhira, Khulna, Bagerhat, Barguna, Patuakhali, Bhola and Cox's Bazar. These districts have been selected based on their high exposure, vulnerability and the feasibility of introducing a crab hatchery supply chain in the regions. The following figure shows the proposed project districts.

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[6] CCC (2016), "Assessment of Sea Level Rise on Bangladesh Coast through Trend Analysis", Climate Change Cell (CCC), Department of Environment, Ministry of Environment and Forests, Bangladesh.

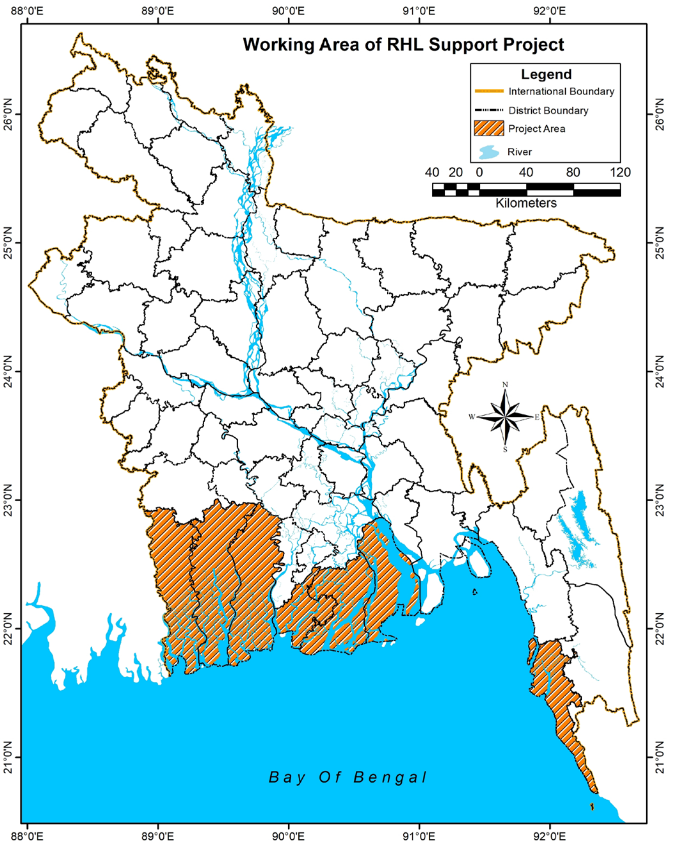


Figure 2: Proposed project area

The total area of these seven districts is 24,691 sq.km and total population is 12,275,996. The average poverty rate is 26.13per cent, which is higher than the national average of 24.3per cent [7]. The dependency ratio is also high in these districts which is 68.59 (detailed in table 1). Moreover, 52.01per cent of household in Khulna division is katcha (temporary) and 26.86per cent are semi-pucca [8], which means that more than three-fourths of the households are vulnerable to intensive precipitation, cyclones and storm surges, etc.

Table 1: Area, population and poverty ratio of the selected districts

Name of district	Area (sq.km)	Population (2011)	Poverty situation (2016) (%)	Dependency Ratio (per cent)
Satkhira	3,358.33	1,985,959	18.6	56.58
Khulna	4,394.46	2,318,527	30.8	54.07
Bagerhat	5,882.18	1,476,090	31	62.77
Cox's Bazar	2,491.86	2,289,990	24.1	85.63
Barguna	1,939.39	892,781	25.7	65.99
Bhola	3,403.48	1,776,795	15.5	83.28
Patuakhali	3,221.31	1,535,854	37.2	71.83
Total	2,4691	12,275,996	Avg.=26.13	68.59

[7] BBS, 2016: Household Income and Expenditure Survey, 2016, Bangladesh Bureau of Statistics, Ministry of Planning, Government of Bangladesh.
[8] BBS, 2015: Bangladesh National Disaster Statistics 2009-2014, Bangladesh Bureau of Statistics, Ministry of Planning, Government of Bangladesh.

Sea level rise and associated salinity intrusion in Bangladesh coastal zone

3. The IPCC sixth assessment report stated with high confidence that Asian coastal countries are likely to incessantly experience relative sea-level rise, which will result in the coastal flooding of many low-lying areas[9]. Bangladesh is considered a hotspot for sea level rise impacts as mentioned in a number of scientific studies[10]. Increasing sea level rise is one of the most critical climate change issues for coastal areas of the country. Currently, because of the low topography in these coastal areas, about 50per cent typically become inundated during the annual monsoons. A study by the SAARC Meteorological Research Center (SMRC)[11] through observing tidal gauge records between 1977-1998, found that the tidal levels in Hiron Point, Char Changa and Cox's Bazar rose by 4.0 mm/year, 6.0 mm/year and 7.8 mm/year respectively. The rate of the trend is almost double that of the eastern coast compared to the western coast. Center for Geographic Information System (CEGIS) and Department of Environment (DOE)[12] have analysed 30 years of tidal gauge data for estimating observed sea level change in the Bangladesh coastal zone without considering any land-sea interaction, i.e., sedimentation, subsidence, erosion, accretion, etc. It shows trends in water level in the Ganges tidal floodplain of 7-8 mm/year. On the other hand, the trend is 6-10 mm/year in the Meghna Estuarine floodplain and 11-21 mm/year in the Chittagong coastal plain areas. The sea level in the 37 coastal stations of Bangladesh is predicted to rise between 0.53 and 0.97m by 2100[13], while the IPCC states in AR6 that under the intermediate GHG emissions scenario (SSP2-4.5), global sea level will elevate 0.44-0.76 m by 2100[14]. These figures are supported by the NASA sea level projection tool (https://sealevel.nasa.gov/ipcc-ar6-sea-level-projection-tool), which suggests that under SSP2-4.5, by 2100, sea level at Hiron Point will rise by 0.42-0.91 m and sea level at Cox's Bazar will rise by 0.24 – 0.73 m (see Figure 3b). It is predicted that a 45 cm rise in sea level may inundate 10-15per cent of the land by the year 2050, resulting in over 35 million climate refugees from coastal districts.[15] The sea level rise may also influence the extent of the tides (currently the lower one-third of the country experiences tidal effects) and alter the salinity quality of both surface and groundwater[16].

4. During the dry season (December to March), saltwater intrusion occurs through various inlets in the western part of the coastal zone (Satkhira, Khulna, Barguna, Patuakhali, etc.) and through the Meghna estuary. Moreover, as freshwater flows decrease in the Lower Meghna during the waning of the monsoon, the saline front can move by as much as 30-40 km from the coast. During the monsoon, about 12per cent of the total area is under high salinity levels, which increase to 29per cent during the dry season. With increased sea level rise, drainage gradients may reduce, thereby decreasing the flow to the Bay of Bengal and allowing riverine salinity to move further inland.

5. The Soil Resource Development Institute (SRDI) of Bangladesh has presented the soil salinity status of Khulna region in their annual report for 2018-2019. The annual investigation of SRDI demonstrated the salinity changes in the different locations of Khulna region pertaining to different months in recent

[9]IPCC (2021) Sixth Assessment Report, Physical Science Basis

[10]Nicholls, R. J., Hutton, C. W., Lázár, A. N., Adger, W. N., Allan, A., Whitehead, P. G., ... &Payo, A. (2018). An integrated approach providing scientific and policy-relevant insights for South-West Bangladesh. Ecosystem Services for Well-Being in Deltas: Integrated Assessment for Policy Analysis; Nicholls, RJ, Hutton, CW, Adger, WN, Hanson, SE, Rahman, MM, Salehin, M., Eds, 49-69.

[11]SMRC (2003). The vulnerability assessment of the SAARC Coastal Region due to sea level rise: Bangladesh case study. Dhaka, SAARC Meteorological Research Center.

[12]CEGIS and DoE (2011). Final report on programmes containing measures to facilitate adaptation to climate change of the second national communication project of Bangladesh. Dhaka, Department of Environment.

[13] Haque, A., Rahman, M. H., Rahman, D., & Rahman, D. (2019). An evaluation of sea level rise vulnerability and resilience strategy to climate change in the coastline of Bangladesh. *International Journal of Environmental Sciences & Natural Resources*, *18*(2), 56-70.

[14] IPCC (2021) Sixth Assessment Report

[16] MOEF (2009): National Adaptation Programme of Action (NAPA), 2009. The Ministry of Environment and Forests, Government of Bangladesh, Dhaka.

[16] CCC (2016): Assessment of Sea Level Rise on Bangladesh Coast through Trend Analysis", Climate Change Cell (CCC), Department of Environment, Ministry of Environment and Forests, Bangladesh.

years. Figure 3a shows that the soil salinity of Batiaghata, Dumuria, Mongla and Paikgacha, (these are sub-districts, locally known as *upazilas*. under Khulna district) was found to vary throughout the year[17].

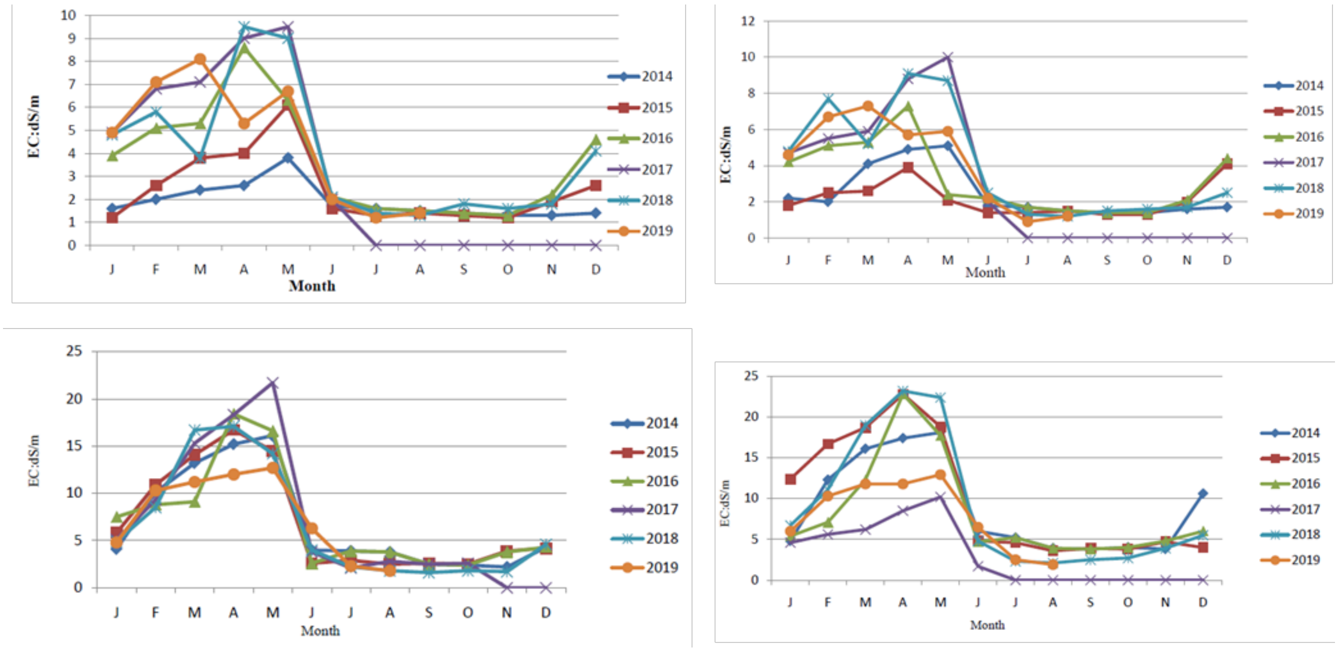
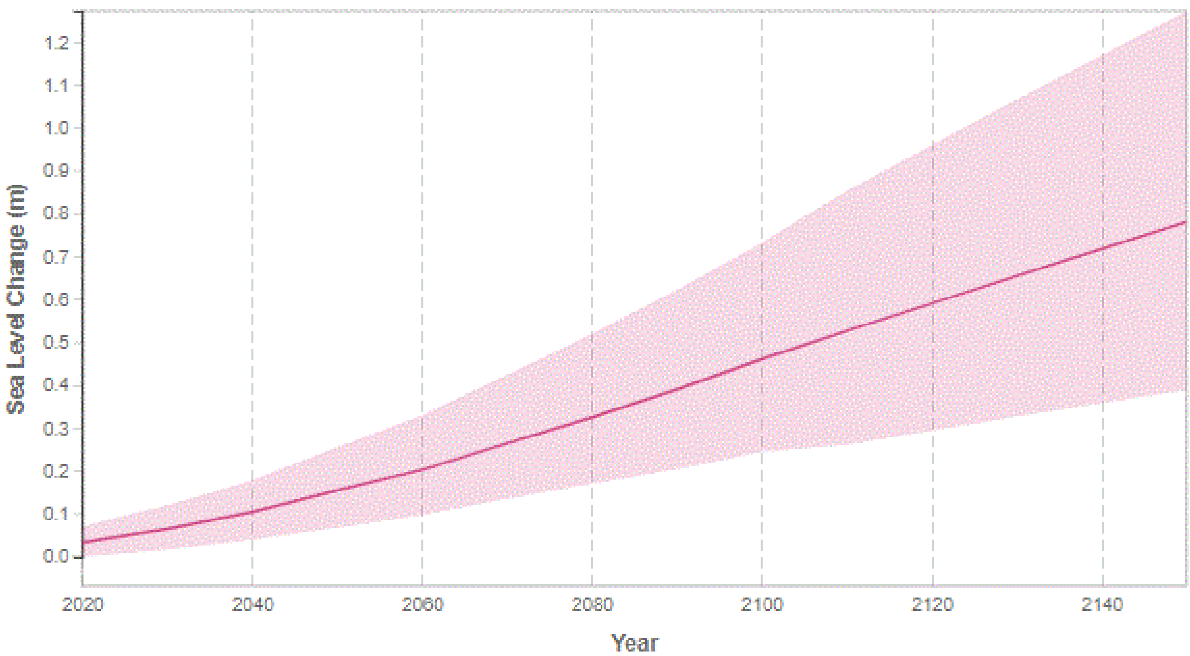


Figure 3a: Temporal distribution of salinity at different locations in Khulna Region (clock-wise location of maps from upper left: Batiaghata, Dumuria, Mongla and Paikgacha).



[17]Soil Resource Development Institute (2019): Annual Report, 2018-2019, Soil Resource Development Institute, Ministry of Agriculture, Dhaka.

Figure 3b. Sea level projections for Cox's Bazar. These projections are for 2100 relative to a 1995-2014 baseline under future scenario SSP2-4.5. The bold line shows the median projections, and the shading represents the 17th-83rd percentiles.

6. It is observed in the SRDI study that over 1 million hectares of cultivable land in the country are affected by salinity intrusion caused by slow- and rapid-onset events, including sea level rise, cyclones and storm surges and consequential coastal flooding. The net cultivated area in Satkhira district decreased by about 7 per cent from 1996 to 2008 and production of the principal rice crop in Satkhira district decreased from about 0.3 million tons in 2008 to 0.2 million tons in 2010 (SRDI, 2010)[18]. During the last 35 years, salinity-affected area have increased by almost 26per cent in the country and are spreading over the non-coastal areas of Bangladesh (Reliefweb, 2019)[19].

7. The salinity front will move about 60 km to the north in about 100 years (CCC, 2016)[20]. It is seen that a sea level rise of 27 cm causes a 6 per cent increase in brackish water area compared to base conditions. About an additional area of 327,700 ha would become a high-saline water zone (>5 ppt) during the dry season due to a 60 cm sea level rise. In the monsoon, about 6 per centper cent of the sweet water area (276,700 ha) will be lost (CCC, 2016).

Tropical Cyclone and Storm Surges

8. The coastal areas of Bangladesh are frequently hit by cyclones that form in the Bay of Bengal. Over the last 50 years, 15 severe cyclones with wind speeds ranging from 140 to 225 km/hour have hit the coastal area of Bangladesh of which 7 hit in the pre-monsoon and rest in the post-monsoon season[21]. The highest frequency of intense tropical cyclones has been reported to occur between the months of October and November. In recent years, the lives and livelihoods of more than a million coastal people have been affected by cyclones in Bangladesh. Cyclone-induced storm surges are more devastating and often result in colossal damage[22]. In 2007, the tropical cyclone Sidr affected Barguna, Patuakhali, Bhola and some parts of Khulna, Bagerhat, and Satkhira districts. It was a category 4 cyclone that struck the south-west coast of Bangladesh i.e., Satkhira, Khulna, Bagerhat, Barguna, Patuakhali districts at 240 km per hour with a surge height of about 6 meters. Of the 2.3 million households affected to some degree by the effects of Cyclone Sidr, about one million were seriously affected[23]. It caused a financial loss of USD 1.7 billion according to an estimate by the Government of Bangladesh (GoB). The number of deaths caused by Sidr is estimated at 3,406, with 1,001 still missing, and over 55,000 people sustaining physical injuries[23]. Super cyclone Aila in 2009 damaged the coastal polder (embankment) in Satkhira, Khulna and Bagerhat districts. As a result, saline water entered the vast area and remained for a longer period, causing widespread damage to agriculture, livestock, and livelihoods. The worst two affected districts are Satkhira and Khulna followed by Bagerhat, Pirojpur, Barisal, Patuakhali, Bhola, Laxmipur, Noakhali, Feni, Chattogram, and Cox's Bazar[24]. According to the official statistics, 545,954 people from 118,757 families are affected in Khulna, while death toll stands on 45[24]. Loss of livestock

[18] Soil Resource Development Institute (SRDI) (2010): Saline Soils of Bangladesh, Soil Resource Development Institute (SRDI), Ministry of Agriculture, Dhaka.

[19]Reliefweb (2019) Climate Change-Induced Salinity Affecting Soil Across Coastal Bangladesh.https://reliefweb.int/report/bangladesh/climate-change-induced-salinity-affecting-soil-across-coastal-bangladesh

[20]CCC (2016), "Assessment of Sea Level Rise on Bangladesh Coast through Trend Analysis", Climate Change Cell (CCC), Department of Environment, Ministry of Environment and Forests, Bangladesh.

[21] MOEF (2009). National Adaptation Programme of Action (NAPA), Ministry of Environment and Forests, Government of Bangladesh.

[22]Reliefweb (2020) Bangladesh: Cyclone - Final report early action EAP2018BD01. https://reliefweb.int/report/bangladesh/bangladesh-cyclone-final-report-early-action-eap2018bd01

[23]GoB (2008).Cyclone Sidr in Bangladesh: Damage, Loss, and Needs Assessment for Disaster Recovery and Reconstruction.A Report Prepared by the Government of Bangladesh Assisted by the International Development Community with Financial Support from the European Commission.

[24] Roy, K., Kumar, U., Mehedi, H., Sultana, T. and Ershad, D. M., 2009. Initial Damage Assessment Report of Cyclone AILA with focus on Khulna District. Unnayan Onneshan Humanity Watch- Nijera Kori, Khulna, Bangladesh, June 23, 2009. pp-31.

and poultry has been reported as 2,080 and 24,505, respectively24. In total 367 km of road has been fully damaged while 1,065 km of road is partially damaged24. About 7,392 acres of agricultural land has been damaged24. About 594 km of embankment has been damaged which is still letting the river water to freely flow on land and lengthening the stay of waterlogged conditions[24]24. Diarrhoea has broken out in the district as 3700 and 4500 people are reported as affected in Dacope and Koyra, respectively24. The storm surge of the 1991 cyclone caused the death of a substantial number of cattle and destroyed all freshwater sources in the affected coastal areas[25]. A summary of recent cyclones can be seen in Table 2.

Table 2. Loss and damage by cyclones in the past 15 years in Bangladesh

Name of cyclone	**Year**	**Wind speed (Km/hour)**	**Affected areas (districts or upazilas)**	**Loss and damage (Loss in USD Million (Gross/Total)**	**Source**
Sidr	2007	240	12 Districts	1703.1	ERD, April 2008
Rashmi	2008	85	9 Districts	1.5	
Aila	2009		11 Districts/64 Upazilas	221.8	Reliefweb, June 2009
Roanu	2016	110	11 Districts/56 Upazilas	508.83	Christian Adi
MORA	2017	110-130	132 Upazilas	1700	Foyjonnesa et al., April, 2019
Bulbul	2019	120-130	16 Districts	30.9	IFRC, May 2020
Amphan	2020	60-90	19 Districts/76 Upazilas	131	Reliefweb, May 2021
Sitrang	2022	74	6 Districts	41.2	The Daily Star, 31 October, 2022

9. There is another dataset for tropical cyclones that have hit Bangladesh since 1961. A total of 61 tropical cyclones hit Bangladesh and its close neighbouring territories during the period 1961-2013. This result is an update of the analyses[26]. The update has been accomplished by using JTWC Best Track tropical cyclone data. According to current statistics, an average of 1.15 tropical cyclones hit Bangladesh per year. The spatial distribution of these cyclones is shown in Table 3. The south-western and south-central coastal zones of Bangladesh were hit by 28 per cent and 16 per cent respectively, by reported cyclones between 1961 and 2013.

Table 3: Distribution of land-falling cyclones to different coastal regions during 1961-2013

[25]Farukh, M. A., Hossen, M. A. M., & Ahmed, S. (2019). Impact of extreme cyclone events on coastal agriculture in Bangladesh. Progressive Agriculture, 30, 33-41.

[26]Quadir, D. A. and Md. Anwar Iqbal. 2008. Tropical Cyclones: Impacts on Coastal Livelihoods, IUCN Working Paper.

Coastal region	No. of tropical cyclones hit the coast	%of the tropical cyclones	Remarks
South-eastern coast (Southern Chattogram, Cox's Bazar and Teknaf)	18	29.5	Cox's Bazar is the project area
Sundarbans (south-western) coast (Satkhira, Khulna and Bagerhat)*	17	27.9	Satkhira, Khulna and Bagerhat are the project areas
Meghna estuary, east central coast (Eastern Bhola*, Noakhali and Chattogram)	16	26.2	Bhola is the project area
Central coast (Barguna, Potuakhali, Pirozpur, Barisal, Bhola)*	10	16.4	Barguna and Patuakhali are the project areas
Total	**61**	**100**	

Proposed project area

10. The Sixth Assessment Report of the IPCC stated that the number of intense tropical cyclones is assumed to increase (High Confidence). However, it is narrated with medium confidence that total number of cyclones throughout the world might decline or remain constant (IPCC, 2021). In AR6, the IPCC has firmly stated that coastal hazards in Asian countries will rise throughout the 21st century. In addition, it is asserted with high confidence that anthropogenic climate change results in increase in heavy precipitation and accompanying tropical cyclones. However, it is unlikely to ascertain trends in the frequency of all-category tropical cyclones due to the unavailability of authentic data.[27] However, it is **impossible** to ascertain the trends in the observed frequency of all-category tropical cyclones due to **insufficient** data.

11. Irrespective of changes in tropical cyclone properties, the unambiguous sea level rise (see paragraph 4) will result in all storm surges reaching a higher threshold and therefore causing increased coastal impacts. Using the IS92a scenario, Unnikrishnan (2006)[28] argues that there will be a significant increase in the frequency of high storm surges in the Bay of Bengal even if there is no substantial change in the frequency. Emanuel (2005) also projects increased intensity of tropical storms by 2100 for the North Indian Ocean[29]. These findings are fully aligned with the latest IPCC Sixth Assessment Report (AR6). Combined with expected rises in sea-level, these higher intensity cyclones are predicted to cause increased tidal surge heights. Bangladesh has already experienced about 40 per cent of the total impact of storm surges in the world[30].

Coastal Flooding

12. Under natural conditions, areas of the coastal zone are often flooded at high tides, either throughout the year at high spring tides or only in the monsoon season when rivers entering the region from inland run at high levels[31]. Coastal flooding by saline water was found on the extreme south-west throughout the year and in the north-west and south-eastern coasts during the dry season[32]. Cyclones and storm surges have a direct impact on increasing coastal flooding. An estimated 3.45 million people were

[27] IPCC Sixth Assessment Report, 2021

[28] Unnikrishnan A.S., R. Kumar, S.E. Fernandez, G.S. Michael and S.K. Patwardhan, 2006. Sea Level Changes along the Indian Coast: Observations a nd Projections Current Science India, 90: 362-368

[29] Emanuel, K. (2005). Increasing destructiveness of tropical cyclones over the past 30 years. *Nature* **436**, 686–688 (2005). https://doi.org/10.1038/nature03906.

[30] Murty TS. Storm surges meteorological ocean tides. Can J Fish Aquat Sci. 1984;212:897.

[31] Hug Brammer, (2013): Bangladesh's dynamic coastal regions and sea-level rise, Climate Risk Management, ELSEVIER, 2013.

[32] Hug Brammer, (2013): Bangladesh's dynamic coastal regions and sea-level rise, Climate Risk Management, ELSEVIER, 2013.

exposed to coastal flooding because of Cyclone Sidr[33]. Compared to the no-climate change baseline scenario, an additional 7.8 million people would be affected by floods higher than one metre in Bangladesh because of a potential 10-year return cyclone in 2050 (an increase of 107 percent)[32]. A total of 9.7 million people (versus the 3.5 million in the baseline scenario) are projected to be exposed to severe inundation of more than 3 meters under this same scenario[34].

13. Dasgupta et al. (2014)[35] modelled cyclone storm surge impacts under a changing climate scenario in 2050. This model, in a bid to determine potential future inundation zones by 2050 under the climate change scenario, was run for five cyclone tracks (covering the entire coastal area), incorporating a 27-cm rise in sea level, a 10 per cent increase in wind speed, and the landfall of cyclones during high tide. It is predicted that by 2050 an additional 15 per cent of the coastal area of Bangladesh will be inundated with storm surges during cyclones. Figure 4 shows the additional area that will be impacted by inundation in 2050. Not only will the areas of Khulna, Bagerhat and Satkhira Districts be newly exposed, but tidal surges of 3 meters will inundate 69 per cent more land area than they do at the present time.

14. Coastal flooding due to tidal surge, along with the huge amount of sediment that is carried by the three major rivers through a steep flow gradient, coupled with the increase in sea level, results in extensive flooding in coastal regions. The Bangladesh Climate Change Strategy and Action Plan (BCCSAP) stressed the need to understand, prepare for and respond to these emerging challenges so that the wellbeing of the people is ensured (MoEF, 2009)[36].

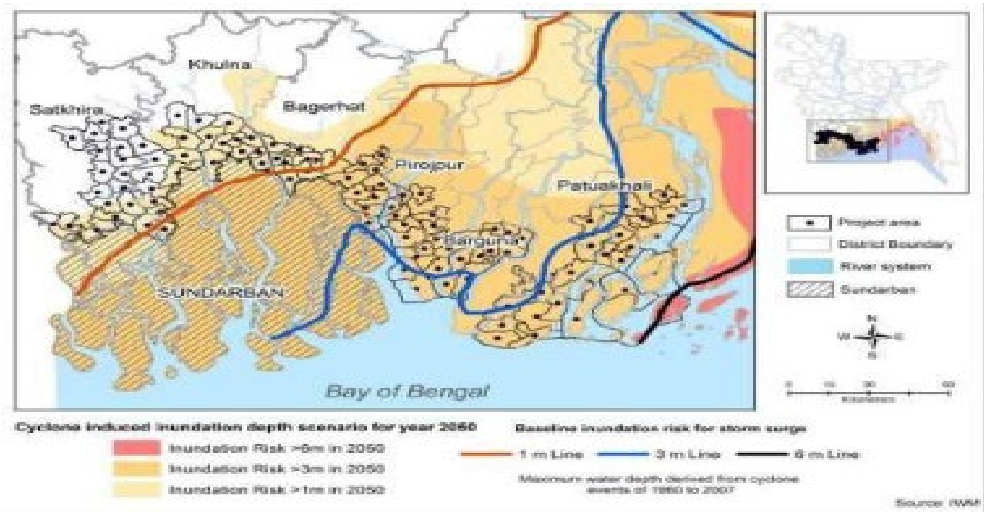


Figure 4: Cyclone inundation map in 2050 under projected climate change scenarios

Impacts of Climate Change

15. Coastal communities are particularly vulnerable to the aforementioned climatic dangers because of their poverty and other challenges, which are explained below.

Vulnerability of homesteads to climate hazards

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[33]https://www.worldbank.org/en/news/press-release/2013/06/19/warming-climate-to-hit-bangladesh-hard-with-sea-level-rise-more-floods-and-cyclones-world-bank-report

[34]https://www.worldbank.org/en/news/press-release/2013/06/19/warming-climate-to-hit-bangladesh-hard-with-sea-level-rise-more-floods-and-cyclones-world-bank-report

[35]Dasgupta, Susmita & Huq, Mainul & Khan, Zahirul & Murshed, Ahmed & Mukherjee, Nandan & Khan, Malik & Pandey, Kiran. (2014). Cyclones in a changing climate: the case of Bangladesh. Climate and Development.

[36]MOEF (2009). National Adaptation Programme of Action (NAPA), Ministry of Environment and Forests, Government of Bangladesh.

16. Houses along the coastal zone are subject to coastal flooding, with more than two-thirds of the houses in coastal communities made of mud and *goal pata* that are easily and severely damaged during cyclones, storm surges and high tides. Affected people spend a significant share of their earnings and time on an annual basis for house repairs or reconstruction due to climate related damage.

17. The construction of communal cyclone shelters, among others, is the main technique used today to protect coastal people from SLR, tidal surge, coastal flooding, and cyclones. These local-level disaster response infrastructures have a number of drawbacks. Due to the lack of adequate sanitation and hygiene facilities and the increased likelihood of sexual harassment incidents, women and girls are often reluctant to use such facilities. Many families opt to seek safety in their own homes rather than using these shelters, which leaves them extremely vulnerable. The multipurpose use of cyclone shelters as elementary schools will be limited in the near future as a result of the rapid demographic changes in Bangladesh.

Livelihoods and Nutrition

18. Livelihoods mostly depend on the climate-vulnerable agriculture sector through seasonal subsistence agriculture and farm labour. This is expected to be further aggravated by degrading agro-ecological landscapes and resources[37] in part due to saltwater intrusion and environmental damage caused by storms.

19. The gradual increasing scenarios of salinity intrusion into the coastal areas (soil, river, and groundwater) of Bangladesh threaten the primary production system, coastal biodiversity, and human health[38]. According to the Soil Resources Development Institute (SRDI), the total amount of salt-affected land in Bangladesh was ~83.3 million ha in 1973, which increased to ~102 million ha in 2000, to ~105.6 million ha in 2009, and is still continuing to increase[39]. Salinity intrusion directly affects the livelihoods of farmers, including rice cultivators and fisherfolk. The net cropped area in coastal Bangladesh has been decreasing over the last few years due to several factors and studies have identified salinity as the main cause for yield reduction in coastal agriculture[40]. Other statistics that point to the trauma of salinity include: (a) salinity has decreased the production of wheat by 4.42 million ton per year in coastal Bangladesh[41], (b) 19 of 40 local rice varieties are already extinct and about four to five varieties have become rare in coastal areas, and (c) between 1975 and 2006 the number of cultivated winter vegetables has declined in coastal areas[42].

20. Livestock, including small ruminants, are primarily affected by coastal inundation associated with tidal surges, monsoon precipitation, and cyclones. As stated earlier, livestock deaths are evident. Moreover, increased salinity has significant negative impacts on livestock products in the low-lying coastal floodplain. In the last few decades, the area of grazing land and the intensity of grazing have decreased due to increased salinity and the loss of agricultural land in coastal areas[43]. This situation challenges the raising of large ruminants by causing a scarcity of fodder and animal feed, particularly in the south-western coastal area. Small ruminants require less fodder and feed, which may be easily arranged in the locality. Besides, SLR-induced salinity, coastal floods, cyclonic storm surges, and waterlogging will cause damage to livestock shelters and sometimes deaths. Despite these challenges, the marginal farmers raise small ruminants and poultry, as this has been traditionally practiced to ensure their livelihood and nutrition security. The slatted houses for goat or sheep will protect them from flooding, cyclones and storm surges and thus reduce their death.

21. Productive land is being lost due to saltwater intrusion, causing a loss of agricultural yield and fodder, while cyclones and storm surges are responsible for many livestock deaths. Ultimately, this is leading

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[37]Agriculture and Food Security Sector Guideline of GCF, 2021.

[38] Islam, M. A., N. Warwick, R. Koech, M. N. Amin, L. L. D. Bruyn. 2020. Science of the Total Environment 727, 138674.

[39] Chen, J., and V. Mueller. 2018. Nature Climate Change 8, 981–985.

[40]Baten, M. A., L. Seal, K. S. Lisa. 2015. American Journal of Climate Change 4, 248.

[41]Habiba, U., M. A. Abedin, R. Shaw, A. W. R. Hassan. 2014. Salinity-induced livelihood stress in coastal region of Bangladesh. Water insecurity: A social dilemma, United Kingdom: Emerald Group Publishing Limited

[42]Rahman, M., M. S. Uddin, M. Uddin, S. A. Bagum, N. Halder, M. Hossain. 2004. Journal of Biological Sciences 4, 1–4.

[43]Rahman A. and Udinn N. 2022. Challenges and Opportunities for Saline Agriculture in Coastal Bangladesh. In The future of Sustainable Agriculture in Saline Environments. DOI: 10.1201/9781003112327

to protein and other deficiencies among coastal populations. Twenty-eight per cent of children under five years old are stunted (have low height for age) and 10 per cent are acutely malnourished or wasted (have low weight for height)[44]. It is expected that this situation would worsen in the coastal area of the country as a direct result of climate change-induced SLR.

22. Increased coastal salinity has affected fish yields, leading to substantial reductions in the inland open-water fishery. Shrimp farming occurs on about 138,600 ha of coastal land in Satkhira (42,550 ha), Khulna (36,500 ha), and Bagerhat (49,550 ha) districts. Apart from these, the production of native freshwater fish species (e.g., rui, katla, carp, boal, tengra, golshatengra, koi, shing, taki, khalisha, potka, kanimagur, is gradually declining due to increased salinity[45].

23. A study in the highly saline Paikgacha sub-district in Khulna and moderately saline Rampal sub district in Bagerhat found that freshwater fish species had decreased by 59 per cent and 21 per cent respectively, during 1975-2005. The slight increase in salt-tolerant species did not compensate for the loss in diversity, which is a serious threat to the local ecosystem and food supply.

24. Increases in salinity by 2050 and other stressors are predicted to significantly negatively impact the Sundarbans mangrove ecosystem[46] which in turn, will impact the livelihoods of poor communities who depend on its resources, especially the poorest *upazilas* (sub-district) via loss of standing timber and honey production. Salinity has resulted in the loss of 12 percent of marine fish and 25 percent of shrimp species in south-western Bangladesh.[47]Local people used to earn cash income by selling ecosystem products, such as fish, honey, and *golpata* at local markets. Due to restrictions from the government as a part of the conservation management of the Surdarbans, people couldn't fully rely on the ecosystem services of Sundarbans. For example, to ensure that wildlife during the mating season had an undisturbed environment, following the trend of 2021, the government prohibited admission into the forest for three months, from June 1 to August 31, 2022, for purposes of fishing, travel, or resource collection. The demand for locally produced livelihoods is therefore more crucial than the need for livelihoods reliant on these fragile natural resources.

Drinking and Sanitation Water

25. Coastal surface and ground water, from which the majority of the population rely for water, is increasingly contaminated with saline water due to sea level rise, cyclones and tidal surges. Coastal people, particularly women, spend otherwise productive time collecting drinking water from long distances, which exacerbates their lack of productivity and ultimately income.

26. The result of community engagement reveals that local people are aware of water scarcity issues, and nearly all of them perceive that salinity is the main reason behind them. Even though socioeconomic factors affect their ability to adapt, communities have their own adaptation technologies to cope with the problem, such as household-based temporary rainwater harvesting, and the informal use of freshwater ponds. In the coastal belt of Chattogram, people were found to suffer both physically and financially due to the freshwater shortage. Inhabitants of those salinity-affected areas spent a significant portion of their income on buying freshwater bottles[48].

Coherence and complementarity with other past and ongoing interventions

27. **Crab hatchery-PKSF:** After successful experimentation with crab hatcheries, PKSF established four small-scale hatcheries. Three of them are at the entrepreneur level. All these hatcheries are in

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[44] Bangladesh Bureau of Statistics (BBS) and UNICEF Bangladesh (2019). Progotir Pathey, Bangladesh Multiple Indicator Cluster Survey 2019, Survey Findings Report. Dhaka, Bangladesh: Bangladesh Bureau of Statistics (BBS).

[45]Alam, M. Z., Carpenter-Boggs, L., Mitra, S., Haque, M., Halsey, J., Rokonuzzaman, M., ... &Moniruzzaman, M. (2017). Effect of salinity intrusion on food crops, livestock, and fish species at Kalapara Coastal Belt in Bangladesh. Journal of Food Quality, 2017

[46] The World Bank (2015): Salinity Intrusion in a Changing Climate Scenario will Hit Coastal Bangladesh Hard.

[47]Alam, M. Z., Carpenter-Boggs, L., Mitra, S., Haque, M., Halsey, J., Rokonuzzaman, M., and Moniruzzaman, M. (2017). Effect of salinity intrusion on food crops, livestock, and fish species at Kalapara Coastal Belt in Bangladesh. Journal of Food Quality, 2017.

[48]Jabed, M. A., Paul, A., and Nath, T. K. (2020). Peoples' perception of the water salinity impacts on human health: a case study in south-eastern coastal region of Bangladesh. Exposure and health, 12(1), 41-50.

operation. A total of US $0.66 million was invested in hatchery development. In addition, PKSF has been promoting crab farming and has invested a total of about US$2.15 million so far to transfer this now proven adaptive technology. The RHL project will scale PKSF's existing efforts to promote pro-saline livelihood options for the coastal vulnerable people.

28. **World Bank Project for national housing market:** PKSF has been implementing the World Bank funded LICHSP to improve the living conditions of low-income community members residing in selected municipalities and city corporations since 20 October 2016. As of September 2022, PKSF, through its seven Partner Organizations (POs) has disbursed a total of about US $24.40 million as loans to 11,641 borrowers for the construction of new houses and the extension or repair of existing ones in 13 towns under 'Shelter Lending & Support' component of this project. This project focused on urban and semi-urban areas, whereas the RHL project will focus on rural housing, which is arguably more vulnerable and where communities have less capacity to afford housing loans. The beneficiaries under the LICHSP are in the medium- to low-income category and are not necessarily climate-vulnerable. However, the LICHSP has developed a guideline for implementing housing loan activities for medium- to low-income groups, which will help the RHL project prepare a guideline for developing a guideline for building climate-resilient homesteads. PKSF's experience in building or repairing houses will also be useful for implementing the RHL project.

29. **Climate Resilient Infrastructure Mainstreaming (CRIM):** This is the first GCF-financed project in Bangladesh. KfW is the accredited entity, and the Local Government Engineering Department (LGED) is the Executing Entity (EE) of the project. The total estimated budget is USD 80 million, of which GCF is contributing USD 40 million as a grant. The project has been implemented in Satkhira, Barguna and Bhola districts. The project has three intervention areas, i.e., a) Institutional Development, b) Pilot Climate Resilient Rural Infrastructure and c) Pilot Climate Resilient Urban Infrastructure. The project will enhance the capacity of LGED for addressing climate change in the infrastructure sector of the country by establishing a "Climate Resilient Local Infrastructure Centre ("CReLIC")" in the organization under the first intervention area a. Under intervention b, it will build 45 new multipurpose community cyclone shelters with innovative design in climate change aspects and rehabilitate 20 existing shelters to a climate-proof standard in the three project districts (Districts Bhola, Barguna and Satkhira). Under intervention area c, the project will improve drainage, flood protection, sanitation, water supply, and transport infrastructure, with priority given to the most vulnerable, such as the inhabitants of the city slums of Satkhira Pouroshova (municipality). The CRIM and RHL projects geographically overlap in three districts, but there is no overlap in project activities. Rather, these two projects will complement each other. For example, the CRIM project will increase capacity of government ogransations whereas the RHL project will increase the capacity of non-government organisations. Similarly, the CRIM project will focus mainly on urban infrastructure except some community-based cyclone shelters, whereas the RHL project will focus on rural livelihood and homestead development activities. Most importantly, the CRIM project will develop resilient community infrastructure, whereas the RHL project will increase resilience of individuals. To avoid duplication of resources, the RHL will additionally conduct formal and informal consultations with the CRIM in the selection of the project sites and beneficiaries. The project will review the training manuals of the CRIM project while preparing the guidelines and training manual for this project.

30. **Enhancing adaptive capacities of coastal communities, especially women, to cope with climate change induced salinity:** The GCF has financed UNDP Bangladesh to enhance the capacity of coastal communities, especially women, to cope with climate change-induced salinity. The total project budget is USD 33 million, of which GCF grant is USD 25 million and the Ministry of Women and Children Affairs contributes USD 8 million. The project will provide assistance to 39,000 women and girls in Satkhira and Khulna to adopt resilient livelihoods, while ensuring reliable, safe drinking water for 130,000 people through community-managed rainwater harvesting solutions. It will also seek to strengthen the participation of women in the last-mile dissemination of gender-responsive early warnings and the continued monitoring and adaptation of livelihoods to evolving climate risks. The UNDP's project implements salinity-resilient livelihoods in both off-farm and on-farm sectors for vulnerable people. Whereas the RHL will intervene only in the on-farm sector, specifically crab farming and goat or sheep rearing. In addition, the RHL will develop climate-resilient homestead for vulnerable people living in low-

lying areas. Moreover, the UNDP project will promote crab fattening, which is mainly nature-based. The RHL project will support the crab fattening activity by supplying hatchery-produced crablets. On the contrary, the UNDP project will promote hydroponic fodder. The RHL project will use this fodder for goat orsheep to be raised under this project. Thus, these two projects will complement each other. The two districts are expected to overlap, but measures will be taken to avoid activity overlap and ensure the complementarity of the two projects.

31. **Extended Community Climate Change Project- Flood (ECCCP-Flood):** The GCF-financed project is to support increased resilience in flood-vulnerable communities in the northern region of the country. It is a four-year project with a total budget of USD 13.68 million, of which GCF grant contribution is USD 9.32 million. The project covers the riverine char areas of Jamalpur, Gaibandha, Kurigram, Lalmonirhat, and Nilphamari districts. Major project activities include raising homestead plinths, provisioning safe water and sanitation, promoting flood-adaptive crop production, and goat or sheep rearing in slatted houses. While some activities are similar, the ECCCP will cover the Brahmaputra-Teesta basin in the northern part of Bangladesh, and the RHL will support coastal areas in the south of the country.

32. Goats and sheep rearing in slatted house is a common activity of the ECCCP-Flood and RHL project. It is a proven pro-poor livelihood option in rural areas of Bangladesh. It requires low investment and a small space, and the operation cost is also very low. It is also critical for household nutrition. In addition, women can take care of goats and sheep at home. The other family members can also look after sheep in parallel with their main duties. The project will add improved technology and management systems, i.e., slatted housing systems, to the conventional goat and sheep rearing process to make it climate-resilient. The slatted house will protect goats and sheep from coastal flooding, storm surge and cyclone. PKSF has demonstrated several hundred slatted houses under Community Climate Change Project (CCCP), which was completed in 2016. The experience of PKSF found that goats and sheep are highly resilient to increased salinity. They are less affected by disease and require a minimum amount of feed for survival. On the other hand, their productivity is very high. Women can easily get involved. The response of the community to the technology in terms of adaptive capacity and economic benefit is significant.

33. The government has established the **Bangladesh Climate Change Trust (BCCT)** and allocated USD 480 million to implement various adaptation projects that include construction of embankments and river bank protective work, building cyclone-resilient shelters, excavation or re-excavation of canals, construction of water control infrastructures including regulators or sluice gates, waste management and drainage infrastructure, introduction and dissemination of stress-tolerant crop varieties and seeds, afforestation, installation of solar panels. In particular, the project has allocated BDT 40.0 million in Khulna and BDT 170.7 million in Satkhira for housing and rehabilitation of cyclone affected people; BDT 4.0 million in Satkhira for rainwater harvesting infrastructure; and another BDT 10 million for housing and pond sand filter (PSF) construction support in Satkhira Upazila. Although these activities are closely related to this proposed project, the shelters constructed under this project are mainly community-based cyclone shelters, whereas the proposed project will implement resilient housing for individual households with raised homesteads.

34. The government has also established and implemented the **Bangladesh Climate Change Resilience Fund (BCCRF)**. Most of the projects under BCCRF are related to the improvement of climate-resilient livelihoods through agricultural interventions, i.e., promotion of stress-tolerant crop varieties, cyclone shelter reconstruction, reconstruction of embankments, etc. In addition to these funds, the government has been implementing **Pilot Program for Climate Resilience (PPCR); Strategic Programme for Climate Resilience (SPCR) fund, Coastal Embankment Improvement Project, Coastal Climate-resilient Infrastructure Project, Coastal Towns Environmental Infrastructure Project, Climate-resilient Agriculture and Food Security, Feasibility Study on Climate- resilient Housing for Low-Income Communities and Climate Change Capacity Building and Knowledge Management projects and Least Development Countries Fund (LDCF)** for implementing adaptation activities including resilient livelihoods, construction and reconstruction of embankments, safe drinking water, and afforestation etc.

35. The Global Environment Facility (GEF) supports **‘Building climate resilient livelihoods in vulnerable landscapes in Bangladesh (BCRL)’** is to improve the resilience of people, communities, and ecosystems to climate change, and improve livelihoods through increased value addition in agricultural food systems. The BCRL is a landscape-based action research project. The BCRL project will develop climate-resilient high-value crop advisories in collaboration with Department of Agricultural Extension - DAE’s Bangladesh Agro-meteorological Information System Development (AMISDP) Project and disseminate the advisories using the same system adopted by AMISDP. It will also help improve the overall effectiveness of the advisories through process evaluation surveys of end-users, which will seek feedback on, for example, the understandability, timeliness, and relevance of information provided (activity supported from project budget, Climate Smart Agriculture Expert along with DAE’s Deputy Project Director and Project Implementation Coordinator). The project has been implemented in three landscape areas: coastal zone, hill tracts, and Barind tracts. In terms of geographical coverage, the BCRL and the RHL are coherent, i.e., two districts such as Khulna and Satkhira districts are common. The complementarity of these two projects is that the BCRL project focused on crop sector, whereas the RHL project focused on fisheries and livestock sectors. In addition, the RHL project will introduce climate-resilient homesteads for the poor and vulnerable community. In the future, the projects like BCRL will have opportunities to incorporate resilient houses for the farmers. So, the RHL project will complement this BCRL project in terms of climate-resilient homesteads for the farmers and promote climate-adaptive aquaculture, i.e., crab farming, which is one of the important sub-sectors of agriculture for avoiding duplication. In addition, a coordination mechanism has already existed at the central level, which is the Technical Advisory Committee (TAC) of the NDA (Economic Relations Division under the Ministry of Finance) to the GCF. The proposed project has already been presented to the TAC and it has received No Objection by incorporating its comments and suggestions. Again, the project will coordinate with the Department of Environment (DoE) which is the Executing Entity (EE) of the BCRL project, so that these projects are complement to each other rather than overlap.

36. The Adaptation Fund (AF) has approved ‘Adaptation Initiative for Climate Vulnerable Offshore Small Islands and Riverine Charland in Bangladesh in 2019’. The United Nations Development Programme (UNDP) is the implementing entity, and the Department of Environment (DoE) of the Ministry of Environment, Forests and Climate Change (MOEFCC) is the executing entity. The AF grant amount is USD 9,995,369. The project is being implemented in Rangpur and Bhola districts. Of these, only one upazila within the Bhola district is where the AF’s project and RHL’s potential activity, i.e., homestead development, will overlap. To avoid it, the RHL will collaborate with representatives of the AF project, DoE and UNDP in the site selection process and promote knowledge and lesson sharing with the AF project for higher effectiveness and efficiency.

Contribution to Policies and Strategies related to climate change

37. The proposed RHL is in the Bangladesh country programme for GCF under the title “Increase resilience to climate change in southwest coastal zones of Bangladesh through adaptive livelihoods, housing, and safe drinking water supply.” The title was slightly revised while maintaining the same project concept.

38. Bangladesh Climate Change Strategy and Action Plan (BCCSAP) 2009 is the key strategy document of the government of Bangladesh combating climate change. The strategy document includes six thematic areas that are: 1) food security, social protection and health, 2) comprehensive disaster management, 3) infrastructure, 4) research and knowledge management, 5) mitigation and low carbon development and 6) capacity building and institutional development. The proposed RHL is aligned with thematic areas 1, 3, 4 and 6. Additionally, it is aligned with the country’s National Adaptation Programme of Action (NAPA) 2005/2009 and also with National Adaptation Plan (NAP), which was published in November 2022. However, the BCCSAP document does not incorporate resilient homestead for the coastal communities. Theme 3 focuses mainly on the large-scale infrastructure development like coastal embankments and multipurpose cyclone shelters.

39. The government prepared the Climate Change and Gender Action Plan in 2013 to mainstream gender issues in all types of climate change-related development. It is expected that the project will involve mostly women, who make up more than 80per cent of the targeted community. Besides, the government of Bangladesh mainstreamed climate change issues in short-, medium- and long-term plans including the 8th Five Year Plan, Vision 2021, and Delta Plan. In addition, the government prepared a coastal zone policy in 2005 in order to utilize coastal resources in sustainable ways. The project will contribute to this policy by restricting crab extraction from nature. It will also comply with the National Fish Policy (1992), National Agriculture Policy (1999), and National Livestock Development Policy (1992).

40. The government has established BCCTF to be financed from the state budget to address climate change impacts (see Para 33). The fund is operated by Climate Change Trust Act 2010. It has also enacted the Environment Conservation Act (ECA)-1995, Biodiversity Conservation Act-2016, and Disaster Management Act 2012.

41. The Department of Environment (DoE), under the Ministry of Environment, Forests and Climate Change is the key institution for climate change issues in Bangladesh. The DoE has established the Climate Change Cell (CCC), which is dedicated to coordinating, facilitating, negotiating and implementing climate change issues. Climate change focal points have been set up in 13 ministries so far. The Economic Relations Division (ERD) under the Ministry of Finance is the National Designated Authority (NDA) of Bangladesh to GCF. Recently, PKSF and IDCOL have been accredited as Direct Access Entities (DAEs). PKSF as a DAE and also one of the leading development organisations in Bangladesh, always works in coordination with ERD, DoE, and other public and private sectors of the country.

Crab related policy gaps

42. The industry is taking preliminary steps in the right direction, with an annual historical increase in both demand and supply. While still in its nascent stage, the sector relies heavily on the collection of crabs from the wild for business, and with the increasing annual demand, this dependency on nature needs to be diverted to hatchery-based production in order to maintain ecological stability and supply-end sustainability. The government of Bangladesh acknowledges the potentiality of the crab sector, but it also means that proper policies need to be developed in order to nurture the industry from its nascent stage into a mature one. Presently, crab and crab-based production and export are controlled by the Government of Bangladesh under the Crab and crab-based production management rules 2019 (under the Wildlife Conservation Act 2012). Under these rules, the government controls crab collection from nature (wild), crab farming, and export. But there are still some gaps in this rule. Primarily, policies need to be reformed regarding the ban on wild collection. Secondly, a policy should also be formulated on the types of crabs that can be collected from the wild in order to protect the population from depleting. Lastly, there needs to be a formulation of export policy mandates for the exporters in order to ensure quality and brand are not compromised in the international market.

43. ***Breeding Policy****:* It is important that the Government of Bangladesh finalize a particular window regarding the ban on wild crab collection. While the current ban is in effect in January and February of every year, researchers claim this ban period is inaccurate, resulting in business losses during the peak export season.

44. ***Wild Crab Collection:*** While there are policies on the type of crab that can be collected from the wild, these policies have yet to be properly implemented. It needs to prepare a separate policy or manual for the wild collection in order to control the population depletion, particularly of the young crabs that are being collected from nature for the soft-shell farms, before these crabs can reproduce.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

45. The coastal communities of Khulna, Bagerhat, Satkhira, Barguna, Patuakhali, Bhola, and Cox's Bazar are enduring increasing climate challenges that are disabling their ability to develop sustainably. These communities live in highly vulnerable homesteads that put themselves and their livelihoods at risk during every slow and sudden extreme-climate event that occurs, leading to the loss of homes, livestock, and arable land, and crops. It is not only life threatening but also financially catastrophic for these communities.

46. Although the GoB has implemented several settlement projects in coastal areas, the activities of the proposed project will not overlap as these are urban housing projects (undertaken by Khulna Development Authority; infrastructure development of Patuakhali *Pourashava; Climate Change Adapted Urban Development (CCAUD) Programme for Barisal City Corporation* project, etc.). And understandably, vulnerable rural people do not have access to these projects.

47. Salt intrusion brought on by flooding and sea level rise is expected to continue, leading to more losses of productive lands, decreased land productivity and degradation of natural resources. Low elevation, and higher salinity levels in coastal zones are already strongly correlated with lower incomes and the migration of working-age adults, leaving behind women, the elderly and disabled, who have limited resilience to climate hazards. Without adaptation, the number of people facing coastal inundation and migration is estimated to grow by 2–7 million by the 2070s[49].

48. The ideal state is one where rural families and communities can withstand cyclones and storm surges so that there is no loss of life or livelihood; livelihoods can persist even with slow-onset changes to soil salinity and higher temperatures; and families are not forced to leave traditional villages for a life of poverty in cities because of financial loss and a lack of access to finance. But to get to this state, there are key barriers that must first be addressed.

Key barriers to a resilient developmental pathway for coastal communities in Bangladesh

Barrier 1: Lack of climate planning and implementation capacity among communities for addressing climate change

49. Severe climate change impacts are today being felt by Bangladesh's communities that are least able to protect themselves. Communities vulnerable to climate change are not organised, lacking knowledge of climate impacts and the capacity and skills to adapt to them. Institutions, both governmental and non-governmental, lack coordinated climate planning and implementation capacity, ultimately leading to communities that have very low adaptive capacity with little hope of improvement.

50. Although local-level institutions, including government and non-government organizations, understand development activities, there is a latency in mainstreaming climate adaptation into them. These institutions, especially those offering extension or advisory services at the local level, have limited understanding of appropriate adaptive responses in relation to their conventional activities, and in the case of aquaculture, there is very little technical expertise. As such, there is a gap both in sectoral guidance and in integrating climate change into regular development activities.

Barrier 2: Lack of access to finance to invest in large capital expenses

51. Despite the substantial expansion of bank branches and the increase in the membership of MFIs and other financial institutions, a large number of the country's adult population still remains excluded from appropriate financial services.[50] Target communities represent the poor and ultra-poor of Bangladesh, living on subsistence agriculture and wage labour. These communities suffer from a lack of awareness, are of low income or assets, suffer social exclusion, and have limited financial literacy, which act as barriers to accessing financial services[51]. As a consequence, they are highly risk-averse to investment and largely fail to obtain the means to minimise additional threats from climate change. They have the least access to loans from commercial banks. While some NGOs provide loans to them, the loan sizes are inadequate when compared to their needs, especially for relatively expensive investments in climate-resilient homestead infrastructure and capital investments for their livelihood diversification. For this reason, most houses of the rural poor in the coastal zone are made of mud and sticks, which are no match for the cyclones and storm surges that plague this region, leading to total loss of assets and sometimes lives.

Barrier 3: Lack of livelihood diversification and limited capacity to adopt adaptation technologies

52. Communities lack the know-how and capacity to identify and implement alternative resilient technologies and practices that would otherwise mitigate climate threats, upscaling traditional farming

[49]Climate Risk Country Profile: Bangladesh (2021): The World Bank Group

[50]Bangladesh Bank: 2019. Survey on Impact Analysis of Access to Finance in Bangladesh

and diversifying activities are key to increasing their adaptive potential. However, target communities lack knowledge and are inherently risk-averse to switching to new practices (e.g., adapting to saline-tolerant crop varieties), technologies (e.g., drip irrigation, hydroponics, aqua-geoponics), and sectors (such as, crab farming). The barrier is further exacerbated by a lack of technical knowledge and support owing to limited capacities among peers and local extension staff. Extremely poor households are often unable to make informed risk management strategies, given that they lack information, and are continuously coping with new shocks and stresses, and have limited space and ability to think about and act on choices that will make a positive difference to their future (relative to households that are relatively more resource-endowed).

Barrier 4: Capacity, standards, supply and market access in developing a community-based crab supply chain

53. Soft- and hard-shelled crab meats are in high demand both locally and internationally. Crab exports from Bangladesh are increasing over time, and so is local consumption among crab producer and non-producer communities. According to World Trade Organization data, Bangladesh is the 7th largest crab exporting country. Between 2010 and 2018, Bangladesh's crab exports increased three times. China is the top export destination country, followed by Thailand. Other market destinations are the USA, Malaysia, Singapore, Japan, and Australia.

54. To meet this demand, wild crabs have been harvested mostly from the Sundarbans by poor communities. However, wild crab stocks are falling, posing great risk to the industry and to the livelihoods of the vulnerable poor. In addition, wild crab harvesting poses risks from tiger attacks and kidnapping by bandits, especially of women. The markets often disadvantage the poor and women, who find it hard to obtain a fair price. Demand for crab, however, is growing both in domestic and international markets. The species is resilient to increased salinity, which poses a livelihood and nutritional opportunity for coastal communities. To maintain a sustainable supply of crab eggs, a hatchery is needed, but technology and technology transfer have been lacking to do this. In addition, the lack of extension capacity, investment shortfalls, and poor bargaining capacity of small-scale farmers lacking direct access to markets limit their ability to establish hatcheries and robust but sustainable supply chains in the target regions.[51]

55. The current crab industry suffers from a lack of quality and food safety standards that hamper its growth, especially as an export commodity. There is a lack of training and guidelines on husbandry standards and activities that mitigate contamination and the spread of disease.

56. In addition, crab feed relies on the importation of *Artemia* (a copepod used as feedstock for crabs) from Vietnam, which has a limited supply and could limit the growth of a community-based industry.

Project Goal and Change Theory

57. Therefore, **if** target coastal communities in Bangladesh employ weather-proof construction, have access to resources and capacity to implement climate-adaptive livelihoods, and communities with the support of institutions are knowledgeable of adaptive strategies and planning and can coordinate responses, **then** coastal communities in Bangladesh will be able to safeguard assets and livelihoods into the future **because** homes will provide protection during future extreme weather events, livelihoods will be maintained in the face of climate hazards and local institutions will be able to provide appropriate and timely support.

58. This project's principal aim is to reduce the risk to the lives and assets of the most vulnerable people living in the coastal zones of Bangladesh and provide them with improved and sustainable alternative livelihood options in the face of extreme climate challenges. It will directly benefit approximately 362,475 people in the target region and have transformative impacts.

59. To meet the needs of the people, at the beginning of the project, communities will be organised into climate change adaptation groups with gender equity to support and provide valuable local knowledge

[51]Lahiri, T, Nazrul, KMS, Rahman, MA, et al. Boom and bust: Soft-shell mud crab farming in south-east coastal Bangladesh. *Aquaculture Research*, 2021; 52: 5056– 5068.

for the implementation of the proposed interventions (outcome 3). Capacity will be built within them so they can seek finance, coordinate and develop further adaptation ideas, and disseminate their knowledge to their peers and communities. Local implementing entities and local governments will also be trained and included throughout the project tenure in order to build coordination, planning, and implementation capacity amongst them that will last beyond the life of the project. This project will transform the way institutions are able to support local communities by tackling development through a climate change lens. Coordination, planning, and implementation will become more efficient for meeting the needs of communities for this and future projects.

60. Together, these actions will create empowered and gender-sensitive adaptive communities throughout the coastal region of Bangladesh that will have greater capacity for climate planning and implementation (barrier 1).

61. The project will combine a mixture of loans and grants that will provide beneficiaries access to otherwise inaccessible resources and sources of finance (barrier 2), addressing a major barrier to the development and security of coastal rural communities. The GCF will provide the grants, whereas the PKSF will provide co-finance in the form of loans and in kind. The loan will be used to implement activities under outcome 2{activity 2.1.2, 2.2.2, 2.2.3 (partial), and 2.2.4 (partial)} whereas the in-kind will be used as a partial cost of PMU (salary of the Project Coordinator) and office rent.

62. This project will implement activities that will provide life and asset protection during extreme climate events, which are a priority for all coastal dwellers (outcomes 1 and 2). The construction of climate-resilient housing to the highest standards for the poorest families will protect dwellers from flooding, tidal surges, and cyclones (output 1.1). In the process of homestead development and house construction, community and institutional groups will be involved to build their capacity by exchanging knowledge. Homesteads will include additional important features, such as rainwater harvesting facilities (1000 litre capacity) to buffer the loss of freshwater.

63. The construction of resilient houses will provide multiple benefits in addition to the protection of homeowners and their families. These include the reduced need for rebuilding and repairs of existing houses in depleted condition after extreme events, saving money and time; greater security, especially for women and girls (co-benefit 1); greater asset protection from looting with valuables being held on site; and the pride of owning a concrete home. In this context, these homes will be transformative for the lives of the beneficiaries.

64. Furthermore, this project will upskill local masons, through supervision by qualified engineers employed by the project, in the design and materials used in construction and provide capacity building to government personnel so that institutional knowledge, advisory and oversight can eventually be provided to ensure safe standards, ensuring that communities will have access to the building of safe, and resilient homes after the end of the project. The project will ensure that the innovative building designs will be freely available for future use, while PKSF will continue to work with the National Housing Authority and House Building Research Institute to sensitize them through seminars and policy dialogue to shape government policy on resilient design standards.

65. Since livestock are valuable assets and are highly vulnerable to extreme climate events (livestock shelters are especially vulnerable to climate events) this project seeks to introduce innovative design to protect them (output 2.1). Hence, this project will combine the innovative housing design with livestock protection through scaling the ECCCP-Flood project (see above) by constructing climate resilient animal shelters and providing locally adapted capacity building for building methods and design for target communities so these important family assets can be protected during storm surges and cyclones. The building design also avoids lingering moisture within the pens, minimising the risk of disease often associated with such climate hazards. The provision of loans to buy livestock will enable small farmers to invest in these crucial assets while also being able to protect those assets through resilient shelters. PKSF, as an AE, will co-finance the loan to buy livestock animals through the selected IEs (mainly Partner Organisations of PKSF).

66. Although vulnerable families and assets will be protected from the immediate effects of climate extremes through resilient homesteads, slow-onset SLR and frequent flooding still directly affect their rural livelihoods. In order to comprehensively affect the lives of these communities, they must also have knowledge and access to resilient livelihoods (outcome 2). To address the loss of production due to soil salinity issues and knowledge and access barriers, training and resources will be provided to target communities for them to employ climate-resilient homestead agriculture (output 2.1). Salt-tolerant varieties and methods will be used to diversify outputs and maintain yields even in the face of increasing soil salinity (barrier 3).

67. To ensure replication, the project will engage respective local government offices while implementing livelihood activities and providing training and technical inputs. The project will arrange exchange visits between communities and institutions to share the results and knowledge of the activities. In addition, PKSF will continue financing best practice activities through its core programmes to partner organisations (POs) that will support replication in other areas.

68. Identifying and implementing climate-appropriate diversified livelihood opportunities for the coastal communities of Bangladesh is a great challenge. The crab industry, however, poses a particular opportunity due to the salt tolerance of the species and the relative ease of rearing them. The sector currently provides livelihoods to about 0.4 million people, the majority of whom are engaged in unsustainable wild crab and crablet collection and trading. PKSF's successful development of a crab hatchery in Satkhira district in 2016, which has been producing crablets ever since, now means that there is a unique opportunity to create a sustainable rural crab industry and supply chain (barriers 3 and 4) if the challenges of investment and technology transfer and capacity building can be met. This project aims to do just that by developing micro-hatcheries throughout the target regions that will develop crablets and sell them to secondary and tertiary crab farmers supported by the project until they reach full size (output 2.2). To ensure food safety and husbandry standards, clear growing and safety guidelines and standards will be promoted and a supply chain enhanced that meets the needs of rural poor populations while meeting the demands of internal and external markets (output 2.2). In addition, this project will maintain strong research links and provide input into the development of local crab feed production as well as monitoring and mitigation of disease risks (output 3.2). The provision of support for technical capacity and supply chain development will ensure that the market will endure after the project ends. Close links with government fisheries departments will be used to advocate for national crab production standards that will standardise practices in health and safety.

69. Outcomes 1 and 2 will not only deliver climate benefits but also a number of significant co-benefits attributable to the project. These are increased security for women and girls by being able to stay at home during an extreme climate event and avoid harvesting wild crabs (co-benefit 1), decreased urban migration since the project will reduce economic push factors (co-benefit 2), increased local secondary businesses that service the livelihoods created from this project (co-benefit 3), and an increase in wild crab populations due to the reduced pressure put on them from harvesting, which will ultimately lead to a balanced and more resilient ecosystem (co-benefit 4).

70. Finally, it will be necessary to provide ongoing support to current and future beneficiaries during and after the project, and to do this, this project will build knowledge and capacity amongst beneficiaries and institutions through the training of technicians and local government, especially in aquaculture techniques, supply chains, and standards, while knowledge products will be created so that information and practices can be continued and applied locally after the project ends (outcome 3).

71. Livelihood interventions supported through this project, using both grants and loans, as clarified in Para 62, will lower the threshold for adoption of resilient livelihoods for vulnerable and poor communities that are traditionally risk-averse to change. Through the demonstration of new livelihood strategies employed by target beneficiaries, the project will enable other indirect beneficiaries to replicate climate adapted activities, thereby increasing the project's impact. Real-time evaluation will ensure that data is captured throughout the project and fed back into adaptive management processes that will improve implementation during and after the project end. The knowledge products and their dissemination will ensure that the project's best practices are replicated throughout the coastal zone region of Bangladesh.

72. The activities and results developed in this proposal will therefore create communities and systems that are transformed into a sustainable and climate-resilient development pathway.

Risks to the project

73. The project relies on the implementation of structures and the planting of trees and growing crops. All of these are vulnerable to extreme climate events, such as cyclones and storm surges when they are immature or partially built. This project will take precautionary measures to protect these assets during their most vulnerable periods and provide training on care and maintenance of those assets to beneficiaries.

74. Shrimp farming is in decline due to a number of reasons, with disease being one of them. Crab diseases that affect farm-raised crabs have not been identified, but it is essential that consistent monitoring and safety guidelines are adhered to. This project will mitigate any risk by implementing monitoring protocols, training beneficiaries in husbandry and food standards, and maintaining research links to diagnose disease quickly so that there is no chance of an outbreak.

75. There is always a risk that a project's capacity and knowledge will only available to those who have received direct training. Such project knowledge must stay inside the institutions and be easily transferable to other institutions if scalability and replication are to be achieved. By creating, gathering, and disseminating knowledge products, this project will ensure that institutions, communities, and not just individuals, are building their capacity.

Assumptions

76. Women will be able to participate as equal partners in the project. All activities that include beneficiaries will include women in equal proportions.

77. Concessional loans will provide enough of an incentive for crab farmers and goat rearers to invest in these climate-resilient livelihood options.

78. Climate models provide an accurate indication of future SLR and cyclonic activity and ensure that homesteads are designed appropriately for actual future climate.

79. Community groups have the desire to work together and address climate challenges together. All indications suggest this is true.

80. Crab farmers will find and maintain market linkages after the project. Through training and the development of guidelines, standards and network, this project will create an environment that will make it simple to maintain market linkages and sell crabs to both in domestic and international markets.

81. Knowledge creation and dissemination will lead to replication. PKSF will do all it can to make sure that knowledge of the project and best practices lead to replication.

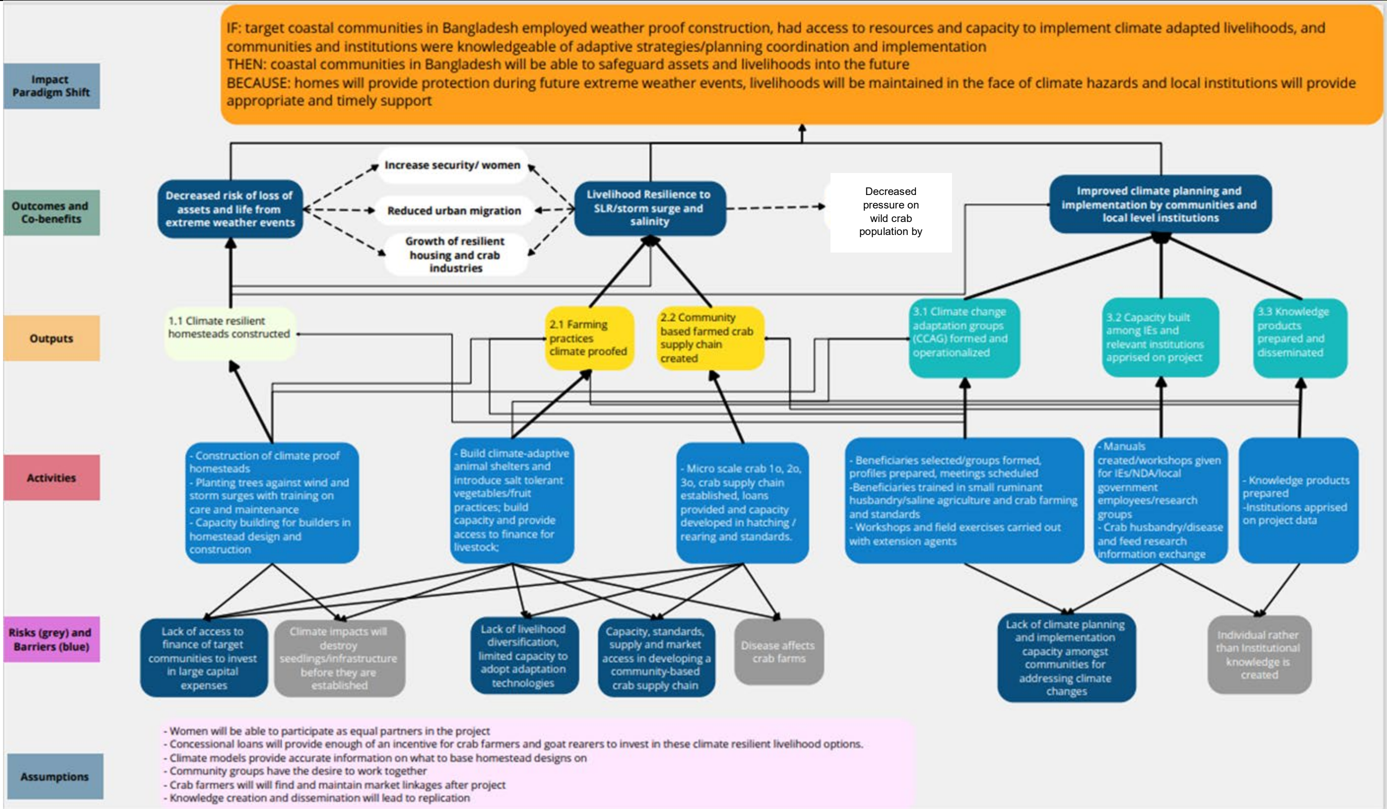


Figure 5: Theory of Change Diagram (broken lines represent co-benefit pathways; heavy lines represent direct attribution, thin lines indirect or weak attribution**)**

Co-benefits

Co-benefit 1: Decreased risk of assault on women

82. Crablet harvesting is a dangerous activity due to large land predators (tigers) and the presence of bandits that regularly kidnap women and girls, while they are out in the forest. This project will help women and girls avoid having to enter the forest for crab harvesting. In communal shelters, women and girls face various difficulties, including cooking, drinking water collection, sanitation, sexual harassment, etc. This project will mitigate this concern by constructing private family homes.

Co-benefit 2: Reduced Urban Migration

83. Vulnerable people from flood- and cyclone-affected regions are leaving their livelihoods and their communities at a faster rate than others from other regions, often because of grave financial losses from the destruction of livelihoods and assets from climate events. This project will reduce urban migration by reducing the push factors, namely the loss of and inability to develop livelihoods, involved in this phenomenon.

Co-benefit 3: Growth of crab and resilient housing industries

84. All communities in the coastal regions of Bangladesh are vulnerable to cyclones and storm surges. This vulnerability also has a negative effect on non-agricultural livelihoods and local economies. This project will indirectly support local service providers who provide building, transportation and labour to beneficiaries of this project.

Co-benefit 4: Decreased pressure on wild crab populations by industry

85. Wild crabs are being overharvested due to high demand and a lack of ability to create a full in vitro life cycle that is appropriate for poor communities. This project will transfer in vitro technology to beneficiaries so that wild crabs do not need to be harvested. This will reduce pressure on wild populations and allow them to recover to baseline population sizes.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

As presented in the diagram above, the project expects three outcomes and five co-benefits (broken line boxes). The table below presents outcome mapping as well as the types of co-benefits. Co-benefit 1 is related to environmental co-benefit, co-benefit 2 is relevant to social and gender issues, and co-benefit 3 is aligned with economic co-benefit.

Outcome	Aligned with result area
Outcome 1: Decreased risk of loss of assets and lives from extreme weather events	The GCF's Adaptation Result Area (ARA) 1: Most Vulnerable People and Communities, is closely related to this outcome. However, it will also indirectly aid in achieving ARA 2: health, wellbeing, food security, and water security because safe housing and access to clean water will improve the beneficiaries' general health. This outcome is also aligned with ARA3 because the RHL project will build climate-resilient homesteads.
Outcome 2: Livelihood resilience to SLR / storm surges and salinity	Outcome 2 is aligned with ARA 2: Health, well-being, food and water security, because it will promote climate-resilient livelihood options that will ensure well-being and food security.
Outcome 3: Improved climate planning and implementation by communities and local level institutions	Outcome 3 is aligned with ARA 1 because trainings and meetings under this outcome will help people and institutions to adopt climate-adaptive technologies and practices, which would reduce the vulnerabilities of coastal people.

Co-benefit	Aligned with result area
Co-benefit 1: Decreased risk of assault on women	Social and Gender co-benefits. Women and girls will be able to avoid communal shelters and harvesting crabs from the forests.
Co-benefit 2: Reduced urban migration	Social co-benefit. The poor will be able to protect economic assets and livelihoods from climate events and hence not be “pushed” to urban areas
Co-benefit 3: Growth of crab and resilient housing industries	Social and Economic co-benefits. Increased flow of money across society benefitting local non-agricultural livelihoods.
Co-benefit 4: Decreased pressure on wild crab populations by industry	Environmental co-benefits. Crab populations will increase due to reduced harvesting – rebalancing ecosystems.

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	**MRA 2** Low-emission transport	**MRA 3** Building, citiies, industries, appliances	**MRA 4** Forestry and land use	**ARA 1** Most vulnerable people and communities	**ARA 2** Health, well-being, food and water security	**ARA 3** Infrastructure and built environment	**ARA 4** Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☐	☒	☒	☒	☐
Outcome 2	☐	☐	☐	☐	☐	☒	☐	☐
Outcome 3	☐	☐	☐	☐	☒	☐	☐	☐

Co-benefit number	Co-benefit					
	Environmental	**Social**	**Economic**	**Gender**	**Adaptation**	**Mitigation**
Co-benefit 1	☐	☒	☐	☒	☐	☐
Co-benefit 2	☐	☒	☐	☐	☐	☐
Co-benefit 3	☐	☒	☒	☐	☐	☐
Co-benefit 4	☒	☐	☐	☐	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

86. The primary goal of the project is to develop climate adaptive coastal communities in Bangladesh through the adoption of climate-resilient livelihood options, technologies and housing. To achieve this, the project will invest in the following outcomes:

(i) decreased risk of loss of assets and lives from extreme weather events;
(ii) livelihood resilience to SLR / storm surge and salinity; and
(iii) improved climate planning and implementation by communities and local-level institutions.

Outcome 1: Decreased risk of loss of assets and lives from extreme weather events

87. Studies have found that more than three-fourths of households in coastal areas are vulnerable to intensive precipitation, cyclones and storm surges, and coastal flooding due to perishable materials. To sustain the livelihoods, the proposed project will provide support to construct climate-resilient housing. The concept of climate resilient housing under the project includes raising homestead plinths above flood or tidal surge level, constructing and/or reconstructing houses with concrete pillars that are resilient to climate change and associated shocks (i.e., cyclone, storm surge, tidal surge, coastal flooding, etc.), construction of climate-resilient sanitary latrines, a rainwater harvesting system, a homestead gardening system, and tree plantations around the homestead area. Resilient housing is very important for building the resilience of the affected community because they have to spend much of their income on repairing their houses each year during post-monsoon period, compromising their income, food and nutrition security.

Output1.1: Climate resilient homesteads constructed

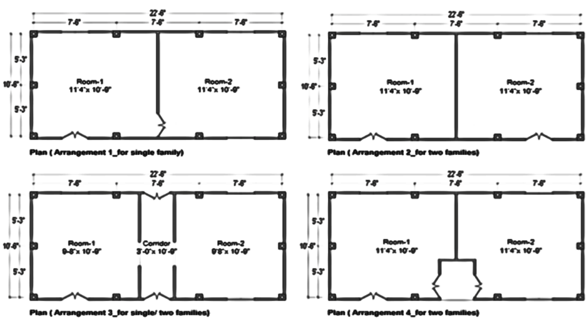
Activity 1.1.1: Design and building of homesteads

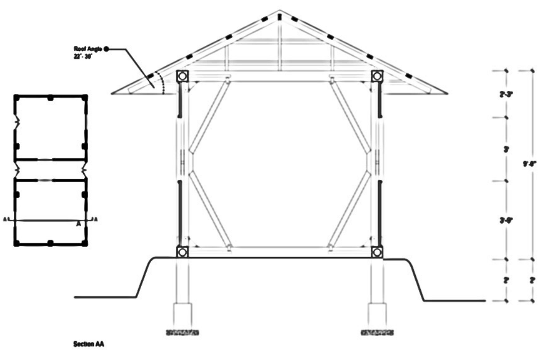
88. The project will select 13,500 beneficiaries based on the selection criteria stated in Para 126, constructing 3,000 climate-proof houses. The IEs staff will carry out consultation meetings in the selected areas to select the beneficiaries. They will prepare a draft list of beneficiaries and submit it to

PKSF for approval. The PMU representatives from PKSF will visit to verify the beneficiary list and provide approval for the final beneficiaries. The same process will be followed for all the activities. The core structure will be made of bricks and concrete. The house will be rectangular in shape, and the shorter edge will face the direction of the wind, which will reduce overall wind exposure. Secondly, the side of the roof with the greater slope will face the wind, and the roof will not extend beyond the wall, which will allow the wind to pass over the roof without affecting the core structure. Moreover, the roof will only be attached to the core structure and will not be supported by any walls, so even if the roof is affected, the walls will remain safe. Furthermore, there would be some gaps between the wall surface and the roof, so when the wind hits the roof, the thrust would be reduced.

89. In coastal areas, salinity-contaminated water is used in the brick field to mix sand and soil for making bricks. Similarly, during the construction, sometimes people use salinity-contaminated water. As a result, salt molecules remain in the construction materials. It absorbs moisture from the air during the monsoon and causes damage to brick walls. The project will ensure that the bricks and concrete are not made with saline water or soil. Fresh water will also be ensured during the construction of the house. In addition, there is local and indigenous knowledge in the coastal community about how to remove salt from water. The use of tamarind (*Tamaindus indica*) and some weak acids in treating saline water are examples of these types of local practices. PKSF has reviewed the recently carried out CTCN model of coastal housing in Bangladesh, the post-cyclone Sidr rehabilitation project of the Government of Bangladesh, and a climate resilient housing study carried out by ADPC with financial support from ADB. PKSF also reviewed coastal housing models suggested by the Bangladesh National Housing Research Institute (details can be seen in the pre-feasibility report). The review results show that most of the models put emphasis on local materials, wind speed, and tidal surge when considering the design. But the key gap in the models is that they did not consider raising the plinth of the house to make more resilient against coastal flooding due to storm surge, sea level rise, etc. But it requires raising the whole homestead area, along with the tree plantation, against the wind. This project will follow the existing design of the selected houses and use materials suggested by the House Building Research Institute (HBRI) for the low-income community, along with raising the whole homestead area. The cost of the activity will be contributed by the participating household in kind, i.e., their own labour, and the value of land that is not specified. Each beneficiary would contribute at least 5per cent of the total costs.

Draft engineering plan and structural design is presented below:





Description of materials

a) For reinforced concrete (RCC) building/housing: The whole frame structure (footing, column, grade beam, slab, and lintel) will be established.

b) If the building will be built by RCC, the exterior and interior walls will be made of brick masonry and finished off with cement plaster.

c) For semi-pucca buildings: The foundation and all walls will be constructed with the help of brick masonry. The roof will be made of CGI sheet roofing with wooden or MS angle trusses. Finally, all civil works will be finished off with cement plaster.

d) CGI House: Strong RCC pillars having capacity to withstand wind velocity at high speeds will be used in CGI (Corrugated Galvanized Iron) houses. The plinth will be made of brick masonry, PCC (Plain Cement Concrete) casting will be done on floor, and walls. The roofing will be made of CGI sheet.

The CGI-roofing will be above wooden or MS angle trusses. In this type of house, all sorts of strengthening techniques will be taken into consideration, like-horizontal and diagonal stiffeners on all walls.

e) During the construction phase, local builders will receive on-the-job training on resilient housing standards, designs, and materials. Through this capacity building, local skills will be ensured, while standards will be updated to meet the demand for new builds.

f) In rural areas, the union parishad (the lowest administrative tier of the government) is the approval authority for house plans. The beneficiary will get the approval from union parishad. The IE will facilitate the beneficiaries getting the approval, and PKSF as the EE will ensure that the design is approved by the local authority.

90. This component is closely linked with the livelihood component. This climate-adaptive homestead will reduce their income erosion and help them increase their resilience against climate change by avoiding significant amount of their earnings and time for house reconstruction due to damage from extreme events. The houses are easy to maintain at almost zero cost. The maintenance is related to occasional checks of the reinforcing materials and the daily cleanliness of the houses, which is largely a routine or daily task of family members in the house. This activity will use grant funding from the GCF.

Activity 1.1.2: Homestead tree planting

91. Tree plantations around the selected house are an approach to making the house resilient to the storm and tidal surge. Trees slow the wind speed of the storm, causing the least damage to the house. It will also improve the ecosystem and biodiversity of the selected areas. The project will promote planting trees around the proposed supported homesteads. Trees have the capacity to reduce the impacts of storms and tidal surges. The tree species will be selected in consultation with the local community. The project will put emphasis on local species that are resilient to strong winds.

92. The project will provide tree saplings (mangrove species) to the selected 20,000 HHs (90,000 beneficiaries), including 13,500 of the most vulnerable people (see outcome 3 below for selection criteria in para 122). More than 400,000 windbreak trees (or 20 saplings per HH) will be planted.

93. The IEs will be responsible for ensuring tree plantations. The procurement of tree saplings will be done in close consultation with the beneficiaries. The GCF grant will be used for tree plantation activities.

Outcome 2: Livelihood resilience to SLR/storm surge and salinity

94. A large portion of the coastal population is highly exposed to climate change impacts due to increased sea level rise, salinity in water and soil, the intensity of cyclones and coastal flooding. These pose a significant threat to agriculture, brackish aquaculture, and open-water fishing. The majority of them are poor or very poor. A recent study by UNDP shows that 16 and 35 percent of those living in Khulna and Satkhira respectively are extremely poor [52] whereas the national average is 12.9 percent. Gender inequality prevails in these districts through various societal and cultural norms that shape women's day-to-day activities as well as their capacity to adapt to climate change. For example, women have less decision-making power within the household and the workplace and are expected to manage the household and care for the family. Compounding these factors, climate change aggravates the burden of unpaid care work, and create a cycle that undermines their climate-resilient livelihoods.

95. The UNDP study has identified eight livelihood options, which are gender-responsive and climate-resilient. These are: a) crab farming and trading; b) crab rearing; c) aqua-geoponics; d) hydroponics; e) plant nursery; f) sesame cultivation; g) homestead gardening; and h) crab and fish feed processing. The other important pro-poor, gender-responsive and climate-resilient livelihood option is goat or sheep rearing in slatted houses, which has been implemented by the Community Climate Change Project (CCCP) of PKSF and the World Bank. These livelihood options are salinity-friendly, suitable for women's

[52] UNDP, 2017, Climate resilient drinking water infrastructure based on demand-supply and gap analysis for 39 unions of 5 upazilas under Khulna and Satkhira district, Submitted to UNDP by Water Aid.

participation and empowerment, and appropriate for the local market system. However, communities require additional support in the form of capacity development and acquiring assets to allow a smooth transition to these climate resilient livelihoods.

96. The proposed RHL will implement goat or sheep rearing in slatted houses and the fruit-fish-fibre model, which considers a combination of crab hatchery and farming, homestead vegetable cultivation, and fruit trees, and mangrove plantations. Critical elements for success of the proposed interventions would be: a) capacity building of participants particularly women, b) adequate and suitable access to resources for beneficiaries and other value chain actors, c) collaboration between government and non-government institutions, d) private sector engagement and improved climate change adaptation knowledge, and practices. The project will provide technological support and capacity training to the selected beneficiaries in promoting saline-resilient technologies and practices, particularly in the agriculture sector. This outcome will be supported through a combination of grant and loan financing. The loans from PKSF as co-finance to beneficiaries will be used for50 purchasing goats or sheep and GCF grants will support building goat/sheep houses. The project will provide technical support and seeds to the selected households for vegetable cultivation, while the cost of production will be financed by the households as per current practice.

Output 2.1: Traditional farming practices climate proofed

Activity 2.1.1: Construction of slatted houses for goat/sheep rearing

97. There is a scarcity of baseline information on the homestead-level goat and sheep population throughout the country. But it is observed that almost every household in rural areas rears either goat or sheep, or both. Small ruminant rearing requires a small space and is considerably less expensive making it a good option for poor households. Most importantly, women are mainly involved in goat and sheep rearing because their management is easier than that of large ruminants like cows or buffaloes. However, farmers often face recurrent economic losses through the deaths of livestock from floods and storms.

98. The RHL will support 20,000 HHs (90,000 beneficiaries) through sheep/goat rearing in slatted houses. The IEs' staff will be responsible for selecting the beneficiaries based on the selection criteria described in Paragraph 121 (Those who do not have financial capacity to purchase livestock and have capacity to rear livestock; and women-headed households and households with disadvantaged members will be given priority; poor and ultra-poor households (as defined in the Household Income and Expenditure Survey (HIES 2016) of the Bangladesh Bureau of Statistics (BBS-2017)); and have access to homestead land).They will carry out consultation meetings in the selected areas to select the beneficiaries. They will prepare a draft list of beneficiaries and submit it to PKSF for approval. The PMU representatives from PKSF will visit to verify the beneficiary list and provide approval for the final beneficiaries. The proposed project will provide financial and technological support for improved goat and sheep management as an alternative livelihood for vulnerable farmers who do not have a sustainable year-round income. The CCAG members will arrange for the builders to construct the slatted houses. Once the houses are completed, PMU staff will visit them for quality assurance. Then the payment will be made on a master roll basis. This activity will be carried out mainly by the female member of the selected household. This will empower them by contributing to family income and reducing their vulnerability to climate change. The slatted house and capacity building and technology part of the activity will be grant-financed from GCF and buying goat or sheep will be financed from PKSF as loans.

The slatted houses will be constructed on raised areas at a height above flood level. It will be constructed with wood or bamboo, and the roof will be made with rust-protected corrugated iron sheets, which are resilient to storms and associated surges. Iron-made angles will be used to align the roof with the walls. Thus, the slatted houses will be resilient to climate change.

Activity 2.1.2: Provide financial support for goat/sheep rearing

99. Most of beneficiaries, selected for this component are poor, ultra-poor and female-headed households. They do not have enough financial capacity to purchase goats and sheep on their own. The project will provide loans for them to purchase the goats and sheep. PKSF will distribute loans among the selected beneficiaries. These beneficiaries are the same as in Activity 2.1.1 and selection criteria are presented in Paragraph 121.

Activity 2.1.3: Introduce the cultivation of saline tolerant vegetables within homestead areas

100. Homestead-based vegetable cultivation is important to meet nutrition demands and enhance income. However, soil salinity hampers the production of vegetables at the homestead level. There are some saline-tolerant species, such as brinjal, carrot, tomato, and cauliflower, but their threshold tolerance is less than 10 ppt salinity whereas in many areas, salinity exceeds 10 ppt, particularly during the dry season. The project will select 90,000 beneficiaries in consultation with the CCAGs for cultivating saline-tolerant vegetables in the homestead areas based on criteria stated in Paragraph 122. The IE staff will facilitate the consultation meetings for selecting the beneficiaries. They will prepare a draft list of beneficiaries and submit it to PKSF for approval. The PMU representatives from PKSF will visit to verify the beneficiary list and provide approval for the final beneficiaries. The project will provide technical support as well as saline-tolerant seeds to the selected households using grant finance from GCF. This activity is especially appropriate for women, who will be able to further contribute to their family's nutritional intake and income through the increased capacity that they acquire through this activity.

Output 2.2: Community-based farmed crab supply chain created

101. Crab farming is relatively mature in Vietnam and practised widely. The Center for Education and Community Development (CECD) found that Bangladesh has natural conditions similar to Vietnam, particularly for crab farming. It is a tropical monsoon climate, hot and humid all year. The average summer temperature is 24- 39^{0}C; and in winter, from 18 - 23^{0}C. With such climatic conditions, the feasibility of crab hatchery production in Bangladesh is high. PKSF and partners have identified locations where small-scale crab hatchery farms could be developed in the coastal districts of Satkhira, Cox's Bazaar and Patuakhali.

102. The project will establish 50 crab hatcheries, which will support a sustainable crab supply chain. The 50 hatcheries will provide services to crab growers by producing juvenile crabs and contributing to a growing crab aquaculture industry in the country. Although the demand for juvenile crabs is high, the project will implement only 50 hatcheries because the capacity of crab-hatchery trainers is limited. The project will follow a "learning-by-doing" approach to develop capacity among hatchery owners and staff. The capacity development will generate knowledge and best practice on crab hatcheries and crab farming, which will be compiled and disseminated among communities and other stakeholders, including policy-makers, through workshops, seminars, consultations, and mass gatherings.

103. The hatchery owner will be provided with the necessary training on crab breeding, hatchery management and maintenance. They will also receive loans from PKSF to develop their hatcheries. Each of the hatcheries will require two staff members leading to the creation of an additional 100 jobs in the selected coastal districts. The project will also develop guidelines and training manuals on food safety and provide training to the various actors in the crab supply chain.

Activity 2.2.1: Development of crab hatcheries (1° stage)

104. The proposed project will establish 50 small enterprises of micro-crab hatcheries. The IEs with guidance from PKSF and in consultation with communities will select the hatchery owners based on the criteria stated in paragraph 123. The hatchery owner will be provided with training on crab breeding, hatchery management, operation, and maintenance. They will also be eligible for loans from PKSF. This loan will partly cover the construction and preliminary costs of operations. A part of the construction costs will be covered by grant funding from GCF. Total initial investment for a hatchery is estimated at US $18,000. Of this amount, the grant from the GCF is estimated at USD10,000 as it is for adaptation technology transfer costs and PKSF will provide co-financing in the form of a loan for operational costs. So, the GCF-PKSF ratio is 10,000: 8,000, i.e., 1:0.8. Each hatchery will require two staff. So, this activity will create an additional 100 jobs in selected coastal districts. Employment will target a gender ratio of at least 50per cent women for both training and crab hatcheries. The salary of the hatchery staff will be paid by the hatchery owners. The hatchery owner may receive a loan from the PKSF from its co-finance proceeds under this project. Beyond the project, the hatchery staff will come from crablet sales. It is to be confirmed that GCF proceed will not be used to pay the salary of hatchery staffs.

105. A crab hatchery at the entrepreneur level has an estimated production capacity 100,000 juvenile crabs per cycle, will have 6-8 larvae rearing tanks (5.68 ft x 5.68 ft x 4.87 ft), 2-4 treatment tanks (5.68 ft x 5.68 ft x 4.87 ft), 2 filter tanks (4.6 ft x 4.6 ft x4.6 ft), 2 sea water treatment tanks (16 ft x 13 ft x 4 ft), and 1 overhead tank (5ft x 5 ft x 5 ft). The filter tank should be set up in a way that its bottom is at least 1 foot higher than the larvae rearing tank. It will ensure a smooth flow of water from the filter tank to the larvae rearing and treatment tanks. The larvae rearing tanks, treatment tanks, and adult female rearing space will be located in one section of the brick house and maintained at an optimum temperature through passive control. Water collected from the deep sea should be stored in a reserve tank. Sea water will be treated with chlorine, and then, after dechlorination, it will be filtered by the filter tank. The filtered water will be transferred to the larvae rearing tank for Zoea rearing (day one larvae).

106. Currently PKSF is planning to establish the hatcheries in Cox's Bazar to avoid the cost of collecting deep sea water. The water quality on Cox's Bazar coast meets the salinity requirement for operating crab hatcheries (around 25 ppt). The project will use small pipelines to collect saline water from the sea.

107. Every larva rearing tank and treatment tank will be connected to the filter tank by plastic pipe to ensure a smooth supply of filtered water. All tanks, including the reserve tank, in this section will have connections with blowers for the continuous supply of oxygen. Every tank will have an electric light hanging from the ceiling in the middle of the tank for a continuous light supply. For an uninterrupted power supply, a generator will be installed in the hatchery area. A microscope will be used for the observation with the zoea, housed in the hatchery, to identify and diagnose problems of the zoea, if they occur. For wastewater drainage and cleaning purposes, a proper drainage system will be constructed inside and outside the hatchery.

108. For installing generators, storing different equipment and other purposes, there will be a space outside of the hatchery section but not far from the hatchery. There needs to be a living room if the hatchery is far away from the residence of the entrepreneur.

The design of the hatchery is presented below:

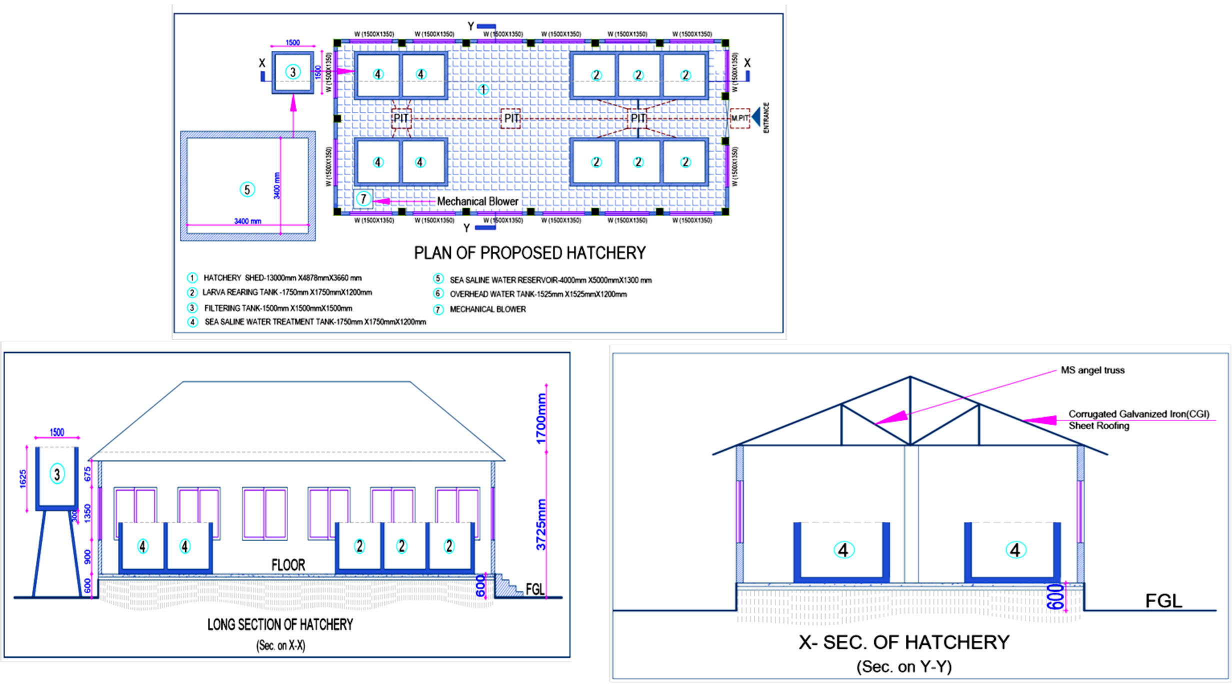


Figure: Engineering design of proposed crab hatchery

109. The hatcheries will be established by private entrepreneurs through PKSF loans. The issues related to the utilization of GCF proceeds and PKSF's loans are described in Paragraph 104. The selection criteria will prioritize capable entrepreneurs with relevant knowledge and experience to ensure sustainability of the activity. A set of the selection and eligibility criteria is described in Outcome 3.

The hatcheries will be established on raised areas so that they will not be flooded by tidal surges or coastal floods. Moreover, it will be semi-*pucca* house to make it resilient to cyclones and storm surges. Thus, the hatcheries will be more resilient to climate change. The land for the building of hatcheries and houses is owned by the beneficiaries. No public land will be used for these structures.

110. Hatchery owners will begin their operations with wild gravid female crabs, although they could also be obtained from their own or other hatcheries. After hatching, they will rear the zoea for up to 25-30 days in the hatchery, following the technical protocol to transform zoeas into juvenile crabs. By this time, each juvenile crab will have gained 0.1gm in weight. At this stage, they will sell the juvenile crabs to nurseries. The hatchery owners may also nurse the juvenile crabs up to their adolescent stage. The hatchery of PKSF at Shyamnagar, Satkhira will be used as a resource centre for capacity development for all crab hatchery related activities. PKSF is also working to make the required raw materials available for the crab hatchery. There is also a continuous effort to make the technology easier for entrepreneurs at the PKSF level.

111. There will be a mechanism to neutralise the wastewater before releasing it from the hatcheries. The wastewater produced from the hatchery will be stored in a reserve tank. Then this water will be treated with chlorine, and then, after de-chlorination, it will be filtered through the filter tank. The filtered water will be released into nearby canals.

Activity 2.2.2: Financial support for producing crablets.

112. Crab hatching technology is comparatively new in Bangladesh. The PKSF has capacity in hatchery management and in the production of juvenile crabs, or crablets. The PKSF has already established three microhatcheries as described above. The PKSF will provide technical support and loans to construct and develop the hatchery as an initial operational cost.

Activity 2.2.3: Technical and financial support for "crab nursers" (2º stage)

113. The project will engage a secondary group between hatchery owners and crab farmers. The project termed this group as "crab nursers". The 2,250 crab collectors (50per cent of them women) will be engaged in juvenile crab rearing. The IEs' staff will select the nurseries based on the selection criteria stated in Paragraph 125. The IEs staff will carry out consultation meetings in the selected areas to select the beneficiaries. They will prepare a draft list of beneficiaries and submit it to PKSF for approval.

The PMU representatives from PKSF will visit to verify the beneficiary list and provide approval for the final beneficiaries. Some of them may also be involved in crab fattening, which is the third part of the production chain. The crab nursers will be provided with the necessary training along with financial support for transporting and nursing juvenile crabs. They will buy juvenile crabs from the hatchery and rear them for one and a half to two months. At this stage, the juvenile crabs will be 40-50 gm in weight and able to grow in a pond. The nursers will sell the juvenile crabs to the crab nursers and fatteners. This activity will use both grant and loan instruments. The budget will be used from GCF proceeds for the pond preparation and the purchase of equipment, such as nets, and cages. PKSF will co-finance the loan as an operational cost, including the purchase of crab-lets, and hiring workers.

114. A nurser will have more than 400 square meter of land for nursing juvenile crabs. Considering five juvenile crabs per square meter, a nurser will rear 2,000 juvenile crabs. For pond preparation, lime will be used to balance ammonia and pH. Then, the pond is fenced with a meshed net and dykes covered with Tripoli (a type of mat). This activity will use both grants and loans. PKSF will provide loans as co-financing for purchasing crablets and other operational costs of nursery, while grant from the GCF will be used to prepare ponds, purchase cages, and procure other materials.

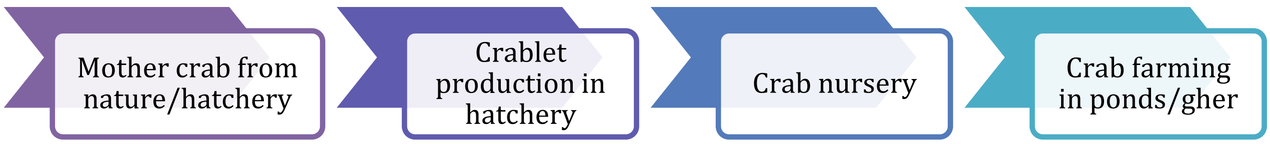


Figure 6: Crab production chain

Activity 2.2.4: Technical and financial support to "crab farmers" (3º stage).

115. The project will select 90,000 tertiary level crab farmers (50per cent women) to produce export-quality crab (both hard shell and soft shell). The IEs staff will carry out consultation meetings in the selected areas to select the beneficiaries. They will prepare a draft list of beneficiaries and submit it to PKSF for approval based on the selection criteria stated in paragraph 125. The PMU representatives from PKSF will visit to verify the beneficiary list and provide approval for the final beneficiaries. They will be provided with the necessary training and financial support. They will buy juvenile crabs from the nurseries and rear them for 30 to 40 days before marketing them. The financial instruments used will be both loans and grants. PKSF will provide the loan for purchasing juvenile crabs from hatcheries, while GCF grants will be used to purchase materials and logistics for preparing ghers or ponds.

116. The project will promote fruit-fish-fiber (3-F model) integrated crab farming and tree plantations in and around the crab-pond or gher. Each crab farmer will plant at least 20 mangrove tree saplings in the crab-pond or gher as well as fruit trees around the ponds. The trees will provide timber, fuel, and fruit, in addition to the crabs that will grow in the pond, leading to increased additional income for the crab farmer.

117. Two types of crab are exported from Bangladesh. One is a live crab (hard shell), and the other is a soft-shell crab. For live crab, the crab collectors collect different sizes of crab from nature and take them to the depot or sell them to the local middleman called *Faria*. There is an agent at the depot. The agent scrutinises crabs before buying them from the collectors, or *Farias*. The exportable crabs are packaged and sent to Dhaka (the capital city), and the non-exportable crabs are sold again to the farmers and consumers. Here, central-level buyers re-grade and repackage them before sending them to the international buyers. Crabs that are not exportable are sold in the local market. For soft-sell crabs, the chain is a little different (Figure 8). For soft-shell crabs, local agents buy small crabs from collectors or hatcheries in a minimum quantity. These agents sell them to large-scale farm owners. There is a processing plant. The farmers sell them to the processors after rearing. On the other hand, the collectors themselves rear for crabs for producing soft- shell crabs and they also sell them to processing plants. The processors package and grade them before exporting them to international markets (Figure 7).

118. For the export market, the buyers ensure food safety standards. PKSF will provide training to the crab growers and other value chain actors on how to maintain these standards. PKSF will also conduct advocacy for the central government to adopt a policy for crab exports that maintains these standards.

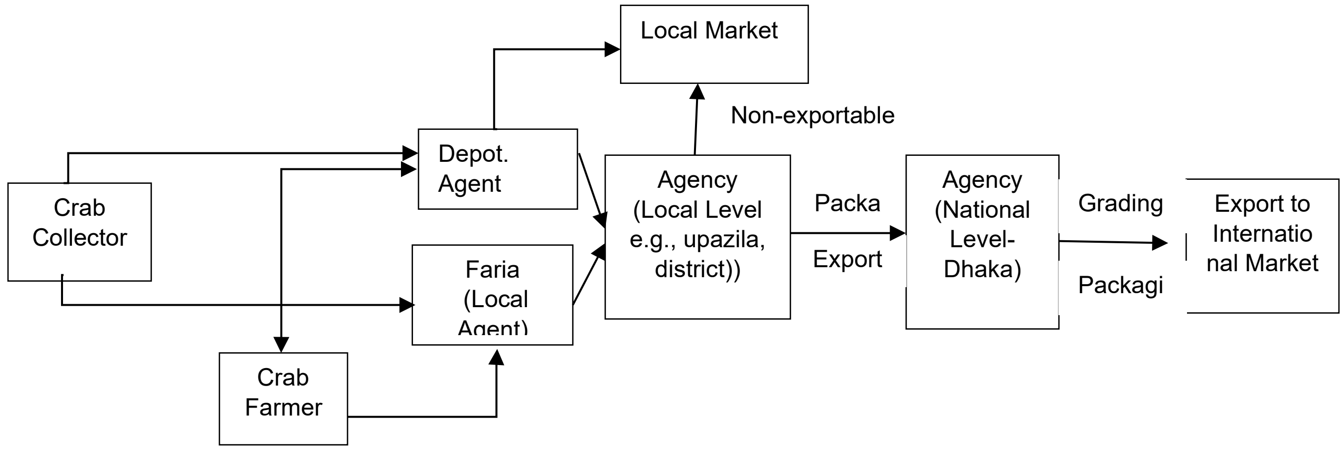


Figure 7: Crab market chain for live crab (hard shell)

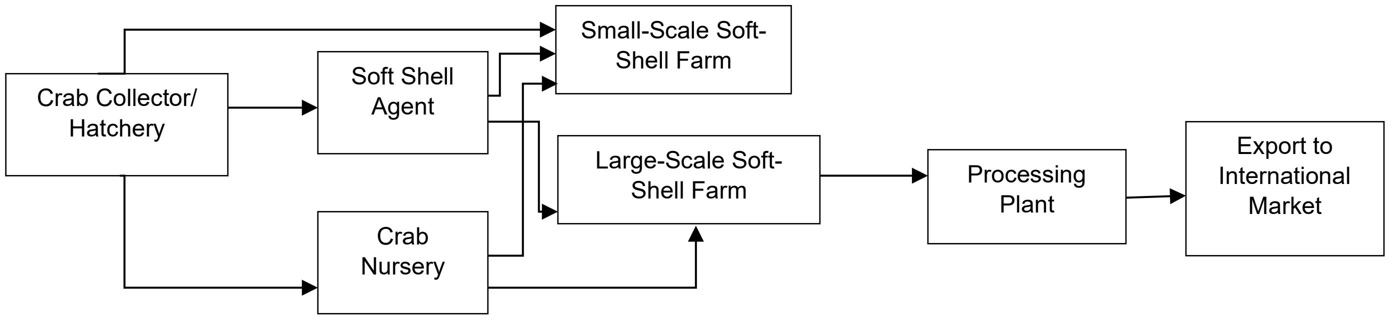


Figure 8: Crab market chain for soft shell crab

Outcome 3: Improved climate planning and implementation by communities and local-level institutions

119. Addressing climate change impacts at the community level requires specialized institutions. Local government institutions in Bangladesh mainly deal with regular development activities. Besides, there are experienced NGOs that have strong and long-term relationships with local communities due to loan programmes. These organizations would play a crucial role in promoting climate change adaptation activities at the community level. The proposed project will select at least 15 NGOs as Implementing Entities in the proposed working areas and enhance their capacity through training and practice adaptation activities. The project will also engage local government representatives in various activities, including area selection, beneficiary selection, and livelihood activities. The project will also carry out policy advocacy with the government, particularly for crab sector development and marketing to contribute to the achievement of the project objectives. Local government departments and institutions will play a role in the decision-making process at the community level by participating in meetings and workshops during implementation. The Union Parishad (UP) chairman will be the focal point of the local GRM process.

Output3.1: Climate change adaptation groups (CCAG) formed and operationalized

Activity 3.1.1: Beneficiary selection and group formation

120. The project will support 362,475 people through a set of selection and eligibility criteria as well as close consultation with local government institutions and community. Available government data will be used to identify the poor and extreme poor in the project area. Direct beneficiaries will be selected based on eligibility and selection criteria. The selection criteria, which are listed below, will be complemented with additional criteria in case of the number of interested and eligible households exceed the target numbers.

a. Those who live in saline-prone coastal vulnerable areas;
b. Priority to women-headed households and other disadvantaged groups;
c. Poor and ultra-poor households (as defined in the Household Income and Expenditure Survey (HIES 2016) of the Bangladesh Bureau of Statistics (BBS-2017)[53]);
d. Per capita daily income is less than USD 1.90 adjusted with PPP as an alternative criterion of HIES's definition of poverty;
e. Those who are not receiving any support from other projects or organizations;
f. Those who have salinity-affected land particularly away from agricultural land;
g. Interested in participation in the project and adoption of the project-promoted technologies and practices; and
h. Willing to contribute to the project through loan, cash and in-kind contributions, as necessary.

121. Selection criteria for goat/sheep rearing in slatted houses (Activity 2.1.1):

a. Those who do not have financial capacity to purchase livestock and have capacity to rear livestock;
b. Women-headed households and households with disadvantaged members will be given priority;
c. Poor and Ultra-poor Households (as defined in the Household Income and Expenditure Survey (HIES 2016) of the Bangladesh Bureau of Statistics (BBS-2017));
d. Have access to homestead land[54]

The selection will be based on the order of the selection criteria from amongst the targeted households in the project area. When the selection process reaches 90,000 beneficiaries, it will be stopped.

122. Selection criteria for homestead tree plantation and saline tolerant vegetable cultivation (Activity 1.1.2 and Activity 2.1.3):

a. Poor and ultra-poor people
b. Having homestead land of at least 60 square meters
c. Women headed households and physically challenged people will be given preference.

The project will select 90,000 beneficiaries for tree plantations and 90,000 beneficiaries for saline-tolerant vegetable cultivation.

123. Selection criteria for crab hatchery establishment (Activity 2.2.1):

a. Those who have the financial capacity to set up a crab hatchery through their own initiative;
b. Minimum or good educational and technological knowledge on crab or shrimp hatcheries;
c. Previous experience with microenterprise operations;

[53]This document defined extreme poor as the person having purchasing power parity (PPP) below 1.25 USD a day and PPP below 1.90 a day is called poor.
[54]Here land rights mean, a person owned land or properties by means of buying or inherited from father/mother or other persons.

d. A minimum of 20 decimals of owned land, but not adjacent to agricultural land to set up a crab hatchery in a small scale;

124. Only barren and fallow land will be considered to set up a hatchery plant, so there will be no chance of losing productive land.

The selection will be based on the order of the selection criteria from amongst the targeted households in the project area. When the selection process reaches 225 beneficiaries, it will be stopped.

125. Selection criteria for crab nursers and farmers (Activity 2.2.3 and Activity 2.2.4)

a. Having cultivable land or capacity to rent [55] at least 0.20 hectare;

b. Women-headed households and households with disadvantaged members will be given priority;

c. Have the ability to run an manage crab farming activities;

d. Poor and ultra poor households (as defined in the Household Income and Expenditure Survey (HIES 2016) of the Bangladesh Bureau of Statistics (BBS-2017).

126. Selection criteria for a resilient homestead (Activity 1.1.1)

a. Those who have their own homestead area at or above tidal flood level with high exposure to climate shock events;

b. Women-headed households and households with disadvantaged members will be given priority;

c. Poor and ultra poor households (as defined in the Household Income and Expenditure Survey (HIES 2016) of the Bangladesh Bureau of Statistics (BBS-2017)); and

d. Those who do not have financial capacity to construct resilient houses.

The selection will be based on the order of the selection criteria from among the targeted households in the project area. When the selection process reaches 13,500 beneficiaries, it will be stopped.

127. Thus, the total beneficiaries will be as follows:

Beneficiary type	**Number of households**	**Number to total direct Beneficiaries**	**Comments**
Activity 1.1.1: Design and building of homesteads	3,000	13,500 (3,000X4.5)	Average family size is 4.5 (Population Census, 2011). These beneficiaries will overlap with tree plantation (Activity 1.1.2)
Activity 1.1.2: Homestead tree planting	20,000	90,000 (20,000X4.5)	
Activity 2.1.1: Construction of slatted houses for goat/sheep rearing	20,000	90,000 (20,000X4.5)	
Activity 2.1.2: Provide financial support for goat/sheep rearing	-	-	The same beneficiaries of Activity 2.1.1

[55] Capacity to hire land means financial capacity of the beneficiary to have rent land.

Activity 2.1.3: Introduce the cultivation of saline tolerant vegetables within homestead areas	20,000	90,000 (20,000X4.5)	
Activity 2.2.1: Development of crab hatcheries	50	225 (50X4.5)	Hatchery owners
Activity 2.2.2: Financial support for producing crablets	-	-	The same beneficiaries of Activity 2.2.1
Activity 2.2.3: Technical and financial support for "crab nursers"	500	2,250 (500X4.5)	
Activity 2.2.4: Technical and financial support to "crab farmers"	20,000	90,000 (20,000X4.5)	
Total	81,450	362,475*	

128. The project will form 3,200 groups with 25 (+/-) persons in each group for delivery support. These groups are named Climate Change Adaptation Groups (CCAGs). A total of 80,000 members will be selected from 362,475 selected beneficiaries to form these CCAGs. The members will be selected based on intensive consultations with selected communities on the basis of pre-defined criteria that include: a) potential to play role in decision-making at the household level, b) less involvement in income-generating activity, c) understanding on the local socio-economic context, and d) willingness to participate in the CCAG meetings. Each group will meet at least once a month and to discuss climate change and its impacts on their lives and livelihoods. They will adopt and practice climate-adaptive technologies that will be promoted by the project. The objective of forming this group is to deliver the support services in groups in order to minimize the delivery cost and to ensure the participation and collective decisions of the affected community in implementing the proposed interventions. It will help transfer knowledge on climate change issues among the society because they will discuss the impacts of climate change at a regular interval, typically monthly, in groups. Thus, they will be able to internalise the impacts of climate change on their lives and livelihoods. The groups will receive training on climate change issues and how to deal with these problems. They will be able to identify climate change problems on their lives and livelihoods and prepare plan accordingly to reduce the impacts of climate change in the future. They will also look after community infrastructures beyond the project. Besides, the group approach will reduce the management costs of the project. The IEs' staff will conduct necessary consultations at the local level and select potential beneficiaries. They will prepare a draft list of beneficiaries and submit it to PKSF for approval. The PMU representatives from PKSF will visit to verify the beneficiary list and provide approval for the final beneficiaries. Thus, the PMU of EE will oversee the process of beneficiary selection and group formation. Grant financing from GCF will be used to carry out this activity.

129. The female headed households and other disadvantaged groups will receive priority while selecting the beneficiaries (see below for detailed information on gender priority). As per experience of CCCP, there will be female headed households. The consultation meetings during beneficiary selection will identify female headed households. Level of vulnerability is the main distinction between a woman in married households and a female head of the households. The female heads are more vulnerable because their income source is very limited; they cannot go outside of their locality, and the rate of women's participation in labour force is also low. These limitations made them more vulnerable the women in married households.

Activity 3.1.2 Prepare Beneficiaries' socio-economic profile

130. A detailed socio-economic profile of the selected households will be prepared by IEs before providing support. The purpose of the profile is to keep a monitoring record from baseline to after project finalisation. This information will be used to compare short term progress achieved by project interventions. The PMU, at the EE level, will provide necessary technical support including developing tools and guidelines. This activity will use grant financing from GCF funds.

Activity 3.1.3 Arrange monthly group meetings on climate change issues for CCAG

131. The IEs will have field-level staff to directly coordinate with the beneficiaries. S/he will assist the groups in organizing meetings, discussions on climate change and other environment and health issues. The meeting notes will be preserved in a register book. The groups will take necessary decisions in addressing climate change impacts through the project interventions. They will decide who will get what types of support from the project, based on their needs. Thus, community-level informal institutions will be shaped and carried forward by these group members. PKSF will co-finance this activity through in-kind contributions. The co-financing budgetary items will involve facilitation of meetings (i.e., staff cost), local travel, brown papers, register books, sign pens etc.

132. This staff will go to the CCAGs and discuss selected climate change issues, for example, climate change basics, climate change and crab farming, climate resilient housing, climate and poverty, and climate change and gender. The PMU at PKSF will provide technical support to IEs for producing posters or papers for discussion at the meetings.

133. The experience from the CCCP shows that CCAGs are functional where they are engaged in financial services. Most importantly, the CCAG members are continuing most of the activities as they are getting benefits from it. This project plans to engage the CCAG members in financial services (credit, savings, enterprise loans etc.) by the partner organizations of PKSF beyond the project. This will ensure the sustainability of the CCAGs.

Output 3.2: Capacity built among IEs and relevant institutions apprised on project

Activity 3.2.1 Prepare training manuals on adaptation technologies and crab value chain

134. The PMU of EE will prepare a training manual to deliver ToT to the IEs' staff on climate change issues and project management (IE selection process is described in para 165 to 168). Approximately 150 staff from at least 15 selected IEs will receive this ToT. This will significantly contribute to strengthening institutions in addressing climate change issues at the community level. Besides, another training manual on climate change will be prepared to provide training to the CCAG members and traders across the value chain. PMU will prepare the training manual and IE staff will deliver the training. On the other hand, PMU will prepare training modules for crab farming as well as crab nurseries. The project will also prepare necessary guidelines including activity implementation guideline, monitoring and evaluation guideline, environmental and social management guideline, procurement guideline, accounting and financial manual etc. This activity will use grant financing from GCF.

Activity 3.2.2 Prepare guidelines on project management

135. The project will also prepare necessary guidelines including food safety standards, activity implementation, monitoring and evaluation, environmental and social safeguards, procurement, accounting and financial management. The guidelines will be utilized by the project implementing entities. This activity will use grant financing from GCF.

Activity 3.2.3 Organize training for beneficiaries and stakeholders

136. Each selected IE (IE selection process is described in para 165 to 168) will prepare a training plan to deliver training to the selected CCAG members. This training plan will require approval from the PMU. PMU staffs will closely monitor the training sessions as per plan.

137. This project will provide classroom as well as on-the-job training to about 100 beneficiaries and value chain activities on crab hatching technology and safety standards so that they are able to provide services to the proposed hatcheries. The present shrimp hatchery owners could be prioritized as they are already experienced in hatching technology that is almost similar to crab hatching. So, involvement of shrimp hatchery owners is an advantage for this project. At the same time, the project will organize training for crab nurser and farmers for effective implementation and crab production. About 50 percent of women participants will be assured of receiving training. The IEs with guidance from PKSF and in consultation with communities will select the hatchery owners based on the criteria stated in Paragraph 123.

138. The project will provide training to the selected participants on improved management and technology of goat/sheep rearing (selection process is described in Paragraph 98). Local level livestock officers will be hired as resource persons for providing this training. About 100 percent women Beneficiaries will be ensured. Grant financing from GCF will be used to carry out the above activity.

Activity 3.2.4 Organize training for IEs' staff

139. PMU will organize and deliver the training sessions. About 155 staff will receive this training in 7 batches (the number of training sessions will be around 10). This will enhance the capacity of the newly recruited IEs' staff. They will learn about climate change and adaptation as well as management of adaptation project. They will contribute to the organizations in practice climate change-related activities within the organization. This activity will use grant financing from GCF.

140. Training for the staff and extension agents who will deal with crab hatcheries and nurseries will be long term, typically six months on the job, residential training to the crab hatchery of PKSF. It is required because the success of the hatcheries will fully depend on the efficiency of the training. The staff will be recruited through a competitive process by the IEs.

Activity 3.2.5 Implement workshops and seminars

141. The project will give 20 workshops at the national and local levels. The workshop will include project inception, policy advocacy, project closing, quarterly progress review workshops, annual learning sharing workshops, training workshops etc. In addition to NDA, representatives from other relevant government agencies including, Ministry of fisheries and livestock, Ministry of Agriculture (MOA), Department of Agricultural Extension (DAE), Ministry of Environment, Forests and Climate Change (MOEFCC), Department of fisheries, Department of Environment (DOE), Bangladesh Climate Change Trust (BCCT), Ministry of Water Resources (MOWR), Local Government Engineering Department (LGED), Ministry of Women and Children Affairs (MOWCA), Bangladesh Rice Research Institute (BRRI), Bangladesh Agricultural Research Institute (BARI), Bangladesh Institute of Nuclear Agriculture (BINA), Department of Disaster Management will be invited to attend the workshop. PKSF will send invitation to the relevant orgnisations stated in paragraph 41 for requesting participants. The higher authority of the organization will nominate the workshop participants. Besides, AE representatives, IE staff and PMU staff will be invited to these workshops. PMU will organize all these workshops. About 50 per cent of them will be per centwomen participants. All the workshops and seminars will be organized with grant financing from GCF.

142. The project will identify best practices and lessons throughout the project period. These best practices and lessons learned will be shared in these workshops.

Activity 3.2.6 Organize exchange visits for beneficiaries and IE staff

143. The project will organize 15 inter-community visits in vulnerable areas. They will learn from each other and be encouraged to adopt climate-resilient technologies and practices. About 300 participants attended the events where 80 per cent per cent of them will be women. The participants will be selected in the CCAG meetings. The CCAG members with the assistance of field-level staff of the IE will discuss the exchange visits including learning objects, best practices, visit places, time, etc. Based on their discussion, CCAG will nominate participants from them. The PMU of PKSF will guide the IE staff to facilitate the meetings. It is a type of in country training and sharing of knowledge/technology for the beneficiaries and IE's staffs. It will be conducted in the project areas or in other areas of the country under implementation of same kind of activity areas). This type of visit will be helpful for the smooth and successful implementation of project. This activity will use grant from the GCF.

Activity 3.2.7 Improve data for crab research and development

144. The project will sign MoU with reputed public universities namely the Chattogram University, the Khulna University, Patuakhali Science and Technology University and Bangladesh Fisheries Research Institute (BFRI) for improve or strengthen their fisheries research laboratories in order to promote crab research and development in the country. This will help hatchery owners, nurseries and growers to identify diseases and treatments, information on water salinity, pH and other composition, and take decisions on crab management.

145. To ensure quality at the farmer level, the project will use these laboratories to examine quality of food. These laboratories will examine the heavy metals, antibiotics, microbial contamination in addition to research on crab disease and diagnosis. Farmers will do this test before selling it to the processors. The grants from the GCF will be used to implement this activity.

Output3.3: Knowledge products prepared and disseminated

Activity 3.3.1: Prepare and disseminate knowledge products

146. The PMU will develop and publish a quarterly newsletter on project progress and learning. This newsletter will be circulated to different stakeholders including GCF, Bangladesh NDA and other government organizations. The published newsletter will also be uploaded in PKSF's website. Grant financing from GCF will be used to carry out this activity.

147. The project will carry out lessons that have been learnt throughout the project period. Relevant project officer for knowledge management of PMU will carry out the lessons and develop a booklet for publication. The knowledge documents will be distributed among the relevant government agencies (as mentioned above), international and national NGOs including partner organizations of PKSF to inform their future projects and programmes. They will use the information from the knowledge documents in designing their future projects, management of adaptation projects and measuring short- and long-term impacts of adaptation projects. They will consider the effectiveness of slatted houses, crab farming and climate-resilient houses to be documented in the knowledge documents. This will also contribute to strengthening institutions at national and local level in designing and implementing adaptation projects in the country. This activity will use grant financing from GCF.

This activity will help to address Barrier 1 and 4 in the funding proposal. Because the guidelines and tools to be developed under this activity will strengthen capacity of the communities and institutions in preparing climate plans and accordingly implementing the plans. On the other hand, tools and guidelines will help to implement standard adaptation practices in the coastal areas and thus it will remove the capacity gaps of the communities and institutions for implementing standard adaptation practices.

Activity 3.3.2 Real time evaluation study of the project activities

148. Currently, there is limited knowledge and understanding of the effectiveness of adaptation interventions globally, including Bangladesh. This project will promote a periodical evaluation system of the project interventions to understand their effectiveness in terms of adaptation to climate change. Hence, the project will conduct a baseline study of the effectiveness of the proposed interventions at the inception phase and continue the effectiveness study annually throughout the project period. This will help generate adaptation knowledge as well as learning for future projects and programmes. This activity will be implemented with grant financing from the GCF.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Implementation arrangement at PKSF level

149. Role of PKSF: PKSF will play dual role in this project because PKSF is Direct Accredited Entity (DAE) as well as Executing Entity (EE) for the project.

150. Role of PKSF as a DAE: PKSF as a DAE, will be responsible to : a) establish the PMU, b) appraise and finalise project implementation arrangements, including mission travel, c) ensure compliance of fiduciary standards, ESS standards and gender related issues, d) facilitate financial flow to the project, e) submit APR and interim unaudited financial report, f) assist project management to draft TORs and advise on the selection of IEs for implementation, g) conduct supervision missions, including briefing operational focal points on project progress, h) provide technical services during supervision missions to advise government officials on technical matters and provide technical assistance for the project, i) oversee procurement and financial management to ensure implementation is in line with AE's policies and timeline, j) undertake the mid-term review, including possible project restructuring, k) disburse funds to the executing entities/vendors and review financial reports, l) oversee the preparation of the required reports for submission to the GCF Secretariat, m) monitor and review project expenditure reports by internal auditors, n) prepare periodic revisions to reflect changes in annual expense category budgets, o) prepare project closing documents for submission to GCF Secretariat, p) prepare the financial closure of the project for submission to GCF Secretariat, q) oversee the preparation of the Project Completion Report/Independent Terminal Evaluation, and submit the report to the GCF Secretariat, r) report required activities as agreed in the AMA and FAAs and so on.

The PMU of the project will have 13 full time staff and 1 short term consultant for procurement. The team will be headed by the Project Coordinator. The positions are a) Project Coordinator, b) Deputy Project Coordinator (ESS), c) Assistant Project Coordinator (Finance and Accounts), d) Senior Assistant Project Coordinator (Crab Hatchery Operation and Management), e) Assistant Project Coordinator (Communications and Knowledge Management), f) Assistant Project Coordinator (Training and Social Development), g) Assistant Project Coordinator (Civil Engineers), h) Assistant Project Coordinator (Programme-3), i) Assistant Project Coordinator (Value chain specialist), j) Finance & Accounts Officer, k) MIS Officer, l) Procurement management (staff recruitment cost, advertisement etc.) (Short term consultant) and Support staff.

151. Role of PKSF as an EE: PKSF will be a single Executing Entity (EE) in the project and establish a Project Management Unit (PMU) to manage the RHL project. A Project Director/Project Coordinator (PD/PC) will head the PMU and be in charge of the overall project implementation. He/she will directly report to a senior official of PKSF and be the contact person at PKSF for the GCF Secretariat and NDA. The PD/PC will report to the GCF in a manner approved by PKSF. The PMU will be responsible to: a) prepare tools and guidelines for implementing the project and seek approval from the AE, b) prepare procurement documents for goods and services required for implementing the project, c) make agreement with IEs through a rigorous selection process, d) organise necessary trainings and workshops for IEs and beneficiaries, e) supervise, guide and monitor day to day activities implementing by IEs at the field level, f) prepare documents with the help of AE to procure independent consultants mid-term and final evaluations, g) maintain management of project information and publish guidelines, manuals and training modules and learning documents, h) recommend fund transfer to the IEs based on quarterly reports and field visits, i) ensure ESS implementation at the field level, provide training to IEs staffs, j) review and finalize beneficiary selection to be carried out by IEs, monitor progress of activity implementation plan of IEs, k) ensure quality of implementation of project activities by IE, l) prepare APR and get approval from PKSF, as an AE (Deputy Managing Director is the approval authority), m) prepare interim unaudited financial reports and get approval from AE and so on.

152. The PMU will engage project personnel who will liaise with the selected IEs and monitor the implementation of their projects. The project personnel will be the PKSF contact points for IEs and report to the PD/PC.

153. A team of technical reviewers will be engaged in using their services when required to appraise projects. These technical experts will review the project locations.

154. The PD/PC, after ensuring compliance to all fiduciary requirements, will submit the Sub-Projects (SPs) through PKSF as a DAE to the PKSF's Governing Body for final approval. As Member Secretary of the Governing Body, the PKSF's Managing Director will present the proposal.

155. PKSF as EE will bear the responsibility of implementing the project in all aspects. PKSF will delegate field level implementation of activities to the selected IEs through legal agreement. The IEs, by signing legal agreement with the PKSF will be responsible to work with the community people to deliver the project services. PKSF as EE will monitor the implementation activities of IEs through both off-site and on-site monitoring systems. PKSF will adhere to Results-Based Monitoring (RBM) system to ensure reaching the project goals efficiently and effectively.

156. PKSF always implements its projects through its partner organizations, which is selected through PKSF's policies and procedures. All these organizations are pre-qualified and enlisted through a transparent and fair technical procedure. The following five guiding principles will be applied to select the IEs and to guide IEs' activities:

I. The project would adopt a strategic and holistic approach that targets clear climate change scenarios. Each of the demand of investment funded under the project would fit within the abovementioned scenarios;

II. Any organization receiving finance must demonstrate how it will contribute – through the community-level interventions – to advancing the skills and knowledge required to adapt to extreme climate variability and climate change. Organizations requesting funds for community-based adaptation must have an established presence in the relevant areas where the project will be implemented. The climate change programme would preferably build upon the foundation and social capital of other projects that the organization is already implementing;

III. The projects would include community leadership and local government bodies while ensuring gender sensitivity; and

IV. Emphasis would be placed on transparency, information monitoring and learning to ensure sustainability of the programmes and replication of those in other similar regions of Bangladesh.

V. CCAGs will play a significant role in ensuring transparency and accountability of the project at the community level.

157. The project under implementation will be subject to monitoring both by the IEs, and PKSF as AE and PMU (PKSF as EE). The basis for monitoring is the Integrated Results Management Framework (IRMF) of the project as mentioned in section E-7. A detailed Monitoring and Evaluation Manual consistent with **GCF frameworks and policies (IRMF, GCF Evaluation Policy, Monitoring and Accountability Framework, etc.) and as well as** PKSF's overall Results-Based Monitoring System will be developed to guide the monitoring practices of PKSF and IEs.

158. The monitoring process under the PMU will have three functions. First, through monitoring by PKSF and PMU will ensure accountability of the IEs to deliver the outputs and outcomes. This implies that resources are used efficiently for the proposed activities. Second, monitoring will establish proper documentation of the implementation process and achievements at different levels (i.e., outputs, outcomes and impacts). Third, monitoring will help gather learning from the process. Since adaptation experiences are highly contextual, documentation of learning under different contexts will add to the knowledge and subsequently to the wisdom for future actions. In short, the role of accountability is significant in case of outputs, whereas learning becomes a core issue for monitoring at the outcome- and iImpact-level achievements.

159. The following 7 steps flow chart characterizes the project awarding process:

Agreement Signing between GCF & PKSF as AE

↓

Selection of Implementing Entity (IEs) by PKSF as EE

↓

Recommendation made on allocation of project activity and Fund to IE by technical committee

↓

Approval by Board of PKSF as EE

↓

Execution of Contract Signing between IEs& PKSF as EE

↓

Implementation (PKSF as EE) → Monitoring & supervision (PKSF as both AE & EE)

↓

Completion of the Project PKSF as EE → Evaluation by external organization/GCF

Contractual Arrangement for the project

FAA sign between GCF and PKSF (as both AE and EE)

↓

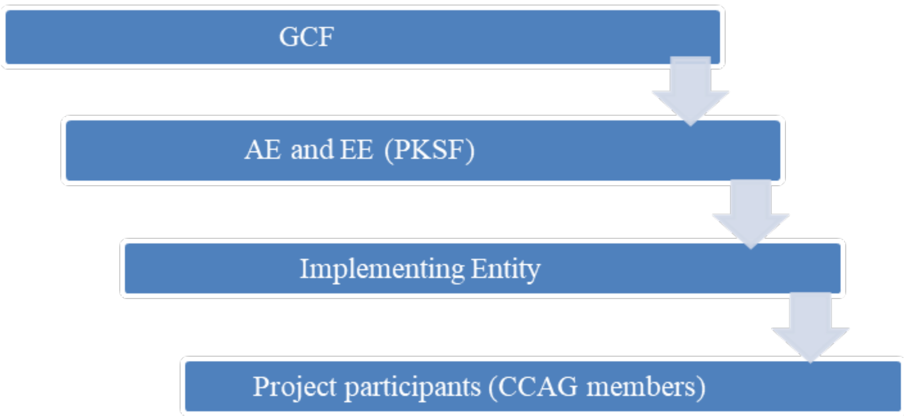
Agreement signed between EE and IE

Fund flow of the project:

160. For this project, PKSF is AE as well as EE. The fund will be directly disbursed to PKSF from GCF. PKSF will receive the GCF Proceeds in the Project specific account held in a commercial bank in USD. In addition to initial advance to IEs from PKSF, PKSF will reimburse the fund to the implementing partners based on satisfactory performance. The PMU will prepare an 'activity implementation guideline', preferably in local language. In the guideline, specific instructions and design will be incorporated for each of the activities. The PMU staffs will visit the field level activities and check, whether the IEs have done any distortion from the assigned guidelines and instructions. If the IE fails to maintain the standards as mentioned in the guideline or other forms, PMU will suggest to take corrective measures. Once, the corrective measures are done, PMU will recommend for reimbursement. The implementing entities will make necessary expenditure for the beneficiaries to increase their resilience. One of the criteria for selecting the IEs is that they must have established offices in the project area. This office mainly operates loan programme financed by PKSF. The loan under the proposed project will be disbursed through **PKSF's mainstream credit programme to the IEs** (as they are also partner organizations of PKSF). In addition, the IE will establish a project management unit at their respective offices. This PMU will operate the grant part and the credit officers will operate the loan part. This is the operational procedure for blending projects and programmes of PKSF.

It is to be noted that PKSF will select at least 15 partner organisations who will be working as IEs at the community level. They will mobilize the beneficiaries, engage local contractors, and ensure project services for the beneficiaries. As such, they will pay to the contractors and other service providers at the local level upfront where necessary. Then they will submit financial reports and bills to PKSF on quarterly basis. PKSF will review the documents and carry out field visits to ensure that the bills are duly prepared and submitted. Then spending amount will be transferred to IE's project account. By re-imbursement, we mean this process.

161. Fund flow diagram

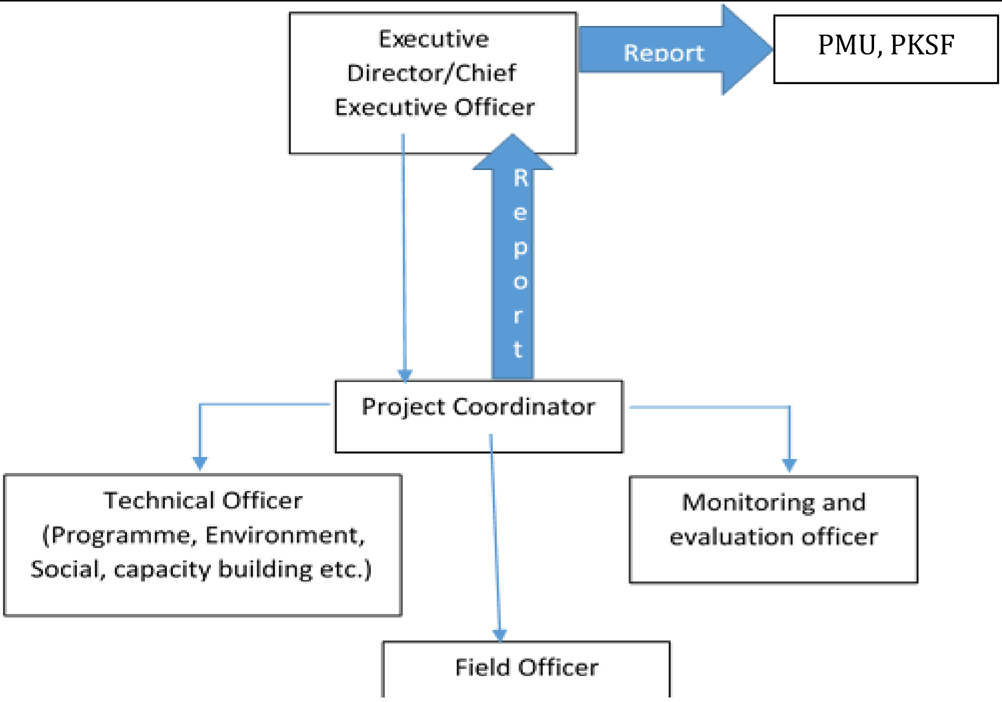


Implementation arrangement at IE level

162. PKSF as an EE will bear the responsibility of implementing the project in all aspects. PKSF will delegate field level implementation of activities to the selected IEs through legal agreement. The IEs, by signing legal agreement with the PKSF, will be responsible to work with the community people to deliver the project services. The work of IEs is always be under monitoring of the EE. The partner organizations who will qualify to implement this project at the community level are the IEs, which have been selected through PKSF's policies, and procedures (for IEs selection criteria see paragraph- 165)

163. The participating entities will mainly implement the activities at the community level. They will monitor the outputs, outcomes and impacts of the activities as well as their impacts on the environment and society. They will also ensure gender participation at the organization and community levels. They will also report to PMU of PKSF on the progress and impacts of the project.

164. The IEs will employ a dedicated monitoring officer who will report to the chief executive or senior officials, not directly entrusted with the implementation of the programme. He or she will implement the monitoring framework as envisaged in the project proposal and will produce quarterly activity monitoring reports based on the 'Activity to Output Monitoring (ATOM)' agreed upon by both parties. The monitoring officer will undertake the outcome-level monitoring half-yearly based on the agreed 'Outcome Assessment Sheet (OAS)' and impact-level monitoring annually based on the agreed 'Impact Assessment Sheet (IAS)', which will be prepared taking indicators of impacts and outcomes into account. He or she will post the information in the assigned fields of the IEs and in the PKSF's online system as well.



165. Selection of Implementing Entity: A selection committee comprised of PKSF's senior officials will be formed. The committee will select the IEs. To be selected, IEs will have, at a minimum:

- Permanent existence of the organization in the project areas.
- At least five years of experience in implementing climate change-related projects or programmes.
- A good track record of financial transaction (i.e., at least BDT one crore, which is around USD 100 thousand, annually for the last three years)
- Must be extra ordinary, excellent or at least good as per PKSF's assessment using defined assessment criteria which include financial efficiency, economic efficiency, operational efficiency, growth indicators, financial strength and risk management, accounting and internal control system, social performance, human capacity and governance.
- Valid legal documents including registration.

166. Organizations will be ineligible on the grounds of involvement in 'Money Laundering and Terrorist Financing'.

167. It is to be noted that PKSF has set criteria for periodic evaluation of the performance of its partner organizations. The criteria include financial efficiency, economic efficiency, operational efficiency, growth indicators, financial strength & risk management, accounting & internal control system, social performance, human capacity and governance. Each of the criteria has several indicators to assess performance of the POs. Based on the performance criteria, the organizations are categorized as extra ordinary, excellent, good, average, and sub-standard and requires special attention (RSA). These criteria will also be considered for eligibility. PKSF will ensure completion of AML/CFT due diligence with satisfactory results in the selection process for IEs and service providers.

168. The selection committee will evaluate the submitted EOIs and prepare a short list of the potential IEs. Thus, the competitive process will be followed to select IEs for this project.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

169. Climate change is an important additional threat to the socio-economic development of Bangladesh. The country is in the process of switching from an LDC to a lower middle-income country by 2025, which requires considerable investments in regular development interventions. Hence, the country is not in a position to invest additional resources for building resilience from climate change shocks. GCF funding is important to build climate resilience for the most vulnerable people.

170. The coastal area of Bangladesh is vulnerable to many natural disasters including climate change variability and extreme events. Primarily, the land and water are affected by increased salinity and sea level rise leading to decrease crop production which is the main livelihood of the coastal people. In addition, many climate-induced hazards, such as cyclones and storm surges, coastal flooding, and erosion also damage to crop agriculture, aquaculture and other livelihoods of the poor community. Establishment of crab sub-sector for the coastal people will be a sustainable adaptation option. Because it requires salinity environment including land and water. If, sea level rises, the brackish aquaculture would be demolished where crab farming would enjoy the increased salinity in the environment. In addition, present system of crab fattening threatens the availability of crab in the nature. Once, crab hatcheries are established, the crab catchers would not need to catch wild crabs. It will help protecting the coastal biodiversity of the country.

171. Most crab growers remain vulnerable to shocks from extreme events of natural disasters, climate change events and salinity intrusions into agricultural land. They lose their assets, fertile land and livelihoods' alternative options, whereby becoming vulnerable. PKSF and IEs with the support of GCF will provide the training, capacity building and financial support for crab hatcheries, crab farming to 362,475 people.

172. GCF involvement is very important and critical in two ways: (i) climate change threat and long- term projections to be mentioned that coastal vulnerability will likely to increase in southern areas of Bangladesh. Coastal floods, salinity intrusion, sea swells, storms and cyclone will likely increase in occurrence and intensity. It will require additional investment to reduce the impacts of and vulnerabilities to these climate variability and extreme events. Therefore, more additional involvement and investment in enhancing knowledge and awareness on climate risk to inform and improve the present government programmes and policy in promoting climate-resilient livelihoods and safe housing is necessary; and (ii) Extreme climatic related threats also require additional finance to increase the scale of climate risk reduction investments to protect the coastal livelihoods and settlements (people and their main assets – the homes) as well as improve the methods and application of a good practice, GCF involvement will considerably enhance the ongoing government programmes, employing best practices and scaling-up achievement of successful pilots and good international practice. As a result, the proposed investment will be transformational.

173. Reflecting the limited resources available to the country as well as the need to trigger systemic change, it is proposed that a significant proportion of the total cost will be covered by the GCF in the form of grant finance. This is essential because Bangladesh is a least developed country with very limited financial capacity to adapt to climate change and having to invest most of its constrained resources to address competing needs. Thus, the financial instruments chosen, which rely heavily on the GCF grants to overcome structural barriers to change, are considered appropriate to the achievement of project objectives.

The project will be financed mutually both by the GCF, the PKSF and the community. The project seeks from GCF US$ 42.20 million as grant for hatchery, capacity building training and building climate-resilient homesteads. PKSF will provide the loan for establishing the hatchery, crab farming and sheep farming for promoting climate-resilient livelihoods and value chain interventions at an estimated US$ 7.79 million.

B.6. Exit strategy (max. 500 words, approximately 1 page)

174. The project will seize the climate change related negative impacts as an opportunity for the coastal people. Crab farming requires a comparatively high level of salinity. The coastal area will experience severe salinity under the changing climate situation. Thus, coastal people will fully use the adverse impacts of climate change by the project interventions for their wellbeing.

175. PKSF has already established a large crab hatchery at a cost of 1 million USD under its technology transfer component of Promoting Agricultural Commercialization and Enterprise (PACE) project. Moreover, PKSF has taken a ‘Value Chain Development’ project for crab rearing. Knowledge gained from these interventions will be used in the proposed project under the GCF. All the technological support for the hatchery and crab culture have been taken from Vietnamese experts. In the meantime, a group of local experts has already been groomed up.

176. PKSF has been implementing goat and sheep rearing in slatted houses for more than a decade in the Chuadanga district of Bangladesh through its partner organization called Wave Foundation (WF). But it was concentrated in very limited areas. The usefulness of slatted houses for goats or sheep in the context of climate change is enormous as experienced by PKSF. The project will promote this learning in the coastal zone of Bangladesh

177. The project interventions will continue in the long run. Because it will develop a profitable economic sub-sector and promote innovative technology for the livestock sub-sector which will be resilient to climate change as well. It will include all classes of people in the society. The entrepreneurs will seek finance from banks and NGOs beyond project periods. As crab farming and trading will be a profitable business, financial organizations and NGOs will finance the sector in the long run. The government will also create favorable environment in promoting the crab sub-sector. Similarly, goat and sheep rearing will also be financed by the local MFIs if it would be explored commercially.

178. The project will have multiple adaptation impacts on the coastal communities as well as contribute to the country’s GDP. It will create employment for vulnerable coastal people and provide livelihood support to more than 362,475 beneficiaries. The project will support entrepreneurs at different stages of the value chain of the crab business. Thus, the project financing will be effective in achieving objectives of the project. In addition, goat and sheep rearing in slatted houses is a profitable livelihood option because slatted houses reduce disease and increase productivity. Besides, women can get easily involved in this activity while staying at home. So, the community will automatically continue this activity in the long run.

179. The project will organize a series of workshops will different stakeholders including government representatives, policy makers, national and local NGOs and financial organizations. The workshops will share lessons of the project to these stakeholders in order to create favourable policy environment for up scaling crab hatchery. The POs and financial organizations will provide financial support to the entrepreneurs for extension of the business.

180. After completing this project, it is expected that participants will be more resilient to any adverse effects of climate change in coastal areas. In addition, to meet their extended demand for livelihoods, they will be involved in existing MFIs. Besides, their involvement with local councils and local government institutions will broaden their social capital and institutional linkage. It is also expected that the Implementing Entities (IEs) will continue their services in communities with other demand-driven services and thus, these IEs will be readily available for the project participants to offer any kind of consulting services on the issues related to climate change.

181. The exit strategy will follow PKSF’s policy on project closure. Primarily, an exit plan is prepared for the project. The implementing entities will be provided necessary guidance on project closure. The project will organize workshops at the national level involving policy makers and actors to share the results of the project interventions. The IEs will organize workshops at the local level involving local government representatives and local actors. Necessary publications will be published during the tenure of the project. An exit plan is presented below.

Sl. No.	Activity	Time of action	Responsibility	Source of budget
1	Prepare guidelines for IEs on the exit plan	Last year of the project cycle	PMU	Project
2	Distribute knowledge materials	Throughout the project period	PMU and IEs	Project finance
3	Organise workshops at central level on lessons that have been learnt	Last quarter of the project	PMU	Project
4	Organise workshop at local level	Last quarter of the project	Respective IEs	Project
5	Undertake maintenance support for the infrastructure	During and post-project period	Beneficiaries	Beneficiaries own fund
5	Engage appropriate stakeholders for the continuation of activities beyond project	Post project period	Local level entrepreneurs and community people	The IEs and other PKSF's partner organizations

182. The project will generate knowledge and learning that will contribute significantly to enhance resilience of rural communities. The learning will be shared with national and international policy and decision makers in order to get policy support for the institutionalising of the sub-sector

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount	Currency
	42.20	million USD ($)

GCF financial instrument	Amount	Tenor	Grace period	Pricing
(i) Senior loans	Enter amount	Enter years	Enter years	Enter per cent
(ii) Subordinated loans	Enter amount	Enter years	Enter years	Enter per cent
(iii) Equity	Enter amount			Enter per cent equity return
(iv) Guarantees	Enter amount			
(v) Reimbursable grants	Enter amount			
(vi) Grants	42.20			
(vii) Results-based payments	Enter amount			

(b) Co-financing information	Total amount	Currency
	7.79	million USD ($)

Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Palli Karma-Sahayak Foundation (PKSF)	Senior Loans	6.60	million USD ($)	5 years Enter years	12%	Options
Palli Karma-Sahayak Foundation (PKSF)	In kind	1.19	million USD ($)	5 years Enter years	Enter%	Options
Click here to enter text.	Options	Enter amount	Options	Enter years Enter years	Enter%	Options
Click here to enter text.	Options	Enter amount	Options	Enter years Enter years	Enter%	Options

(c) Total financing (c) = (a)+(b)	Amount	Currency
	49.99	million USD ($)

(d) Other financing arrangements and contributions (max.250 words, approximately 0.5 page)	

C.2. Financing by component

Outcome	Output	Indicative Cost million USD ($)	GCF financing		Co-financing		
			Amount million USD ($)	Financial Instrument	Amount million USD ($)	Financial Instrument	Name of Institutions
Outcome 1: Decreased risk of loss of assets and lives from extreme weather events	Output 1.1 Climate resilient homesteads constructed	24.41	24.41	Grants			
Outcome 2: Livelihood resilience to SLR/storm surge and salinity	Output 2.1 Traditional farming practices climate proofed	7.23	4.23	Grants	3.00	Senior loans	PKSF
	Output 2.2 Community-based farmed crab supply chain created	12.18	8.58	Grants	3.6	Senior loans	PKSF
Outcome 3: Improved climate planning and implementation by communities and local level institutions	Output 3.1: Climate change adaptation groups (CCAG) formed and operationalized	1.34	0.47	Grants	0.87	In-kind	PKSF
	Output 3.2: Capacity built among IEs and relevant institutions apprised on project	2.20	2.20	Grants			
	Output 3.3 Knowledge products prepared and disseminated	0.62	0.62	Grants			
PMC		2.01	1.70	Grants	0.32	In-kind	PKSF
Indicative total cost(USD)		**49.99**	**42.20**		**7.79**		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

183. The project will establish 50 crab hatcheries for smooth supply of adolescent crab to the crab growing and fattening farmers. Estimated GCF financing for this technology is USD 1,205,000. Initially the project will provide training to two technical staff in each hatchery for crab let production as well as crab rearing for the duration of six months. Through this training, they will learn technical details on the structural design of the hatcheries, crab hatching, nursing and rearing. Each hatchery will have 5-6 staff for operation. They will also learn about crab hatching technology and crab rearing. At the same time, the project provided training to 500 crab nurseries; 20,000 crab farmers. When trained farmers are started farming the neighbour follows them to use improved technology rather than traditional.

184. The slatted house for goats and sheep is also an emerging technology for the selected areas. The estimated GCF financing for this technology is USD 2,825,000. The farmers, particularly the women, will be trained on this technology along with other veterinary services. This will enhance their knowledge and capacity on climate-resilient livestock farming in the changing situation. In addition, climate-adaptive vegetable cultivation on homesteads will be implemented. Bangladesh Agricultural Research Institute (BARI) has invented various salinity-adaptive vegetables for the coastal zone. The project will promote those varieties which will cost USD 1,405,000.

185. Resilient homestead is another important technology of the project to be transferred to the vulnerable coastal people. The estimated GCF financing of this technology is USD 22,785,600. 20,000 households will receive support for planting trees around their homesteads to increase resilience of the houses against storm. The estimated GCF financing of this technology is US$1,625,000.

186. Besides these, IE's staff will receive training on project management, procurement, climate ~~change~~ adaptation and implementation procedures for smooth operation of the project. The estimated GCF financing of capacity building is USD 2,823,500.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

D.1. Impact potential (max. 500 words, approximately 1 page)

187. The proposed project will **contribute to increased climate-resilient sustainable development**. The proposed climate-resilient homesteads will secure lives and livelihood resources of the selected 13,500 beneficiaries. The women beneficiaries will have additional benefits by avoiding difficulties they face when using community shelters (Details can be seen in Annex 8). The poultry and livestock of the beneficiaries will be secured because they will stay at raised and safe place. The trees around the homesteads to be planted will protect them from cyclonic storm as well as provide income. More importantly, other people may take shelter on the resilient homestead during an emergency situation. Thus, the dwellers of the resilient homesteads will have social values and honor by providing shelter to the affected communities.

188. The land-based crop production in the project area has already become a challenging task. Mud-crab, a salinity-tolerant species which is resilient and less affected by natural calamities and diseases. Establishment of crab hatcheries will ensure supply of available juvenile crabs for fattening which is currently collected from nature. The availability of crab in nature is also going to extinct due to over exploitation. So, the project will contribute to adapt with the increasing salinity in the coastal areas of Bangladesh as well as protect the species in the nature.

189. The establishment of the crab hatchery and farming will create opportunities of sustainable employment for the coastal people. Some people will produce juvenile crabs, some will nurse in ponds for fattening, and some other people will fatten the crabs on their farm land. In addition, some people will get involved in transporting crab-lets and mature crabs to local, national or international markets, some other people will get involved in crab processing. These value chain actors will be indirectly benefited by the project interventions. Thus, the project will create a new window into climate adaptive economic development. It is expected that the project will make resilient to 75 per cent of the selected community, generate climate-adaptive employment for 80 per cent and improve overall socio-economic conditions (i.e., income, health, nutrition, housing) of 75 per cent of the selected community.

190. The number of direct beneficiaries will be 3,62,475 . The project will establish 50 micro crab hatcheries in the coastal areas to produce juvenile crabs which will be used by the crab growers. One micro hatchery will support 10 crab nurseries. So, 50 micro hatcheries will provide support to 500 crab nurseries. On an average 2 fulltime workers will work at each crab hatchery. So, the micro crab hatcheries will have 100 direct beneficiaries. 2,250 crab nurseries will support 20,000 crab growers for crab growing. In addition, 900 people will be engaged in different levels of value chain of crab business. Furthermore, the proposed project will provide technological support for sheep rearing to 90,000 beneficiaries, which have been proven as salinity tolerant livelihood options for the coastal zone. The project will support another 90,000 beneficiaries through tree plantations at their homesteads. This will help them earning income, meet their nutrition demand and protect their houses from cyclones and storms. The housing component of the project will protect their health and livelihood resources leading improvement well-being. The project will also support 90,000 beneficiaries in producing salinity-tolerant vegetable cultivations in their homesteads. 13,500 vulnerable and ultra-poor households will receive support for climate resilient homestead. The number of indirect beneficiaries will be around 770,050 people. It is expected that 75 per cent of the selected people will increase their resilience to climate change and related consequences.

191. Women will particularly engage in homestead-based activities i.e., sheep rearing, homestead vegetable gardening and tree plantation around homestead. Hence, all the beneficiaries related to these activities will be women. The project targets that at least 10 per cent percent of the households will be female headed. The project will put priority to the women heads for supporting climate resilient homestead. The project also targets about 20 per cent percent of the crab growers will be women. Thus, roughly estimated 50 per cent of the direct beneficiaries will be women.

D.2. Paradigm shift potential (max. 500 words, approximately1 page)

Potential for scaling up and replication

192. The project will use the negative impacts of climate change related negative impacts as an opportunity for the coastal people. Crab farming requires a comparatively high level of salinity. The coastal area will experience severe salinity under the changing climate situation. Thus, coastal people will fully use the adverse impacts of climate change by the project interventions for their well-being.

193. The homestead of coastal people is highly exposed to sea level rise, coastal flooding and storm surge. The project will promote climate-resilient homestead which is a self-sustained adaptation interventions for the coastal vulnerable communities in the country. This is an innovative approach of making the homestead resilient to climate change in the coastal zone. Because, traditionally people raise only the plinth of house but not the whole homestead area. As a result, home yards inundate during monsoon, even if there are normal tides and associated floods. Through these interventions, the people will learn how to raise the whole homestead area above flood and tidal surge level. They also learn how to construct an affordable climate-resilient house in their locality.

194. As the impact of climate change is reducing opportunities in the agricultural sector due to increase in the salinity level, the whole community should be involved in potential and sustainable land use and production. Crab hatching and fattening is one of the most suitable alternatives for the people in coastal regions.

195. Traditionally, some people are already engaged in crab rearing. The juvenile crabs are collected from natural sources and they are fattening the crabs in traditional ways. The local people informed that they could catch crab in their nearby rivers earlier (about 10-15 years back) but currently they have to travel 3-4 km meters away to catch crab (derived from community consultation in Satkhira district). Providing technological support and training to them will significantly increase the production and minimize the cost. Both the hard- and soft-shell crab interventions will be in the project. Moreover, indiscriminate collection of juvenile crabs from nature is damaging the ecological balance. Establishment of micro crab hatcheries will ensure sustainable collection of juvenile crabs. PKSF will provide micro-enterprise loans to establish the crab sub-sector in the coastal districts. PKSF will nurture the activities through Value Chain Development (VCD) for crabs. Under the value chain components of the project, crab farmers will get technological support, market linkage and microenterprise loan support from the project.

196. The PKSF expects that crab hatchery will rapidly be extended in the coastal community due to its multiple benefits. They will learn about sustainable new technology in crab hatchery and farming through the project intervention. It will create employment opportunities for technical and non-technical people and increase their income earning. It is also expected that project interventions will create profitable business interventions at local, national and international levels which will help rapid scaling up of the activities.

197. The PKSF has demonstrated goat and sheep rearing in slatted houses in Assasuni upazila (sub district) of Satkhira district under CCCP. The demonstration’s effects were courageous. Many people started goat and sheep rearing in slatted houses to increase their income. The project will follow similar technology and management for goat and sheep rearing. This is a women-friendly technology for livelihood adaptation to coastal zones. The women can get easily involved in goat and sheep rearing. It increases their income due to low risk and higher income.

198. The proposed project conceptualizes relatively expensive housing to make it resilient to climate change related shocks. Because low-cost housing or traditional one cannot resist the critical climate sensitivity in the coastal zone. The proposed project will design a model house for demonstration. The design will consider the effects of storm surges, cyclones and coastal flooding. The project also considers challenges of existing the housing system to make it sustainable. Hence, it is expected that other development practitioners and community people in the coastal area will follow the housing model for sustainability.

Potential for knowledge sharing and learning

199. The project will generate information and knowledge on the effectiveness of proposed interventions and share with different levels of stakeholders through meetings, seminars, workshops and other informal communications. The project will also carry out best practices and lessons learned to disseminate the technologies and practices in other communities in the coastal zone.

Contribution to the creation of an enabling environment

200. PKSF has been investing in research and development of hatchery-based crab farming for the last few years. A big hatchery was established in 2016 with technical support from Vietnam experts, which is now in operation. Besides, PKSF has established 3 micro-scale hatcheries at entrepreneurs' level. These micro-scale hatcheries are also in operation. Moreover, commercial banks and MFIs exist in the project area to finance the expansion of hatchery-based crab farming and business.

201. PKSF will implement the project through IEs. They will have expertise on addressing climate change impacts. The communities will increase their knowledge through practice climate resilient livelihood and resilient homestead. This will create demonstration effects in the society. The government will also incorporate these activities in its climate change related policies and strategies. IEs will continue with their microenterprise loan in promoting this livelihood. There is already an established market and it will continue further when the IEs will continue working with their micro enterprise loan component.

Contribution to the regulatory framework and policies

202. Historically, the government of Bangladesh adopts a re-active type of policies and regulations to promote potential economic and development sectors. For example, the government adopted the National Shrimp Policy in 2014 while this sector was developed in 1980's and expanded in 1990's. It is expected that the project will influence the policy makers by contributing to the export earning of the country, local economy, and creating pro-saline livelihood of the coastal vulnerable people. Once, economic activity and income of the crab sector would be noticeable to the government, they would take necessary steps to amend existing fisheries policy or adopt a new policy related to crab farming. PKSF will engage all levels of government officials through meetings, workshops and seminars to sensitize them in this regard. Besides, PKSF as government-owned company can influence the government of Bangladesh's policies and strategies for integrating climate change adaptation livelihood and resilient housing in the coastal areas of the country.

Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans

203. It is expected that the project will influence several policies of the government. The major policies to be influenced by the project interventions are national land use policy, national fisheries policy, national water policy, and jalmahal (wetland) policy. With the growing economic development of the sub-sector, the government will create favorable policy and regulatory environment for production and trading of crab. There is huge potential of exporting of processed meat of goat and sheep. Satisfactory production of meat and other products like milk, skin would encourage the government of Bangladesh to create favourable policy and regulations for production and trading of goat and sheep.

Transformational aspects of the project:

204. Crab hatchery and farming using already salinity affected areas. Private entrepreneurs will be engaged in establishment of crab hatcheries. They will be continuing their business in the long run. Besides, PKSF through its partner organizations will continue financing in this sector based on needs of the entrepreneurs. The market infrastructure and linkages are already established as this business is being run with natural crab.

205. The entrepreneurs, local NGOs and selected Universities will enhance capacity in juvenile crab production, growing crab and promote trades in national and global market.

206. The RHL project will empower the project beneficiaries in terms of awareness raising, increasing income for their families and playing role in decision making of their families.

207. The resilient homesteads will influence government's existing and future housing projects in integrating climate change for greater sustainability. The project will share the results of resilient housing with the National House Building Research Institution so that they integrate climate change in their design.

D.3. Sustainable development (max. 500 words, approximately 1 page)

Environmental co-benefits

208. By nature, the project will benefit the environment of the intervention areas. Presently, crab catchers and farmers collect crab from the natural sources. The crab catchers informed that they do not get crab in their nearby areas, which they used to get 10 years back. Now, they have to go far from their locality into the river or Sundarbans, i.e., the iconic mangrove forest in Bangladesh, for catching crabs. It means, availability of crabs in the nature is decreasing day by day. The project will produce crablets in the hatchery and hence, crab catchers do not need to catch wild crabs. So, it will automatically protect the crab in the nature and improve ecological balance. Litters of goats and sheep are used to produce good quality of composed-fertilizer. It is also used to produce vermi-compost which is very important for improvement the quality of soil.

Social co-benefits

209. Individuals will not only be better able to protect assets as a result of this project (direct benefit) but also it will provide security and well-being co-benefits to women and girls by avoiding the need to use communal shelters during extreme weather events. Increased income of the coastal people and diversification of their farming operations will improve quality of their lives and food and nutrition security, reducing the traditional factors that push rural communities to migrate to urban areas.

Economic co-benefits

210. The project promises to spark growth in local secondary service economic activities, from transport to casual laborers to equipment maintenance and provision. It is anticipated that these secondary service providers will synergistically encourage local growth and employment, increasing local wealth and further decreasing urban migration.

Gender-sensitive development benefit

211. Crab growing and fattening technology is very simple and easily manageable. Women can get involved in crab fattening and other activities of the project. PKSF from its beginning puts emphasis on disadvantaged and excluded segments of the society. 80 per cent per centof PKSF's beneficiaries are women. Like other projects or programmes, the project will focus on gender-based development so that women and other excluded groups can get access to the project intervention. It will enhance the capacity of women in the coastal zone and raise their voice towards their rights. Wages of male and female will be the same under the project intervention.

212. Traditionally, women in Bangladesh look after livestock at household level. The project will select all women participants for implementing goat/sheep rearing in slatted houses. Thus, the interventions selected under this project are women friendly and will empower them economically as well as decision making. The housing component of the project will certainly reduce their physical vulnerabilities as they would not need to go shelter during the emergency situation.

213. The proposed climate resilient homestead will ensure security of women and adolescent girls during disasters because they would not need to go to community shelter. In addition, they can grow vegetables round the year on their raised homesteads that will improve nutrition of both male and female members of the households.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

214. In spite of many challenges including a huge population, poverty, climate change and disaster; Bangladesh has become one of the fast-growing economies in the world. Before COVID-19, the country has been maintaining around 7-8 per cent per cent GDP growth over the last decade. The poverty headcount ratio at USD1.90/day reduced to 5.14 per cent per centin 2021 from 20.38 per cent per centin 2010. Considering poverty level at USD 3.20/day, the poverty headcount ratio is very high through shows a reduction trend from 57.86 per cent per cent to 33.17 per cent respectively per cent (Source: https://dashboards.sdgindex.org/profiles/bangladesh/indicators). Bangladesh has also achieved significant development goals in many indicators. For example, maternal mortality rate has reduced from 434/100,000 birth in 2000 to 173/100,000 birth in 2017; mortality rate of under 5 children reduced to 30.83/1000 live births in 2019 from 86.50/1000 live births in 2000; net primary enrolment increased to 120per cent in 2020 from 105per cent in 2010; per capita income increased to USD 1961.6 from USD 781.2 in 2010; and GDP growth rate increased from 5.57 per cent per cent in 2010 to 8.2 per cent per cent in 2019 with the exception in 2020 when GDP's growth rate reduced to 3.5per cent mainly due to COVID-19 pandemic. However, this development is not equal throughout the country. The rural areas are lagging behind from urban areas and the climate hotspots within the rural areas are far behind this development trajectories. The coastal region of Bangladesh is recognized as one of the most climate hotspot regions of the country.

215. A World Bank Study (EACC, 2010)[56] estimates that under the baseline scenario, the damages and losses stand at USD 4.6 billion from a single cyclone and associated storm surges of a 10 years return period. With climate change, the damages and losses would increase to USD 9.16 billion, including USD 4.5 billion attributable to climate change.

216. The housing of coastal areas is mainly affected by cyclones and storm surges and coastal flooding. Nation-wide, 70.31 per cent households in disaster-prone areas have *katcha* (making by mud) houses and 17.44 per cent have semi-*pucca* houses. These houses are vulnerable to heavy rainfall, flooding and storm surges as they are made with mud and straw or *golpata (Nypa spp.)* (traditional leaves found in the Sundarbans) (BBS, 2015)[57]. This means, around 88 per cent per centhouseholds are vulnerable to climate change and associated disasters in the country. The recent cyclone in May, 2020 has destroyed 55,667 houses completely and around 162,000 houses partially in the southwest coastal zone of Bangladesh. Cyclone Mahasen on 16 May 2013 has damaged 150,000 houses (IFRC, 2013)[58]; Cyclone Aila in May, 2009 has damaged 500,000 houses (DMB, 2009)[59] and Sidr in November, 2007 has damaged 600,000 houses in the coastal areas of Bangladesh. These damages are expected to be increased in future with the increasing intensity of cyclones and storm surges.

217. Considering salinity intrusion due to climate change, crop and vegetable cultivation is facing growing challenges. The salinity intrusion affects coastal agriculture faster than the research outcome reaches the farmers. However, there are some salinity-tolerant varieties including BRRI dhan-47, BINA dhan8, BINA dhan10, sunflower, and sugarcane, but these varieties can tolerate limited levels of salinity i.e., less than 15 ppt. But most of the coastal land experiences more salinity, particularly during the dry season (mostly from September to May). Thus, technologies and practices should consider salinity-based production systems for building resilience of the vulnerable community. In this context, crab fattening a potential resilient livelihood for the coastal poor community.

218. Thus, community consultations carried out during preparatory phase of this project suggested that majority of the coastal population is poor, small and marginal farm families and shrimp workers. They build their houses in low-lying areas which are subject to coastal flooding. Most of the houses are built with mud and *goal pata* which are severely affected by cyclones, storm surges and high tides. These

[56]EACC, 2010: *Bangladesh - Economic of adaptation to climate change : Main report (English).* Washington, D.C. : World Bank Group.

[57]BBS (2015). Bangladesh Disaster Related Statistics, Bangladesh Bureau of Statistics, Government of Bangladesh.

[58]IFRC (2013).Emergecy Appeal Six-Month Consolidated Report – Bangladesh: Tropical Cyclone Mahasen.https://www.ifrc.org/docs/Appeals/13/MDRBD01302.pdf, accessed 10/07/2015

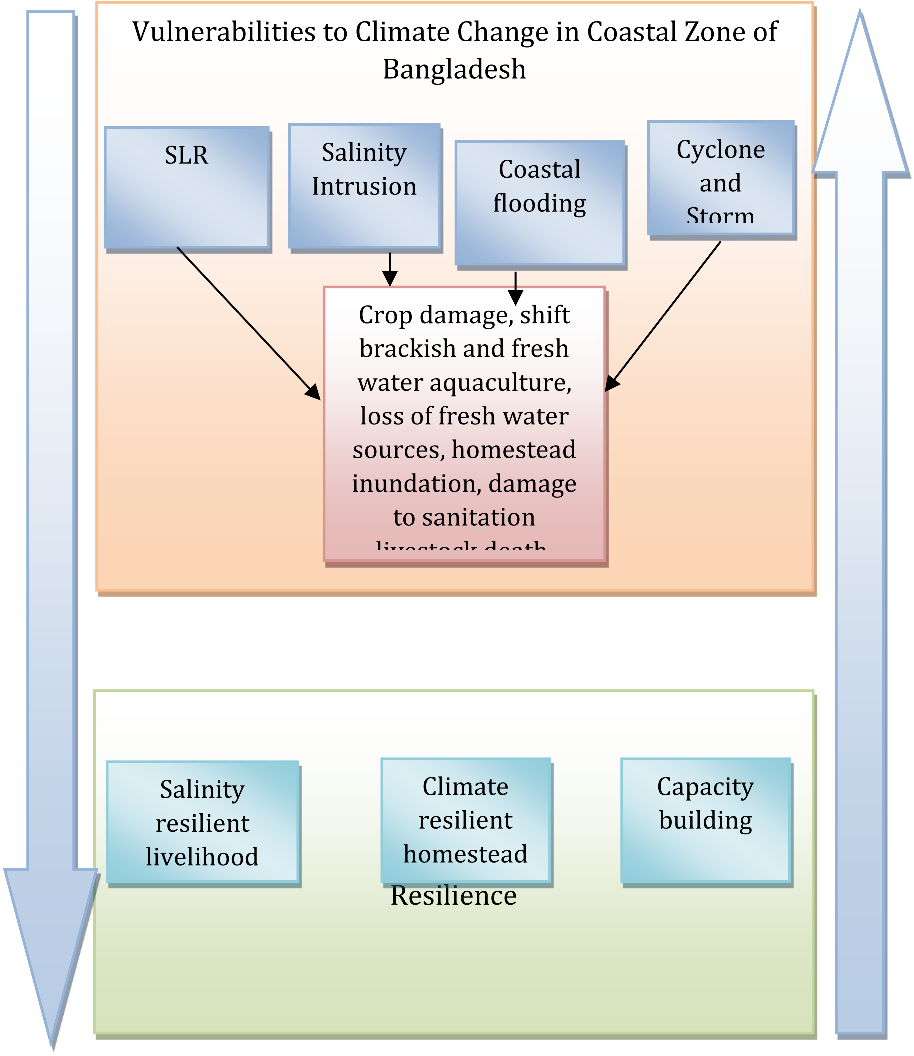
[59]DMB (2009a). District-wise Damage Report. Disaster Management Bureau (DMB). Ministry of Food and Disaster Management (MOFDM). Government of Bangladesh. Dhaka: 3 June 2009.

people have to spend a significant amount of their earnings repairing houses each year. Hence, they could not come out from the economic poverty. The livelihoods of these people primarily depend on seasonal subsistence agriculture and agriculture wage labour which are highly climate sensitive. Cultivable land is shrinking due to increased sea level and expansion of salinity. Thus, agricultural production is hampered. The surface and groundwater in the coastal zone of the country is increasingly contaminated with saline water due to seal level rise, cyclones and tidal surges. The coastal people, particularly the women, collect drinking water far away from their locality, spending a productive long time which also push them back into poverty. The climate scientists argue that this situation will be further aggravated by the consequences of climate change.

219. Physical risks associated with these extreme events in conjunction with the societal exposure risk levies immense pressure upon coastal habitat. Sea level rise comes as a direct result of the adverse influence of the changing climate upon global tidal patterns and is ultimately increasing the risks of the vulnerable population and their livelihoods. Apart from being a catastrophe in itself, abrupt rise in sea level can cause a chain reaction of inimical natural calamities based on cause and effect. Salinity intrusion in the coastal regions is one of the major effects of sea level rise, which is currently evident in not only the various existent freshwater sources but also in the soil, thus threatening agriculture production-based livelihoods, water security and crop production. Intrusion of saline water will continue as the area under 1 ppt salinity line is expected to increase by 18.22 per cent and area under 5 ppt salinity is expected to increase 24 per cent by 2050 (CEGIS and DOE, 2011)[60].

[60]CEGIS and DoE (2011). Final report on programmes containing measures to facilitate adaptation to climate change of the second national communication project of Bangladesh. Dhaka, Department of Environment.

220. Polders are particularly at risk of higher tidal surges. This is because once a polder embankment is breached, the enclosed land often stays waterlogged for long periods of time, making agriculture and other livelihood activities nearly impossible. Due to saline water intrusion, salinity in the soil increases. This salinity further goes down in the subsurface soil. Mainly due to lack of precipitation during the dry season, sometimes sub-surface salinity moves upward. So agricultural and livelihood activities could be hampered even after water recedes.



D.5. Country ownership(max. 500 words, approximately 1 page)

221. The constitution of the People's Republic of Bangladesh has given rights to all citizens to have a secured residence. The project will support most vulnerable coastal people to avail their rights as per the constitution. This will contribute to the government's different policy implementation particularly addressing climate change impacts including Bangladesh Climate Change Strategy and Action Plan (BCCSAP), Bangladesh National Adaptation Programme of Action (NAPA), National Communications, 8th Five Years Plan, and Perspective Plan.

222. The project is listed in the summary table in page 4-5 of the Bangladesh country programme. It is numbered A25 and titled "Increase resilience to climate change in southwest coastal zones of Bangladesh through adaptive livelihoods, housing, and safe drinking water supply." The project will address the Bangladesh Climate Change Strategy and Action Plan (BCCSAP), 2009 and one of its pillars -- food security, social protection and health. It is the first among the six pillars of the BCCSAP. The pillar has nine programmes of which three are directly related to the project interventions. Programme 4 of the pillar is "Adaptation to Fisheries Sector" and Programme 5 is "Adaptation to

Livestock Sector". As the increased sea level rise and salinity will retreat brackish water aquaculture, crab farming would be popular in this changing situation. People will take the opportunity from the adverse situation through crab production, transportation, processing and marketing. The project will also address Programme area 8, i.e., livelihood protection in ecologically fragile areas. The coastal zone of Bangladesh is critical by nature. Because, the poor people who are living in the coastal zone are dependent on the coastal ecosystem services. Thus, over exploitation of resources makes the ecosystem critical in terms of sustainability. The project intervention will create sustainable livelihood for about 0.1 million people. They would no longer depend on natural resources. Thus, the project will help achieving national strategies and action plans particularly related to climate change. The project will also support Programme areas 5 and 6 of the Thematic Area of "Infrastructure" of the BCCSAP, 2009, i.e., adaptation to flood and adaptation against cyclones and storm surges through implementing climate-resilient houses for the vulnerable coastal community.

223. The project will address the Bangladesh National Adaptation Programme of Action (NAPA). The National Adaptation Programme of Action (NAPA) for Bangladesh has been prepared by the Ministry of Environment and Forest (MOEF), Government of the People's Republic of Bangladesh as a response to the decision of the Seventh Session of the Conference of the Parties (COP-7) of the United Nations Framework Convention on Climate Change (UNFCCC). NAPA prepared a list of priority activities to mitigate impact of climate change and most of the activities of the proposed project qualify with NAPA listed activities. NAPA also developed 45 project concept notes and the nature of activity of the proposed project is partially the same with all concept notes. The proposed project is directly similar to Project No. 4 outlined in the NAPA. Climate change and adaptation information dissemination to vulnerable communities for emergency preparedness measures and awareness rising on enhanced climatic disasters, Project No. 6, i.e., mainstreaming adaptation to climate change into policies and programmes in different sectors (focusing on disaster management, water, agriculture, health and industry). Project No. 10, i.e., promotion of research on drought, flood and saline tolerant varieties of crops to facilitate adaptation in future, and Project No. 12, i.e., adaptation to agriculture systems in areas prone to enhanced flash flooding – northeast and central region of the Bangladeshi coast.

224. The Government of Bangladesh under the leadership of Prime Minister Sheikh Hasina adopted the Vision 2021.The Vision 2021 and the associated Perspective Plan 2010-2021 have set solid development targets for Bangladesh by the end of 2021.The vision of the perspective plan is to take effective measures to protect Bangladesh from the adverse effects of climate change and global warming. The plan targets to take all possible steps to protect the vulnerable people from natural calamities. Steps will also be taken to make Bangladesh an ecologically attractive place and to promote tourism in this regard. The objectives of the proposed project comply with that of Vision 2021.

225. This project is strongly aligned with national policies including: Seventh Five Year plan of Bangladesh government, the Bangladesh National Adaptation Programme of Action (NAPA) and Bangladesh Climate Change Strategies and Action Plan (BCCSAP), 2009, National Plan for Disaster Management (NPDM), and Sixth Five Year Plan (SFYP) of GOB boldly articulate the country's commitment to addressing climate change and equitable development, while the National Plan for Disaster Management (NPDM) 2008-2015 addresses Disaster Risk Reduction (DRR) and climate change adaptation (CCA) comprehensively in all development plans, programme and policies. The forthcoming Seventh Five Year Plan (2016-20) put emphasis on Accelerating Growth, Empowering Every Citizen.

Capacity of Accredited Entities or Executing Entities to deliver

226. PKSF will play the role of Executing Entity for the project. It was established by the Government of Bangladesh in 1990 and registered under the Companies Act 1913/1994 as a "not for profit" organization with the vision of "A Bangladesh where poverty has been eradicated; ruling development and governance paradigm in inclusive, people-centered, equitable and sustainable; and all citizens live health, appropriately, educated and empowered and humanly dignified life".

227. PKSF provides a wide range of development services including financial, health, educational, capacity development, technology transfer and business development services to disadvantaged segments of the society through appropriate pro-poor institutions. Mobilization of poor people and provision of necessary training with appropriate financial supports have been the initial and continuous interventions. PKSF has a significant pro-poor strategy that looks at poverty alleviation from a holistic way. PKSF constantly redefines and redesigns its interventions, taking into account the concerns and needs of the poor with the changes of times. PKSF believes that appropriate financial support is not the only answer to alleviate poverty and it cannot be achieved with one component only. This requires addressing the needs of education, training, and healthcare, access to resources and equal opportunities for all. Hence PKSF developed 9 core programmes which include inclusive financial services, people centered holistic development programme, enterprise development programme, social protection programme, capacity building programmes, advocacy and knowledge management, research and development (R&D) and Environment and Climate Change programmes. PKSF has established the 'Environment and Climate Change Unit' for addressing climate change in Bangladesh. The Community Climate Change Project (CCCP) implemented by PKSF has been highly appreciated by participating development partners, the World Bank, the Government of Bangladesh and the civil society and media of the country. The project has successfully established a financial mechanism for channelling climate change fund in LDC country like Bangladesh.

228. The project has been designed in full consultation with the relevant stakeholders of Bangladesh. The concept was also shared other relevant stakeholders including the Department of Environment, Department of Fisheries, Department of Agriculture Extension. In addition, the proposal was also shared with relevant national and international organizations including UNDP, ICCCAD, Action Aid, and WFP. Based on their comments and suggestions, the project proposal was finalized.

229. Several consultation meetings and interviews were carried out in the coastal zone. Local government representatives, and local NGOs and community people participated in discussions on the barriers, challenges and adaptation gaps. The consultations focused on new technology of crab hatchery, potentials of crab sub-sector in terms of climate change, climate resilient livelihood options, crab processing and marketing. The current proposal incorporates the feedback from these consultations. The project will engage further with relevant stakeholders (target communities, NGOs, CBOs, local government) to ensure stakeholder input throughout the implementation period for the proposed activities.

230. Crab hatching technology is an emerging practice in Bangladesh. The project will require high level technical expertise for establishment of hatchery and providing training. PKSF has already gained experience in establishing crab hatchery at enterprise level. Crab hatching technology is very critical in terms of water management (maintaining water salinity, PH, temperature etc.), feeding, disease identification etc. It also established three hatcheries in Cox's Bazar and Satkhira districts at the entrepreneur level. The PKSF is now capable of hatchery management and producing crab from the hatchery.

231. Women participation is the most important element to ensure the views and captured, specific efforts were made to consult with women groups, and to collect information regarding the impacts of climate change on women, in the design of this project proposal. PKSF mostly preference to work with women at projects and programmes and was consulted at both the national and local level, and field missions took care to consult with both women and men regarding lessons learned to date. The project also benefits from important lessons learned in previous implemented projects and programs that have specifically aimed to increase the participation of women, senior citizens, youth and other vulnerable groups. Feedback and lessons learned from previous project and programs reviews and policy reviews have been applied in the design of activities. The application of community-based approaches during implementation will also ensure that regular communication is maintained throughout implementation with commune level representatives.

D.6. Efficiency and effectiveness (max`. 500 words, approximately 1 page)

232. It is envisioned that through proposed interventions, the project will address several crucial sustainable livelihood and resilient related issues pertinent to the coastal inhabitants of Bangladesh. Specifically, the project will employ interventions related to (i). inadequate quality of human settlements in low-lying areas of Bangladesh, (ii). climate sensitivity of the population in selected regions, (iii). scarcity of safe drinking water.

233. While it is expected that all proposed project interventions will provide significant benefits to intended beneficiaries and the entire economy, the benefits stemming from some of these interventions cannot be monetized due to the issues related to the lack of readily available valuation literatures in Bangladesh. Considering time and budget constraints the benefits have been guesstimated. On the other hand, interventions that consist of activities expected to positively influence the incomes of intended beneficiaries (e.g., interventions proposing the establishment of crab hatcheries and fattening crab activities, etc.) were assessed in financial and economic terms.

234. It is envisioned that the project will help establish micro crab hatcheries and crab culture/fattening facilities in the coastal areas of Bangladesh. PKSF's piloting experience setting up crab hatcheries will help facilitate the process. The project will provide micro-enterprise loans to the beneficiaries (micro hatchery and crab growers) and technological and value chain support. PKSF will continue these project's activities through the same IEs with microenterprise loans to the crab sub-sector. The necessary technical know-how and value chain development are expected to be passed on to all intended beneficiaries.

235. The goal of proposed project interventions in the crab sector is to jump-start seafood production by providing the necessary financial support for establishing environmentally conscious and economically sustainable crab production. This sub-sector will additionally help promote innovative and climate-friendly technological solutions in other branches of seafood production. Furthermore, it is expected that the private sector will catch up in the longer run, and private entrepreneurs will start investing in the sector, helping it grow beyond the project's life. The Government of Bangladesh will help create a favorable policy and business environment to promote the sector. It is expected that when the sector develops further and shows investment potential, the availability of commercial funding will increase.

236. Furthermore, it is envisioned that similarly to crab sector interventions, the other proposed activities in the livestock sector (e.g., construction of slated houses for goat and sheep rearing or delivery of proper inputs and training for saline-tolerant vegetable production) and saline-tolerant vegetable production will also positively influence the incomes of intended beneficiaries. Once initiated by the funding stemming from this project in the long run, these activities will be financed by the local MFIs, once they become commercially operational and viable.

237. As to adaptation benefits, overall, the project is expected to deliver multiple positive adaptation impacts in the coastal communities of Bangladesh and will contribute to the national GDP. The project will help in developing a base of various entrepreneurs at different stages of the value chain of the crab, livestock, and vegetable sub-sectors. The project is also expected to create employment for vulnerable coastal people and provide livelihood support to more than 362,475 beneficiaries. The project envisions a significant impact on the women and youth population as many of the proposed activities can be pursued by females or youth. Thus, the project financing will effectively obtain the project's objectives and help diminish gender disparities in coastal communities by empowering women. By implementing project activities that are gender inclusionary, the community will automatically continue this activity in the long run.

238. Lastly, the project will organize workshops and meetings with stakeholders, including government representatives, policymakers, national and local NGOs, and financial organizations. The workshops will share project lessons with these stakeholders to create a favorable policy environment for scaling proposed interventions. The project's-facilitated communication and knowledge pass-through is expected to help create an enabling environment for the future development of supported sectors. It is anticipated that the NGOs and financial organizations will be more willing to provide financial support to the entrepreneurs to extend the business co-financed by this project.

Economic and Financial Analysis

239. The RHL project proposes implementing several activities to increase livelihood resilience to climate change in the coastal region of Bangladesh. As the coastal areas of Bangladesh are the most vulnerable to climatic shocks, e.g., cyclones, and the frequency and intensity of these shocks are expected to increase over time, the proposed project's activities will help address population vulnerabilities to these shocks.

240. The appraised project's interventions are geared toward the most vulnerable communities in the coastal areas of Bangladesh with highest poverty rates and significant levels of social and economic disadvantage. The intended beneficiaries are characterized by limited scope to increase their incomes and livelihoods. It is certain that without any external support, these communities will be unable to internalize on their own the costs of improved adaptation technology due to low incomes and lack of access to commercial financing (e.g., lack of collateral or credit worthiness to obtain a commercial loan).

241. Financial Analysis was pursued on interventions that are expected to provide income increase for intended beneficiaries. The analysis was conducted separately for all five (05) activities (goat rearing, traditional vegetable farming, crab hatchery, crab nursing and crab fattening), under two different scenarios; one with RCP 4.5 (medium case scenario) and another with RCP 8.5 (worst case scenario), with and without adaptation measures, and compared with the BaU case.
The results of models assessed in the financial analysis are presented in Table 6.

Table 6. Economic and Financial Analysis

1. Goat rearing

Aggregate Results of Financial Analysis									
Activities	NPV			IRR			Payback Period		
	BaU	RCP 4.5	RCP 8.5	BaU	RCP 4.5	RCP 8.5	BaU	RCP 4.5	RCP 8.5
	Present Climate	Moderate	Extreme	Present Climate	Moderate	Extreme	Present Climate	Moderate	Extreme
Without GCF (No grant/100% PKSF loan)	-$51,951	-$1,162,565	-$1,361,291	9%	6%	5%	More than 14 years	More than 14 years	More than 14 years
With GCF (GCF grant+ PKSF Loan)	$2,309,726	$1,199,112	$1,000,386	17%	13%	12%	More than 9 years	More than 10 years	More than 11 years
With GCF (100%GCF grant+0%PKSF loan)	$5,472,686	$4,362,072	$4,163,346	35%	30%	29%	More than 6 years	More than 6 years	More than 6 years

2. Traditional vegetable farming

Aggregate Results of Financial Analysis									
Activities	NPV			IRR			Payback Period		
	BaU	RCP 4.5	RCP 8.5	BaU	RCP 4.5	RCP 8.5	BaU	RCP 4.5	RCP 8.5
	Present Climate	Moderate	Extreme	Present Climate	Moderate	Extreme	Present Climate	Moderate	Extreme
Without GCF (No grant/100% PKSF loan)	-$939,211	-$954,994	-$966,357	-	-	-	More than 9 years	More than 9 years	More than 9 years
With GCF (GCF grant+ PKSF Loan)	-$453,640	-$469,423	-$480,785	-	-	-	More than 9 years	More than 9 years	More than 9 years
With GCF (100%GCF grant+0%PKSF loan)	$31,932	$16,149	$4,786	24%	17%	11%	More than 1 years	More than 1 years	More than 1 years

3. Crab Hatchery

Aggregate Results of Financial Analysis									
Activities	NPV			IRR			Payback Period		
	BaU	RCP 4.5	RCP 8.5	BaU	RCP 4.5	RCP 8.5	BaU	RCP 4.5	RCP 8.5
	Present Climate	Moderate	Extreme	Present Climate	Moderate	Extreme	Present Climate	Moderate	Extreme
Without GCF (No grant/100% PKSF loan)	-$262,666	-$361,450	-$442,454	5%	5%	4%	More than 24 years	More than 24 years	More than 24 years
With GCF (GCF grant+ PKSF Loan)	$227,060	$128,276	$47,272	12%	11%	10%	More than 16 years	More than18 years	More than 19 years
With GCF (100%GCF grant+0%PKSF loan)	$618,841	$520,056	$439,052	22%	20%	18%	More than 9 years	More than 10 years	More than 10 years

4. Crab Nursing

Aggregate Results of Financial Analysis									
Activities	NPV			IRR			Payback Period		
	BaU	RCP 4.5	RCP 8.5	BaU	RCP 4.5	RCP 8.5	BaU	RCP 4.5	RCP 8.5
	Present Climate	Moderate	Extreme	Present Climate	Moderate	Extreme	Present Climate	Moderate	Extreme
Without GCF (No grant/100% PKSF loan)	-$27,989	-$173,262	-$272,351	8%	6%	4%	More than 24 years	More than 24 years	More than 24 years
With GCF (GCF grant+ PKSF Loan)	$172,725	$27,452	-$71,637	13%	10%	7%	More than 16 years	More than 21 years	More than 24 years
With GCF (100%GCF grant+0%PKSF loan)	$273,082	$127,808	$28,720	18%	13%	10%	More than 14 years	More than 18 years	More than 24 years

5. Crab Fattening

Aggregate Results of Financial Analysis									
Activities	NPV			IRR			Payback Period		
	BaU	RCP 4.5	RCP 8.5	BaU	RCP 4.5	RCP 8.5	BaU	RCP 4.5	RCP 8.5
	Present Climate	Moderate	Extreme	Present Climate	Moderate	Extreme	Present Climate	Moderate	Extreme
Without GCF (No grant/100% PKSF loan)	-$728,730	-$5,126,099	-$6,094,944	8%	4%	3%	More than 24 years	More than 24 years	More than 24 years
With GCF (GCF grant+ PKSF Loan)	$5,266,466	$869,098	-$99,748	17%	10%	9%	More than 11 years	More than 18 years	More than 24 years
With GCF (100%GCF grant+0%PKSF loan)	$8,363,984	$3,966,616	$2,997,770	29%	18%	16%	More than 7 years	More than 12 years	More than 13 years

The overall result shows that, with proposed grant support from GCF (both proportionate GCF grant and proportionate PKSF loan), respective NPV is positive for four (04) individual activities (goat rearing, crab hatchery, crab nursing and crab fattening) under RCP 4.5. But under RCP 8.5, there are both positive and negative NPVs. In case of traditional vegetable farming, NPV is positive under all three scenarios (BaU, RCP 4.5 and RCP 8.5) only with 100% grant support from GCF (No PKSF Loan). The analysis represents the necessity of GCF funding for increasing resilience under vulnerable climatic situation.

In case of IRR, its generally low under RCP 8.5 regardless with and without adaptation measures due to the intensity/scale of cyclones of this category. In general, payback period increases under both RCP 4.5 & 8.5, in comparison to BaU situation. Under RCP 8.5, payback period is more than respective project life for all activities.

242. In the EFA process, Economic Analysis was also pursued on interventions that are expected to provide income increase for intended beneficiaries. The analysis was conducted separately for all five (05) activities (goat rearing, traditional vegetable farming, crab hatchery, crab nursing and crab fattening), under two different scenarios; one with RCP 4.5 (medium case scenario) and another with RCP 8.5 (worst case scenario), with and without adaptation measures, and compared with the BaU case.

The results of models assessed in the economic analysis under BaU scenario are presented in Table 7. The results under RCP 4.5 and RCP 8.5 are not shown in funding proposal due to the large volume of data.

Table 7: Economic Analysis of RHL Project (Without GCF)-Aggregate

Goat/sheep rearing

	Economic Benefits in Present Value	Economic Costs in Present Value	ENPV	EIRR	Benefit to Cost ratio
Without GCF (No grant/100% PKSF loan)	$33,185,624	$31,964,842	$1,220,782	9%	1.04
With GCF (GCF grant+ PKSF Loan)	$37,301,085	$31,475,670	$5,825,415	23%	1.19
With GCF (100%GCF grant+0%PKSF loan)	$37,301,085	$27,949,811	$9,351,274	44%	1.33

Traditional Vegetable Farming

	Economic Benefits in Present Value	Economic Costs in Present Value	ENPV	EIRR	Benefit to Cost ratio
Without GCF (No grant/100% PKSF loan)	$3,222,915	$4,272,338	-$1,049,424	9%	0.75
With GCF (GCF grant+ PKSF Loan)	$13,074,602	$13,475,872	-$401,269	23%	0.97
With GCF (100%GCF grant+0%PKSF loan)	$13,074,602	$12,950,941	$123,661	44%	1.01

Crab Hatchery_Nursery_Fattening (as a value chain activity)

	Economic Benefits in Present Value	Economic Costs in Present Value	ENPV	EIRR	Benefit to Cost ratio
Without GCF (No grant/100% PKSF loan)	$112,162,819	$99,983,407	$12,179,411	13.88%	1.12
With GCF (GCF grant+ PKSF Loan)	$112,164,387	$92,812,976	$19,351,411	26.59%	1.21
With GCF (100%GCF grant+0%PKSF loan)	$112,164,387	$88,951,166	$23,213,221	48.91%	1.26

243. ENPVs are positive and benefit to cost ratios is more than 1.00 in all the proposed funding scenarios under this proposal. Crab hatchery, crab nursery and crab fattening have been considered as a chain activity in order to conduct economic analysis..

244. The obtained results in the above economic analysis include the valuation of economic benefits like biodiversity gains due to the environmentally friendly production of cablets suggest that the RHL project will be economically viable and sustainable. Economic benefits will be increased if the disability adjusted life year, savings from avoidance of injuries, number of days in hospitals, avoidance of labour day etc. factors are also considered.

Cost effectiveness and efficiency

245. The total budget of the RHL project is proposed at USD 49.99 million, with the GCF grant financing of USD 42.20 million (GCF grant), which is 84.42 percent of the total funding and USD 7.79 million, which is around 15.58 per cent co-financing from PKSF. The project is expected to deliver direct benefits to an estimated 362,475 direct beneficiaries and 770,050 indirect beneficiaries.

246. The project's concessionally is rationalized through: (i). the project's provision of goods that are largely public in nature (e.g., interventions in mangroves and climate adaptive homesteads), (ii). the delivery of interventions to communities that might initially be private in nature (e.g., funding of hatcheries for crabs or crab fattening or funding improved production of saline resistant vegetables) but in the longer run due to biodiversity co-benefits might become public in nature.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

☐ Reduced emissions (mitigation)

☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	At time of formulation of this proposal approximately 66per cent of houses (very approximate estimation) of the most vulnerable in the target area are currently not built to withstand flood or cyclone risks. Repeated disasters have left this population with even fewer financial means to rebuild using resilient designs/materials, reducing their adaptive capacity. Livestock assets are also at risk and highly vulnerable to climate hazards leading to drastic loss of wealth after an extreme weather event.	Low	The project has initiated scaling so that all coastal communities are knowledgeable and engaged with resilient livelihood options that cover infrastructure, small scale agriculture and livestock husbandry as well as small business approaches that are adapted to current and future climatic and environmental conditions. Collaboration between institutions and communities will be pro-active and built on scientific evidence and human development will proceed in a gender sensitive manner.	Components 1 and 2 will directly benefit 362,475 people and indirectly benefit a further 770,050. This level of engagement with coastal communities is expected to provide the critical mass of expertise and interest from decision makers and end users that will allow this project to increase quantifiable results within and beyond the scope of the project quickly and widely. In addition, specific communications activities in Component 3 have been incorporated into the design of the project to build even greater momentum for scaling. ill develop communication outputs that will market the results and The project will enable communities and institutions to better serve each other through technical assistance and extension,

	There are few livelihood options that are resilient to increased salinity Local communities and support institutions do not have the capacity to plan or implement climate adaptation interventions.			demonstrating a model that through its success is likely to scale.
Replicability	At time of formulation of this proposal no climate resilient homesteads programmes have yet been developed in the coastal area. So, baseline of replicability is considered zero The PKSF-established hatchery is fully functioning. Three entrepreneurs established micro-hatcheries. Two of them are producing juvenile crabs and the remaining one is yet to complete the establishment.	Low	If an appropriate model of resilient homestead and livelihood can be developed, then learning could be replicated in the similar geographical areas in the country as well as internationally. The project outputs will be increased both in terms of population and geographical region. In addition, the local institutions and commercial banks are in place to finance these activities at the local level. So, these activities will be replicated beyond the scope of the targeted interventions.	The project interventions will generate new knowledge and through its compilation and dissemination it will be possible to replicate best practices developed here in similar climate/environmental regions.
Sustainability	The House Building Research Institute (HBRI) has been attempting to promote resilient housing for the last few decades. They however lack an appropriate demonstration of how this would be implemented	Low	A paradigm shift would see the formation of community groups, strong support of non-government and government institutions (including financial), access to supply chain networks.	The project will closely work with local NGOs and government institutions to provide continued technical support and finance for housing, livestock and hatchery-based crab farming and trading.

	from a technical and financial enabling environment. The government and donors have built several communal shelters although these save lives, they do not address asset loss. The crab sector is currently unsustainable. Efforts to develop a fully in vitro supply chain was achieved by PKSF in 2016 which is still functioning well. A few private micro-hatcheries are also functioning and producing juvenile crabs.		A paradigm shift would see national housing directives that incorporated climate-resilient homestead building standards and finance options; as well as adoption of national standards in hatchery-based crab farming and trading. The project would accelerate a transformational change in crab farming and trading by replacing wild caught with hatchery-based farming.	PKSF housing programme will continue funding through partner organisations (POs) in building resilient homesteads. PKSF will work with government to develop policies that would enhance the enabling environment created by this project.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final[61]	
Total project beneficiaries	Core 2: Direct and indirect beneficiaries reached	Baseline reports, Quarterly monitoring report, mid-term, final evaluation report and household survey reports. Government	0	Direct: 150,000 Male: 75,000 Female: 75,000	Direct: 362,475 Male: 181,238 Female: 181,237	All beneficiaries will be reachable / appreciate the risks of non-participation are higher than participation. Please refer to annex 23 for more details.

[61] The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

		and non-governmental reports.		Indirect: 323,420 Male: 161,710 Female: 161,710	Indirect: 770,050 Male: 385,025 Female: 385,025	
ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	Baseline reports, Quarterly monitoring report, mid-term, final evaluation report and household survey reports	0	Direct: 150,000 Male: 75,000 Female: 75,000 Indirect: 309,000 Male: 154,500 Female: 154,500	Direct: 362,475 Male: 181,238 Female: 181,237 Indirect: 770,050 Male: 385,025 Female: 385,025	All beneficiaries will be reachable / appreciate the risks of non-participation are higher than participation. Please refer to annex 23 for more details.
	Supplementary 2.1: Beneficiaries (female/male) adopting improved	Baseline reports, Quarterly monitoring report, mid-term and	0	Direct:150,000 Male: 75,000	Direct: 362,475 Male: 181,238	Selected community members, particularly women, appreciate the risks of climate change on their livelihoods

	and/or new climate-resilient livelihood options	final evaluation report, household surveys. Government and non-governmental reports.		Female: 75,000 Indirect: 309,000 Male: 154,500 Female: 154,500	Female: 181,237 Indirect: 770,050 Male: 385,025 Female: 385,025	and are able to practice the prescribed adaptation options. Adaptation options designed are viable and applicable to communities. Please refer to Annex 23 for indirect beneficiary calculation Baseline is considered 0 because there are very few people currently practising resilient livelihoods. The targets have been considered in consultation with the local communities and institutions. Please refer to Annex 23 for more details.
ARA2 Health, well-being, food and water security	Core 2: Direct and indirect beneficiaries reached	Baseline reports, Quarterly monitoring report, mid-term and final evaluation report, household surveys	0	Direct: 150,000 Male: 75,000 Female: 75,000 Indirect: 323,420 Male: 161,710	Direct: 360,000 Male: 180,000 Female: 180,000 Indirect: 770,000 Male: 385,000	The selected communities particularly the women members of the community understand climate change impacts and practice the prescribed adaptation options Adaptation options designed are viable and applicable to communities Baseline is considered 0 because very few people have

				Female: 161,710	Female: 385,000	been implementing planned adaptation for building resilience to climate change. The target is considered in consultation with the local communities and institutions. Please refer to Annex 23 for more details.
	Supplementary 2.2: Beneficiaries (female/male) with improved food security	Baseline reports, Quarterly monitoring report, mid-term and final evaluation report, household surveys. Government and non-governmental reports.[62]	0	Direct: 150,000 Male: 75,000 Female: 75,000 Indirect: 323,420 Male: 161,710 Female: 161,710	Direct: 362,475 Male: 181,238 Female: 181,237 Indirect: 770,000 Male: 385,000 Female: 385,000	Selected community members, particularly women, appreciate the risks of climate change on their livelihoods and are able to practice the prescribed adaptation options. Adaptation options designed are viable and applicable to communities. Please refer to Annex 23 for more details.
	Supplementary 2.3: Beneficiaries (female/male) with	Baseline reports, Quarterly monitoring report, mid-term and	0	6,750 Male: 3,375 Female 3,375	13,500 Male: 6,750 Female 6,750	Selected community members, particularly women, appreciate the risks of climate change on their livelihoods

[62] This indicator will be measured through the (FIES) methodology (https://www.fao.org/in-action/voices-of-the-hungry/fies/en/)

	more climate-resilient water security	final evaluation report, household surveys [63]		Indirect: 14,400 Male: 7,200 Female: 7,200	Indirect: 36,000 Male: 18,000 Female: 18,000	and are able to practice the prescribed adaptation options. Adaptation options designed are viable and applicable to communities. Please refer to Annex 23 for more details.
ARA3 Intrastructure and built environment	Core 2: Direct and indirect beneficiaries reached	Baseline reports, Quarterly monitoring report, mid-term and final evaluation report, household surveys, local government statistics	0	Direct: 51,888 Male: 25,944 Female: 25,944 Indirect: 54,420 Male: 27,210 Female: 27,210	Direct: 103,775 Male: 51,888 Female: 51,887 Indirect: 136,050 Male: 68,025 Female: 68,025	Topographical and other physical attributes exist for specific construction type. Training is successful. Local craftsmen are skilled in this construction type and are available. Assumes that homestead building starts immediately on project initiation and continues at even pace. Direct beneficiaries reached includes 13,500 for homesteads, 90,000 for goat sheds and 225 for crab hatcheries. Please refer to Annex 23 for more details.

[63] Evaluated based on sufficient available fresh water for basic needs during a flooding/surge event without intervention versus that with intervention

	Supplementary 2.6: Beneficiaries (female/male) living in buildings that have increased resilience against climate hazards	Baseline reports, Quarterly monitoring report, mid-term and final evaluation report, household surveys. Government and non-governmental reports.	0	Direct: 6,750 Male: 3,375 Female: 3,375 Indirect: 14,400 Male: 7,200 Female: 7,200	Direct: 13,500 Male: 6,750 Female: 6,750 Indirect: 36,000 Male: 18,000 Female: 18,000	Willing participants. Local government administration and departments are supportive. Soil of raised plinths are available without disturbing topsoil. Assumes that homestead building starts immediately on project initiation and continues at even pace. Includes only beneficiaries of homesteads. Please refer to annex 23 for more details.
	Core 3: Value of physical assets made more resilient to the effects of climate change and/or more able to reduce GHG emissions	Baseline reports, Quarterly monitoring report, mid-term and final evaluation report, household surveys, local government statistics	0	USD 9,016,000	USD 22,540,000	Topographical and other physical attributes exist for specific construction type. Local craftsmen are skilled in this construction type and are available. Total value combines homesteads, goat sheds with crab hatchery infrastructure. Assumes that homestead building starts immediately on

						project initiation and continues at even pace. Please refer to feasibility study and annex 23 for more details.
	Supplementary 3.1: Change in expected losses of economic assets due to the impact of extreme climate-related disasters in te geographic area of the GCF intervention	Baseline report, quarterly monitoring report, mid-term and final evaluation report, government / agency reports where possible	Local loss baseline TBD at inception	USD 9,016,000	USD 22,540,000	Topographical and other physical attributes exist for specific construction type. Local craftsmen are skilled in this construction type and are available. Total value combines homesteads, goat sheds with crab hatchery infrastructure. Assumes that homestead building starts immediately on project initiation and continues at even pace. Please refer to feasibility study and annex 23 for more details.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and	The government has adopted a national fisheries policy which does not specifically cover the crab	low	Local government departments particularly department of fisheries, universities and NGOs have	Due to the close working nature of government with this project and through PKSF advocacy it is expected that the	Single sub-national area within a country

regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	hatchery-based production system. The Environmental Conservation Rules (ECR) of the government also does not address crab hatchery technology and production.		increased knowledge and capacity to support in vitro crab production, juvenile crabs nursing, fattening. General guidelines for food safety and husbandry are developed at national level.	government will implement national housing policy standards that include climate resiliency and in addition amend fisheries policy and ECR for integrating hatchery-based production system and trade. Financial resources from PKSF will be used to continue to promote regulatory frameworks within the housing and fisheries sectors. Coordination mechanisms between beneficiaries, civil society organisations and local government will be created and mainstreamed through this project. Civil society organisations that will be executing this project will understand the important contribution that they will make to climate challenges.	

Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation	PKSF have initiated a small pilot programme after developing their crab hatchery production unit. Although operational these are not yet of any significant scale.	low	All people have access to climate resilient house/shelter technology to protect lives and assets in coastal Bangladesh. Crab hatchery technology (and financing) is available to any entrepreneur wishing to adopt this climate resilient livelihood option.	The project will introduce new building design technology that is resilient to climate extremes, through providing 3000 very poor vulnerable families with homesteads. The project will disseminate crab hatchery technology and establish 50 micro scale crab hatcheries to promote pro-salinity production in the coastal areas of Bangladesh, directly benefitting 90,000 people. PKSF will continue to commit funding to promote technology transfer in these sectors.	Single sub-national area within a country
Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level	Current markets are based on the unregulated / unsustainable harvesting of wild crabs. An (export) market is already established but supply	medium	The crab sector reaches market maturity that includes people of all levels, is transparent, supported and guided by institutions and fair prices are attainable at	The investment in crab sector under this project will transform the market by having a fully in vitro life cycle. The project will allow small producers to enter supply chain and obtain fair prices.	Single sub-national area within a country

	chain is largely unregulated.		all levels of the supply chain.	The project will enable a shift in market dynamics from an informal wild based resource to a standardised, safe and formal supply chain.	
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	There is very little knowledge of resilient housing standards/materials and designs. There is no general knowledge by communities or institutions on crab hatchery technology	low	Knowledge is generated compiled and easily accessible by everyone. That the knowledge accrued is used and best practices continuously refined through practice.	Valuable knowledge will be created, compiled and disseminated nationally through this project. The project will develop a swathe of best practices that will be refined through a real time evaluation methodology which will improve adaptation management of the project and across organisations. The project will generate quarterly monitoring reports. Mid-term and final evaluation report, quarterly newsletters, e-newsletters, carried annual studies under IRMF process etc.	Single sub-national area within a country

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	**Project/programme specific Indicator**	**Means of Verification (MoV)**	**Baseline**	**Target**		**Assumptions / Note**
				Mid-term	**Final**	
Outcome 1: Decreased risk of loss of assets and lives from extreme weather events	Share of earnings and time spent on repairs from climate hazard event	Baseline report, monitoring report, evaluation report, household surveys	Local baseline TBD at inception	60per cent	30per cent	New construction is adequate against predicted weather events, not necessarily un-predicted ones. An extreme weather event will occur during the M&E phase to measure this indicator.
Output 1.1Climate resilient homesteads constructed	Number of new homesteads constructed that include house, water harvesting facilities and green windbreaks	Quarterly monitoring report, evaluation report, government reports	0	1,500	3,000	Local communities and institutions agree to new constructions. Adequate soil and other physical conditions for construction/planting

						No extreme event destroys trees before establishment
Outcome 2: Livelihood resilience to SLR/storm surge and salinity	Number of beneficiaries that reduce their coping strategies to climate events compared with pre-intervention years[64]	Baseline survey report of PKSF, Quarterly monitoring report, evaluation report, household surveys	0	180,000 Men: 90,000 Women: 90,000	360,000 Men: 180,000 Women: 180,000	Beneficiaries, particularly women recognise value in interventions, are available to participate and adhere to training. Local community groups and institutions are able to support individuals in resilient livelihoods.
Output 2.1: Traditional farming practices climate proofed	Number of slatted houses constructed for sheep/goat rearing	Quarterly monitoring report, evaluation report, household surveys	0	8,000	20,000	Beneficiaries, particularly women recognise value in interventions, are available to participate and adhere to practising their training.

[64]Coping Strategies: are mechanisms that people choose as a way of living through difficult times such as climate events. Some coping strategies are not damaging to livelihoods and are easily reversible: for example, short-term dietary changes, migration of individuals for work, use of savings or solidarity networks. Other strategies may be damaging and tend to be harder to reverse: for example, sale of land, sale of 'productive' assets, early weddings, intensive use of wood from nearby areas causing deforestation, taking children out of school to make them work (child labour) or prostitution.

	Number of households cultivating saline tolerant vegetables	Quarterly monitoring report, evaluation report, household surveys	0	8,000	20,000	Beneficiaries, particularly women recognise value in interventions, are available to participate and adhere to practising their training.
Output 2.2: Community-based farmed crab supply chain created	Number of beneficiaries involved in crab supply chain using new in vitro model of production with 3-F	Quarterly monitoring report, evaluation report, household surveys	0	Crab hatchers: 90 (men 90, women 0) Nursers: 900 (men 450, women 450) Farmers: 40,000 (men 20,000, women 20,000) Other supply chain: 2000 (men 1000, women 1000)	Crab hatchers: 225 (men 225, women 0) Nursers: 2250 (men 1125, women 1125) Farmers: 90,000 (men 45,000, women 45,000) Other supply chain: 4050 (men 2025, women 2025)	Beneficiaries, particularly women recognise value in interventions, are available to participate and adhere to practising their training.
Outcome 3: Improved climate planning and implementation by	Percent of target with embedded personnel with disaster preparedness and	Quarterly monitoring report, evaluation report, community surveys	0	40per cent	100per cent	Local communities and institutions are enthusiastic / have time to take part in

communities and local level institutions	response knowledge					the project interventions
Output 3.1: Climate change adaptation groups (CCAG) formed and operationalized	Number of groups formed and operationalized	Quarterly monitoring report, evaluation report, community surveys	0	2,000	3,200	Willingness and time to participate
Output 3.2: Capacity built among IEs and relevant institutions apprised on project	Number of trainees whose knowledge and capacity has been improved	Testing results prior to and after trainings. Quarterly monitoring report, evaluation report	Low	Moderate	High	The beneficiaries regularly attend the meetings.
Output 3.3: Knowledge products prepared and disseminated	Number of knowledge products disseminated	Quarterly monitoring report, evaluation report, household survey	0	Guidelines and training manuals/ focus notes/ other documents: 3 Newsletters: 10 Lessons Learned Document: 0 Best practices documents: 2	Guidelines and training manuals/ focus notes/ other documents: 5 Newsletters: 20 Lessons Learned Document: 1 Best practices documents: 5	New information is created that will be of use for adaptive management and future projects.
Project/programme co-benefit indicators						
Co-benefit 1: Decreased risk of assault on women	Number of women and adolescent girls	Baseline report, monitoring report,	0	3,500 women and girls of	7,000 women and girls of	Resilient homes offer increased

	not having to go to communal shelters	evaluation report, community survey		resilient homesteads	resilient homesteads	safety for women and girls
Co-benefit 2: Reduced Urban Migration	Number of beneficiaries migrating away due to livelihood issues	Baseline report, monitoring report, evaluation report, community survey	TBD at project inception	30per cent reduction from baseline	75per cent reduction from baseline	Livelihood options are being practiced as expected
Co-benefit 3: Growth of crab and resilient housing industries	Number of service providers providing new services to resilient housing market and crab supply chain	Baseline report, monitoring report, evaluation report, community survey	0	200	500	Livelihood options practiced as expected.
Co-benefit 4: Decreased pressure on wild crab populations by industry	Percent of collectors collecting from the wild from target villages	Baseline report, monitoring report, evaluation report, community survey	100per cent	50per cent	0per cent	Wild crab collection is more expensive, dangerous and time constraining than the alternative proposed here.

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 1.1.1 Design and building of homesteads	The proposed project will provide designs and implement construction of climate resilient houses. The house will be as simple as possible and include rainwater harvesting provision. Moreover, the plinth height will be raised more than 1 meter (may	Sub-activity 1.1.1.1 Selection of beneficiaries during CCAG meeting Sub-activity 1.1.1.2 Validation of land ownership Sub-activity 1.1.1.3 Selection of vendors	• A list of 3,000 household beneficiaries • Deed verification database • Vendor list • Photo books of 3000 homesteads • Titles and photographs of construction

	vary based on location) to reduce risk of flooding and water logging.	Sub-activity 1.1.1.4 Construction of house with on-the-job training for local builders Sub-activity 1.1.1.5 Handover the house to its owner	
Activity 1.1.2 Homestead tree planting	Wind resistant native trees planted as wind barriers for houses.	Sub-activity 1.1.2.1 Selection of beneficiaries during CCAG meeting Sub-activity 1.1.2.2 Validation of land ownership Sub-activity 1.1.2.3 Selection of vendors Sub-activity 1.1.2.4 Planting of trees	• A list of 20,000 Beneficiary households • A report on tree plantation
Activity 2.1.1 Construction of slatted houses for goat/sheep rearing	IE field officers in consultation with CCAG members will select women interested in rearing goat/sheep. In addition, IEs will procure work for construction of goat/sheep shelters as per the approved procurement plan	Sub-activity 2.1.1.1 Selection of beneficiaries for goat/sheep rearing Sub-activity 2.1.1.3 Selection of vendors for slatted house Sub-activity 2.1.1.4 Distribute slatted house for goat/sheep to the selected beneficiaries’	• A list of 20,000 beneficiary households • Vendor list • A report of completed slatted houses
Activity 2.1.2 Provide financial support for goat/sheep rearing	IEs’ staff will select the Beneficiaries’ during CCAG meeting for goat/sheep rearing. After selection IEs’ staffs provide training and distribute slatted house. The AE will provide loan to the beneficiaries through the IEs if required for purchase goat.	Sub-activity 2.1.2.1 Selection of appropriate beneficiaries during CCAG meeting Sub-activity 2.1.2.2 Provide training on goat/sheep rearing	• Beneficiaries list • Annual training reports • Annual loan disbursement report / agreements

		Sub-activity 2.1.2.4 Distribute loan to the beneficiaries	
Activity 2.1.3: Introduce the cultivation of saline tolerant vegetables within homestead areas	IE staff will select the Beneficiaries who have minimum required space to employ this activity. S/He will be trained to cultivate saline tolerant vegetable gardening within the homestead. IE staff will provide technical support, seeds and training.	Sub-activity 2.1.3.1 Select the appropriate beneficiaries Sub-activity 3.1.3.2 Provide training / resources and technical support	• A list of 20,000 beneficiary households • Annual training reports
Activity 2.2.1: Development of crab hatcheries (1° stage)	Hatchery technology will be transferred to local entrepreneurs that have experience in aquaculture. The hatchery owner will be provided necessary training on crab breeding, hatchery management and maintenance.	Sub-activity 2.2.1.1 Selection of crab hatchery entrepreneur Sub-activity 2.2.1.2 Provide training on crab hatchery management Sub-activity 2.2.1.3 Partially financial support to establish crab hatchery from Project	• A list of 50 crab hatchery entrepreneurs • Annual training reports / management guidelines • Photo book of 50 hatcheries with brief description and record of financial support provided and agreements
Activity 2.2.2 Financial support for producing crablets	The project will provide both grants and loans to establish the crab hatcheries for initial operational costs.	Sub-activity 2.2.2.1: Assessment of loan demand for hatchery operation and its approval by PKSF Sub-activity 2.2.2.2: Loan disbursement to the hatchery owners	• Loan approval documents • Annual loan and grant disbursement report/ agreements
Activity 2.2.3 Technical and financial support for “crab nursers” (2° stage)	The crab nursers will be provided necessary training along with financial support on transporting and nursing crablets. They will buy crablets from the hatchery (1°) and	Sub-activity 2.2.3.1 Selection of Beneficiaries (crab nurser) from CCAG Sub-activity 2.2.3.2 Provide training	• A list of 500 beneficiary households as crab nursers • Annual training report

	rear them for two months. At which time the juvenile crabs will be adolescent weighting 90 gm. and able to grow in a pond. The nursers (2°) will sell the adolescent crab to the growers (3°).	Sub-activity 2.2.3.3 Provide partial financial support from project Sub-activity 2.2.3.4 Provide financial support (Loan) from PKSF Sub-activity 2.2.3.5 Establish crab nursery ponds with mangrove and fruit trees	• Annual loan / grant disbursement report / agreements • A report on crab nursery.
Activity 2.2.4 Provide technical and financial support to crab farmers (3° stage)	The project will select crab farmers for producing export quality crab (it is already practiced in coastal districts of Bangladesh). IE staff and extension agents will be provided necessary training and support. Farmers will buy adolescent crabs from the nurseries and rear them for 30 to 40 days. Then they will sell it to the market.	Sub-activity 2.2.4.1 Selection of Beneficiaries (crab fattener) from CCAG Sub-activity 2.2.4.2 Provide training Sub-activity 2.2.4.3 Provide partially financial support from project Sub-activity 2.2.4.4 Provide financial support (Loan) from PKSF Sub-activity 2.2.4.5 Establish crab fattening ponds with mangrove and fruit trees	• A list of 20,000 beneficiary households • Annual training report • Annual loan / grant disbursement report / agreement • A report on establishment of 20,000 ponds for crab growing with mangrove tree plantation
Activity 3.1.1: Beneficiary selection and group formation	The project will select 362,475 beneficiaries in consultation with local government institutions and community members. The field officers of the IEs will carry out this activity. IEs will require approval of the list of selected beneficiaries form the PMU of Executing Entity.	Sub-activity 3.1.1.1 Beneficiaries selection and group formation through PRA	A report on 3,200 PRA exercises by the CCAGs

Activity 3.1.2: Prepare beneficiaries' socio-economic profile	After selection, the IE field level staff will visit door to door of the selected beneficiaries and collect their socio-economic information before providing supports from the project. The PMU will prepare the format for collecting socio-economic profile. PMU will share this format with GCF relevant staff before execution	Sub-activity 3.1.2.1 Prepare socio-economic profile template Sub-activity 3.1.2.2 Field test by the IEs' and EEs' staff Sub-activity 3.1.2.3 Collection of information from the individual HH beneficiaries Sub-activity 3.1.2.4 Prepare report	• Socio economic profile template • A report on field test of socio-economic template • Information database with data • Socio economic profile report
Activity 3.1.3: Arrange monthly group meetings on climate change issues for CCAG	The PMU will prepare content for monthly group discussion in Bangla. Relevant field staffs of the IEs will facilitate this content in the group meetings.	Sub-activity 3.1.3.1 Prepare contents for session with CCAG Sub-activity 3.1.3.2 Arrange CCAGs' monthly group meeting Sub-activity 3.1.3.3 Take decision and put it to register for further planning	• A document on meeting content for the CCAGs • Annual CCAG meeting reports
Activity 3.2.1: Prepare training manual on adaptation technologies and crab value chain	The PMU will prepare this training manual. It will be shared with GCF before finalizing this manual. The manual will be used to provide training to the staffs of IEs. Bangla version of this manual will be developed to provide training to the CCAG members	Sub-activity 3.2.1.1 Prepare ToR for selection of consultant Sub-activity 3.2.1.2 Selection of consultants Sub-activity 3.2.1.3 Prepare training manuals on adaptation and crab value chain for Beneficiaries' Sub-activity 3.2.1.4 Validation consultation meeting on training manual	• ToR for consultant • Consultant agreements • Training manual • Meeting notes

		Sub-activity 3.2.1.5 Delivery of training manuals	
Activity 3.2.2: Prepare guidelines on project management	The PMU will prepare the guidelines on project management. It will be shared with GCF before finalizing this manual. The guidelines will be used to smooth operations of the project and activities for the PMU and IEs'. A Bangla version of this guidelines will be developed.	Sub-activity 3.2.2.1 Prepare ToR for selection of consultant Sub-activity 3.2.2.2 Selection of consultants Sub-activity 3.2.2.3 Prepare guidelines on project management and different activities Sub-activity 3.2.2.4 Validation consultation meeting on guidelines Sub-activity 3.2.2.5 Delivery of guidelines	• ToR for consultants • Consultant agreements • Meeting notes • Training plan document • 5 guideline documents
Activity 3.2.3: Organize training for beneficiaries and stakeholders	IE's staffs will prepare this training plan and get approval from the PMU. The IE staffs will organize the training session as per approved plan. PMU will physically monitor the training activities on sample basis.	Sub-activity 3.2.3.1 Prepare training plan by the IEs' Sub-activity 3.2.3.2 Selection of appropriate participants Sub-activity 3.2.3.3 Selection of venue Sub-activity 3.2.3.4 Conduct training to the Beneficiaries Sub-activity 3.2.3.5 Report to PMU	• Training plan document • Training participants' list • Training presentation / notes • Reports prepared by IEs
Activity 3.2.4: Organize training for IE staff	IEs' staff and extension agents will carry out the implementation of project activities in the field level. They need to know the procedure of the implantation as well as	Sub-activity 3.2.4.1 Prepare training plan	• Training plan document • Training participants' list • Training presentationTraining reports prepared by PMU

	climate change adaptation. PMU will carry out this training.	Sub-activity 3.2.4.2 Selection of venue Sub-activity 3.2.4.3 Conduct training to the IEs' staff Sub-activity 3.2.4.4 Reporting on training session	
Activity 3.2.5: Implement workshops and Seminars	PMU will carry out these activities. Government representatives, development partners, civil society representatives, IEs etc. will take part in these workshops and seminars	Sub-activity 3.2.5.1 Selection of venue Sub-activity 3.2.5.2 Invite stakeholders Sub-activity 3.2.5.3 Hire local-consultant Sub-activity 3.2.5.4 Organize workshop Sub-activity 3.2.5.5 Report on workshop	• Invitation letters • Workshop presentation / agenda / photos • Workshop report
Activity 3.2.6: Organize exchange visits for beneficiaries and IE staff	PMU will carry out this activity. IE staffs and beneficiaries will take part in this activity.	Sub-activity 3.2.6.1 Selection of appropriate activities for visit Sub-activity 3.2.6.2 Selection of participants Sub-activity 3.2.6.3 Organize exchange visit	• Activity list • A list of 5,000 Participants • Exchange visit plan document and report
Activity 3.2.7: Improve data for crab research and development	Exchange of information on field / lab data with university collaborators	Sub-activity 3.2.7.1 Organize periodic meetings	• Meeting notes

Activity 3.3.1: Prepare and disseminate knowledge products	The knowledge products will include newsletters, evaluation reports, lessons learned documentation, and narrative monitoring reports.	Sub-activity 3.3.1.1 ToR of consultant Sub-activity 3.3.1.2 Prepare newsletter, booklet and related publications Sub-activity 3.3.1.3 Selection of vendors Sub-activity 3.3.1.4 Printing of knowledge documents	• Consultant agreement • 5 guidelines and training manuals; 20 newsletters; 1 lessons learned booklet, 5 best practice publications • Physical copies of documents
Activity 3.3.2 Realtime evaluation study of project activities	Annual impact study will be carried out to measure short-term outcome and impacts of the project	Sub-activity 3.3.2.1 Develop ToR for national level consultants for evaluation study Sub-activity 3.3.2.2 Procure Consultants Sub-activity 3.3.2.3 Follow up and monitor the study	• ToR for consultants • Study tools including questionnaire and checklists • FGD and KII reports • Study reports

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

247. The monitoring and evaluation (M&E) under this project will have three functions. First, thorough monitoring by PMU will ensure accountability of the IEs to deliver the outputs agreed in the project proposal, which implies that the resources are used efficiently for the proposed activities. Secondly, monitoring will establish proper documentation of the implementation process and achievements at different levels (outputs, outcomes, and impacts). Third, monitoring will help gather learning from the process. In short, the role of accountability is significant in the case of outputs, whereas learning becomes a core issue of monitoring at the outcome level and impacts level achievements. In order to analyze and guide the project strategy, maintain efficient operations, satisfy internal and external reporting obligations, and inform future programming, project design, implementation, and completion, the RHL's M&E system should always be aligned with the major stages of a project's lifecycle (see Figure 9).

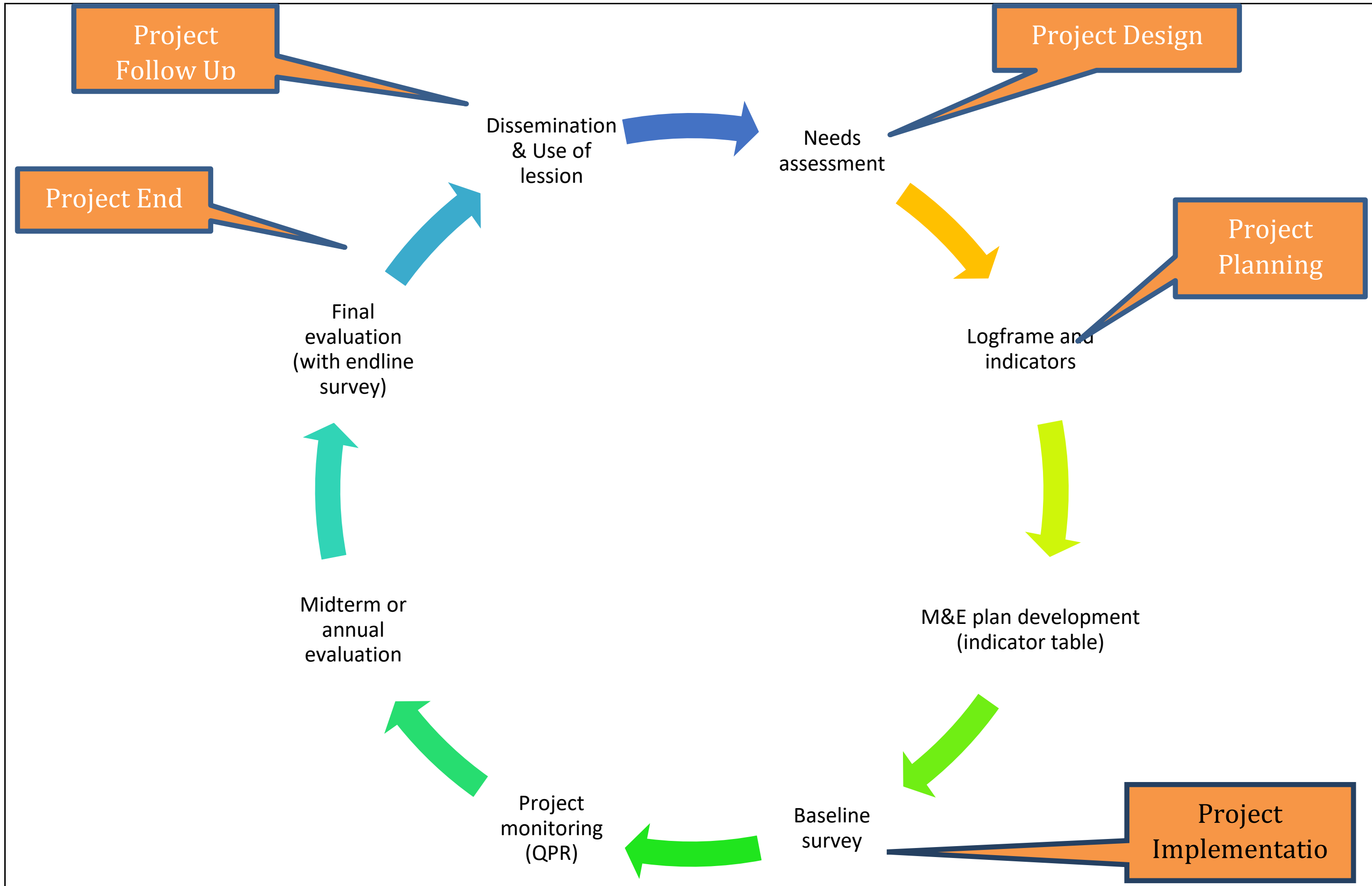


Figure 9: Major stages of project's lifecycle.

248. Monitoring is conceptualized in this project as the continuous assessment of the interventions and its implementation processes. It will take place at all levels of management and uses both formal reporting and informal communication. A minimum common format for progress and monitoring reports will be established, on which each concerned partner organization will develop its own internal monitoring system, which will be centrally consolidated from the project at the PMU level. Mainly, four types of monitoring will be done in this project using different instrument. They are: i) **Progress Monitoring**: In this monitoring system, progress monitoring generally refers to the activity monitoring of the project, ii) **Process Monitoring**: This justifies the delivery mechanism of the project and monitors whether the deliverables are being supplied properly, iii) **Budget**

Monitoring: Budget monitoring refers to proper book keeping of the account of the project. This ensures the transparency in financial matters of the project and iv) **Impact Monitoring**: The overall impact of the project is monitored through impact monitoring. This monitoring is conducted at various stages of the project such as periodically using RBM and finally, by conducting independent final evaluation.

249. Monitoring and Evaluation will be done, at different stages of the project implementation, through following instruments or systems such as i) On-site Field Monitoring, ii) Off-site Monitoring, iii) Results-Based Monitoring (RBM), and iv) Independent Evaluation. Monitoring is an important part for any projects implemented by PKSF. The main question is: “Who monitors what to whom, how, when and where, and under what conditions?” Many stages are followed in PKSF for monitoring any project activities. **First,** the **on-site monitoring** will be conducted once in every three month for each EE by member of the PMU of the project. The on-site monitoring report will be one of the crucial monitoring systems of the project which will be done in customized systematic format. This report will be subjective in nature and provides valuable insight about the projects at the grass root level. Senior management from PKSF will oversee the PMU and participate in field level operations as part of AE's objective (PKSF as the Executing Entity). The AE will review evaluation reports and audit reports carried out by individual consultants. It will oversee procurement and financial management of the project; prepare a project completion report; and submit the report to the secretariat of GCF. AE will ensure that all compliance requirements are met, including the gender action plan, ESMF, and IP. The AE will also review the annual budget in order to enhance the efficient use of resources. The AE will monitor the audit process, fund disbursement, prepare the financial closing report of the project, and project closing documents for submission to the secretariat. **Second,** the quarterly MIS and training evaluation is also conducted once in every three months and is based on monthly data of the EEs which is accumulated over time and evaluated after consolidating to assess the trend and progress of project activities. **Third**, Results-based Monitoring (RBM) is mainly used for measuring the output, outcome and impact of the project. A distinct set of indicators have been developed based on log frame as well as literature for assessing the project overall objective over the time. For implementing RBM, a sample of households among all project participants will be chosen to collect required data using formatted questionnaire based on an agreed result framework. The sample size will be then adjusted for addressing the heterogeneity problem and statistical software will be used to analyze the data to find out the results. The IEs should hire a specialized monitoring officer who will report to the Chief Executive or any other authorized person. He/she will implement the Monitoring Framework as envisaged in the project proposal and will produce quarterly activity monitoring reports based on the Activity to Output Monitoring (ATOM) agreed upon. The Monitoring Officer will undertake outcome-level monitoring half-yearly based on an agreed Outcome Assessment Sheet (OAS) and impacts-level monitoring annually based on an agreed Impact Assessment Sheet (IAS) which were prepared taking indicators of impacts and outcomes into account. He/she will post the information in the assigned fields of the IEs and on the PKSF server online as well. **Fourth,** to gauge the unbiased impact of the project, various independent studies will also be undertaken at different project period to have the true picture of the project’s impact on the project participants. Evaluation is defined in this project as the systematic and objective assessment of an on-going or completed project, including its design, implementation and results. It generally aims to determine the relevance and fulfillment of objectives of the project, development of efficiency, effectiveness, impact and sustainability. The objectives of evaluation process of this project is to explain how and why progress is causing about (or not), and identify recommendations to strengthen the project, determine the outcomes and impacts of the project and assess their value, and re-examine the project rationale in light of subsequent developments.

250. Different types of data is used in the evaluation process such as secondary data, on-demand sample surveys, field reports, qualitative studies and external assessment to name a few among others. In this project, a systematic evaluation will be done by conducting base-line study, mid-line study and end-line study. It has a plan to use Randomized Control Trial (RCT) to understand the effectiveness of different or a package of interventions on the project participants. For this, there might have a control group to determine a representation counter factual. A brief description is given below on base-line, mid-line and end-line studies.

a) **Base-line Survey:** The independent consulting firm will perform a baseline survey at the start of the project to learn more about the existing state of the indicators that will be monitored throughout the project. It will be completed prior to the project's official start. It is the procedure or action performed to find out how the targeted families were doing in terms of various indicators previous to the project's introduction. The main reason for doing a baseline survey is to determine what change may be directly attributed to the project intervention.

b) **Mid-line Survey:** The mid-line study, also known as a mid-term review or mid-term evaluation, is described as a thorough exercise typically held at the halfway point of a project to re-evaluate the original development objectives of the project, their applicability in light of new circumstances, and their likelihood of realization. In the midst of the project, a midline study will be done. It will serve two immediate purposes: decision-making and taking stock of initial lessons from experience. Specifically, a mid-line study will provide the project managers with a basis for identifying appropriate actions to: (a) address particular issues or problems in design, implementation and management, and (b) reinforce initiatives that demonstrate the potential for success.

c) **End-line Survey:** Following the completion of the project, an end-line study will be carried out. The outcomes of the end-line will be evaluated in comparison to some comparative data, ideally the results of the baseline investigation. The action will be taken as part of an impact assessment. By comparing the same individuals before and after the project intervention, an end-line research will be conducted with the aim of estimating the impact of the project intervention. Every single possible variable will be compared with same group. The project has a plan to use one of the most scientific methods of impact evaluation of certain interventions is Randomized Control Trial (RCT), where programme interventions are placed randomly at the individual or cluster level. Recently, this method has been used in the context of social research. Impact evaluation, in general requires comparing the outcome of 'treated' group with 'control' group. However, if there is 'selection bias', then comparing the treatment group with the control group provides biased estimate of the intervention impact. Selection bias arises particularly for two reasons: (1) intervention is placed non-randomly (by choice) by the programme implementers and (2) programme participants self-select them to participate in the programme. In RCT method, interventions are placed randomly which remove the possibility of the selection bias and provide unbiased estimation of intervention impact.

251. Finally, for a successful M&E system, the many capacities of an organization need to be taken care of, for example, human capacities, such as knowledge, skills, attitudes, relationships, resources and motivations; management information system (MIS), such as data storage, processing, access and retrieval; organisational structure, for example, how do M&E functions fit in organisational information flows, and do they support efficient communication?; M&E plans, i.e., articulating information needs, how they will be addressed, what resources are committed, who is responsible.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

252. The project design has carefully considered all social, financial, operational and environmental risks. The environmental risks from crab hatchery and farming and house construction are considered low to medium. To mitigate Environmental risks, the project has already developed an Environmental Management Framework. Two types of tools will be used in the project considering nature of intervention and magnitude of impact on the environment. These are Environmental Screening and ES impact assessment. Social risk may arise due to the establishment of hatchery because existing crab collectors may lose their income as crab growers will buy crab-let from the hatcheries. These will be mitigated by involving them as project beneficiaries. Financial risk may arise at the procurement and beneficiary selection stage, and the project design includes adequate safeguard measures to minimize this risk. The major operational risk, at medium probability, is the reduction of coverage of beneficiaries due to currency exchange loss or domestic inflation.

The project has been screened based on the exclusion criteria presented in Section 4.3 of the annex 6: Environmental and Social Management Framework. The screening results found that the project will have some impacts that will require some mitigation measures. For example, the project will construct crab hatcheries that will require water treatments facilities. The labour may require to wear personal protective equipment including masks, hand-gloves, helmet etc. For reducing dust, the construction activities will require water-spray etc. In this consideration, the project is categorized B. The project was screened based on the exclusion criteria of GCF's ES safeguard policy. As per the screen results, the project is categorized B. The rationale of categorization B is presented below:

a) The project will have small construction work like houses for vulnerable people as well as hatchery infrastructure. There is least possibility to become injured at a scale that requires hospitalized or cause significant treatment cost. No children or vulnerable women will be engaged in this activity. The project will not require any force labour.
b) The project will promote crab hatchery and crab culture. Hence, it has potential to increase soil and water salinity around the hatchery and crab ponds. But it will be implemented in already salinity-affected land.
c) The project will have small scale construction activities like crab hatchery, resilient homestead, slatted houses for goat/sheep etc. But the project will not require critical infrastructure like dam, embankment, coastal infrastructure etc. So, the activities will not require further technical assessment and safety studies.
d) The project will not propose any activities which may include resettlement and dispossession, land acquisition, and economic displacement of persons and communities. Moreover, there is no temporary displacement issue. The households/families will be living in the same houses as the houses have alternative rooms. The rooms will be constructed one after another. So, families/HHs just have to be shifted from one room to another.
e) The project will be implemented in existing settlement and farming areas.
f) There are insignificant ethnic minorities found in the selected project area. However, the project will avoid those areas where these people are living.
g) In the project area, there is no archaeological paleontological, historical, cultural, artistic, and religious values or contains features considered as critical cultural heritage.
h) The activities will not affect human rights because these are selected in consultation with the affected communities. Besides, these will be implemented on beneficiaries' own land to avoid any dispute related to land.
i) The activities may involve women labour in the crab farm and construction of crab hatcheries. So, they may be affected by other local people or co-workers during on the way to work place or vis-à-vis. However, as external workers will not be involved in the implementation process, the possibility of sexual harassment, abuse or exploitation is very limited.

There is potential to pay lower wage to the female than male workers. But there is limited possibility of wage discrimination due to the nature of the project. Because the project will involve mostly women members of the community who will make decisions on wage rate, bargain with contractors etc. At the institutional level, HR policies of PKSF and IEs will be applicable to ensure equal wage of male and female staffs.

253. Awareness of these risks is well integrated in the project design, and resources have been allocated to mitigate them. The environmental risk will be mitigated by following the principles of environmental sustainability and standards for biodiversity conservation. Stringent safeguard measures will be in place in the Sundarbans area, following GoB's guidelines. For the financial risk, the project will include representatives from relevant department and/or sections in major procurement stages; undertake audits by the internationally reputed audit farm. Coverage risk will be mitigated by careful forecasting of price and exchange rate and building-in of appropriate contingencies. The beneficiary selection risk will be mitigated by introducing official gazette selection criteria, utilization of participatory methodologies and complaints and grievance redress mechanism to be regularly monitored by the project, and drawing guidance from the project board and steering committee.

Selected Risk Factor 1		
Category	**Probability**	**Impact**
Technical and operational	Low	Low
Description		
Timely implementation of the planned interventions due to cyclone and storm surge		
Mitigation Measure(s)		
(i) Preparation and execution of effective implementation and monitoring plan ; (ii) Periodical progress review plan and timely initiatives of corrective measures; and (iv) Ensure suitable human resources with adequate skills and capacities, Raising awareness on early warning and Build cyclone resilient house and construction		
Selected Risk Factor 2		
Category	**Probability**	**Impact**
Technical and operational	Low	Low
Description		
Unavailability of mother crabs		
Mitigation Measure(s)		
Alternative source will be ensured ffom Bangladesh deep sea areas; (ii) Linkage with fisherman who are catching fish from deep sea and (iii) Liaison with other area/district of Bangladesh		
Selected Risk Factor 3		
Category	**Probability**	**Impact**
Technical and operational	Low	Low
Description		
Technical person not available		
Mitigation Measure(s)		

i) Local people will be provided training on implementation and operation on crab hatchery and (ii) trained the local expert for hatchery operations.

Selected Risk Factor 4		
Category	**Probability**	**Impact**
Technical and operational	Low	Low
Description		
Disease infested		
Mitigation measure(s)		
i) the project will strengthen laboratories of three universities for carrying out research on crab development including disease, ii) Linkage between crab farmers and service providers including department of fisheries will be established.		

Selected Risk Factor 5		
Category	**Probability**	**Impact**
Operational	Low	Low
Description		
Remoteness of the project areas		
Mitigation measure(s)		
• Remoteness will be solved by using local transport system • Partners will be selected based on the establishment in the project areas		

Selected Risk Factor 6		
Category	**Probability**	**Impact**
Operational and Technical	Low	Low
Description		

Selected Risk Factor 7		
Category	**Probability**	**Impact**
Physical	Low	Low
Description		
Availability of appropriate land to establish hatchery		

Mitigation measure(s)		
Consultation meetings and reconnaissance visit will be carried out while selecting sites for hatchery establishment.		
Selected Risk Factor 8		
Category	**Probability**	**Impact**
Physical	Low	Low
Description		
Availability of appropriate land to establish hatchery		
Mitigation measure(s)		
Before establish the hatchery feasibility survey will be conducted to overcome the unavailable suitable land		
Selected Risk Factor 9		
Category	**Probability**	**Impact**
Physical	Low	Low
Description		
Cyclone and storm surge may make the implementation process delay.		
Mitigation measure(s)		
• Raising plinths of hatcheries and homesteads • Execute early warning information and decision • Construction of storm resilient structure of house and hatcheries • Raised dykes of ponds of crab nursery and farm		
Selected Risk Factor 10		
Category	**Probability**	**Impact**
Physical	Low	Low
Description		
Unplanned extension of crab farms		
Mitigation measure(s)		
• Critical minimum interventions will be ensured • Ensure implementation of EMF of the project		

Selected Risk Factor 11		
Category	**Probability**	**Impact**
Social	Low	Low
Description		
Conflicts on inherited land resources for housing		
Mitigation measure(s)		
• House will be built on existing land • Legal documents of land ownership must be ensured • Selection criterion on proof of land and property ownership will be used.		

Selected Risk Factor 12		
Category	**Probability**	**Impact**
Social	Low	Low
Description		
Conflict with the local influential persons		
Mitigation measure(s)		
The project will carry out consultations with different stakeholders including shrimp farmers, local money lenders, crop famers, traders etc. before establishment of hatcheries.		

Selected Risk Factor 13		
Category	**Probability**	**Impact**
Social	Low	Low
Description		
Political influence		
Mitigation measure(s)		
The project will involve local government institutions and divisions to sensitise them about the project interventions.		

Selected Risk Factor 14		
Category	**Probability**	**Impact**
Environmental	Low	Low
Description		

Loss of productive land due to establishment of hatcheries		
Mitigation measure(s)		
Only barren & fallow land will be considered to install the hatchery plant, so there will have no chance of loss of productive land.		
Selected Risk Factor 15		
Category	**Probability**	**Impact**
Environmental	Low	Low
Description		
Salinity increased at adjacent agricultural land.		
Mitigation measure(s)		
Wastewater treatment system will be built in with the hatchery through de-chlorination. At the farm level, only salinity affected lands will be used for crab farming.		
Selected Risk Factor 16		
Category	**Probability**	**Impact**
Financial (ML/FT)	Low	Low
Description		
ML and FT could hamper the implementation of the project activities		
Mitigation Measures(s)		
Given that both ML and FT are typically committed through abuse of financial institution, thereby it will be mitigated through: i) Maintain strong KYC database for the POs ii) Regular audit the bank statement and expenditure of the POs iii) Strongly monitor the POs' activities grading project implementation and expenditure balance sheet.		

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

254. The project design has carefully considered all social, financial, operational and environmental risks. The environmental risks from crab hatchery and farming and house construction are considered low to medium. To mitigate Environmental and social risks, the project has already developed an Environmental and Social Management Framework (ESMF). Two types of tools will be used in the project considering nature of intervention and magnitude of impact on the environment. These are Environmental and social Screening and ES impact assessment. Some environmental and social risks are presented below:

Identified environmental risks

255. Climate resilient homestead development will require earthen works that will have potential to use top soil materials. Taking soil from the top soil layer of agricultural land diminishes agricultural production because it deprives the land of fertile nutrients. By gathering soil from waste land or storing topsoil and restoring it later, such an impact can be avoided. Additionally, surrounding borrow pits and ponds can be used to collect soil, which will assist those regions produce more fish.

256. None of the project activities will require cutting trees or forests. As the plinth of the homesteads will be raised, some grasses and herbs may be damaged. The crab hatchery will reduce the capture of wild crabs and thus increase crab stock in nature, which is the most significant impact of the project interventions on biodiversity.

257. Dust pollution occurs as a result of soil handling during construction and, more specifically, a lack of watering of the earth's surface. Such pollution is also a function of weather conditions—in the dry season, nuisance is greater; during the rainy season, dust nuisance subsides. Pre-construction and construction stages are more susceptible to dust accumulation. Noise pollution is normally due to some construction-related activities.

258. The project will promote crab hatcheries and farming, which will require saline water. The hatchery will use deep sea water that contains more than 25 ppt. In addition, crab ponds will also require saline water. These two activities may cause changes in the present soil and water salinity condition of the specific sites.

Identified social risks

259. Social risk may be arisen due to establishment of hatchery because existing crab collectors may lose their income as crab growers will buy crab-let from the hatcheries. The project is only undergoing minor construction. As a result, the workers' risk of injury is quite low. The project intervention can be carried out with the help of local labour. The recipients typically carry out the suggested tasks themselves. Construction of hatcheries, dirt cutting for constructing plinths, the construction of goat or sheep barns, and the reconstruction of homes will all demand labour. Local resources exist for this labour. Therefore, no outside labour will be required for the project. Additionally, there is a very slim probability of engaging in forced labour or child labour.

260. According to the Bangladesh Population and Housing Census 2022, there are approximately 1.65 million indigenous people living in Bangladesh, divided into 52 categories. Of these, approximately 49.98% are men and 50.02% are women. Every ethnic indigenous group in Bangladesh has its own unique tradition, culture, legacy, and sense of self. They also have a variety of ways to make a living, most of which are based on survival techniques and strategies that have been tailored to the various agro-ecological zones where they dwell. Twelve coastal districts in Bangladesh's south-west are included in the project. According to population census, there are roughly 13.92 million people living in the project area. Only 26,047 of them, or an insignificant 0.19% of the overall population, are indigenous people. The indigenous people mostly are from the Rakhyain community and concentrated mainly in the Patuakhali, Barguna and Cox's Bazar districts. If any indigenous people comes under the project, necessary action will be take according to the IPPF of the project.

261. The project interventions are community based and will not require external labour. Hence, risks associated with violation of human rights, particularly women, disable and other ethnic minorities (if any) are very limited or absent from external source. But, there are some internal risks to engage child labour in project interventions particularly crab farming and goat/sheep rearing. This may be occurred by the family members.

262. There is less possibility to occur Sexual Exploitation, Abuse and Harassment (SEAH) by the project interventions. Because, selected women members will directly involve many of the activities as these will be implemented at their homestead and neighbouring areas. In addition, some women may require to away to work in the crab farm. They might have risks to SEAH while working in the farm or traveling from home to farm and vice versa. However, the risk is very limited or negligible. But the challenge is that if any woman is affected, she does not want to disclose due to either shyness or fear of loss of dignity. At the IE level, there may be female staffs who will have to travel frequently in the selected villages. These staffs may also face similar types of difficulties. Similarly at the PMU level, female staffs may be recruited. These female staffs would require to frequent field visits in the remote areas of the country. Thus, they might be exposed to SEAH. High Court's guideline on sexual harassment will be applied in this project to SEAH.

263. The project interventions have limited risk to make discrimination between male, female and other disadvantaged group of people in the project area due to gender sensitive design. The contractor may provide low wage to the female labours. Besides, children and adolescent girls may engage in the project interventions. These problems will be solved by implementing community based approached. The CCAG members are the key actors at the local level for implementing the proposed activities, the project will engage 80per cent female while forming the CCAGs. They will actively participate in the project. So, discrimination should not take place.

Financial risk

Financial risk may arise at the procurement and beneficiary selection stage, and the project design includes adequate safeguard measures to minimize this risk. The major operational risk, at medium probability, is the reduction of coverage of beneficiaries due to currency exchange loss or domestic inflation.

To ensure internal control and prevent money laundering, PKSF has sufficient and verified information about client - "know your client" (KYC) - and make use of that information that underpin the most effective defense against being used to launder the proceeds of crime. PKSF has very strong relation with its partner organization and maintain the KYC.PKSF, having long experience in managing projects supported by the donor and other development partners, has developed efficient financial management and internal control systems. PKSF has an established finance division headed by a Deputy Managing Director for Finance and Administration. In order to monitor the activities of it's partner organizations. PKSF has also have its an effective internal audit cell headed by a General Manager who directly reports to the Managing Director of the organization. PKSF has built up a system and capacity for disbursing fund to POs based on efficient review procedures in coordination with field level monitoring. An internationally acclaimed audit firm carries out external audit of PKSF every year, and will continue to do the same for the PKSF's financial management of the project. Detail is described in Operation Manual of PKSF. (For details, please visit https://pksf.org.bd/policy-documents/). PKSF has also adopted a Whistle Blower Protection Policy. If any occurrence is identified or complaints are received, an investigation committee is formed. The committee generates an investigation report with recommendation. The Managing Director of PKSF takes the final decision (about punishment) about the offence.

In addition, in line with the international standards and initiatives, Bangladesh has passed Money Laundering Prevention Act (MLPA), 2002 (Amended, 2012). The Government has also enacted Anti-Terrorism Act (ATA) in 2009 (Amended in 2012) aiming to combat terrorism and terrorism financing. Both the Acts have empowered Bangladesh Financial Intelligence Unit (BFIU), Bangladesh Bank (BB) to perform the anchor role in combating ML/TF through issuing instructions and directives for reporting agencies and building awareness in the financial sectors. There are no United Nations Security Council (UNSC) sanctions and any other restrictions against Bangladesh. In compliance with its accreditation to the GCF, PKSF has Anti-money laundering /counter financing of terrorism measures according to the above-mentioned acts to prevent these prohibited practices. PKSF conducts due diligence to its partner organisations before signing any financial agreement. The due diligence includes physical visits of the organisations, review financial transactions, accounts, background check and book keepings etc. The same process will be followed for this project

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

264. The project is well aware about gender mainstreaming in the project activities. Hence, the proposed project has taken a gender responsive and transformative approach to climate change vulnerability, considering gendered differences in access to resources, ability to pursue adaptive livelihoods and institutional support and capacity building, and this has fundamentally shaped all of the activities and outputs of the project. The proposed project recognizes women's essential contributions as leaders and agents of change in the face of a changing climate and resource constraints. The project will select 362,475 direct beneficiaries for transferring knowledge and adaptation technologies proposed under this project. More than 50 percent of the total direct beneficiaries i.e., 181,237 will be women. It is already mentioned that the project will form 3,200 CCAGs. Considering the gender sensitivity of the proposed project, 80per cent of the CCAG members will be women beneficiaries (among the beneficiaries under PKSF, around 90per cent are women). 10per cent of the CCAG members will be women heads. The project will select mostly women because they usually teach their children at home. They will also teach their children about climate change issues what they will learn through training and meetings. Thus, climate change concepts and practices would transmit to the next generation which will have long term implication of addressing climate change in this country. Besides, necessary female staffs will be ensured at the field level so that women members can easily express their opinions and actively take part in the project activities. Considering gender integrity, the project proposes more crab farming and slatted housing for goat and sheep to outreach maximum number of women. Allocated budget for female beneficiaries also very high which is estimated USD 22.56 million which is more than 50 percent of GCF proceed.

265. Besides, CCCP experiences showed that women were benefitted both economically and socially due to engage them with CCAG. In coastal areas, women usually are not willing to go to cyclone shelter during cyclone or tidal surge. They prefer to stay at home because they feel safe at home than at shelter. Most important lesson was that they could talk about climate and disaster in their locality. They also felt empowered because they contributed to family income through vegetable cultivation, goat rearing etc. We expect the similar outcomes in the proposed project. The CCCP faced some challenges to engage the CCAG with the women members of the families. Initially, the women in the vulnerable areas were not much supportive due to shyness and hesitation. Besides, climate change was new issue to them. However, motivation through disseminating proper information helped to overcome this challenge.

266. The activities are designed in a way that the women will be mostly benefited economically and socially. The important livelihood option selected for the proposed project is goat and sheep rearing in slatted houses. The proposed project will select only women participants for implementing the activity because traditionally, all most all women in rural areas of Bangladesh including saline zone commonly rear livestock animal including goat and sheep. But the traditional process of management is a constraint of achieving expected benefit of rearing goat and sheep. The proposed project will provide support technological support and capacity building training to make it climate resilient and sustainable livelihood adaptation to climate change.

267. The RHL project considers not only the benefits of women, but also considers the inter-sectional vulnerability to changing conditions, of those beneficiaries facing additional marginalization due to poverty, and social exclusion. The project design recognizes to build adaptive capacity in regards to changing climatic conditions, by supporting climate resilient livelihoods, resilient homestead and better integration into local value chains, in which women are already playing a growing role.

268. The RHL project will accommodate GoB's policies and strategies on women's resilience and their critical role in preparedness and recovery from disasters and the necessity of shifting livelihoods towards adaptive options, efforts remain limited compared to the actual and acute needs of women. The Gender Assessment expands on the information provided throughout the proposal, by providing additional information on the national and local gender context, particularly in regards to women's access to resources, their role in decision-making and the gendered aspects of local livelihoods, and provides the basis for, and lessons on which, the Gender Action Plan (which is reflective of the overall project design) has been built. The activities of the proposed project have been selected considering that women can easily implement to enhance their capacity and increase their resilience to climate change.

269. For promoting women's empowerment through the project interventions, we will consult not only with the women members of a family, but also with the male members and other guardians. This will help eliminate hesitation and shyness. Besides, IEs will build good rapport by disseminating appropriate information with the vulnerable community.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

Financial Management and Audit

270. The project will follow AE's guidelines on Financial Management and Public Procurement Act 2006 (PPA2006) and Public procurement Policy 2008 (PPR2008) for procurement. PKSF, having long experience in managing projects supported by the donor and other development partners, has developed efficient Financial Management and internal control systems. PKSF has an established finance division headed by a Deputy Managing Director for Finance. In order to monitor the activities of its partner organizations, it also has in place a properly staffed internal audit cell headed by a General Manager who directly reports to the Managing Director of the organization. PKSF has built up a system and capacity for disbursing fund to IEs based on efficient review procedures in coordination with field level monitoring. An independent audit firm carries out external audit of PKSF, and will continue to do so for the PKSF's financial management of the project. As per GCF guideline PKSF will be provided financial management report and letter to GCF within 06 months of audited period of the AE's fiscal year end. Both internal and external audit is carried out once in a year. External audit is carried out using International Accounting Standard (IAS). PKSF as the AE will oversee, supervise, and approve the annual work plans of the 15 implementing entities and executing entities.

Disbursement and Funds Flow:

271. The funds for the project will flow through a Segregated Designated Account (DA) to be opened by PKSF in a commercial bank. The disbursement is report based; i.e., an advance to the DA is made on submission of Quarterly Financial Reports (QFRs), including a forecast of projected expenditures for the next, two calendar quarters. Further advances as required would be made to the DA based on updated expenditure forecasts for the subsequent two quarters having the balance available in the Designated Account including the balances in the bank accounts of the IEs at the end of the reporting quarter. The amount spent from the DA on eligible expenditures is documented as project expenditures based on claims for documentation in the QFRs, and the advances to the DA is adjusted accordingly.

Books of Accounts and Financial Reporting

272. The accounts of PKSF are prepared in accordance with International Accounting Standard (IAS) as adopted by the Institute of Chartered Accountants of Bangladesh (ICAB), on a going-concern basis under Generally Accepted Accounting Principles. PKSF's accounts are maintained on accrual basis under historical cost convention. The accounting manual of PKSF has been consistently followed for preparation of software-based final accounts.

273. PKSF follows Public Procurement Act 2006 and Public Procurement Rules 2008 for procurement goods and services. As per the PPA 2006 and PPR 2008, IEs will prepare and submit a procurement plan and get approval from PKSF. The first package of goods and services will be subject to prior review. The rest of the packages will be reviewed by the PMU staff at the field office of the IEs after procurement is over. This way, PKSF will ensure that all procurements are conducted as per policy. To ensure appropriate procurement, the first package of goods and services will be subject to prior review, and the rest will be subject to post-review. **It is observed from the previous experience that providing rigorous support for the first package, the IEs are equipped with proper knowledge of the procurement process. Nevertheless, the PMU will provide necessary support during the project period. For this,** IEs will submit their annual procurement plan; therefore, PMU will be aware of the procurement. In addition, IEs will submit a quarterly financial report which PKSF's audit team will check. Besides, the PMU staff will visit IEs field office at least quarterly to verify the procurement and financial transaction. **On t**op of that**,** PKSF**'s internal a**udit team will also visit the IEs office frequently for auditing. Any mis-procurement noticed by either PMU or the core audit team, immediate action will be taken as per policy.

274. PKSF recruits audit firms as per PPA 2006 and PPR 2008 for auditing all accounts and management of PKSF including each project. This audit firm submits a separate audit report for each project. The procurement committee of PKSF carries out the whole procurement activities. The committee calls for Expression of Interest (EOI) in the national dailies in both Bangla and English. Only the enlisted firm can submit the EOI. The procurement committee evaluates the applications and makes recommendations to select the firm. The Governing Body of PKSF select the firm in its Annual General Meeting based on the recommendation. A detailed procurement plan as per GCF format is presented in Annex 10.

275. Tax Exemption: Value Added Tax and Income Tax will be applicable as per PPA 2006 and PPR . PKSF enjoys Income Tax exemptions. Hence, GCF proceeds will not be used to finance any taxes in relation to the Goods and Services to be procured under the Project.

G.4. Disclosure of funding proposal

☒No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

☒	Annex 1	NDA no-objection letter(s) **(template provided)**
☒	Annex 2	Feasibility study - and a market study, if applicable
☒	Annex 3	Economic and/or financial analyses in spreadsheet format
☒	Annex 4	Detailed budget plan **(template provided)**
☒	Annex 5	Implementation timetable including key project/programme milestones **(template provided)**
☒	Annex 6	E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): **(ESS disclosure form provided)** ☐ Environmental and Social Impact Assessment (ESIA) or ☐ Environmental and Social Management Plan (ESMP) or ☐ Environmental and Social Management System (ESMS) ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
☒	Annex 7	Summary of consultations and stakeholder engagement plan
☒	Annex 8	Gender assessment and project/programme-level action plan **(template provided)**
☒	Annex 9	Legal due diligence (regulation, taxation and insurance)
☒	Annex 10	Procurement plan **(template provided)**
☒	Annex 11	Monitoring and evaluation plan **(template provided)**
☒	Annex 12	AE fee request **(template provided)**
☒	Annex 13	Co-financing commitment letter, if applicable **(template provided)**
☒	Annex 14	Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

☐	Annex 15	Evidence of internal approval **(template provided)**
☒	Annex 16	Map(s) indicating the location of proposed interventions
☐	Annex 17	Multi-country project/programme information **(template provided)**
☐	Annex 18	Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
☐	Annex 19	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☐	Annex 20	First level AML/CFT (KYC) assessment
☐	Annex 21	Operations manual (Operations and maintenance)
☐	Annex 22	Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)[65]
☒	Annex 23	Other references: Methods of beneficiary calculation
☒	Annex 24	IPPF_RHL_PKSF

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

[65] Annex 22 is mandatory for mitigation and cross-cutting projects.

No-objection letter issued by the national designated authority(ies) or focal point(s)

Secretary
Economic Relations Division
Ministry of Finance
Government of Bangladesh

D.O. No-09.00.0000.105.05.011.17-374

08 February 2018

Subject : Funding proposal for the GCF by PKSF regarding the project, 'Resilient Homestead and Livelihood Support to the Vulnerable Coastal People of Bangladesh (RHL)'

Dear Executive Director,

As the National Designated Authority of Bangladesh, I would like to refer to the 'Resilient Homestead and Livelihood Support to the Vulnerable Coastal People of Bangladesh (RHL)' project as included in the funding proposal submitted by one of the NIEs, PKSF to us on 29 January 2018.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project 'Resilient Homestead and Livelihood Support to the Vulnerable Coastal People of Bangladesh (RHL)' as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The Government of Bangladesh has no-objection to the project 'Resilient Homestead and Livelihood Support to the Vulnerable Coastal People of Bangladesh (RHL)' as included in the funding proposal;
(b) The project 'Resilient Homestead and Livelihood Support to the Vulnerable Coastal People of Bangladesh (RHL)' as included in the funding proposal is in conformity with Bangladesh's national priorities, strategies and plans;
(c) In accordance with the GCF's environmental and social safeguards, the project 'Resilient Homestead and Livelihood Support to the Vulnerable Coastal People of Bangladesh (RHL)' as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project 'Resilient Homestead and Livelihood Support to the Vulnerable Coastal People of Bangladesh (RHL)' as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Kazi Shofiqul Azam 8.2.2018

GOVERNMENT OF THE PEOPLE'S REPUBLIC OF BANGLADESH
MINISTRY OF FINANCE
ECONOMIC RELATIONS DIVISION
DHAKA

Mr. Howard Bamsey
Executive Director, Green Climate Fund
Songdo Business District, 175 Art Center-daero
Yeonsu-gu, Incheon 22004, Republic of Korea
Copy to: Chairman, PKSF, E-4/B, Agargoan, Dhaka-1207

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Resilient Homestead and Livelihood support to the vulnerable coastal people of Bangladesh (RHL)
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	Palli Karma-Sahayak Foundation (PKSF)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Districts of Bangladesh: Satkhira, Khulna, Bagerhat, Barguna, Patuakhali, Bhola, and Cox's Bazar.
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Friday, June 9, 2023
Language(s) of disclosure	English and Bangla
Explanation on language	English is a widely used language in Bangladesh and Bangla is the official language of Bangladesh.
Link to disclosure	English: https://pksf.org.bd/wp-content/uploads/2023/06/Annex-6-ES-RHL-08062023.pdf Bangla*: https://pksf.org.bd/wp-content/uploads/2023/06/Draft-Bangla-ESRM.pdf
Other link(s)	https://pksf.org.bd/pksf-submits-indigenous-peoples-planning-framework-to-gcf
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the "Environmental and Social Management Framework (ESMF)".
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Friday, June 9, 2023
Language(s) of disclosure	English and Bangla
Explanation on language	English is a widely used language in Bangladesh and Bangla is the official language of Bangladesh.
Link to disclosure	English: https://pksf.org.bd/wp-content/uploads/2023/06/Annex-6-ES-RHL-08062023.pdf Bangla*: https://pksf.org.bd/wp-content/uploads/2023/06/Draft-Bangla-ESRM.pdf
Other link(s)	https://pksf.org.bd/pksf-submits-indigenous-peoples-planning-framework-to-gcf

Remarks	An ESMP consistent with the requirements for a Category B project is contained in the "Environmental and Social Management Framework (ESMF)".
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity's website	Indigenous Peoples Planning Framework (IPPF) / Friday, June 9, 2023
Language(s) of disclosure	English
Explanation on language	English is a widely used language in Bangladesh and Bangla is the official language of Bangladesh.
Link to disclosure	English: https://pksf.org.bd/wp-content/uploads/2023/06/Annex-24-IPPF_RHL_PKSF_clean-08062023.pdf Bangla**: https://pksf.org.bd/wp-content/uploads/2023/06/%E0%A6%86%E0%A6%87%E0%A6%AA%E0%A6%BF%E0%A6%AA%E0%A6%BF%E0%A6%8F%E0%A6%AB-%E0%A6%AC%E0%A6%BE%E0%A6%82%E0%A6%B2%E0%A6%BE.pdf
Other link(s)	https://pksf.org.bd/pksf-submits-indigenous-peoples-planning-framework-to-gcf
Remarks	An IPPF consistent with the requirements for a Category B project is contained in the "Annex 24: Indigenous People Planning Framework (IPPF)".
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, June 9, 2023
Place	Meeting places of the Community Climate Change Groups (CCAGs) of the project PKSF will also display the documents in its office information Board. The project yet to select the Implementing Entity(ies) (IE), once the IEs is selected, the documents also be displayed in project office and field office information board. As the meeting places of Community Climate Change Groups (CCAGs) of the projects are not yet to be confirmed, the documents will be disclosed in: (1) Village: Dumuria, Union: Gabura, Upzilla: Shyamnagar, District: Satkhira (2) Village: Gab-bunia, Union: Uttor bedkashi, Upzilla: Koyra, District: Khulna (3) Village: Shuravi, Union: CharPatila, Upzilla: Char Kukri Mukri, Char Fashion, District: Bhola

	(4) Village: Charpara, Upzilla: Cox's Bazar sadar, District: Cox's Bazar (5) Village: Chela Bazar, Union: Chela, Upzilla: Mongla, District: Bagerhat (6) Village: Horingahta, Upzilla: Pathorghata, District: Borguna (7) Village: Auliapur, Union: Auliapur, Upzilla: Patuakhali Sadar, District: Patuakhali
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, July 10, 2023

* Subsequent to the disclosure of this form to the Board and active observers on 9 June 2023, the following updates have been made: The Environmental Social Management Framework has been disclosed in Bangla on 16 June 2023.

** Subsequent to the disclosure of this form to the Board and active observers on 9 June 2023, the following updates have been made: The Indigenous People Planning Framework has been disclosed in Bangla on16 June 2023.

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP206

Proposal name:	Resilient Homestead and Livelihood support to the vulnerable coastal people of Bangladesh (RHL)
Accredited entity:	Palli Karma-Sahayak Foundation (PKSF)
Country/(ies):	Bangladesh
Project/programme size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The proposal presents a good example of thinking ahead. First, the proposal treats the climate-linked increased salinity intrusion as an opportunity to create an alternative livelihood option. In response to increased salinization and salinity fluctuations that are leading to declines in paddy productivity and losses of agricultural land, and shrimp production, the project proposes development of a crab farming system due to its higher salinity tolerance and promotes adoption of salt tolerant livestock forage and horticulture varieties. Second, it proposes rebuilding existing houses of the ultra-poor and poor according to a design that is adapted to current and future projected inundation levels. The house design adopts the concept of "building back better": not only in terms of resilience but by incorporating water tanks, latrines and raised slatted units for livestock. Third, the resilient house designs, which will constitute a new building code for coastal areas, will be scaled up through the Palli Karma-Sahayak Foundation (PKSF) loan programme (under development) and government housing programme. Finally, the fruit-fibre-fish livelihood model to be promoted under the project aims at a balanced resilience–income–nutrition nexus that will be a critical element for sustainability of the project.	Replication and scaling up of the project will require external sources of funding, and the proposal does not present a strategy or demonstrate an ability to catalyse or mobilize financial resources in the future.

The proposal builds on existing investments and knowledge of PKSF and government on the development of the crab farming sector and resilient house designs. In terms of the crab sector development, PKSF invested (with the support of the Government of Bangladesh and its research partner from Viet Nam) in research on the suitability of crab production in coastal zones of Bangladesh, establishing crab hatcheries, and research into local production of crab feed. In terms of the resilient house design, PKSF developed the proposed design options jointly with the Housing and Building Research Institute and the designs will constitute a new building code for the coastal zones.	The accredited entity (AE) will work with 15 partner organizations that will serve as implementing entities (IEs) to implement the field work. IEs play a crucial role in ensuring the impacts of the project and their continuation after the project closure. The AE will be utilizing its strong partnership with the IEs, and the monitoring and capacity-building activities are central to the project. The AE will need to ensure monitoring and evaluation of the capacity and implementation skills of IEs so that project impacts will be ongoing in the long term.
The proposal presents a strategic linkage with government priorities for addressing climate change impacts. Adoption of a new building code for the coastal region based on resilient house design is expected to be scaled up by PKSF's new housing loan programme (in progress) and the government housing programme.	
The proposal represents a social protection project, which the **Intergovernmental Panel on Climate Change emphasizes in its latest report.** It supports the climate vulnerable ultra-poor, poor and women-headed households to access resilient housing and have a resilient livelihood option that will generate income and also provide security and nutrition.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XVII, titled "List of proposed conditions and recommendations", respectively.

II. Summary of the Secretariat's assessment

2.1 Project background

3. Bangladesh is a low-lying river delta with a network of more than 230 major rivers and thousands of tributaries and canals, with high population densities of 160 million people in an area of 145,570 square kilometres. Coastal zones are only 2 metres above sea level. Its geographic location, population density and low adaptation capacity make tit one of the

countries most vulnerable to disaster risk in the world, with an annual loss to disasters of around USD 3 billion, which is approximately 1–2 per cent of the gross domestic product, according to the United Nations Office for Disaster Risk Reduction.

4. The vulnerability of the coastal people is characterized in three ways: climate sensitive livelihoods; vulnerable settlements in low-lying areas; and scarcity of safe drinking water. The majority of the coastal population is ultra-poor or poor,[1] with small and marginal farm families and shrimp workers. They primarily depend on seasonal subsistence agriculture and agriculture wage labour, which are highly climate sensitive. Moreover, poor coastal communities build their houses in low-lying areas that are subject to coastal flooding. Most of the houses, which are built of mud and goal pata (leaves of an indigenous coastal plant, are severely affected by cyclones, storm surges and high tides. Coastal inhabitants have to spend a significant amount of their earnings on repairing houses each year, making it extremely difficult for them to escape economic poverty.

5. The Resilient Homestead and Livelihood support to the vulnerable coastal people of Bangladesh (RHL) project aims to develop climate-adaptive coastal communities through the adoption of climate-resilient housing and livelihood options in seven coastal districts (Satkhira, Khulna, Bagerhat, Barguna, Patuakhali, Bhola and Cox’s Bazar) with the highest exposure to sea and tidal surge and intense salinity, where it is feasible to develop crab hatcheries.

6. The project addresses the following barriers:

(a) Lack of climate planning and implementation capacity among communities for addressing climate change;

(b) Lack of access to finance to invest in large capital expenses;

(c) Lack of livelihood diversification, limited capacity to adopt adaptation technologies; and

(d) Capacity, food safety and quality standards, supply and market access in developing a community-based crab supply chain.

7. The total project funding amount is USD 49.9 million, of which the requested GCF funding is USD 42.2 million in grants. The accredited entity (AE) is providing co-financing of USD 6.6 million in loans and USD 1 million in the form of in-kind.

8. The project’s environmental and social safeguards (ESS) risk is categorized as B.

2.2 Component-by-component analysis

Outcome 1: Decreased risk of loss of assets and life from extreme weather events (total cost: USD 24.4 million; GCF cost: USD 24.4 million)

9. This outcome will be achieved by construction of climate resilient homesteads for climate vulnerable ultra-poor and poor households the majority of which will be women headed. The housing design, which is developed jointly with the Public Housing Research Agency, will consider present and future inundation projections and raising homestead plinths above coastal flood level. The design is also based on the building back better concept and incorporates building rainwater harvesting facilities, livestock rearing space, sanitary latrines, homestead gardening, and tree planting. The activity will benefit 13,500 beneficiaries.

10. Additionally, homestead tree planting around the selected houses will be supported to contribute to resilience to storm and tidal surges. About 90,000 beneficiaries will receive tree saplings, including mangrove species.

[1] Extreme poverty is defined as purchasing power parity (PPP) below USD 1.25 a day and poverty is defined as PPP below USD 1.90 a day by the Household Income and Expenditure Survey of 2016.

<u>Outcome 2: Livelihood resilience to sea level rise/storm surge and salinity (total cost: USD 19.4 million; GCF cost: USD 12.8 million)</u>

11. Under this outcome, the project sees salinity intrusion as an opportunity for building climate-resilient livelihood options while maintaining the resilience-income-nutrition nexus. To achieve this, the project will support the following outputs and activities:

(a) Output 2.1: Traditional farming practices climate proofed. Under this output, the project will promote climate resilient homestead-based livestock and vegetable and fruit tree farming. Both are traditionally pro-poor livelihood activities and their resilience ensures diversification of household incomes and nutrition. Main activities will include: building slatted houses for small ruminants that can resist storms and tidal surges, improving livestock management practices, and adoption of livestock feed varieties that could be grown in saline conditions. Additionally, this output will support adoption of saline tolerant varieties and resilient farm management practices; and

(b) Output 2.2: Community-based farmed crab supply chain created. Under this output, the project intends to capitalize on crab farming along with mangrove tree plantations because crab farming is tolerant to a wide range of salinity levels in comparison to paddy and shrimp farming which are declining due to rapid salinity fluctuations. Current practices of crab farming are based on capturing crablets and crabs from nature, with a negative impact on biodiversity. The output will support the shift from unsustainable paddy and shrimp farming as well as exploitative crab-capture farming, to culture crab farming through investments in mini-scale crab hatcheries, upscaling farmers' technological skills, and market inclusion of small and marginalized crab farmers. This is expected to lead to increases in the natural stock of crab species and mangrove plantations, thereby enhancing biodiversity.

<u>Outcome 3: Improved climate planning and implementation by communities and local-level institutions (total cost: USD 4.16 million; GCF cost: USD 3.3 million)</u>

12. As climate change is an emerging issue, local public and grassroot institutions lack climate related knowledge and experience. This component will focus on strengthening the capacity of institutions and communities to address climate change.

13. Under this component, the activities foresee working with the government to develop a crop sector policy and create housing loans programmes for the poor.

14. Main outputs to be achieved under outcome 3 will include the following:

(a) Output 3.1: Climate change adaptation groups (CCAGs) formed and operationalized. Under this output, the project will focus on: (i) selection of project beneficiaries based on socioeconomic profiling as well as technical assessments, particularly for the beneficiaries targeted for the housing activities; and (ii) raising climate change awareness among implementing entities (IEs);

(b) Output 3.2: Capacity built among IEs and relevant institutions involved in the project. Key activities under this output will include development of training materials for adaptation technologies and crab value chain development; preparation of the project management and implementation guidelines; strengthening the capacities of beneficiaries and stakeholders (including training of construction companies and firms); and support data systems for crab sector development and research; and

(c) Output 3.3: Knowledge products prepared and disseminated.

<u>Project management (total cost: USD 2 million; GCF cost: USD 1.7 million)</u>

15. The GCF portion of the project management cost is less than 5 per cent of the total requested GCF funding and is compliant with the GCF policy on fees.

2.2.1. Implementation arrangements

16. The Palli Karma-Sahayak Foundation (PKSF) is the AE and also the executing entity (EE). PKSF will work with at least 15 non-governmental organizations, which will serve as IEs responsible for working with the communities. The IEs will be selected by a Selection Committee established for the project, and based on the following selection criteria.

17. IEs will have at a minimum:

(a) Permanent existence of the organization in the project areas;

(b) At least five years of experience in implementing climate change-related projects or programmes;

(c) A good track record of financial transaction (at least Bangladeshi taka (BDT) 1 crore annually for the last three years);

(d) Must be extraordinary, excellent or at least good as per PKSF's assessment using defined assessment criteria which include financial efficiency, economic efficiency, operational efficiency growth indicators, financial strength and risk management, accounting and internal control systems, social performance, human capacity and governance; and

(e) Valid legal documents including registration. Organizations will be ineligible on the grounds of involvement in money laundering and terrorist financing.

III. Assessment of performance against investment criteria

3.1 Impact potential *Scale: N/A*

18. The project will directly benefit 362,475 climate, socioeconomically vulnerable beneficiaries, of which 50 per cent will be female. Additionally, it will support 770,050 indirect beneficiaries, of which 50 per cent will be female.

19. Through the climate resilient homestead provision, the project will secure lives and livelihood resources of 13,500 ultra-poor and poor beneficiaries. Current government practices focus primarily on securing human lives through strengthening embankments and building communal cyclone shelters where women and girls often face difficulties due to limited sanitation and hygiene facilities as well as the high risk of sexual harassment. Securing household assets has not yet been a primary focus or practice in Bangladesh, therefore by protecting dwellers and household assets, the climate resilient housing activities would address risks faced by communities and vulnerable populations, particularly women and girls.

20. The establishment of the crab hatchery and farming will create opportunities of sustainable employment for coastal people. The project will establish 50 micro crab hatcheries in the coastal areas to produce juvenile crabs which will be used by the crab growers. One micro hatchery will support 10 crab nurseries. Fifty micro hatcheries will provide support to 500 crab nurseries. On an average, two full-time workers will be employed at each crab hatchery. The micro crab hatcheries will therefore support 100 direct beneficiaries and 2,250 crab nurseries will support 20,000 crab growers. In addition, 900 people will be engaged at different levels of the crab production value chain. Furthermore, the proposed project will provide technological support to 90,000 direct beneficiaries practising sheep rearing, which is a proven salinity tolerant livelihood option for coastal zones.

21. The project will support another 90,000 direct beneficiaries through tree planting around homesteads. This will help them to earn income, meet nutrition requirements and protect houses from cyclone and storm damage. The housing component of the project will protect health and livelihood resources and lead to improved well-being. The project will also

support 90,000 direct beneficiaries in developing saline-tolerant vegetable cultivation on their homesteads.

22. The number of indirect beneficiaries will be 770,050 people. It is expected that 75 per cent of selected community will increase their resilience to climate change and related consequences.

3.2 Paradigm shift potential *Scale: N/A*

23. The project will promote paradigm shift towards resilient and sustainable development pathways by seizing climate-induced salinity intrusion as an opportunity for creation of resilient livelihoods and promotion of saline-tolerant housing and farming technologies and practices. Technologies and innovations that will be promoted to achieve the paradigm shift are built on PKSF's experience and investments in crab hatcheries and nurseries, and research on culture crab farming and saline-tolerant fodder and horticulture crops.

24. The project will also contribute to the creation of an enabling environment by influencing several government policies. The project interventions will primarily influence national land-use policy, national fisheries policy, national water policy and *jalmahal* (wetland) policy. With the growing economic development of the subsector, the government will create a favourable policy and regulatory environment for the production and trading of crab. The resilient homesteads will influence government's existing and future housing projects by integrating greater sustainability in the context of climate change. The project will share the results of resilient housing with the Housing and Building Research Institute so that it will be able to integrate sustainability in future designs.

3.3 Sustainable development potential *Scale: N/A*

25. The project will address several crucial sustainable livelihood and resiliency-related issues pertinent to the coastal inhabitants of Bangladesh. Specifically, the project will employ interventions related to (i) inadequate quality of human settlements in low-lying areas of Bangladesh; (ii) climate sensitivity of the population in selected regions; and (iii) scarcity of safe drinking water.

26. By providing protection for the individual assets of marginalized and climate vulnerable households, such as housing and livestock, the project will also provide security and well-being co-benefits to women and girls. By increasing income and production of nutritious food crops, and by creating new job opportunities in the crab production value chain, the project will also improve the quality of lives of poor communities, and their food and nutrition security.

27. By supporting the development of climate-resilient crab farming, the project is also expected to contribute to the development of secondary service-economy activities with a positive impact on local economies, reducing climate migration.

28. The project activity on crab farming supports the shift from unsustainable shrimp farming and exploitative uncultured crab farming with expected positive impacts on the growth of natural stocks of crab species and an increase in biodiversity.

29. The project puts emphasis on disadvantaged, marginalized and excluded segments of society. Its investments primarily focus on livestock, horticulture and crab farming activities that are traditionally undertaken by women and people in poor communities.

30. The project will contribute to achieving the Sustainable Development Goals (SDGs), including SDG1 (no poverty: targets 1.1, 1.4 and 1.5); SDG 2 (zero hunger: targets 2.1, 2.3, 2.4 and 2.5); SDG 6 (clean water and sanitation: targets 6.4 and 6.6); SDG 13 (climate action: target 13.2); and SDG 15 (life on land: targets 15.1, 15.2, 15.3, 15.4 and 15.9).

3.4 Needs of the recipient *Scale: N/A*

31. Bangladesh is one of the least developed countries, although it is scheduled to graduate from this status in 2026. The project targets the coastal rural areas, particularly the selected districts with communities most vulnerable to climate change.

32. Institutional needs are addressed through the training and capacity-building of the partner organizations that will serve as the IEs. Training and capacity-building of IEs will support the project to achieve the target of increased resilience of beneficiaries and communities in the long term by providing necessary guidance on maintenance after the project ends. The project places a strong emphasis on knowledge-sharing and skill enhancement at the local level.

33. The eligibility criteria for direct beneficiaries have been carefully designed based on the experience of PKSF and targeted beneficiaries will be selected according to this set of criteria. The selection criteria will be further informed and validated during the first year of implementation through a socioeconomic baseline survey and technical assessment of target households' locations (proximity to storm/cyclone affected zones) to assess any additional metrics.

3.5 Country ownership *Scale: N/A*

34. The project will contribute to implementation of the Bangladesh Climate Change Strategy and Action Plan, the national adaptation programme of action and the national communications, the Eighth Five-Year Plan and the Perspective Plan.

35. PKSF will work with at least 15 IEs that will work closely with the communities to develop skills and capacity to deliver project impacts beyond project closure. This is crucial in order to support appropriate maintenance.

36. Stakeholder consultation has been undertaken extensively, including with local communities, government and non-government representatives.

3.6 Efficiency and effectiveness *Scale: N/A*

37. The project activities have been tested and adapted to best practices and technologies to address sustainable livelihood options, and increase resilience of communities. The activities build on the existing PKSF investment and knowledge base, with a strong emphasis on capacity-building and knowledge-sharing with the local beneficiaries and organizations.

38. The project delivers public goods, for example through interventions in mangrove and climate-adaptive homesteads. It also delivers an increase in income through diversification of livelihood options offered by the interventions in hatcheries for crabs and saline-resistant vegetables. The potential of reducing beneficiaries' economic losses is estimated to be 60 per cent.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

39. **Project background.** The project aims to decrease the risk of loss of assets and lives from extreme weather events; enhance livelihood resilience to sea level rise, storm surge and salinity; and improve climate planning and implementation by coastal communities and local-

level institutions. Expected environmental co-benefits include protection of crabs in nature due to crab hatcheries supported by the project. Social co-benefits are expected from increased incomes through diversification of farming operations.

40. **Environmental and social risk category.** The confirmed and assigned environmental and social risk category of the project is equivalent to GCF's medium/category B, which is within the AE's accreditation level. This categorization is due to the actions which include renovations or construction of houses by raising plinths and providing slatted houses for household livestock rearing, and establishing crab microhatchery facilities that are expected to have risks and impacts that are generally reversible and site-specific, and which can also be readily managed with mitigation measures.

41. **Safeguard instruments.** The AE provided an environmental and social and management framework with the funding proposal as the safeguards document. The document provides an overview of the national context for environmental and social risk assessment, potential environmental and social risks likely to be associated with the project activities, including environment and social impact assessment and management plans for subprojects categorized as medium risk during the project's implementation.

42. **Compliance with GCF's environmental and social safeguards (ESS) standards.** The paragraphs below identify the key risks and/or impacts relating to the GCF ESS standards and how they are addressed in the proposal:

43. **ESS 1: Assessment and management of environmental and social risks and impacts:** Subprojects will be screened and categorized as low/minimal and/or moderate environmental and social risk. Screening will be carried out by the project's implementing entities. Subprojects categorized as moderate environmental and social risk will be required to have site-specific environmental and social impact assessments and management plans. A management plan for potential risks and impacts is presented in the safeguards document. In addition, an assessment of the significance level of these risks has been provided, with most risks being assessed as low and a few as medium (e.g. those related to construction activities). Key potential risks are associated with labour and working conditions, pollution, community health and safety, and biodiversity. The description of baseline environmental and social conditions provided in the document is minimal and will need to be elaborated at the subproject level.

44. **ESS 2: Labour and working conditions:** Construction-related activities under components 1 and 2, which include raising houses and livestock rearing barns respectively, have potential occupational health and safety risks for the workers involved. This will be remediated by providing workers with personal protective equipment to wear during construction. The AE will also ensure the safety of the workers involved in the project by requiring them to follow its environmental health and safety guidelines. Procurement of materials for use in the project will be local, as far as possible. The AE is recommended to establish a process for assessing risks related to labour in supply chains and mitigating them at the subproject level, including for solar power components.

45. **ESS 3: Resource efficiency and pollution prevention:** Construction activities may create temporary noise and air pollution for households and communities in the vicinity of such operations. Sanitary latrines constructed at homesteads by the project will be positioned to avoid potential contamination of water sources. Waste is expected to be generated from construction activities, crab hatcheries and livestock rearing. Additionally, the operational phase of the project also generates waste from water used in crab hatcheries and solar power components, including panels, after their lifespan has been reached. Manure from livestock rearing will be used in agricultural fields as fertilizer for crops. Mechanisms for the treatment of saline water will be built into the hatchery design for the saline water required to produce

crabs, in addition to limiting crab farming to areas that are already salinized. The AE is advised to ensure that it has a process for safe disposal of solar power components used by the project.

46. **ESS 4: Community health, safety and security**: Construction may also pose a safety risk to households and local communities. This will be mitigated by preventing access to rooms during construction to prevent injuries. Building materials that are not considered to be of good quality will not be used by the project. The AE is advised to also employ the use of signage in the local language to communicate safety risks at construction sites. Additionally, the AE is advised to ensure training for food safety and management for crab production.

47. **ESS 5: Land acquisition and involuntary resettlement**. The project excludes activities that will result in physical or economic displacement of people and communities. Households will reside in other rooms of the house while the buildings undergo renovations to raise plinths, thus avoid temporary displacement during construction activities. For any land that may be required to implement project activities, such as the establishment of crab production facilities, the AE will ensure that land rights are obtained.

48. **ESS 6: Biodiversity conservation and sustainable management of living natural resources**. The project aims to prevent the collection of juvenile crabs in water sources by ensuring that beneficiary crab collectors use crablets and juveniles from hatcheries and crab nurseries produced in the facilities that it will establish. Any loss of vegetation from construction of structures in homesteads is expected to be minimal.

49. **GCF Indigenous Peoples Policy and ESS 7: Indigenous peoples**: The AE has screened for indigenous peoples in line with the GCF Indigenous Peoples Policy and found no indigenous peoples in the areas targeted by the project. If, during the implementation process, indigenous peoples are found, the AE has provided an indigenous peoples planning framework as a source of guidance for the development of indigenous peoples plans and consultation processes. In line with its roles and functions, the GCF Indigenous Peoples Advisory Group is available to provide advice to the accredited entity and executing entities.

50. **ESS 8: Cultural heritage**: The AE's due diligence indicates that there are no archaeological, palaeontological, historical, artistic or religious sites considered as cultural heritage in the project area; therefore the risk is low.

51. **Sexual exploitation, sexual abuse and sexual harassment (SEAH)**: The revised GCF Environmental and Social Policy adopted by decision B.BM-2021/18 requires safeguarding from SEAH in GCF-financed activities. The environmental and social management framework of the RHL programme highlights key SEAH safeguarding in alignment with GCF policies, and the provisions outlined in the country's penal code. The AE envisages that there is less possibility of SEAH occurring under the interventions planned for this project, on the basis of PKSE's first-level due diligence. However, it is recommended that the AE ensure prevention of SEAH, including putting in place clear and adequate measures to address this risk. In order to assess any SEAH risks associated with the project activities, the AE will engage different stakeholders throughout the project implementation period. PKSF will establish a project management unit (PMU) where any SEAH (and/or its equivalent) will be assessed and mitigation measures put in place. The GCF recommends that the AE put in place, in advance, SEAH grievance and reporting mechanisms in the event that SEAH occurs even though the AE's assessment is that the risk of SEAH is low. In addition, the focus on SEAH under this project should include all genders, although women are the main focus under this project: the GCF SEAH safeguarding applies to all people who may be at risk of SEAH in GCF-financed activities.

52. **Implementation arrangements.** The project will provide capacity-building training to staff of IEs with roles and responsibilities for implementing ESS procedures, including screening. The PMU will have an ESS coordinator to guide the implementation of matters relating to ESS, including training staff of IEs, ensuring due diligence is undertaken and conducting site visits to monitor subprojects. A third party will be engaged to evaluate the

effectiveness of safeguards measures, and templates for monitoring plans are included in the safeguards document. The AE will provide oversight and supervise the management of ESS matters throughout the project's implementation. For the items that require financial resources, the AE is required to set aside funds from the project's budget to execute the measures during implementation, such as the preparation of impact assessments and management plans for subprojects classified as moderate environmental and social risk, category B.

53. **Stakeholder engagement and information disclosure**. Consultations were undertaken with various stakeholders, including local communities, crab catchers, crab traders and representatives of government agencies and civil society organizations, although it is not clear to the Secretariat when these consultations were conducted during the preparation of the project. However, issues raised by participants in the meetings are outlined in the safeguards document. A stakeholder engagement plan is also included in the document. It identifies the different types of stakeholders, their roles in the project, and proposed strategies for engaging them during implementation. Environmental and social impact assessments and management plans for medium risk subprojects will be disclosed by the AE in accordance with the GCF Information Disclosure Policy.

54. **Grievance redress mechanism**. The project will make use of the AE's and GCF redress mechanisms and contact details of both are included in the safeguards document. Complainants may also contact a project-level grievance mechanism that will be established by the PMU. Grievances received and registered by the project management unit will be handled by the ESS coordinator. In line with the GCF Indigenous Peoples Policy, if relevant, the GCF indigenous peoples focal point will be available for assistance at any stage, including before a claim has been made.

4.2 Gender policy

55. The AE has provided a gender assessment and gender action plan therefore complies with the requirements of the GCF Gender Policy.

56. The gender assessment was conducted through desk reviews and field visits. Consultations at this stage did not include women focus groups, organizations and institutions. A detailed gender gap analysis and annual action plan will be developed after further consultation with the project beneficiaries and women and women's groups, which will be done at the inception phase of the project.

57. Bangladesh has developed several policies and strategies to address gender inequality and women's empowerment. It has developed a women's development policy and nation action plan to ensure the overall safety and security of women and children to deal with disasters, engage and benefit equally in the rehabilitation process as well as getting access to necessary inputs. However, like many countries, implementation and addressing the gender inequality issues in the country falls short and needs more work. This includes the efforts of the government to mainstream gender issues in climate change policies as well. The agriculture policy acknowledges the critical role that women play in the sector particularly for food processing and nutrition, and also notes women's contribution as producers and in value chains need to be improved through capacity building, increased access to information, knowledge, financial assets, technology other resources and markets. Overall, policies themselves and their implementation have gaps in the level of analysis of the differentiated impacts of climate change on women and men. This along with the paucity of sex dis aggregated data has led to gaps in measures that address the differentiated needs related to access to land, finance, services, and inputs.

58. The gender assessment shed light on the linkages and differentiated impacts of climate change and particularly climate related disaster on women and men. The gender assessment

which was undertaken based on desk review indicates many differentiated impacts and clearly shows the different roles of women and men, and the related challenges in mitigating and adapting to disasters and climate change. It for examples illustrates that mobility of women in Bangladesh varies depending on social status, religious affiliation and whether they live in urban or rural areas. Socio-cultural norms shape the perceptions of the value of women and largely designates unpaid domestic responsibilities to women such as household and care for the family, looking after children and the elderly, collecting water, and cooking. Their contribution remains socially invisible and has little exchange value thus reinforcing women's weak decision-making power within the household and the workplace. Many women do not own land or other property, have limited access to cash, livestock, and poultry, and may have access to but have no control over agricultural land or homestead areas for vegetable gardening. Lack of access and ownership disempowers women economically and makes them socially subordinate. Illiteracy, limited access to resources, limited access to information and skills, limited decision making power, little access to paid labor and unequal pay renders women vulnerable, with little or no leverage to manage and overcome disasters. These factors lead to women being poorly represented in planning and decision-making processes in climate change policies, limiting their capacity to engage in political decisions that can positively impact their specific needs and vulnerabilities.

59. Crab harvesting, which is one of the project activities, is quite a recent economic activity in which women are involved. Crab harvest time is the busiest period adding to the already full schedule that women have. However, it provides valuable income while maintaining their ability to fulfill their roles as home makers, and their roles and responsibilities within households. Women are limited to certain roles within the crab industry that limits their access to innovative technology and approaches. These roles include specific tasks in the crab production value chain such as farmed crabs, which is less labor intensive. Only a few women are engaged in marketing, which could be a result of mobility issues and cultural norms that frown upon women working outside of the home. The low level of education of women may also restrict their ability to adopt and use technologies.

60. The gender action plan provided is based on the findings of the gender assessment and includes activities, indicators, targets, timeline, budget, and gender expertise to implement, supervise and report on the gender action plan. In the core team of PKSF there is a gender expert who is currently involved in project formulation and will be available during the project implementation. This expert will be involved in decision making aspects of the project. The gender action plan provided is at a program level and will be further enhanced with detailed gender gap analysis and action plan to be developed after more extensive consultations with the project beneficiaries at the inception phase of the project. This revised action plan will be submitted to the secretariat.

61. Currently activities identified are targeted at addressing a variety of challenges that women experience. The project will strengthen participation of women in the crablet production, crab nursery, crab farming and marketing. Women will also be supported to engage in the goat/sheep rearing in slatted houses, fruit-forest-fisheries model including crab hatchery and farming, homestead vegetable cultivation livelihood options including fruit trees and mangrove plantation. Women and women headed households will be targeted to benefit from the construction of improved climate resilient homes. To fill the technical gaps as well as address the time constraints of women, women trainers and women-exclusive training sessions, including flexible times, and provision of household training for women-headed households will be undertaken. The presence of households headed by women will be a key factor in the selection process for vulnerable women. Men and community elders, will be involved and integrated where necessary, in community trainings that address women's participation and norms around appropriate work for women, as well as mobility outside the homestead on Gievance Redress Mechanism. Community and women's groups will receive training on climate change issues and how to deal with these problems. Trainings will help women identify climate

change related problems on their lives and livelihoods and prepare/plan accordingly to reduce their vulnerability to the impacts of climate change.

62. The AE is recommended to include actions that further enhance mainstreaming of gender once consultations have been undertaken and thus improve on expected outcomes. The AE is also encouraged to set a specific target for women-headed households that will benefit from the construction of climate-resilient homes related to outcome 1.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

63. This funding proposal is an adaptation programme to build resilient homesteads and provide livelihood support for vulnerable coastal people in Bangladesh.

64. GCF is considering a grant of USD 42.2 million together with co-financing from PKSF in Bangladesh of USD 6.6 million senior loan and a USD 1.2 million of in-kind contribution. The co-financing ratio is 1:0.18 and accounts for 15.5 per cent of total financing.

4.3.2. Accredited entity/executing entity capability to execute the current programme (medium risk)

65. PKSF will act as both AE and EE for the project. As the AE, PKSF will be responsible for high-level implementation tasks including establishing various policies and standards, as well as reviewing and reporting. In addition, PKSF will engage in day-to-day project implementation for the different components. The Secretariat views positively the selection of PKSF as a partner, given its track record of over 30 years in poverty alleviation through socioeconomic development and environmental protection work in Bangladesh.

4.3.3. Programme-specific execution risks (medium risk)

66. The funding proposal's largest allocation will be for implementing component 1, which is the design and building of climate-resilient homesteads in selected coastal areas in Bangladesh. The execution risk is likely to be limited, given PKSF's experience in building and repairing houses. However, component 2 will involve several livelihood subprojects including development of the crab-raising industry. Crab raising is a nascent industry requiring a high level of technical expertise to succeed. This is partly mitigated by the experience and knowledge gained by PKSF in crab hatching technology on a small scale. The next step is to replicate that experience to an industrial level of operation.

4.3.4. Compliance risk (medium risk)

67. The project has controls to identify risks of money laundering and terrorism financing. During the implementation phase of the project, appropriate due diligence will be necessary to ensure that funds are used properly and that the project avoids violating sanctions or engages in prohibited practices. The implementers are committed to monitoring the project and to undertaking actions to reduce any significant risks to financial integrity.

68. The project will have a whistle-blower facility for the reporting of any irregularities or prohibited practices.

69. Based on the commitments of the implementers and the anticipated risk profile for the programme in terms of money laundering, sanctions or prohibited practices, the risk is rated as medium.

4.3.5. Project viability and concessionality

70. The proposed project aims to build climate-resilient homesteads and provide livelihood projects to selected coastal communities in Bangladesh. The business models for climate-resilient buildings and most of the livelihood projects will rely on existing technology. Financing will be provided by non-governmental organizations and microfinance institutions. The livelihood project on crab raising is the only aspect that will involve new technology, but this is expected to be viable, based on PKSF's experience. Requested participation from GCF will represent 85 per cent of total programme size in a wholly grant-funded project. The proposed concessionality level is justified to help vulnerable coastal communities in the country adapt to climate change, given the limited resources of government. The project is expected to generate significant adaptation benefits to the selected coastal areas in Bangladesh.

4.3.6. GCF portfolio concentration risk (low risk)

71. In case of approval, the impact of this proposal on the GCF portfolio concentration in terms of results area and single proposal is not material.

4.3.7. Recommendation

72. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall programme	Medium
Accredited entity (AE)/executing entity (EE) capability	Medium
Project-specific execution	Medium
GCF portfolio concentration	Low
Compliance	Medium

4.4 Fiduciary

73. PKSF will act as the AE and the EE for the project.

74. As the AE, PKSF will be responsible for activities including but not limited to: (i) establishing the PMU; (ii) appraisal of and finalizing the project implementation arrangements, including mission travel; (iii) ensuring compliance of fiduciary standards, ESS standards and gender-related issues; (iv) facilitating financial flow to the project; (v) submitting annual performance reports and interim unaudited financial reports; and (vi) assisting project management to draft terms of reference and advising on the selection of entities for implementation.

75. As the EE, PKSF will bear the responsibility of implementing the project in all aspects. PKSF will delegate field-level implementation of activities to the selected IEs through legal agreements. Each IE, by signing a legal agreement with PKSF, will be responsible for working with communities to deliver the project services, and will be monitored by the EE. The partner organizations who will qualify to implement the project at the community level will have been procured through PKSF's applicable procurement policies and procedures.

76. In terms of the flow of project funds, these will be directly disbursed to PKSF from GCF. PKSF will receive the GCF proceeds in the project-specific account held in a commercial bank in USD. In additional to initial advance to IEs from PKSF, PKSF will reimburse the fund to the implementing partners based on satisfactory performance.

77. PSKF will select at least 15 partner organizations that will be working at the community level. The PKSF loan under the proposed project will be disbursed through PKSF's mainstream credit programme to the partner organizations. In addition, a PMU will be established at their respective offices. The PMU will operate the grant part and the credit officers will operate the PKSF loan part, as per PKSF's operational procedure for blending projects/programmes.

78. PKSF always implements its projects through partner organizations, which it selects via its own procurement policies and procedures. These partner organizations will mobilize beneficiaries, engage local contractors and ensure project services for the beneficiaries. As such, it will pay the contractors and service providers at the local level upfront where necessary. The PMU will engage project personnel who will liaise with the selected IEs and monitor the implementation of their projects.

79. As AE, PKSF will approve and oversee activities of the PMU, which is responsible for monitoring day-to-day activities of the 15 entities and reporting to PKSF, as well as approving the entities' workplans with the guidance of PKSF. The AE will review evaluation reports and audit reports complied by individual consultants. It will oversee procurement and financial management of the project; prepare a project completion report; and submit the report to the GCF Secretariat. The AE will ensure that all compliance requirements are met, including the gender action plan, environmental and social management framework, and Indigenous Peoples Policy. The AE will monitor the audit process and fund disbursement, as well as preparing the project's financial closing report and closing documents for submission to the GCF Secretariat.

80. As per GCF guidelines, PKSF will provide a financial management report and letter for the audit period to GCF within 6 months of the AE's fiscal year end. Both internal and external audits are carried out once a year. External audit is carried out using the International Accounting Standard. PKSF recruits the audit firm as per PKSF procurement rules for auditing all PKSF accounts and management including of each project. This audit firm will submit separate audit reports for each project. Specifically, the audit firm will undertake external audits on expenditures for all activities implemented/delivered by all partner organizations/implementing entities.

4.5 Results monitoring and reporting

81. As an adaptation initiative, the project aims at generating various adaptation results across multiple GCF result areas. It is expected that 362,475 direct beneficiaries and 770,050 indirect beneficiaries will receive adaptation benefits. The AE has provided a methodology for defining project adaptation beneficiaries in the separately attached annex, and it provides a basis for target-setting in the logical framework.

82. The theory of change diagram adequately captures various levels of expected changes and logical linkages between them from the goal statement to the proposed activities. It is helpful that the theory of change diagram identifies co-benefits that can be achieved along the way towards climate results as well as capturing the causal relationship between outcomes and co-benefits. The barriers and risks that may hinder the achievement of these changes are identified and linked to pertinent activities that will tackle them accordingly.

83. The logical framework is structured in alignment with the GCF integrated results management framework (IRMF). It shows how the project can contribute to three GCF adaptation result areas with three project outcomes and four co-benefits. IRMF indicators are chosen to monitor and report the project results in the respective GCF result areas. The AE has been very responsive in addressing Secretariat comments to improve the logical framework.

84. The project has identified relevant project-specific indicators for project outcomes and outputs, and this is expected to enable strong monitoring and evaluation. The indicators are assessed to be specific, measurable, achievable, relevant and time-bound (SMART) and have

considered both the qualitative and quantitative aspects of results. It is worth noting the importance of the baseline survey, because its results will determine performance levels and targets. If any changes are made in baselines and targets of the logical framework after the survey, the AE must report them to GCF and finalize them at as early a stage as possible of project implementation.

85. The allocated project monitoring and evaluation (M&E) budget does not fully meet the requirement of 2–5 per cent of the total budget as per the GCF Evaluation Policy. However, in addition to the fee, the AE has provided funds to supplement the M&E budget and the Secretariat considers this to be sufficient to secure M&E activities.

4.6 Legal assessment

86. The Accreditation Master Agreement (the "AMA") was signed with the Accredited Entity on 19 November 2018 and became effective on 21 December 2018.

87. The Accredited Entity has provided a legal certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

88. The proposed project will be implemented in the People's Republic of Bangladesh (the "Host Country"), a country in which GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. The Secretariat submitted the first draft of the privileges and immunities agreement and the background note to the Government of Bangladesh on 30 September 2015, which is under review by the Government of Bangladesh.

89. The Heads of the Independent Redress Mechanism (IRM) and Independent Integrity Unit (IIU) have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where the GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by the GCF are made only after the GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities.

90. In addition, as the GCF is not provided with privileges and immunities in the Host Country, the legal due diligence submitted by the Accredited Entity in connection with the Funding Proposal indicates that approval from the Central Bank of Bangladesh will be required in order for the Accredited Entity to transfer funds outside of the Host Country to the GCF. These funds include investment income accrued on GCF proceeds and GCF proceeds which have not been used for the implementation of the project. To the extent that such approvals are not obtained timely or at all, the Accredited Entity may not be able to fully comply with its obligations under the AMA and the funded activity agreement (the "FAA") to transfer funds to the GCF at the times specified in those agreements.

4.7 List of proposed conditions (including legal)

91. In order to mitigate risk, it is recommended that any approval by the Board is made subject to the following conditions:

(a) Signature of the FAA in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and

(b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP206

Proposal name:	Resilient Homestead and Livelihood support to the vulnerable coastal people of Bangladesh (RHL)
Accredited entity:	Palli Karma-Sahayak Foundation (PKSF)
Country/(ies):	Bangladesh
Project/programme size:	Small

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. **Target country: Bangladesh.** The proposed project is located in the country of Bangladesh and targets the seven exposed districts of Bangladesh's coastal zone. These are Satkhira, Khulna, Bagerhat, Barguna, Patuakhali, Bhola and Cox's Bazar. These districts have been selected based on their high exposure and vulnerability to climate change and the feasibility of introducing a crab hatchery supply chain in the regions as part of a climate-resilient livelihoods strategy. The total area of these seven districts is 24,691 square kilometres, and the total population is 12,275,996. The average poverty rate is 26.13 per cent, which is higher than national average of 24.3 per cent.[1] The dependency ratio is also high in these districts at 68.59.[2] Moreover, 52.01 per cent of households in Khulna division are *katcha* (temporary), and 26.86 per cent are *semi-pucca,*[3] which means that more than three quarters of the households are vulnerable to intensive precipitation, cyclones and storm surge.

2. **Type of finance: blended finance.** Funding requested from the GCF is USD 42,201,200 in the form of a non-reimbursable grant for the implementation of the project. The Palli Karma-Sahayak Foundation (PKSF) will provide a total co-financing amount of USD 7,788,000, of which USD 6,600,000 is to be provided in the form of a loan. The total project cost is USD 49.99 million, and the project will be implemented over a period of five years and will continue for 20 years.

3. **Context of vulnerability.** The geographical location and low elevation of the coastal zones of Bangladesh make it susceptible to disasters, with climate change worsening the impacts of such disasters on lives and livelihoods in the region.[4] The Sixth Assessment Report of the Intergovernmental Panel on Climate Change (AR6) argues that global sea level will be elevated by 0.44–0.76 metres by 2100 under the intermediate greenhouse gas (GHG) emissions scenario Shared Socioeconomic Pathways (SSP) 2–4.5. The National Aeronautics and Space Administration (NASA) sea level tool (see section B.1) confirms these projections for the entire coastline, but with an even higher maximum value of 0.91 metres to the west of the country by 2100 under SSP 2–4.5. Even in the low-GHG emission scenario SSP 1–2.6, sea levels

[1] Bangladesh Bureau of Statistics (BBS), 2016: Household Income and Expenditure Survey, 2016, BBS, Ministry of Planning, Government of Bangladesh.
[2] See funding proposal, table 1, for details.
[3] BBS, 2015: Bangladesh National Disaster Statistics 2009-2014, BBS, Ministry of Planning, Government of Bangladesh.
[4] Climate Change Cell (CCC) (2016), "Assessment of Sea Level Rise on Bangladesh Coast through Trend Analysis", CCC, Department of Environment, Ministry of Environment and Forests, Bangladesh.

will rise by 0.32–0.62 metres by the end of the twenty-first century.[5] It is predicted for Bangladesh that a 45 centimetre rise in sea level may inundate 10–15 per cent of the land by the year 2050, resulting in over 35 million climate migrants from the coastal districts.[6] The proposed project areas that include Khulna, Bagerhat, Satkhira, Barguna, Patuakhali, Bhola and Cox's Bazar districts are particularly vulnerable to sea level rise, salinity intrusion, coastal flooding, cyclones and storm surges because of low elevation ranging from 1–2 metres above average sea level.

4. The vulnerability of coastal people is characterized as follows: (i) poor human settlements in low-lying areas; (ii) climate-sensitive livelihoods; and (iii) scarcity of safe drinking water. The majority of the coastal population is poor, small and marginal farm families and shrimp workers. The poor coastal communities build their houses in low-lying areas, which are subject to coastal flooding. Most of the houses are built with mud and *goal pata* (leaves of indigenous coastal plants) and are severely affected by cyclones, storm surges and high tides. These people have to spend a significant amount of their earnings on repairing houses each year. Besides, the coastal communities primarily depend on seasonal subsistence from climate-sensitive agriculture and agriculture wage labour. Drinking water is highly vulnerable to sea level rise and salinity in the coastal zone of the country. The communities and local-level organizations lack understanding on the present and future climate change impacts, which is one of the key barriers to promoting climate-resilient development in the coastal zones.

5. **Theory of Change**. IF target coastal communities in Bangladesh employ weather-proof construction, have access to resources and capacity to implement climate-adapted livelihoods, and, with the support of institutions, are knowledgeable of adaptive strategies/planning and can coordinate responses; THEN coastal communities in Bangladesh will be able to safeguard assets and livelihoods into the future. BECAUSE homes will provide protection during future extreme weather events, livelihoods will be maintained in the face of climate hazards, and local institutions will provide appropriate and timely support.

6. **Funding proposal outcomes, outputs and activities**. The funding proposal has the primary goal of developing climate-adaptive coastal communities in Bangladesh through the adoption of climate-resilient housing and livelihood technologies. The project will also enhance the capacities of communities and organizations in addressing climate change impacts in their localities.

7. The funding proposal has three immediate outcomes. Outcome 1 is decreased risk of loss of assets and life from extreme weather events. Outcome 2 is livelihood resilience to sea level rise, storm surge and salinity. Outcome 3 is improved climate planning and implementation by communities and local-level institutions.

(a) For outcome 1, the output is related to the construction of climate-resilient homesteads (output 1.1). This entails the design and building of homesteads (activity 1.1.1) and homestead tree planting (activity 1.1.2).

(b) For outcome 2, the project interventions are to climate-proof traditional farming practices (output 2.1) and the creation of a community-based farmed crab supply chain (Output 2.2). The activities for output 2.1 cover the construction of slatted houses for goat/sheep rearing (activity 2.1.1), the provision of financial support for goat/sheep rearing (activity 2.1.2), and the introduction of the cultivation of saline-tolerant vegetables within homestead areas (activity 2.1.3). For output 2.2, the activities focus on

[5] Intergovernmental Panel on Climate Change, 2021: Summary for Policymakers. In: Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Masson-Delmotte, V., P. Zhai, A. Pirani, S. L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M. I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J.B.R. Matthews, T. K. Maycock, T. Waterfield, O. Yelekçi, R. Yu and B. Zhou (eds.)]. Cambridge University Press. In Press.

[6] National adaptation programme of action, 2009. Ministry of Environment and Forests, Government of Bangladesh, Dhaka.

the development of crab hatcheries (stage 1) (activity 2.2.1), the provision of financial support for producing crablets (activity 2.2.2), the provision of technical and financial support for a group between hatchery owners and crab farmers, or [7]"crab nursers" (stage 2) (activity 2.2.3), and the provision of technical and financial support to "crab farmers" (3º stage) (activity 2.2.4).

(c) For outcome 3, the interventions focus on the formation and operationalization of the climate change adaptation groups (CCAGs) (output 3.1), building capacities among implementing entities (IEs) and relevant institutions apprised on the project (output 3.2), and the dissemination of knowledge products prepared (output 3.3). The activities for output 3.1 pertain to beneficiary selection and group formation (activity 3.1.1), the preparation of beneficiaries' socioeconomic profile (activity 3.1.2), and arrangement of monthly group meetings on climate change issues for CCAGs (activity 3.1.3). The activities for output 3.2 cover the preparation of training manuals on adaptation technologies and the crab value chain (activity 3.2.1), the preparation of guidelines on project management (activity 3.2.2), the organization of training for beneficiaries and stakeholders (activity 3.2.3), the organization of training for IE staff (activity 3.2.4), the implementation of workshops and seminars (activity 3.2.5), the organization of exchange visits for beneficiaries and IE staff (activity 3.2.6), and the improvement of data for crab research and development (activity 3.2.7). For output 3.3, the activities include the preparation and dissemination of knowledge products (activity 3.3.1) and the real-time evaluation study of project activities (activity 3.3.2).

1.2 Impact potential *Scale: N/A*

8. The project will impact about 362,475 direct beneficiaries (181,238 males and 181,237 females), representing 0.002 per cent of the total population of Bangladesh. Indirect beneficiaries total 770,050 (335,025 males and 335,025 females), and this represents 0.005 per cent of the population. The beneficiaries selected are the most vulnerable and were selected according to the following criteria: (a) those who live in vulnerable, saline-prone coastal areas; (b) women-headed households and other disadvantaged groups; (c) those who belong to poor and ultra-poor households;[8] (d) those who have a daily income of less than USD 1.90; (e) those who do not receive any similar support from other projects or organizations; (f) those who have salinity-affected lands, particularly away from agricultural land; and (g) those who are interested in participating in the project and adopting the project-promoted technologies and practices, and are willing to contribute to the project through loan, cash and in-kind contributions, as necessary.[9]

9. The AE also clarified that the selection criterion "willing to contribute to the project either through loan, cash, or in-kind, as necessary" has been set to ensure the beneficiaries' ownership of the project activities. However, as per the AE, the project will not expect all three contributions but rather encourage any form of contribution. This contribution will be applied only to the grant support. In this case, PKSF mostly encourages in-kind contributions from the beneficiaries. Moreover, the AE also emphasized that "there will be no exclusion of a poor farmer based on in-kind contribution requirement, as the farmer's in-kind contribution will come from the engagement of productive land/other resources (e.g. a goat, a ,etc.), farming time, or the use of the homestead in order to achieve the specific objectives of the project. So, there is virtually no option for rejecting a farmer due to the non-fulfilment of this criterion". Thus, the

[7] See FP206 para 114
[8] Note: as defined in the Household Income and Expenditure Survey (2016) of the BBS (BBS-2017).This document defined 'extreme poor' as a person having purchasing power parity (PPP) below USD 1.25 a day. A PPP below USD 1.90 a day is called 'poor'.
[9] See FP206 annex 14 'Term Sheet'. Annex 2 –Initial Eligibility Criteria for Selection of Beneficiaries

"project will not exclude anyone". As a part of the selection process, there will be a number of assessments during the inception stage. A socioeconomic assessment and vulnerability assessment will be done. Based on the indicative set of criteria, a long list of potential beneficiaries will be prepared. An additional set of criteria could be added and will be developed together with participating communities in case the list of people meeting the initial set of criteria is very long. This is highly likely to happen, as the actual needs in the target areas exceed the project's financial scope to support all eligible communities. While not all community members will receive housing and investment/grant support, the majority of the community members are expected to benefit indirectly and/or through learning about promoted technologies and practices (housing design, crab nurseries/hatcheries, and other topics).[10]

10. **Reduced losses and damages from cyclones**. In the past 15 years in Bangladesh, loss and damage has been mainly due to the increasing intensity and frequency of tropical cyclones in the target project areas (i.e. Cox's Bazar (29.5% of tropical cyclones); Sundarban coast (Satkhira, Khulna and Bagerhat) (27.9% of tropical cyclones); Meghna Estuary, east central coast (Eastern Bhola) (26.2% of tropical cyclones); and Central coast (Barguna, Patuakhali, Pirozpur, Barisal, Bhola) (16.4% of tropical cyclones). As a natural consequence, coastal flooding, storm surges, and high tides frequently damaged more than two thirds of the houses of coastal communities made of mud and *goal pata*, thus significantly adversely affecting annual resources (earning, savings, time) for house repairs and rebuilding due to climate-related damage. Outcome 1 will reduce the risk of loss of life and assets from these extreme weather events via the construction of climate-resilient homesteads (output 1.1).

11. **Enhanced livelihood resilience of coastal communities**. Similarly, tropical cyclones, along with coastal flooding, storm surges, high tides and sea level rise, continue to degrade agro-ecological landscapes and resources, in part due to saltwater intrusion, and lead to environmental damage. According to the Soil Resources Development Institute, the total amount of salt-affected land in Bangladesh was approximately 83.3 million hectares (ha) in 1973, which increased to around 102 million ha in 2000, and to an estimated 105.6 million ha in 2009, and is still continuing to increase.[11] The net cropped area in coastal Bangladesh has been decreasing over the last few years due to several factors, and studies have identified salinity as the main cause for yield reduction in coastal agriculture.[12] Outcome 2 will enhance the resilience of livelihoods in project areas in the face of sea level rise and increased storm surges and salinity via the climate-proofing of traditional farming practices and the creation of a community-based farmed crab supply.

12. **Enhanced access to finance.** The project will combine a mixture of loans and grants that will provide beneficiaries access to otherwise inaccessible resources and sources of finance (barrier 2), thus addressing a major barrier to the development and security of coastal rural communities. GCF will provide the grants, and PKSF will provide co-financing in the form of loans and in-kind support. The loan will be used to implement activities under outcome 2 (activities 2.1.2, 2.2.2, 2.2.3 (partial), and 2.2.4 (partial)), whereas the in-kind support will be used to partially cover the project management unit (salary of the project coordinator) and office rent.

13. Overall, iTAP's assessment for the adaptation impact potential is medium to high.

1.3 Paradigm shift potential *Scale: N/A*

[10] AE responses to iTAP questions from May 2023.
[11] Chen, J., and V. Mueller. 2018. Nature Climate Change 8, 981–985.
[12] Baten, M. A., L. Seal, K. S. Lisa. 2015. American Journal of Climate Change 4, 248.

14. Access to resources enables the climate change-vulnerable coastal farmers to transition towards a more resilient community of farmers, coastal ecosystems, livelihoods, and built infrastructure/shelters.

15. The construction of climate-resilient houses (i.e. raised plinths and mangrove tree plantations as protection) will limit the asset losses and damages in lower-category storm surges that occur more frequently (i.e. category 1, 2 and 3). This will enable coastal farmers to save, plan and invest more in livelihood opportunities that will generate more income streams. The increased protection from the impact of extreme events will finally give the farmers a chance to extricate themselves from the vicious cycle of climate loss and damage.

16. The introduction of new adaptive farming techniques (i.e. salt-tolerant vegetables, crab farming, sheep/goat rearing) for the coastal farmers will enable them to transition towards climate-smart agriculture practices. Given that the old traditional farming methods are already becoming challenging in the face of a changing climate, the new techniques should motivate the farmers and entrepreneurs alike to save, plan and reinvest more, thus enabling these crab entrepreneurs and farmers to move up the business value chain.

17. **Access to finance enables the paradigm shift.** The derisking interventions of the coastal farmers and farmer activities will enable coastal farmers to access finance in support of their coastal livelihood activities.

18. The CCAGs also enable the paradigm shift among the coastal communities. The planned organization of communities into CCAGs will further increase the resilience of communities via the sharing of climate-relevant information affecting their communities' livelihoods. The CCAG will essentially act as a climate advisory body that will enable the mainstreaming of climate risk management into the daily lives of the coastal communities. As per the funding proposal, "The project will form 3,200 groups having 25 (+/-) persons in each group for delivery support. A total of 80,000 members will be selected from 362,475 selected beneficiaries to form these CCAGs. The members will be selected based on intensive consultations with selected communities on the basis of pre-defined criteria that include (a) potential to play role in decision-making at household level; (b) less involvement in income-generating activity; (c) having an understanding of the local socioeconomic context; and (d) willingness to participate in CCAG meetings. Each group will meet at least once a month and discuss climate change and its impacts on their lives and livelihoods. They will adopt and practice climate-adaptive technologies that will be promoted by the project. The objective of forming this group is to deliver the support services in groups in order to minimize the delivery cost as well as ensure the participation and collective decisions of the affected community in implementing the proposed interventions. It will help transfer knowledge on climate change issues among the society because the groups will discuss climate change at regular intervals (typically monthly). Thus, they will be able to internalize climate change impacts in their lives and livelihoods."[13]

19. The project has high potential for paradigm shift. The combined interventions, from resilient housing and livelihoods to increased capacities to manage climate risks, will greatly enable the coastal farmers to extricate themselves from the vicious cycle of loss and damage due to climate change.

1.4 Sustainable development potential *Scale: N/A*

20. The funding proposal cuts across several Sustanable Development Goals (SDGs). Outcome 1 supports SDGs 6 (Clean water and sanitaiton), 10 (Reduced inequalities within and among countries), 11 (Sustainable cities and communities), 13 (Climate action), 14 (Life below water), and 15 (Life on land). Outcome 2 supports SDGs 1 (No poverty), 2 (Zero hunger), 3

[13] See funding proposal, para. 129.

(Good health and well-being), 8 (Decent work and economic growth), 9 (Industry, innovation and infrastructure), 10 (Reduced inequalities within and among countries), 11 (Sustainable cities and communities), 12 (Responsible consumption and production), 13 (Climate action), 14 (Life below water), and 15 (Life on land). Outcome 3 supports the same SDGs as outcomes 1 and 2 plus the addition of SDG 4 (Quality education), 5 (Gender equality), 16 (Peace, justice and strong institutions), and 17 (Partnerships for the goals).

21. **Socioeconomic and gender co-benefits.** Project interventions to support crab hatching, growing and fattening will create new business opportunities for poor members of communities that will allow them to increase and diversify household income, while also improving nutritional status. Crab fattening and improved sheep and goat-rearing livelihood opportunities will be available to women and will improve incomes for women-headed households. The project promises to stimulate growth in local secondary economic activities in support of the new livelihoods, including transport, construction, maintenance and equipment provision. More resilient housing will mean that women and girls are less likely to need to retreat to community shelters where their safety can be at risk.

22. While these interventions in the coastal areas enhance the resilience of communities and livelihoods against the impacts of extreme climate events(i.e. categories 1, 2 and 3), there is still the ever-present risk of major extreme events (i.e. categories 4 and 5) and sea level rise, which may happen and erode all of these developmental gains in future.

23. Based on the above discussions, iTAP rates the sustainable development impact potential as medium to high.

1.5 Needs of the recipient *Scale: N/A*

24. The target beneficiaries live in the coastal areas of Bangladesh (i.e. Khulna, Bagerhat, Satkhira, Barguna, Patuakhali, Bhola and Cox's Bazar). The housing of coastal areas is mainly affected by cyclones, storm surges and coastal flooding. Nation-wide, 70.31 per cent of households in disaster-prone areas have *katcha* houses(constructed with mud), and 17.44 per cent have *semi-pucca* houses. These houses are vulnerable to heavy rainfall, flooding and storm surge as they are made with mud and straw or *golpata* (Nypa spp., a traditional leaf found in Sundarban). This means that around 88 per cent of households are vulnerable to climate change and associated disasters in the country. The recent cyclone in May 2020 completely destroyed 55,667 houses and partially destroyed around 162,000 houses in the southwest coastal zone of Bangladesh. Cyclone Mahasen in May 2013 damaged 150,000 houses; Cyclone Aila in May 2009 damaged 500,000 houses, and Cyclone Sidr in November 2007 damaged 600,000 houses in the coastal areas of Bangladesh. These damages are expected to increase in future with the increasing intensity of cyclones and storm surges.

25. In terms of climate impacts on livelihoods among the target beneficiaries, the coastal areas, polders, homestead gardens and small livestock cages are exposed to increased soil and water salinity, frequent coastal flooding, sea-level rise, and tropical storms (categories 1, 2 and 3). This leads to land salinization, crop damage, shifts in brackish and fresh water aquaculture (impacting crab production), loss of freshwater, and loss of livestock.

26. The proposed funding proposal interventions directly address these needs of the beneficiaries. Among the barriers that the funding proposal will address are (i) lack of access to finance to invest in large capital expenses; (ii) lack of livelihood diversification; (iii)limited capacity to adopt adaptation technologies, capacity, standards, (iv) constrained development on supply and market access for a community-based crab business value chain; and (v) lack of climate planning and implementation capacity among communities to address climate change.

27. The approach of the AE to (i) carry out an assessment of "best practices" in individual (farming-related) adaptation modalities based on available evaluation reports/records and

published literature; and (ii) triangulate the results by conducting public consultation in areas where such modalities have been tried ensures a clear understanding of the stakeholders' needs in the project areas.[14]

28. Old traditional farming practices are increasingly becoming very challenging due to limitations posed by increasing salinity and risks associated with sea level rise and other factors; there is thus a need for alternative means of climate-smart farming methods. When farmers are adequately informed about the new alternatives, they are responsive, especially those involved with Sub-Assistant Agriculture Officers employed by the government for extension and who have strong communication with the communities. Moreover, the non-governmental organization (NGO) workers have easy access to farmers, who act as change makers to popularize alternative farming options. The project will consider these avenues to popularize alternative farming practices for adaptation.

29. The needs of the recipients are assessed as high.

1.6 Country ownership *Scale: N/A*

30. **Full alignment with country devleopment plans and climate initiatives.** The funding proposal is aligned with the Bangladesh's Climate Change Strategy and Action Plan, national adaptation programme of action (NAPA), national communications, 8th Five Year Plan, and Perspective Plan. The proposed project is also aligned with various aspects of the concept notes prepared under the NAPA.

31. **Strong local partne**r. PKSF is the executing entity (EE) and accredited entity (AE)[15] for the project and has a vision of "a Bangladesh where poverty has been eradicated; ruling development and governance paradigm in inclusive, people-centred, equitable and sustainable; and all citizens live a healthy, appropriate, educated and empowered and humanly dignified life"[16]. PKSF is a not-for-profit organization established by the Government of Bangladesh in 1990 and registered under the Companies Act 1913/1994. It has a wide range of development services, including financial, health, educational, capacity development, technology transfer and business development services, for disadvantaged segments of the society through appropriate pro-poor institutions. PKSF has a demonstrated track record in Bangladesh; it has established an Environment and Climate Change Unit, implemented the Community Climate Change Project; facilitated the inflows of climate change funds in Bangladesh as a least developed country; developed nine core programmes (e.g. inclusive financial services, a people-centred holistic development programme, an enterprise development programme, a social protection programme, capacity-building programmes, advocacy and knowledge management, research and development, and environment and climate change programmes). PKSF also has strong experience in crab hatchery enterprises, especially in the management and production of crabs, having established three hatcheries in the Cox's Bazar and Satkhira districts, thus facilitating PKSF's understanding of the crab hatchery technology needed in terms of water management (maintaining water salinity, pH, temperature, etc.), feeding, and disease identification, etc.

32. **Extensive local stakeholder engagement.** Local coastal stakeholders (e.g. local governments, NGOs and communities) have been closely consulted to identify local issues, challenges, risks and barriers and formulate the funding proposal. The funding proposal has undergone thorough reviews by key organizations (i.e. Department of Environment, Department of Fisheries, Department of Agriculture Extension, United Nations Development Programme,International Center for Climate Change and Development , Action Aid and the World Food Programme).

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[14] AE responses to iTAP questions in May 2023.
[15] See FP para 161
[16] See FP para 229

33. When asked if the AE has already secured partnership agreements with the target beneficiaries in relation to their participation in the activities of the funding proposal, the AE replied that PKSF has not yet signed a memorandum of understanding(MOU)/memorandum of agreement (MOA) with the target project beneficiaries, as the beneficiaries have not been selected yet. The MOU/MOA will be confirmed after the final selection of the beneficiaries. The conditions of the MOU will be incorporated in the activity implementation guidelines to be prepared at the inception phase of the project in consultation with the beneficiaries, IEs and PKSF. Once the beneficiaries are selected, the roles and responsibilities of communities will also be detailed out in the activity implementation guidelines. Upon request by iTAP, the AE has provided sample agreements between PKSF and IEs, and between IEs and beneficiaries, the latter of which cover ongoing maintenance and in-kind or financial contributions which are necessary to ensure the long-term sustainability of project interventions.

34. The country ownership is assessed as high.

1.7 Efficiency and effectiveness *Scale: N/A*

35. **Strong complementarity of funding proposal with GCF projects.** The funding proposal optimizes the benefits of FP069, which provides potable drinking water for women who have been suffering from saline drinking water and domestic drudgery. This has mutual benefits for the funding proposal programme on resilient housing and livelihoods (RHL). As per the AE, the RHL project also proposes a community approach through organizing communities into CCAGs. The RHL project will apply lessons learned from the FP069 into implementing the rainwater harvesting system of the proposed resilient houses.

36. **Integrating lessons learned and best practices from other GCF projects.** As per the AE, the funding proposal initiatives on goat and sheep-rearing and resilient homesteads are also being implemented by the Extended Community Climate Change Project-Flood (ECCCP-Flood) project; the RHL funding proposal is focused on southern coastal systems while the ECCCP-Flood funding proposal is focused on the semi-arid north-western part of Bangladesh. Given that the implementation of slatted houses will be similar under both projects, potential cross-learning opportunities from ECCCP-Flood can be adopted for the RHL funding proposal, thus ensuring the integration of best practices in each project.

37. **Project concessionality.** The project's concessionality is rationalized through: (i) the project's provision of goods that are largely public in nature (e.g. interventions in mangroves and climate-adaptive homesteads); (ii) the delivery of interventions to communities that might initially be private in nature (e.g. funding of hatcheries for crabs/crab fattening or of improved production of saline-resistant vegetables), but over the longer term, due to biodiversity co-benefits, might become public in nature.

38. **Rational use of GCF funds.** As presented by the AE, an analysis of the planned investments, under three scenarios, was conducted separately for all five activities (i.e. goat rearing, traditional vegetable farming, crab hatchery, crab nursing and crab fattening). One scenario used representative concentration pathway (RCP) 4.5 (medium-case scenario), another used RCP 8.5 (worst case scenario) (both of these scenarios were assessed with and without adaptation measures), and these were compared with the business-as-usual (BAU) case. All the components under benefits and costs have also been modelled for all scenarios.[17]

39. **Impact on financial net present value (NPV), internal rate of return (IRR) and payback period.** The overall results show that, with the proposed grant support from GCF (both proportionate GCF grant and proportionate PKSF loan), the respective NPVs are positive for the four individual activities (i.e. goat rearing, crab hatchery, crab nursing and crab

[17] See revised funding proposal annex 3A and 3B as per the request by iTAP during the meeting of the AE and iTAP.

fattening) under the RCP 4.5 scenario. However, under the RCP 8.5 scenario, there are both positive and negative NPVs; in the case of traditional vegetable farming, the NPV is only positive under all three scenarios (i.e. BAU, RCP 4.5 and RCP 8.5) if there is a 100 per cent grant support from GCF. With respect to the IRR, the results show that these agriculture activities generally have low IRRs under the RCP 8.5 scenario, be it with or without adaptation measures due to the intensity/scale of cyclones of this category. As with the payback period, it generally increases under both RCP 4.5 and 8.5 scenarios when compared to the BAU situation; under the RCP 8.5 scenario, the payback period is much longer than the project life for all activities.[18]

40. Impact **on economic benefits and cost present values, economic net present value (ENPV), economic internal rate of return (EIRR), and benefit-to-cost ratio.** An economic analysis was also conductedon the proposed interventions that are expected to provide income increases for the intended beneficiaries. The analysis was conducted separately for all five activities (i.e. goat rearing, traditional vegetable farming, crab hatchery, crab nursing and crab fattening) under two different scenarios: one with RCP 4.5 (medium-case scenario) and another with RCP 8.5 (worst-case scenario) (both of these scenarios were assessed with and without adaptation measures), and these were compared with the BAU case. The ENPVs are all positive, and the EIRR is highest under the GCF case of 100 per cent grant coverage. With the exception of traditional vegetable farming (for the BAU case and the 50:50 GCF/PKSF funding case), the benefit-to-cost ratios are all greater than 1.00, and the highest ratio under the GCF case of 100 per cent grant funding in all the proposed funding scenarios under this proposal. Crab hatchery, crab nursery and crab fattening have been considered as a chain activity in order to conduct economic analysis.

41. Through these analyses, the AE is able to demonstrate the rational necessity of GCF funding for increasing the resilience of these agriculture activities under the various vulnerable climatic situations.

42. **Resilient housing and livelihoods versus category 1, 2 and 3 storms.** The design of resilient homestead and hatchery activities has been planned to withstand a category 3 cyclone and should functionally work as storm shelters. The raised plinth housing design with mangrove protection is intended to prevent loss and damage from the frequent lower category storms (1, 2 and 3); to a certain extent, as per the AE, some of the structures might withstand a category 4 or 5 cyclone.

43. The AE is mindful that there is that an unfortunate occurrence of a 'once in a 25-to-30-year' category-5 cyclonic event might force the tidal surge above the adjusted height of the plinth. In such a case; the eventual loss cannot be completely avoided because each adaptation practice or intervention has its limits in the face of extreme events. When the extreme situation crosses these limits, it causes loss and damage. This is mitigated by the existing redundancy measures already put in place by the government and local communities and was very well demonstrated by the most recent category 5 storm that hit Bangladesh. The AE refers to the previous cyclones in 1991 and 2009, and notes that the most recent cyclone (Mocha) on 14 May 2023 did not bring severe damage to lives and property due to the combination of early warning systems (EWS), cyclone shelters, the timing of the cyclone (during low tide) and the location in the Sundarbans. The strong government response prior to the storm enabled the evacuation of 750,000 people and the housing of these people in cyclone shelters.[19] As highlighted by the AE, "Bangladesh has a very strong EWS in place, along with a unique capacity to mobilize about 76,000 trained human resources to reach almost everybody with an effective early warning with a five-day lead time. The strong legal measures through the implementation of Standing Orders on Disasters present clear roles and responsibilities for all stakeholders, from local-level service providers all the way up to the Prime Minister's office, in the event of the issuance of an

[18] See footnote 14.
[19] BBC. 2023. Cyclone Mocha: Deadly storm hits Myanmar and Bangladesh coasts.

EWS. The Cyclone Preparedness Programme is a globally awarded programme, fully functional and fully integrated with local-level development programming in Bangladesh."

44. **Cost efficiency**. The total budget of the RHL project is proposed at USD 49.99 million, with GCF grant financing of USD 42.20 million (85 per cent of the total funding) and USD 7.77 million (around 16 per cent) in co-financing. The project is expected to deliver direct benefits to an estimated 360,000 direct beneficiaries and 200,000 indirect beneficiaries.

45. The efficiency and effectiveness of the project is assessed as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

46. The urgency of resilience measures cannot be overstated. Bangladeshi citizens living in "kutcha" mud embankments are always at risk of the impacts of fast and slow-onset climate extreme events, thus the ever pressing need for effective adaptation in the form of resilient housing and livelihoods as they are always the first casualties of the climate change-intensified extreme events in the coastal zones.

47. While building resilient houses and livelihoods along the coastal zones where poor citizens have lived for generations is indeed an opportunity to increase the resilience of these coastal farmers, there is also the risk of a one-in-25-to-30-year category 5 cyclonic event, an ever-present risk that could immediately wipe out the lives, properties and development gains in the project area. Even with the strong government EWS and pre-storm evacuation experience and response, as demonstrated in the very recent category 5 Cyclone Mocha storm in May 2023, iTAP still recommends the following as redundancy measures:

(a) Conduct awareness-raising activities informing the project beneficiaries of the limitations of the project interventions versus the impacts of fast-onset (category 4 and 5) and slow-onset (e.g. sea level rise) climate events;

(b) Conduct risk mapping as the basis for building resilient housing and livelihoods;

(c) Conduct regular disaster risk preparation and evacuation drills targeting the funding proposal project beneficiaries; essentially, the provision of regular climate community drills on category 4/5 storm evacuation and timely and effective dissemination of category 4/5 storms (T-5); warnings as early as 5 days before impact.

(d) Use redundant/back-up communication systems in the communities, climate-proof wireless technology infrastructure against wind and intense precipitation, and integrate climate services information and disseminate it via other wireless technology infrastructure.

48. The funding proposal is endorsed by iTAP.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP206)

Proposal name:	Resilient Homestead and Livelihood support to the vulnerable coastal people of Bangladesh (RHL)
Accredited entity:	Palli Karma-Sahayak Foundation (PKSF)
Country/(ies):	Bangladesh
Project/programme size:	Small

Impact potential

The AE appreciates iTAP Assessment, especially for duly acknowledging particular strengths of the RHL Project against the urgency of vulnerable communities along the coastal zone of Bangladesh. The AE also extends its deep appreciation for highlighting a few relatively weaker points, with recommendations to improve upon during the course of the implementation of the project subject to the Board’s affirmative decision.

In addition to reduction of L&D and increased resilience of selected communities, the AE affirms that the project will enhance capacity of 3 public universities and 1 research institutes by supporting their laboratory-based research and development of crab.

Paradigm shift potential

Additional to iTAP assessment, the project has plan to engage public universities and research institutions in crab research and development for future capacity building of the institutions and communities in crab aquaculture section. The project intends to sign MoU with several (public financed) universities located in Chattogram, Khulna, Patuakhali and also the Bangladesh Fisheries Research Institute to improve their laboratories for conducting crab research focusing identification of crab diseases, breeding, production efficiency etc. This will extend the support services to the crab farmers in a sustainable manure, while contributing to the management of phytosanitary standards of the exportable item.

Sustainable development potential

In terms of sustainable development potential, the environmental co-benefits of the project would be significant. The hatchery-based crab farming will reduce the number of people currently going to Sundarban for crab catching/crablet collection. By supplying crablets from outside the natural system, free extraction of naturally available crab resources will be reduced significantly. It will, in turn, increase mud-crab stock in the nature, which is currently in decreasing trend. In addition, integrated fruit-fish-fibre practice will enhance the quality of the coastal ecosystem and biodiversity. Thus the current rating on this investment criteria appears conservative.

Needs of the recipient

No response

Country ownership
No response
Efficiency and effectiveness
No response
Overall remarks from the independent Technical Advisory Panel: 47(a) The project will form 3,200 Climate Change Adaptation Groups (CCAGs). These groups will arrange monthly meetings to discuss various climate change related issues including slow and rapid onset, gradual sea level rise, salinity intrusion, disaster preparedness, emergency response, early warning, rehabilitation, livelihood recovery etc. This activity is already proposed in the project (activity 3.1.3: Arrange monthly group meetings on climate change issues for CCAGs). This activity will help the CCAG members to frequently discuss the issues that will ultimately increase their level of understanding and awareness on climate change and associated disasters. 47(b) PKSF confirms that the project will carry out the risks mapping at the inception phase of the project for building resilient housing and livelihoods. 47 (c) The project will arrange annual mock drills with the targeted beneficiaries on category 4/5 storm evacuation and timely and effective dissemination of early warning information as early as possible (typically having 5-days lead time). 47(d) The project will consult with the Bangladesh Meteorology Department (BMD) and local level disaster management committees for identifying and establishment of wireless infrastructure that would be resilient to cyclonic storm and intense precipitation to integrate climate services information and dissemination system.

Gender Assessment and Action Plan

Resilient Homestead and Livelihood Support to the Vulnerable Coastal People of Bangladesh (RHL)

February, 2018
Updated: 2023

Contents

Abbreviations & Acronyms

BCAS	Bangladesh Centre for Advance Studies
BCCSAP	Bangladesh Climate Change Strategy and Action Plan
CCAG	Climate Change Adaptation Group
CCCP	Community Climate Change Project
ccGAP	Climate Change and Gender Action Plan
CEDAW	Convention on the Elimination of all Forms of Discrimination against Women
EDA	Enhanced Direct Access
ESMF	Environmental and Social Management Framework
GBV	Gender Based Violence
GCF	Green Climate Fund
GoB	Government of Bangladesh
GRM	Grievance Redress Mechanism
HH	Household
IPCC	Intergovernmental Panel on Climate Change
MCH	Maternal Care Health
MDG	Millennium Development Goal
M&E	Monitoring and Evaluation
MoEF	Ministry of Environment and Forest
NGO	Non-Government Organization
PACE	Promoting Agricultural Commercialization and Enterprises
PIP	Project Implementing Partner
PKSF	Palli Karma-Sahayak Foundation
RHL	Resilient Homestead and Livelihood Support to the vulnerable coastal people of Bangladesh
SEAH	Sexual Exploitation, Abuse and Harassment
UDMC	Union Disaster Management Committee
UzDMC	Upazilla Disaster Management Committee
UN	United Nations
UNDP	United Nations Development Programme
UNFCCC	United Nations Framework Convention on Climate Change
VGD	Vulnerable Group Development
VGF	Vulnerable Group Feeding
WDP	Women's Development Policy
WHO	World Health Organisation

1. Introduction

The Green Climate Fund (GCF) recognizes the importance of gender considerations in terms of both impact of and access to climate funding, and requires a Gender Assessment and Gender Action Plan to be submitted as part of the project-funding proposals that it assesses. The main objective of the Gender Assessment is to screen the gender aspects of the projects to be financed by GCF, and to subsequently strengthen the gender responsive actions within the project. The current project aims to reduce this vulnerability by addressing their adaptive capacity from multiple levels. The gender assessment report should be considered as an integral part of the project proposal including other annexure like the Feasibility Study, Stakeholder Engagement and Environmental and Social Management Framework (ESMF) etc.

2. An overview of the proposed project

Considering the vulnerability to climate change of the coastal people of Bangladesh, PKSF has decided to submit the project proposal to GCF on "Resilient homestead and livelihood support to the vulnerable coastal people of Bangladesh (RHL)". A brief description of the proposed project is presented below:

Goal: The primary goal of the project is to develop climate-adaptive coastal communities in Bangladesh through the adoption of climate-resilient housing and livelihood technologies. The project will also enhance the capacity of communities and organisations to address climate change impacts in their localities.

The objectives of the project are:

1. To develop climate-resilient homesteads for vulnerable communities in the southwest coastal zone of Bangladesh;
2. To develop climate-adaptive livelihoods for vulnerable coastal communities; and
3. To enhance the capacity of vulnerable communities and institutions so that they are able to plan and implement adaptation interventions.

Major activities

The project will directly involve 362,475 people. 770,050 more people will be indirectly benefitted from the project. Total budget is estimated USD 49,989,200 and project period is 5 years. Major activities are design and building of homesteads, homestead tree planting, construction of slatted houses for goat/sheep rearing, provide financial support for goat/sheep rearing, introduce the cultivation of saline tolerant vegetables within homestead areas, development of crab hatcheries, financial support for producing crablets, technical and financial support for "crab nursers" and provide technical and financial support to crab farmers. The project will also provide training to the local institutions and communities on climate change issues and adaptation technologies and transfer knowledge through practicing and documentation.

Monitoring and evaluation will be an integral part of the proposed project. It is expected that the project will improve wellbeing of rural coastal community, create employment opportunities, improve value chain system, improve ecosystem services and influence policies of the government.

Components of the proposed project

The activities to achieve the above goal and objectives are divided into three components such as 1) Decreased risk of loss of assets and lives from extreme weather events, 2) Increased livelihood resilience to SLR/storm surges and salinity and 3) Improved climate planning and implementation by communities and local level institutions

Component 1: Decreased risk of loss of assets and lives from extreme weather events

Studies found that more than three-fourth of the households in coastal areas are vulnerable to intensive precipitation, cyclone and storm surge and coastal flooding due to perishable materials. For sustaining the livelihood, the proposed project will provide support to construct climate resilient housing. The concept of climate resilient housing under the project includes raising homesteads plinth above flood or tidal surge level, constructing and/or reconstructing houses with concrete pillars that are resilient to climate change and associated shocks (i.e., cyclone, storm surge, tidal surge, coastal flooding etc.), construction of climate resilient sanitary latrines, rainwater harvesting system, homestead gardening system, and tree plantation around the homestead area. Resilient housing is very important for building resilience of the affected community; because they have to spend much of their income in repairing their houses each year during post-monsoon period.

Component 2: Increased livelihood resilience to SLR/storm surges and salinity

A large portion of coastal population is highly exposed to climate change impacts due to increased sea level rise, salinity in water and soil, intensity of cyclones and coastal flooding. These pose a significant threat to agriculture, brackish aquaculture and open water fishing. A recent study by UNDP shows that 16 and 35 percent of living in Khulna and Satkhira respectively are extremely poor whereas the national average is 12.9 percent. Gender inequality is prevailed in these districts through various societal and cultural norms that shape women’s day-to-day activities as well as their capacity to adapt to climate change. For example, women have less decision-making power within the household and the workplace, and are expected to manage the household and care for the family. Compounding these factors, climate change aggravates the burden of unpaid care work, creating a cycle which undermines their climate change resilient livelihood. The proposed project will implement the goat/sheep rearing in slatted houses, fruit-forest-fisheries model including crab hatchery and farming, homestead vegetable cultivation livelihood options and fruit trees and mangrove plantation. The elements that are crucial to success of the proposed interventions have been identified as: a) capacity building of participants particularly women, b) adequate and suitable access to resources for the participants and value chain actors, c) collaboration between government and local government institutions, d) private sector engagement and improved climate change adaptation knowledge, attitudes and practices. The project will provide technological support and capacity training to the selected beneficiaries in promoting saline resilient technologies and practices particularly in agriculture sector.

Component 3: Improved climate planning and implementation by communities and local level institutions

Addressing climate change impacts at the community level requires specialized institutions. Local government institutions in Bangladesh mainly deal with regular development activities. Besides, there are experienced NGOs who have strong and long-term relationship with local communities due to credit programmes. These organizations would play crucial role in promoting climate change adaptation activities at community level. The proposed project will select at least 15 NGOs as Implementing Entities (IEs) in the proposed working areas and enhanced their capacity through training and practicing adaptation activities. This will significantly contribute to achieve the objectives of the project. The local government departments and institutions will play role in decision making process at the community level through participating meetings and workshops during implementation. The Union Parishad (UP) chairman will be the focal person of local GRM process.

PKSF always works with poor and vulnerable people in group-based approach. For climate change adaptation projects, these groups are termed as “Climate Change Adaptation Groups (CCAGs).” One representative from each selected HHs will be the members of the group. About twenty five (+/-) participants together will form a group. The objective of forming this group is to deliver the support services in groups in order to minimize the delivery cost as well as to ensure participation and collective decisions of the affected community in implementing the proposed interventions. It will help transfer of knowledge on climate change issues among the society because they will discuss about climate change in a regular periodic interval typically fortnightly or monthly in groups. Thus, they will be able to internalize climate change impacts on their lives and livelihoods. The groups will receive training on climate change issues and how to deal with these problems. They will be able to identify climate change problems on their lives and livelihoods and prepare plan accordingly to reduce the impacts of climate change. They will also look after community infrastructures beyond the project period. Besides, the group approach reduces the management cost of the project.

3. Gender Assessment in the National Context

Although Bangladesh has made significant progress in poverty, human development and gender equality indicators over the last few decades, poverty and inequality remains prevalent, and the social status of Bangladeshi women still need to be improved, especially in rural areas. Central to the issue of gendered inequality, is that Bangladeshi women suffer under a particularly high burden of low-paid work, responsible for a range of essential household functions such as collecting water, providing childcare, and producing half of the food at the household level, yet making up only a quarter of the industrial workforce.

3.1 Social Aspects

The mobility of women in Bangladesh varies depending on social status, religious affiliation and whether they live in urban or rural areas. Socio-cultural norms not only shape perceptions of the value of women, but also restrict a large proportion of women to unpaid domestic responsibilities, further reducing their productive value in the eyes of Bangladeshi society. In 2011, only 54.5% of girls were enrolled in secondary school, while 42% of women aged 15-19 were unable to attend a health center alone. The recent local study indicates that only 12% women travel outside of their village alone, and that when they travel other family members such as children (52%) and other female members (18%) usually accompany them, which has important implications in terms of women's access to markets. Although these social dynamics are in flux, and there have been important shifts due to economic conditions and opportunities, traditional beliefs regarding the role of women in the household and public spheres remain deeply conservative.

Looking after children & old and cooking for all members of the family are seen as the central roles of a woman throughout Bangladesh, particularly in rural areas, and the nature of work a woman performs is principally conducted within the premises of the household. This type of labor remains socially invisible and has little exchange value or impact on woman's decision-making power, reinforcing women's undervalued role in Bangladeshi society. The tradition of dowry still prevails, violence against women and child marriage is decreasing in the area because people are becoming more aware. Promisingly, a recent study carried out in the target districts, indicated a changing awareness in regards to the challenges faced by women, with women reporting that if they are financially empowered, they can do anything.

3.2. Gendered norms and vulnerabilities

Field visit report showed women's work continued from early morning to late at night. Crab harvest time is the busiest period. Women laborers can neither give up their paid labor work nor reduce their unpaid household work as both are important for family wellbeing. Unless women ask for help, men never join in with the household chores as these are taken for granted as women's work. Thus women fall under extreme time pressure which leads to both mental and physical stress and threatens their wellbeing. If women are away for work or visiting a relative's house, men find it difficult to provide food and other care for the family and women must request neighbors to support their men and children. If women fail to organize support, they can suffer mental stress and may have to cancel their time away.

Substantial numbers of women in the poorest groups in the project area engaged in income generating activities, mostly as agricultural laborers, but women from middle income and wealthy families were not encouraged to work outside as it is seen to be a matter of social prestige. Women's decision making and choices are strongly limited by the patriarchal social norms. Even the women who work cannot choose the type of work they do freely; most sectors like fishing, tailoring in an open market, pulling vans and rickshaws, and business are male dominated. Moreover, even those women who are involved in farming and agricultural production are usually prohibited from participating in market-based activities, which is another male dominated sector. Men carryout market-based buying and selling and handle cash and thus have more assets.

When women work as agri-laborers they are paid less than men. Wage discrimination is common in all the project areas, although the actual amount paid varies depending on the working conditions, workload, and approach of the employer. The limitation of women's freedom of choice by patriarchal norms and practices reduces their bargaining power and ability to demand equal wages. Working women suffered both as a result of the natural disasters affecting agriculture, the main occupation open to them, and from male domination in the labor market, both of which result in disempowerment.

3.3. Access and control over resources and opportunities

Vulnerability is multi-dimensional and derives from inequalities and discriminatory practices related to resource distribution and patterns of access to and control over resources that are shaped by a history of social dimensions and marginalization. In the project areas, women are the poorest of the poor and more vulnerable to the impacts of climate than men because of normative gender differentiated access and control over resources (ownership of property and land) and opportunities (education, employment, health services).

3.3.1. Land ownership

Land is usually owned by men, which gives them more power as well as socio-cultural, economic, and political status. In many of the poor and marginalized groups in the project area, women do not own land or other property, have limited access to other resources such as cash, livestock, and poultry, and may have access to but have no control over agricultural land or homestead areas for vegetable gardening. Women's access to and ownership of resources are shaped by the patriarchal norms. Lack of access and ownership disempowers them economically and makes them socially in secure as resources offer the main form of financial security in times of crisis and are powerful assets. As Hertel et al. (2010) stated "People who do not have their own land for their own living are more vulnerable to climate change impacts". The field study found that women are discriminated against and deprived of ownership of family property and access to resources because of poverty and patriarchal norms and practices.

3.3.2. Education

Education is an important component in enabling an individual to acquire skills and become empowered and develop the capability to adapt to extreme situations. Though girls' access to education has significantly improved, they still lag behind the boys. The most recent Population and Housing Census (BBS, 2022) showed that in rural areas, male literacy rate is 73.29% and female literacy rate is 69.93%. Following the last census in 2011, the education rate for girls have increased as a result of the recently implemented government program with free education for girls up to grade 10, stipends for female students, and free distribution of national curriculum books from class 5–12 among all students, which was introduced to meet the Constitutional mandate of free and compulsory education (http://bdlaws.minlaw.gov.bd/print_sections_all.php?id=367). Nevertheless, the practice of dowry and early marriage, and hard-core poverty still act as barriers to girls' education. Disasters also increase the difficulties for girls to access education, both physically – poor families are the most climate affected – and economically – loss of income, livestock, and dwellings can encourage poor families to marry off their daughters before they complete school as a way to reduce food intake in the family and ensure food security, social security, and economic security for their daughters. Thus, notwithstanding the efforts of government, girls' education is hindered by poverty and disaster, both of which encourage early marriage and reduce school attendance.

3.3.3. Access to services

The field visit report showed that poor women in the villages have less access to necessary national, and government (I/NGO) services than men and wealthy groups. The majority of the people in the project area are marginalized, landless, and poor and thus unable to access government and banking services. As a result, NGOs have widened their support for these people to help them survive with dignity and improve women's agency. For example, PKSF is providing support to about 1.5 crore families by more than 200 partner organisations (POs). More than 90% of the fund receivers are women.

3.3.4. Health

Health is another area where women are at a disadvantage due to cultural constraints, with problems worsened by climatic extreme events. Often, if women need medicine, they get it from the local pharmacy without seeing a doctor. For primary health care, women and girls generally seek services from a community clinic, which has limited services, rather than go to a hospital because of poverty. Villagers in general only go to a hospital in extreme cases. Midwives are generally called on for delivery because of the low cost and easy availability, and because married women prefer support and comfort from another woman and feel safer away from a male doctor. During disaster (cyclone, storm surge), villagers are often unable to move outside the local area as a result of remoteness, which further limits access to health services. Although government and other institutions provide saline and other medicines for the affected villages, these are not sufficient and not distributed equally.

3.3.5. Mobility and participation

Men have greater access to power and mobility in village society. They can easily migrate to other places and take decisions without discussion, while women's mobility is restricted by the normative gendered roles and responsibilities. Girls and women are not allowed to go anywhere alone outside of their home and village as a result of considerations of both security and social prestige. Even in extreme events, women are not allowed to move to another place to save themselves due to the lack of security, and cannot leave their family members behind because they are considered to be the providers of family wellbeing. When men migrate to other places to work, women have to take all the responsibility for agricultural and household activities, including taking on some decision-making authority. But due to restrictions on their mobility and the type of work they are allowed to do, women face problems in managing these responsibilities and in coping with extreme climatic events. In addition to the cultural constraints, several factors combine to increase women's vulnerability including poor education, lack of skills, and lack of freedom of choice, and financial stresses become extreme when their livelihood options are reduced by disaster.

3.3.6. Power and decision making

According to traditional practice, men are considered to be the household head and family decision maker, while women, children, and other family members must obey their decisions and respect their choices. This form of patriarchy defines a form of power relations between men and women in which men dominate, oppress, and exploit women. Almost all the families in project area were headed by men and men take the major decisions on family matters, regardless of women's educational or financial status. Poor and marginalized women are even less able to take decisions on family matters and have less personal choice than wealthy and educated women. They cannot break the social rules and exercise their rights to power. The normative practice of a gendered hierarchy is for men to be the guardians of the family and to have the right to dominate women. The field study indicated that the key reasons for women's subordination and limited capacity are low income and education, restricted mobility, unemployment, misconceptions about divorce, financial and social insecurity, and domestic violence or physical assault. The dominant perception among both women and men is that men are more intellectual and socialized and have a better understanding about the outside world than women. Hence, women are not allowed to participate in village arbitration or any decision-making process except as a witness or to attend cultural gatherings. Women can participate when there is a call from the union parishad on special issues. These restrictions limit development of women's mental capability and decision and choice making abilities, and teach them that they are less capable than men. This gender bias perception was visible even in accessing and using modern amenities like mobiles. However, this situation is being gradually increasing over the last decades by the government's initiatives on women's participation in the local government elections. Now, the women have access to open competition in local level elections as well as they have reserve seats at the local government structure.

Women are poorly represented in planning and decision-making processes in climate change policies, limiting their capacity to engage in political decisions that can impact their specific needs and vulnerabilities.[1] There has been increasing recognition in international policy frameworks on the importance of incorporating gender in climate risk reduction efforts. In 2009, the Committee on the Elimination of Discrimination against Women (CEDAW) stated, "all stakeholders should ensure that climate change and disaster risk reduction measures are gender-responsive, sensitive to indigenous knowledge systems and respect human rights. Women's right to participate at all levels of decision-making must be guaranteed in climate change policies and programmes" and the IPCC's report in 2014 highlights vulnerability due to climate change due to gender.[2] The UNFCCC Paris agreement in 2015 also formally recognized the intersection of climate change and gender equality, but women's participation in planning and decision-making on climate protection is still low, even in industrialized countries, and is linked above all to the heavily technical nature and male dominance in key areas of work related to climate risk including energy, transport, and urban planning. This is certainly the case in Bangladesh, where women's perspectives on resilience are sometimes absent from national conversations.

In regards to women's role in the domestic sphere, most household activities are done by women, with the highest participation in activities such as house cleaning, child care, cooking and meal preparation and lower but significant participation in household level activities such as tree plantations, dairy farming, and poultry rearing[3]. Despite this central role in household activities, women's decision-making power remains limited, with a recent study indicating that 31% of household decisions are made by women and that women's participation rate in choice of crop to be grown, and the buying and selling of agricultural products is 19% and 34% respectively and even lower in decision regarding property at 20%.

Regardless, women's central role in household management places them in a pivotal position for adapting livelihood strategies to changing environments. Given that women's roles in decision-making is higher in areas such as food preparation and distribution, resolving food deficits and household work, women are central in assuring household food security as livelihood strategies shift due to slow-onset impacts such as salinity and are assigned higher responsibility in disaster preparedness particularly in storage of food and water, during rapid-onset disasters. Adding nuance, a context-specific view of women's role in household decision-making in the vulnerable coastal districts targeted by the project is also available from the baseline assessment of socio-economic conditions carried out by UN Women, and is presented in Table 1 below. The results clearly indicate that women's decision-making power greatly limited in all spheres, with higher participation in regards to food distribution and household work (including collection of water).

Table 1: Role of women in decision-making

Sl.	Type of Decision	Percent
1	Food related (Meal preparation, distribution etc.)	86.78
2	Meeting food deficit	33.58
3	Selling assets (land, house, livestock, seeds)	9.40
4	Selling agricultural production (crops, seeds)	6.88
5	Buying household assets (livestock, ornament, trees.)	11.10
6	Buying agricultural production (crops, seeds etc.)	7.35
7	Receive credit from mohajon/relatives/bank/NGO/GO	14.50

[1]CCC, 2009
[2]UN Women, 2016
[3] Asaduzzaman, 2016

8	Agricultural work (crop cultivation, land mortgage etc.)	5.84
9	Household work (Collection of Water, Collection of natural resource etc.)	47.91
10	Household decision making (Engage in new income generating activity, Conceiving a baby, Using savings, ownership of VGD/ VGF	11.59
11	Female and children healthcare decision making	16.32
12	Decision making about communication (Female going outside the homestead, going for work, education for children)	11.06
13	Decision making on disaster preparedness/coping/adaptation (Going to a shelter, Engaging in alternative livelihood activity	11.48
14	Other	14.29

Source: UN Women (2014)

4. Position of Women in Bangladesh

The Constitution of Bangladesh (Articles 27, 28, 29 and 31) guarantees equality and non-discrimination on account of sex, religion, ethnicity, place of birth in order to provide scope for affirmative action in favour of the “backward section of citizens”. Article 24 promised to ensure religious freedom within a pluralist, National framework and Article 28 (sections 1, 2 and 3) ensures equality in all spheres of life between women and men. Although the constitution guarantees equality between women and men in public domain but further scope for improvements remains in the private sphere. These have been upheld in differing degrees since independence some 4 decades ago, changes have occurred in some contexts, including in the situation of women. Efforts towards women’s development in Bangladesh are based on a wide array of international commitments including the Millennium Development Goals (MDGs), the CEDAW (1979), and the Beijing Platform of Action (1995), amongst others. Following the declaration of the UN Decade of Women (1976-85), the Government of Bangladesh, national and international non-government organizations and others have undertaken several programs towards the advancement of women in the country. Simultaneously, the women’s movement has also played an important role in raising mass awareness of women issues and enhancing women’s participation in every sphere of life in order to achieve equality. As a result, over the last 40 years, women in Bangladesh, as was the case with women in other developing countries, have gradually become more visible in the labour force, development programs and local institutions such as local government bodies.

Gender parity in primary and secondary education has been achieved and the Government of Bangladesh also established institutions for girls and women at the secondary and tertiary level. However, concerns are raised over the high drop-out rate among girls, especially in rural areas, the gender gap at technical/vocational and the tertiary education levels, and the high number of girls who suffer sexual abuse and harassment both at school as well as on their way there. Barriers experienced by women and girls to quality education, for example, the lack of physical infrastructure, the lack of facilities for girls in schools, the negative impact of early marriages and the lack of access to education by rural women and girls are also of concern. The Bangladesh Labour Act (2006) promotes equality of opportunity in employment and provides for equal pay amongst men and women. However, it does not extend to workers in the informal sector where the largest population of Bangladesh’s women is being employed. The persistence of discrimination against women in the labour market, in particular, occupational segregation, a wide gender wage gap and the exploitation of girls is also prevalent.

With regards to SDG-5 (Gender Equality), it is Noteworthy to mention that the total fertility rate (TFR) has fallen from 7 live births in the mid-70s to 2.01 births per woman in 2019 as the contraceptive prevalence rate increased from about 8% in the early 1970s to 40 % in early 1990s to 62.7% by 2019. The reduction in birth rate is also attributed to education of girls and more women joining the work force. Another positive development is that women’s life expectancy has increased to 74.89 years in 2020 from 46.7 years in 1960. Overall mortality amongst women of reproductive age has consistently declined over the last 10 years. The maternal mortality has decreased from 322 per 100,000 live births in 2001 to 173 in 2017. More needs to be done, however, to meet the SDG-5 target. At primary and secondary level enrolment in educational institutions, girls now account for larger proportions at 1.02% and 1.14% respectively. Girls are also doing better, or no worse, in public examinations at these levels compared to their male counterparts. However, at the tertiary level the proportion of girls is only 39%, which is largely due to social reasons such as the marrying off of girls at that age. Overall, girls lag behind in science education. The World Development Report 2012: Gender Equality and Development mentions that in Bangladesh, a woman earns only 12 cents for every dollar that a man earns, one of the lowest wages earned by women compared to other countries of the world. A major breakthrough has been achieved in the area of education and employment for girls due to affirmative action by the government and employment opportunities in the Ready-Made Garments (RMGs) industries that employ mostly women. Although the wage rates at entry-level within this sector is much lower than in other sectors requiring similar (or less) skill. Other issues such as unsafe working conditions and high levels of harassment also reduce the contribution to women's empowerment and gender equality.

Reasons of Gender Discrimination in Bangladesh

Although there are some initiatives taken by the government to address the problems of gender inequality, the improvement in reducing gender inequality is not satisfactory due to several reasons.

Structural and Social Institutions: Traditionally, women were often discouraged from participating in public life and were mainly recognised only for their reproductive role. The social forces, which are creating gender differentials, are based on the age-old patriarchal traditions and values that still prevail in most of the parts of Bangladesh. Traditional perceptions about the role of women as home-makers still persist.

Lack of Explicit Policy Initiatives: There are policies to ensure women's security at home, educational institutions, road, organizations and the like. Increasing violence against women is now a grave concern. Under these circumstances, parents are more likely to keep their girls inside their home. Hence, they are not able to participate in education, health, employment or other sectors. Therefore, inequality is still persisting.

Preoccupied Mind-set: There are perceptions that men are better off than women are as far as the ability to work is concerned and only men can look after their parents. That is why girls are subject to discrimination from their births. In addition, son preferences in the traditional Bangladeshi society create gender discrimination. In the case of employment, the employers in Bangladesh still tend to employ men first rather than women.

Early Marriage: Early marriage of girls is a very common phenomenon in Bangladesh. Early marriage is one of the vital barriers to women's and girls' education, health and employment. Early marriage has historically limited young women's access to education and thereby to employment opportunities as well as creating a vulnerable situation to their health.

Gender of the head of the family significantly influences the household decision on whether or not to adopt any climate adaptation strategies, and also while choosing individual or combination of adaptation strategies. As female-headed households often tend to suffer from labour shortages in Bangladesh, they are less likely to opt for a change in farming practices as an adaptation strategy. Social restrictions on mobility and the burden of household responsibilities, in addition to cultural and social hegemony, prevent many women from seeking an additional job and diversifying their livelihoods. In these cases, differences in the adoption of climate adaptation strategies can also result from inequities in endowments among male and female-headed households. Large endowment difference between male and female-headed households is possible in Bangladesh because women are mostly involved in unpaid family labour works.

5. Gender and the Women's Development Policy (WDP)

In the context of the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) and the Beijing Platform of Action, Bangladesh has developed several policies and sectoral strategies to ensure gender equality, including the Women's Development Policy (WDP), 2011 and the National Action Plan (NAP) to implement the WDP. The objective of this policy is to take special measures to enhance the overall safety and security of women and children, including helping them deal with disasters, ensuring rehabilitation services of those affected with special consideration for disabled women and ensuring food distribution and assistance to eliminate bottlenecks created due to extreme climate events and disasters. The proposed project will consider the following policies, strategies and action plans regarding gender aspects.

Bangladesh has several policies and strategies to promote gender equality in agricultural development addressing climate change. The government recognizes the importance of having both women and men equally involved in adapting to climate change and other environmental challenges. However, despite affirmation from the government of its intention to mainstream gender in national and climate change policies, such efforts remain inconsistently applied.

No	Key national laws and policies	Gender provisions
1)	Representation of the People Order, 1972	• Focuses on integration of environment, climate change, and disaster management into planning and budgeting with the goal of promoting sustainable development. • EFYP emphasizes on" developing Gender-Inclusive Climate Change Response framework" to harmonize the priorities and strategies among different national documents (policies/ strategies/ visions/plans) related to climate actions. • Directs measures to increase women's knowledge of environmental management and conservation, and make investments in education, capacity-building training, technology transfer, and environmental projects focusing on women.
2)	National Biodiversity Strategy and Action Plan (2016-2021) MoEFCC	• Translate measures set out in the Convention on Biological Diversity. • Recommends inclusion and recognition of women's existing active role in biodiversity conservation to offer them equal opportunity. • Increase capacity of rural women to enable them to engage actively in biodiversity conservation at both household and community levels.
3)	Bangladesh Delta Plan (BDP) 2100 Ministry of Water Resources (MoWR)	• A plan with a long-term vision for "achieving safe, climate resilient and prosperous delta." • Gender reference is minimal in this planning document. It mentions women as "vulnerable", but does not portray them as potential change agents in the process towards building climate and disaster resilient development. • There are no specific strategies or plans that directly relate to gender equality.

No	Key national laws and policies	Gender provisions
4)	National Plan on Disaster Management (NPDM, 2016-2020) MoDMR	• DRR and emergency management are integrated in the disaster management policies. • The plan provides a directive to integrate gender in all its plans and actions.
5)	National Sustainable Development Strategy	Focuses on the constitutional obligations of Bangladesh to have a people-centric approach with a vision for sustainable development.
6)	Perspective Plan, 2021-2041 General Economics Division, Planning Commission	Considers both gender and environment as important perspectives for development by addressing those in separate chapters.
7)	Mujib Climate Prosperity Plan (MCPP) – Decade 2030	Intends to facilitate climate financing for vulnerable communities and to encourage women's empowerment. MCPP is formulated in honor of the Father of the Nation on his birth centenary.
8)	National Adaptation Programme of Action (NAPA), 2009 MoEFCC	Suggests specific strategies for adaptation and recommends 15 projects to strengthen the immediate and urgent adaptation activities to address the current and anticipated adverse effects of climate change, including extreme events. NAPA was the first attempt to guide the coordination and implementation of adaptation initiatives in the country. However, differentiated gender impacts were not recognized.
9)	Climate Change and Gender Action Plan (cc-GAP), 2013 MoEFCC	Prepared with an aim to ensure the integration of gender equality into climate change-related policies, strategies and interventions. The ccGAP integrates gender considerations into four of the six main pillars in the BCCSAP: (i) food security, social protection and health; (ii) comprehensive disaster management; (iii) infrastructure; and (iv) mitigation and low-carbon development. It is in the process of being updated in light of the revised BCCSAP.
10)	Eighth Five-Year Plan, 2020-2025	Acknowledges the role of women in the food and nutrition security of Bangladesh and focuses on removing barriers to productive participation of women in agricultural employment by addressing the following issues, amongst others: 1) Socio-economic backwardness and constraints that women endure in a male dominated society 2) Wage differences between male and female in agriculture 3) Women's access to institutions and facilities including

No	Key national laws and policies	Gender provisions
		extension and credit services and linkages with other services such as health and nutrition 4) Women's access to markets and high value-added agriculture
11)	National Women Development Policy, 2011 *MoWCA*	Highlights the inclusive growth and participation of women in all spheres of national life and fulfils objectives, such as the following: 1) Take steps to ensure that farming women have equal opportunity in obtaining agricultural inputs such as fertilizer, seed, farmer's card and credit facilities 2) Take initiative to ensure equal wages for the same job 3) Put special emphasis on the health of women alongside food during post-disaster emergencies
12)	National Agriculture Policy, 2018 *MoA*	Recognizes the direct and indirect contribution of women in different stages of production. • The main strategies towards enhanced women's participation in the agriculture sector are envisaged as the following: 1) Recognition of women's labor and participation to ensure their social dignity and safety 2) Elimination of the wage differential between men and women labor in agriculture and ensuring equal pay for men and women 3) Homestead gardening and promotion of cash payment 4) Agricultural education and research 5) Encouraging women to participate in the formal economic sphere by providing support to their involvement in agricultural product-based small and cottage industries 6) Training on families' nutritional security, agricultural production, storage, marketing, agricultural businesses and industries to build enhanced capacities 7) Participation of women in food security-related planning, decision making, supervision and distribution activities 8) Adoption of specific extension activities for women farmers
13)	National Agricultural Extension Policy, 2020, *MoA*	• Addresses the conditions that hinder the recognition and effective participation of women in decision-making spaces by engendering those spaces, forming women farmer groups, encouraging women-led SME development in agri-business, developing their confidence in raising their voice through grassroots-level women farmers' organizations, and creating gender awareness in both women and male farmers • The Policy also puts emphasis on homestead gardening as a means to women's economic empowerment, poverty alleviation, and food and nutritional security

After analyzing agriculture policies, it is evident that there are relevant policies in place that acknowledge that women are crucial for food processing and nutrition security. While women are missing from other roles in agriculture sector — as producer and other parts of the value chain — some of these policies recognize women's predominant engagement in seed conservation and post-harvest activities. These policies suggest capacity-building measures, increased access to information, knowledge, financial assets, technology, farming resources and markets. These include initiatives to equal pay for equal work and to remove wage discrimination in agriculture, steps to ensure women have equal opportunity in securing agricultural inputs like fertilizer, seed, farmer's card and credit facilities etc. There is also policy guidance to create alternative options. However, these policies fall short in three aspects. First, the agriculture policies and strategies lack the analysis of differentiated impacts of climate change on women and men. Second, they do not provide specifically climate-gender interlinked measures. Third, lack of gender-disintegrated data on access to land, finance, extension services and agricultural tools act as barrier to better gender-responsive climate actions.

Mainstreaming of gender would continue and all macro-economic and sectoral policies would integrate gender as a crosscutting theme. Action plans should be drawn with a view to reduce inequality and promote an equal relationship between sexes. To ensure results from actions related to gender equality all reporting of national progress including those related to SDG-5 would be based on sex disaggregated data to allow a better understanding the progress in the area of gender equality and women's empowerment. The framework for women's empowerment and gender equality comprises of 4 areas of strategic objectives:

Improve women's human capabilities: This deals with women's and girls' access to health care, life expectancy, nutrition, reproductive health, education, information, training, and other services that enables women to achieve better health and educational outcomes. This also includes women's freedom from violence and coercion.

Increase women's economic benefits: This relates to women's access to or control over productive assets, resources, services, skills, property, employment, income, information, technology, financial services, and other economic opportunities including community resources like land, water, forest etc.

Enhance women's voice and agency: This pertains to women's role as decision makers in public and private spheres including politics and promotion of their leadership is considered here. Changed attitudes on women's and girls' rights, women's enhanced knowledge of their rights and increasing their bargaining power are reflected on.

Create an enabling environment for women's advancement: The socio-political environment, legal and policy support, and congenial social norms are the key in this area. Oversight, enforcement of laws, regular collection of sex-disaggregated data, gender and social analysis skills including the capacity to develop, implement, and monitor gender strategies, understanding of gender issues in the sector are the key areas.

To implement these strategic objectives, seven action areas have been identified that will contribute in achieving results in these four areas.

i. Increase access to human development opportunities
ii. Enhance access to and control over productive resources
iii. Increase participation and decision making
iv. Establish conducive legal and regulatory environment
v. Improve institutional capacity, accountability and oversight
vi. Increase protection and resilience from crisis and shocks
vii. Promote positive social norms

6. Gender Assessment in the Context of the Proposed Project

Gender mainstreaming is an integral part of the RHL project design. Hence, the project has analysed gender-based vulnerabilities to climate change, particularly that of women and girls who are affected more severely than men. The project also tried to identify gender-based violence and SEAH related risks, particularly due to implementation of the project activities.

6.1. Gender and Climate Change Vulnerability in the Project Area

The impacts of climate change are gendered-specific. A wide range of climate-change fallouts (food insecurity, disrupted production, out-migration) are experienced differently by men and women. Women and girls, due to their limited mobility, access to resources and decision-making processes, are disproportionately affected by climate-related disasters. Furthermore, their vulnerability is a result of socio-economic and political factors that are exacerbated by climate change.

Women in Bangladesh are more victims of different types of violence as well as natural disaster, particularly in the coastal areas. In this respect, Zahan (2022) conducted a study in Satkhira in the southwest part of Bangladesh by using structured questionnaire, survey, and observation method in order to assess the challenges of women during a cyclone event. A total of 60 households from Satkhira were randomly selected for better understanding of crab fattening. It is evident from the study that more than three-fourths of people suffer from the effects of cyclones. 43% of affected respondents do not seek cyclone shelter due to a variety of reasons. In this regard, a major part of the respondents expressed their dissatisfaction with the environment as well as safety for women in the cyclone shelter. A large number of respondents stated that a major challenge in the shelter center is insufficient latrine which causes health problems, particularly for women. Around 50% of the respondents suffered from diarrhea and nearly one-third from skin-related diseases. It is a matter of concern that a major part of the respondents suffers a lack of pure drinking water for their necessary uses. More importantly, almost all of the respondents stated that they have no right to take any decision at the time of emergency due to socio-cultural trends.

Ferdoushi (2009) conducted another study in three different areas; Khulna, Bagerhat and Satkhira in the southwest part of Bangladesh by using semi-structured and pre-tested interview schedule in order to assess the role of women in mud crab fattening. A total of 150 households from Khulna; Bagerhat; and Satkhira were randomly selected for better understanding of crab fattening. In this study, about 74% women found to be involved in crab farming. A greater involvement found in Khulna area (78%). However, age, education, training and experience in crab farming did not vary significantly among the women in those three study areas. All of those finding revealed that anyone could practice the farming regardless of their ages and other characteristics. Crab farming in Bangladesh seems to be a recent development and women involvement in this sector has not been so long existence. A number of innovative technical supports are needed to disseminate the proper technologies to the women. In the present study, all women found to involve in feed application to their farmed crabs which is less laborious works, while only 18.92% were involved in preparing their pond before stocking. However, a small proportion of women were involved in marketing (33.33%) might be due to the restriction by some social religious norms and due to the pressure of other household works. Level of formal education is also positively related with adopting any technology. In the present study, more educated and more experienced mud crab fattening farmers found to have more benefit that is in line with others finding established the positive impact of literacy over farm efficiency (Wang et al. 1996). From the correlation results, it is evident that level of involvement positively affected (1% level of significance) the net revenue earned from the crab fattening indicating that higher involvement returned higher revenue. Moreover, the value of correlation coefficient for training indicates that if the women get more and more training through different organizations could increase the revenue. During the period of the survey, Hindus women were found more capable and competent than Muslim women in crab fattening. In the recent year, there is raising awareness among the women through their participation in different income generating activities like aquaculture and other farming. From the study, lack of proper knowledge of crab farming and lack of training are addressed by women as problems for sustainable crab farming. Moreover, they also reported that their movement sometimes restricted by some religious norms.

The field visit report revealed significant gender-related vulnerabilities of coastal areas in terms of unequal wages, unsafe working conditions, violence, and access to food, limited education and inequalities stemming from the patriarchal system. Related to unequal access to employment opportunities is widespread discrimination in wages for paid work.

Due to the increasing economic crisis combined with sickness or loss of the main household breadwinner, anxiety and depression have increased in the proposed project areas. Village women are encountered economic insecurity, which is additionally increasing violence against women. In addition to this socio-ecological effect, violence is promoted as part of the patriarchy. It is used to preserve social order and secure the privileged under the social framework of patriarchy by maintaining its control over resources. This tradition is becoming further entrenched due to increasing climate vulnerabilities and related poverty.

Coastal disaster effects cause damage and loss to food production and resultant food price increases, creating unequal access to food and nutrition. During our filed visit village people informed that cyclone causes an extreme level of food crisis due to failures of crop production, vegetation loss, and declining wild fish." They sometimes eat only once daily because of these climate extremes. The only available food was of low quality, such as traditional bread called roti, partly rotten rice called pantha vat, or watered rive or jao.

Coastal disaster effects linked to failures of effective adaptation programs are responsible for many of these problems, which also impact other rights such as access to education. Due to climate change vulnerabilities, women are unable to receive formal education and could not read and write. Low levels of female education are the foundation of gender-based socialization in village. Participant informed us: "Early marriage requires lesser dowry, reducing the scope for girl's education. Although the government scholarship for primary education is helpful for ensuring their right to primary education, they cannot afford the education for the next level like secondary or higher. Eve teasing on the way to school or sexual harassment causes further challenges, and an older wife is less docile and not helpful for family peace and stability.

It is widely documented climate change impacts on food production and access also disproportionately affects the nutrition and health of poor women[4]. Finally, recent research has also shown that the strenuous economic conditions created by climate change are leading to an increase in child and forced marriages in Bangladesh, as dowries become cheaper[5]. Compelling evidence from this research has shown that child and forced marriages of girls appear to be short term solution designed to ease both the food insecurity and future financial pressures on families exacerbated as a result of climate events. The research concludes that attention to climate challenges must take a much broader focus on social consequences in order to protect the human rights of women and girls in vulnerable communities.

The IPCC suggests that the differentiation of vulnerability to climate change among population groups can be clearly observed in the pattern of vulnerability to natural disasters. In general, women have less access to resources that are essential in disaster preparedness, mitigation and rehabilitation[6] and women and children are 14 times more likely to die than men during disasters.[7] In Bangladesh, as in global estimates, women are more affected and suffer more during and after disasters than men, exemplified by the impacts of cyclones on women in the coastal areas of Bangladesh. During Cyclone Sidr for example, many of the female casualties in coastal Bangladesh occurred because women, the majority of which are homebound, were busy tending the family livestock when the cyclone struck and

[4] IPCC, 2001
5 Alston, 2014
[6] UN Women, 2014
[7] Araujo, 2007

could not leave without prior preparations, others died because their traditional clothing (saris) got trapped in trees and other objects while running, and others perished trying to rescue or search for children who could not evacuate fast enough[8,9]. Furthermore, the cyclone was announced primarily among men, with many women lacking the necessary information to evacuate, remaining at home and facing serious risks.[10] Disaster preparedness requires decision-making and leadership, but in coastal Bangladesh, women are generally excluded from such roles[11]. Post disaster stages also take a toll on women. Often, women find facilities for personal hygiene in shelters are inadequate, and with few alternatives, are exposed to urinary tract diseases, maybe sexually abused while looking for firewood or reconstruction materials, face deteriorating nutrition status as they eat less in order to offer more food to other household members and they lose the natural resources and livelihood assets they depend upon[12]. Regarding early warning and disaster preparedness, women consulted mentioned having been included in village disaster management committees and have been provided training and necessary equipment, such as early warning flags.

The climate variability has pushed women into a vulnerable and marginalized position in Bangladesh. For the analytical purpose, already mentioned that climate change itself does not directly affect the women, but the disasters especially natural disasters and man-made disasters like socially constructed system have made the situation possible where climate change plays a key role in instigating the vulnerabilities. WHO notes, ‘women and children are particularly affected by disasters, accounting for more than seventy five percent of displaced persons. In addition to the general effects of natural disaster and lack of health care, women are vulnerable to reproductive and sexual health problem, and increased rates of sexual and domestic violence. Moreover, gender roles dictate that women become the primary caretakers for those affected by disasters-including children, the injured and sick, and the elderly-substantially increasing their emotional and material workload. Women’s vulnerability is further increased by the loss of men and/or livelihoods, especially when a male head of household has died and the women must provide for their families. Post disaster stress symptoms are often but not universally reported more frequently by women than men’ (Dasgupta et al., 2010, p.6).

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[8] Kabir, 2016
[9] Alam, 2010
[10]Kabir, 2016
[11] Alam, 2010
[12] MoEF, 2012

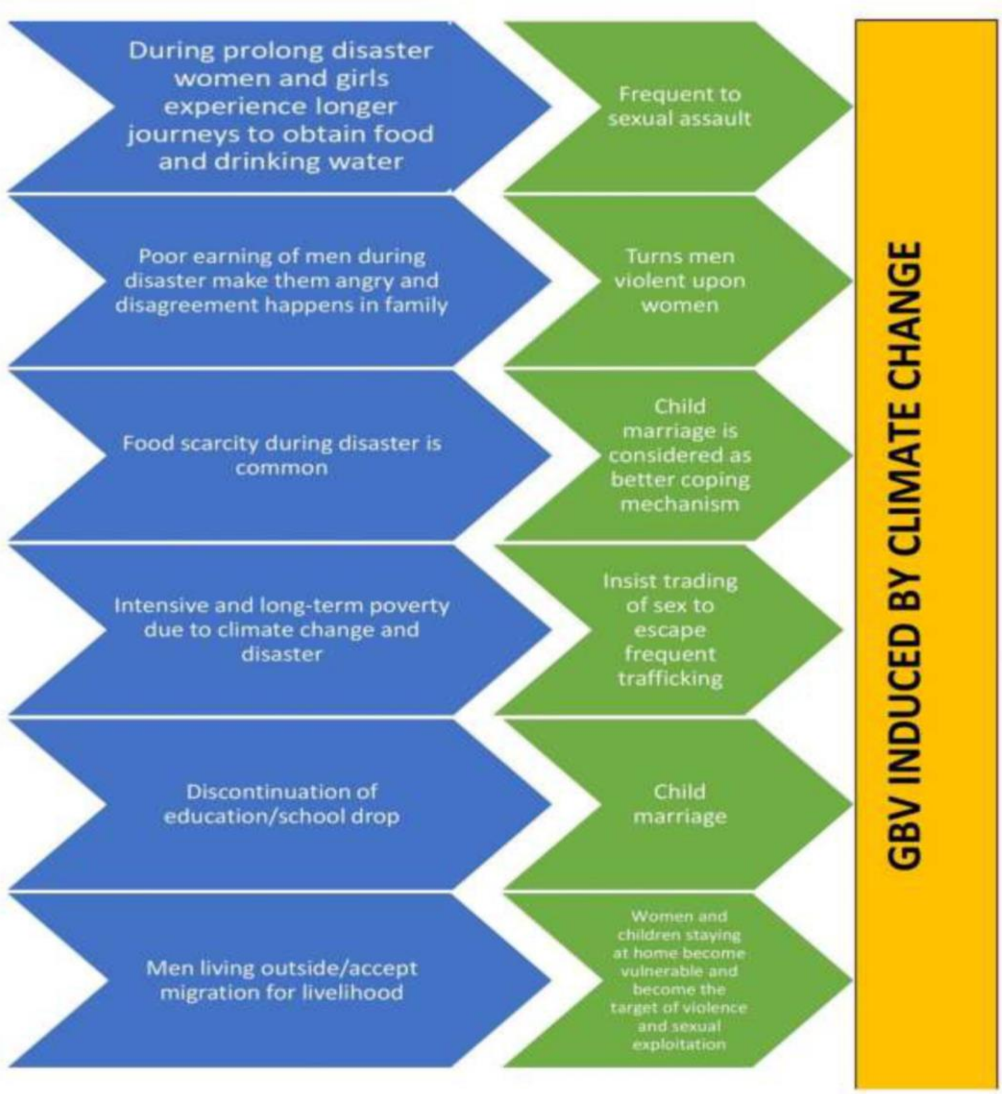

Figure 1: Consolidated perspective of gender differentiated vulnerabilities linked to coastal areas of Bangladesh

Both during disasters and in the face of changing environmental conditions, women's role in communities is not formally recognized or accounted for in mitigation, adaptation and relief efforts. Women's knowledge about ecosystems and their particular strategies, experiences and skills for coping with water shortages, are often ignored[13]. For example, Cyclone Sidr contaminated at least 6,000 surface water ponds with saline water, used primarily by women for small vegetable farming and domestic water requirements but their knowledge was not received enough attention during the rehabilitation of these ponds.[14] Overall, women and girl's vulnerability to climate change generally depends on the interaction of three key functions: - exposure (E), sensitivity (S), and adaptive capacity (AC). The exposure is largely determined by the climatic hazards and the extent the women and girls are exposed to cyclones, salinity and sea level rise. The following table provides a summary of the vulnerabilities of women and girls in the context of climate change in coastal areas in Bangladesh:

Table 2: Women and Girls Vulnerability to Coastal Hazards: Sea Level Rise, Coastal Flooding, Cyclone and Tidal Surge, Salinity Intrusion, Water logging

[13] Dankelman, 2002
[14] UN Women, BCAS (2014)

Critical elements at risk	*Exposure (degree and frequency)*				*Sensitivity (Low to High)*				*Deficit in Adaptive Capacity*
	Cyclone & Tidal Surge	*Coastal Flooding*	*Salinity*	*Water logging*	*Cyclone & Tidal Surge*	*Coastal Flooding*	*Salinity*	*Water-logging*	
Life	Very Likely	Very Likely	Certain	Very Likely	High	High	Low	Low	Less education to understand cyclone early warning; lack of warning system for tidal surge and coastal flooding; less access to early warnings; less places to evacuate during cyclone; tendency to undermine the risks from cyclones and tidal surges; lack of long term predictions of salinity and water-logging, inadequate facilities for women and girls in public cyclone shelters; lack of women volunteers; lack of gender sensitive rehabilitation; lack of water and sanitation in houses and public shelters during cyclone, tidal surge, coastal flooding, salinity and water-logging.
Employment	Very Likely	Very Likely	Certain	Very Likely	High	High	Low	Low	Lack of diversity of livelihoods; lack of off-farm livelihood skills; reduced options for on-farm livelihoods; cultural barriers in employment in industry sector; limited SMEs to absorb women labour; lack of women with diversified skills in urban sector jobs; poor capacity to enter into skilled service sectors; heavy domestic responsibility; lack of incentives in skilled job outside domestic territory; sole responsibility for child care.
Potable Water	Certain	Certain	Certain	Very Likely	High	High	Low	Low	Very limited number of safe and salinity free water-points in public and private spheres; lack of available water sources during droughts; lack of economic ability for poor women and women headed households to install salinity free water sources; forced to spends long hours to collect water from distant sources; insecurity due to sexual harassment during long walks to collect water from distant sources.
Food Production	Very Likely	Very Likely	Certain	Very Likely	High	High	Low	Low	Lack of available varieties of food to produce in salinity and water logging context; lack of means to recover food loss from cyclones, tidal surges and coastal flooding; lack of fresh irrigation options; lack of grasses and other inputs for livestock rearing.

Critical elements at risk	*Exposure (degree and frequency)*				*Sensitivity (Low to High)*				*Deficit in Adaptive Capacity*
	Cyclone & Tidal Surge	*Coastal Flooding*	*Salinity*	*Water logging*	*Cyclone & Tidal Surge*	*Coastal Flooding*	*Salinity*	*Water-logging*	
Food Preparation	Very Likely	Very Likely	Certain	Very Likely	High	High	Low	Low	Lack of fire-wood during coastal flooding, cyclone, tidal surges, salinity and water logging; unsafe and saline water for cooking; lack of hygiene during different hazards; lack of food during cyclone and tidal surge; lack of knowledge on food and nutrition standards; lack of storage facilities during hazard onsets; challenge of food preservation in extreme temperatures.
Sanitation and Hygiene	Very Likely	Very Likely	Certain	Very Likely	High	High	Low	Low	Lack of number of salinity, cyclone, flooding and waterlogged proof/resilient toilets; lack of hazard proof public toilets; poor public health condition; lack of personal hygiene knowledge.
Core Shelter Maintenance	Certain	Certain	Certain	Very Likely	High	High	Low	Low	Poor maintenance of household assets and housing materials safer from salinity, coastal flooding, water logging, and tidal surge; lack of saline free housing materials for durable and cyclone resistant housing; lack of retrofitting materials and capacities to protect house from hazards; lack of financial capacities to prepare hazard proof/resilient house materials.
Child Care	Certain	Certain	Certain	Very Likely	High	High	Low	Low	Lack of means and knowledge to protect children from death, injury, fever, drowning, de-hydration, malaria, pneumonia, and other water-borne diseases.
Reproductive Health	Certain	Certain	Certain	Very Likely	High	High	Low	Low	Lack of knowledge and means for safe births during cyclone, tidal surge, water logging and coastal inundation; lack of trained birth attendants in disasters; lack of easy access to MCH clinic and hospitals in disasters.
Girl's Education	Certain	Certain	Certain	Very Likely	High	High	Low	Low	Challenge to continue education of girls during cyclone, coastal flooding, tidal surge and water-logging; increased role of adolescent girls in domestic spheres during disasters; increasing tendency to early marriage amongst disaster affected households; discontinuation of girl's education; lack of social safety net for girl's continued education.

Source: UNDP Bangladesh, 2015 (modified on the basis of field visit)

6.2 Assessment of SEAH Related Risks

There is less possibility to occur SEAH related risks by the project interventions. Because, selected women members will directly involve many of the activities as these will be implemented at their homestead and neighbouring areas. In addition, some women may require to go away to work in the crab farm. They might have risks to SEAH while working in the farm or traveling from home to farm and vice versa. However, the risk is very limited or negligible. But the challenge is that if any woman is affected, she does not want to disclose due to either shyness or fear of loss of dignity. At the IE level, there may be female staff who will have to travel frequently in the selected villages. These staff may also face similar types of difficulties. Similarly at the PMU level, female staff may be recruited. These female staff would require to frequent field visits in the remote areas of the country. Thus, they might be exposed to SEAH related risks.

Different types of stakeholders will be involved during the implementation of the project. At the central level, PKSF will establish the project management unit (PMU) where the desired number of female staff are expected to be recruited. These staff will be required to travel in the remote areas alone or with male colleagues. In this case, the female staff may be exposed to SEAH risks. On the other hand, selected IEs also may recruit female staff who will also require to travel at the village levels for community mobilization, CCAG activities, monitoring physical interventions etc. They will also be required to travel to Dhaka or other areas for training under this project. They may be affected in various ways that include but are not limited to lack of sanitation facilities at work place, eve teasing, sexual exploitation and harassment, wage discrimination etc. An action plan matrix is developed to reduce the risk.

6.3 Action plan matrix for protection of GVB and SEAH

PKSF strictly follow government's policy on Sexual Harassment-Free Educational and Working Environment. As per the policy, each organisation will form a committee to receive, investigate and remedial measure against complains on sexual exploitation, abuse and harassment.

SL#	Identified risks	Mitigation measures	Responsibility	Source of Budget
1.	Wage discrimination	• Awareness raising through CCAG meetings. • Ensure equal payment to male and female labour during earth work. • Establish grievance redress mechanism at union level (the lowest administrative unit of Bangladesh).	IE and CCAG members	No additional budget is required
2.	Sexual harassment and/or eve teasing due to lack of sanitation facilities at work place	• Temporary separate sanitation facilities at the work place for both male and female members. • Establish grievance redress mechanism at union level (the lowest administrative unit of Bangladesh).	IE and local contractors	Budget is built in the relevant activity.
3.	Sexual harassment and/or eve teasing on the way to and from	• Establish grievance redress mechanism at union level (the lowest administrative	IE under the supervision of	

SL#	Identified risks	Mitigation measures	Responsibility	Source of Budget
	work place	unit of Bangladesh).	EE	
4.	Risks associated with SEAH at PKSF level	• PKSF's guideline will be applicable for this project (Annex 25) • For travel to remote areas, official vehicle will be ensured instead of public transport.	PKSF	No additional budget is required
5	Risks associated with SEAH at IE level	• Training related to project management will incorporate SEAH and GBV related sessions to enhance awareness. • Accommodation of female staff will be arranged separately considering individual requirement of female staff. • Necessary security and privacy will be maintained.	PKSF and IE	Existing training budget

***** In every cases gender policy and GRM of PKSF will be applicable. PKSF strongly follow the zero-tolerance policy on SEAH and GBV. It is applicable both for PKSF and IEs

7. Gender Mainstreaming in the Project

The project is well aware about gender mainstreaming in the project activities. Hence, the proposed project has taken a gender responsive and transformative approach to climate change vulnerability, considering gendered differences in access to resources, ability to pursue adaptive livelihoods and institutional support and capacity building, and this has fundamentally shaped all of the activities and outputs of the project. The impact of the proposed activities is capture in Figure-2. The proposed project recognizes women's essential contributions as leaders and agents of change in the face of a changing climate and resource constraints. The project will select 362,475 direct beneficiaries for transferring knowledge and adaptation technologies proposed under this project. About 50% of the total direct beneficiaries will be women. It is already mentioned that the project will form 3,200 CCAGs for support delivery and community awareness raising. Considering the gender sensitivity of the proposed project, 80% of the CCAG members will be women beneficiaries (among the beneficiaries under PKSF, around 90% are women). The project will select mostly women because they usually teach their children at home. They will also teach their children about climate change issues what they will learn through training and meetings. Thus, climate change concepts and practices would transmit to the next generation which will have long term implication of addressing climate change in this country. Besides, necessary female staffs will be ensured at the field level so that women members can easily express their opinions and actively take part in the project activities. Considering gender integrity, the project proposes more water facilities to outreach maximum number of women. Allocated budget for female beneficiaries also very high which is estimated US$23.89 million.

Besides, CCCP experiences showed that women were benefitted both economically and socially due to engage them with CCAG. In drought prone areas, women usually are not willing to go too far away from their house to collect water. They prefer to collect water from adjacent areas of their house because they can use their time in productive works. Most important lesson was that they could talk about climate and disaster in their locality. They also felt empowered because they contributed to family income through vegetable cultivation, goat rearing etc. using the time they could save from collecting water. We expect the similar outcomes in the proposed project. The CCCP faced some challenges to engage the CCAG with the women members of the families. Initially, the women in the vulnerable areas were not much supportive due to shyness and hesitation. Besides, climate change was new issue to them. However, motivation through disseminating proper information helped to overcome this challenge.

The RHL project considers not only the benefits of women, but also considers the inter-sectional vulnerability to changing conditions, of those beneficiaries facing additional marginalization due to poverty, and social exclusion. The project design recognizes to build adaptive capacity in regards to changing climatic conditions, by supporting climate resilient cropping pattern, adaptive water infrastructure and knowledge transformation.

The RHL project will accommodate GoB’s policies and strategies on women’s resilience and their critical role in preparedness and recovery from disasters and the necessity of shifting livelihoods towards adaptive options, efforts remain limited compared to the actual and acute needs of women. The Gender Assessment expands on the information provided throughout the proposal, by providing additional information on the national and local gender context, particularly in regards to women’s access to resources, their role in decision-making and the gendered aspects of local livelihoods, and provides the basis for, and lessons on which, the Gender Action Plan (which is reflective of the overall project design) has been built. The activities of the proposed project have been selected considering that women can easily implement to enhance their capacity and increase their resilience to climate change.

For promoting women’s empowerment through the project interventions, we will consult not only with the women members of a family, but also with the male members and other guardians. This will help eliminate hesitation and shyness. Besides, IEs will build good rapport by disseminating appropriate information with the vulnerable community.

The project will not only empower the women, but also sensitize the men about the role and responsibility of women in their family and life through CCGA meeting and training during the project period. After sufficiently sensitize, men will have understanding on the role of women, his partner, for boosting family income, social status and sharing responsibilities. In this way men will be assisting women to come to the carb value chain rather protest.

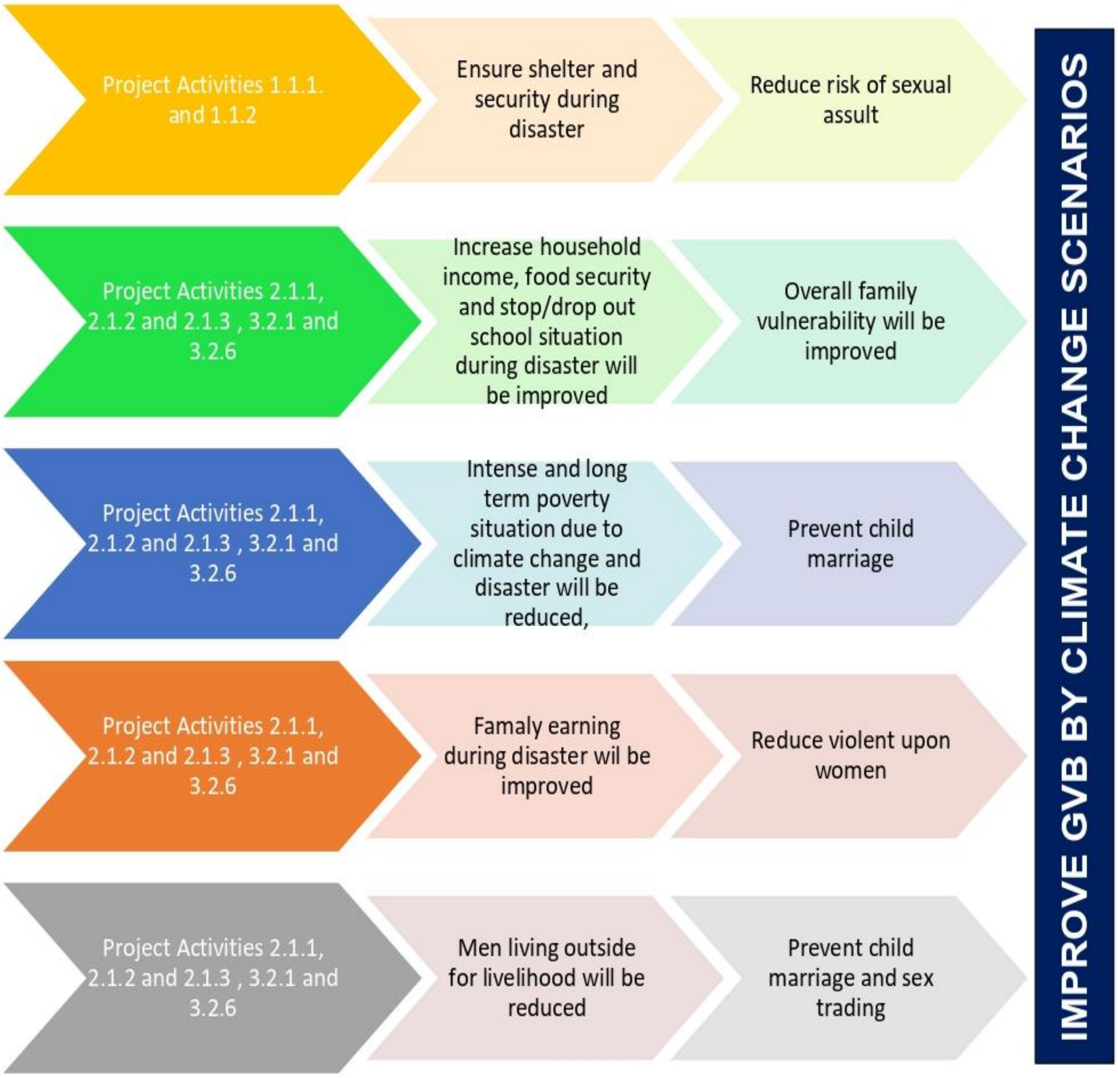

Figure 2: The impact of the proposed activities

Overall:

- Increase women's participation in decision-making (contingency planning for disaster and climate resilient livelihoods, water provision, disaster preparedness committees, early warning system).
- Account for differing needs of women and other marginalized groups in building climate resilience.
- Identify gaps in equality through the use of sex and age disaggregated data, enabling development of action plans to close those gaps, devoting resources and expertise for implementing such strategies, monitoring the results of implementation, and holding individuals and institutions accountable for outcomes that promote gender equality.
- Include all stakeholders involved in the project to develop awareness / trainings aimed at drawing attention to the implication of climate resilience adaptation and gender equality.
- Undertake community dialogue in relation to gender and social inclusion in climate resilience.

For Trainings:

- Provision of women trainers and women-exclusive training sessions, including flexible times, and provision of household training for women-headed households as required.
- Integration of men and community elders in community trainings that addresses women's participation, and norms around appropriate work for women, as well as mobility outside the homestead.

For Livelihoods:

- Analysis of the gendered division of labour (gender-differentiated responsibilities, and needs).
- Ensuring that working conditions are gender responsive at the homestead and community levels
- For hatcheries: segregated sanitation, ensuring equal pay for equal work, and access to the administrative, technical and managerial positions.
- For household and community level agriculture and aquaculture-based livelihoods: Consider Shaded ponds, shallow water depth, and proximity to homestead.

For Ownership of Assets:

- Enhance women's control over productive assets (finance, productive skills, livelihood assets and technology) through women led groups.
- Monitoring of productive assets and revenues, to ensure that revenues are kept in the hands of women and targeted beneficiaries.

For Market Integration:

- Access to finance, financial management training, and market prices for women.
- Conservation training on managing wild stocks, and alternatives to mangrove fuel wood, given that women are often the collectors of wild crab fry and fuel wood from the mangrove areas.

Give preference for employment in crab hatcheries.

For Grievances:

- Adoption of gender sensitive (and marginalized group sensitive) grievance mechanism, which allows women easy and unrestricted access, including provision of female GRM focal points.

For Institutional Strengthening:

- Improve institutional capacity, accountability and oversight (capacity of project executing and implementing agencies in addressing gender issues in climate change programming).
- Build on the projects, structures and initiatives being rolled out by the GoB and GCF in order to maximize the use of resources, and for greatest efficiency and effectiveness.

8. Proposed Gender Logframe

The purpose of a Gender Action Plan is to operationalize the constraints and opportunities for women and men that were identified during the gender analysis, towards fully integrating them into the project design, providing the framework for a gender-responsive and socially inclusive project. For this, particularly considering the disadvantaged position of women, steps will be taken to ensure that the realities of women are taken into account in the activities during the project's planning, executing, and monitoring phases. The project has a plan to undertake a "detailed gender gap analysis and action plan" at the project inception stage. In addition, specific indicators are also proposed to measure and track progress on these actions at the activity level, which can be incorporated into the detailed M&E plan which will be developed at the start of implementation, and provides concrete recommendations on how to ensure that the degree of gender-responsiveness and transformation continues to be measured throughout implementation. The gender expert will be involved in any of the activities for which gender expertise will be required. Furthermore, it is recommended that the project takes into consideration gender and social inclusion measures outlined above and these measures are tailored specifically for a Bangladesh context. Based on the approach a Gender logframe is developed for the project which is given in the Table-3 below.

Table 3: Gender Logframe of RHL Project

Objective	Actions	Targets, Indicators and Timeline	Responsible institutions	Allocated budget (USD) million
Outcome 1: Decreased risk of loss of assets and lives from extreme weather events ***Result 1.1: Climate resilient homesteads constructed***				
Activity 1.1.1: Design and building of homesteads[15]	Selection of beneficiaries, including vulnerable women and provide support of construct/re-construct house[16].	Baseline=0 Target: 50% vulnerable women beneficiaries Indicator: 50% vulnerable women beneficiaries selected. Timeline: It will be started within 6 months of the project's commencement.	IEs and PMU	11.39
Activity 1.1.2: Homestead tree planting	Selection of beneficiaries, including vulnerable women and provide support for tree plantation.	Baseline=0 Target: 50% vulnerable women beneficiaries Indicator: 50% vulnerable women beneficiaries selected. Timeline: It will be started within 9 months of the project's commencement.	IEs and PMU	0.81
Outcome 2: Increased livelihood resilience to SLR, storm surges and salinity ***Result 2.1: Traditional farming practices climate proofed***				

[15] The aspects of inclusion and basic utilities will be kept into considerations within the budget of housing activities.
[16] The presence of households headed by women will be a key factor in the selection process for vulnerable women.

Activity 2.1.1: Construction of slatted houses for goat/sheep rearing	Selection of vulnerable women and provide support for *Construction of slatted houses for goat/sheep rearing.*	Baseline=0 Target: 50% vulnerable women beneficiaries Indicator: 50% vulnerable women beneficiaries selected. Timeline: It will be started within 9 months of the project's commencement.	IEs and PMU	1.4
Activity 2.1.2: Provide financial support for goat/sheep rearing	Selection of vulnerable women and provide support for purchased of goat/sheep for rearing in *slatted houses.*	Baseline=0 Target: 50% vulnerable women beneficiaries Indicator: 50% vulnerable women beneficiaries selected. Timeline: It will be started within 9 months of the project's commencement.	IEs and PMU	1.5
Activity 2.1.3: Introduce the cultivation of saline tolerant vegetables within homestead areas	Selection of vulnerable women and provide support for *cultivation of saline tolerant vegetables within homestead areas.*	Baseline=0 Target: 50% vulnerable women beneficiaries Indicator: 50% vulnerable women beneficiaries selected. Timeline: It will be started within 9 months of the project’s commencement	IEs and PMU	0.71
Result 2.2: Community-based farmed crab supply chain created[17]				

[17] Necessary steps will be taken to ensure gender-responsive working conditions at the homestead and community levels.

2.2.1: Development of crab hatcheries (1o stage)	Select hatchery entrepreneurs, form groups, carry out consultation meetings,	Baseline=0 Target: 50% vulnerable women beneficiaries Indicator: 50% selected vulnerable women beneficiaries. Timeline: It will be started within 9 months of the project's commencement	IEs and PMU	0.61
Activity 2.2.2 Financial support for producing crablets	Assess loan demand of the selected entrepreneurs, carry out due diligence, disburse loan etc.	Baseline=0 Target: 50% vulnerable women beneficiaries Indicator: 50% selected vulnerable women beneficiaries. Timeline: It will be started within 9 months of the project's commencement	IEs and PMU	0.20
Activity 2.2.3: Technical and financial support for "crab nursers" (2º stage)	Selection and trained of vulnerable women and for the *crab nursers"*	Baseline=0 Target: 50% vulnerable women beneficiaries Indicator: 50% selected vulnerable women beneficiaries. Timeline: It will be started within 9 months of the project's commencement	IEs and PMU	0.44

Activity 2.2.4: Technical and financial support to "crab farmers" (3º stage).	Selection and trained of vulnerable women and for the *crab farming and marketing*	Baseline=0 Target: 50% vulnerable women beneficiaries Indicator: 50% selected vulnerable women beneficiaries. Timeline: It will be started within 12 months of the project's commencement.	IEs and PMU	4.45
Outcome 3: Improved climate planning and implementation by communities and local level institutions ***Result 3.1: Climate change adaptation groups (CCAG) formed and operationalized***				
Activity 3.1.1: Beneficiary selection and group formation[18]	Selection of beneficiaries and group formation	Baseline=0 Target: 50% women beneficiaries Indicator: 50% selected vulnerable women beneficiaries. Timeline: It will be started within 2 months of the project's commencement.	IEs and PMU	0.12
3.1.2 Prepare Beneficiaries' socio-economic profile	Prepare format, provide training to IEs staff, interview with households	Baseline=0 Target: 50% women beneficiaries Indicator: 50% selected vulnerable women beneficiaries. Timeline: It will be started within 2 months of the project's commencement in parallel with activity 3.1.1.	IEs and PMU	0.12

[18] There will be provision of women trainers and women-exclusive training sessions, including flexible times, and provision of household training for women-headed households as required and There will be integration of men and community elders, where necessary, in community trainings that address women's participation and norms around appropriate work for women, as well as mobility outside the homestead on GRM.

Activity 3.1.3 Arrange monthly group meetings on climate change issues for CCAG	Selection of beneficiaries, group formation and organize monthly meeting	Baseline=0 Target: 80% women beneficiaries Indicator: 80% selected vulnerable women beneficiaries. Timeline: It will be started within 2 months of the project's commencement.	IEs and PMU	0.70
Activity 3.2.1: Prepare training manual on adaptation technologies and crab value chain	Procure consultant, and contract management	Gender disaggregated baseline and indicator is not applicable.	PMU and IE	Gender disaggregated budget is not applicable
Activity 3.2.2: Prepare guidelines on project management	Prepare guidelines, get approval from PKSF, print the approved guidelines and share with selected IEs.	Gender disaggregated baseline and indicator is not applicable.	PMU and IE	Gender disaggregated budget is not applicable
Activity 3.2.3 Organize training for beneficiaries and stakeholders	Selection of beneficiaries, group formation, organizing training[19] and undertaking quarterly community dialogues engaging gender experts and also a gender gap analysis at the inception stage.	Baseline=0 Target: 80% women beneficiaries Indicator: 80% selected vulnerable women beneficiaries. Timeline: It will be started within 3 months of the project's commencement.	IEs and PMU	1.38
Activity 3.2.4 Organize training for IEs' staff	Recruitment of IE staff, and organize training[20].	Baseline=0 Target: 0% women beneficiaries Indicator: 50% selected vulnerable women beneficiaries. Timeline: It will be started within 2 months of the project's commencement.	IEs and PMU	0.06

[19] The topics for the training sessions will be chosen from a wide range of topics, including market prices, managing wild stocks, access to financing, financial management, and substitutes for mangrove fuel wood. Prior to beginning any training of this project, need assessments will be conducted to make them demand-driven.

[20] IE's focal points need to be trained, particularly on gender issues, since particularly IEs are not in a position to procure gender experts.

Activity 3.2.5 Implement workshops and seminars	Selection of beneficiaries, group formation and organize training	Baseline=0 Target: 50% women beneficiaries Indicator: 50% selected vulnerable women beneficiaries. Timeline: It will be started within 6 months of the project's commencement.	IEs and PMU	0.05
Activity 3.2.6 Organize exchange visits for beneficiaries and IE staff	Selection of beneficiaries, group formation and organize training	Baseline=0 Target: 80% women beneficiaries Indicator: 80% selected vulnerable women beneficiaries. Timeline: It will be started within 12 months of the project's commencement	IEs and PMU	0.07
3.2.7 Improve data for crab research and development	Data collection, data analysis, report preparation, data storage etc. related to crab sector	Gender disaggregated baseline and indicator is not applicable.	PMU and IE	Gender disaggregated budget is not applicable
Activity 3.3.1: Prepare and disseminate knowledge products	Data collection, data analysis, report preparation, data storage etc	Gender disaggregated baseline and indicator is not applicable.	PMU and IE	Gender disaggregated budget is not applicable
Activity 3.3.2 Realtime evaluation study of project activities	Procure consultants, contract management, publish evaluation reports	Gender disaggregated baseline and indicator is not applicable.	PMU and IE	Gender disaggregated budget is not applicable
Total				**23.89**

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